UNITED STATES
SECURITIES AND EXCHANGE COMMISSION
Washington, D.C. 20549

FORM 10-K

(Mark One)

☒ **ANNUAL REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**

For the fiscal year ended December 31, 2023
OR

☐ **TRANSITION REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**

For the transition period from _____ to _____

Commission File Number 001-41325

HF SINCLAIR CORPORATION

(Exact name of registrant as specified in its charter)

Delaware	**87-2092143**
(State or other jurisdiction of incorporation or organization)	**(I.R.S. Employer Identification No.)**
2828 N. Harwood, Suite 1300	
Dallas	
Texas	**75201**
(Address of principal executive offices)	**(Zip Code)**

(214) 871-3555
Registrant's telephone number, including area code

Securities registered pursuant to Section 12(b) of the Securities Exchange Act of 1934:

Title of each class	Trading Symbol(s)	Name of each exchange on which registered
Common Stock $0.01 par value	DINO	New York Stock Exchange

Indicate by check mark if the registrant is a well-known seasoned issuer, as defined in Rule 405 of the Securities Act. Yes ☒ No ☐
Indicate by check mark if the registrant is not required to file reports pursuant to Section 13 or Section 15 (d) of the Act. Yes ☐ No ☒

Indicate by check mark whether the registrant (1) has filed all reports required to be filed by Section 13 or 15 (d) of the Securities Exchange Act of 1934 during the preceding 12 months (or for such shorter period that the registrant was required to file such reports), and (2) has been subject to such filing requirements for the past 90 days. Yes ☒ No ☐

Indicate by check mark whether the registrant has submitted electronically every Interactive Data File required to be submitted pursuant to Rule 405 of Regulation S-T (§232.405 of this chapter) during the preceding 12 months (or for such shorter period that the registrant was required to submit such files). Yes ☒ No ☐

Indicate by check mark whether the registrant is a large accelerated filer, an accelerated filer, a non-accelerated filer, a smaller reporting company, or an emerging growth company. See the definitions of "large accelerated filer," "accelerated filer," "smaller reporting company" and "emerging growth company" in Rule 12b-2 of the Exchange Act.

Large accelerated filer	☒	Accelerated filer	☐	Non-accelerated filer	☐	Smaller reporting company	☐
Emerging growth company	☐						

If an emerging growth company, indicate by check mark if the registrant has elected not to use the extended transition period for complying with any new or revised financial accounting standards provided pursuant to Section 13(a) of the Exchange Act. ☐

Indicate by check mark whether the registrant has filed a report on and attestation to its management's assessment of the effectiveness of its internal control over financial reporting under Section 404(b) of the Sarbanes-Oxley Act (15 U.S.C.7262(b)) by the registered public accounting firm that prepared or issued its audit report. ☒

If securities are registered pursuant to Section 12(b) of the Act, indicate by check mark whether the financial statements of the registrant included in the filing reflect the correction of an error to previously issued financial statements. ☐

Indicate by check mark whether any of those error corrections are restatements that required a recovery analysis of incentive-based compensation received by any of the registrant's executive officers during the relevant recovery period pursuant to § 240.10D-1(b). ☐

Indicate by check mark whether the registrant is a shell company (as defined in Rule 12b-2 of the Exchange Act). Yes ☐ No ☒

On June 30, 2023, the last business day of the registrant's most recently completed second fiscal quarter, the aggregate market value of the Common Stock, par value $0.01 per share, held by non-affiliates of the registrant was approximately $7.0 billion, based upon the closing price on the New York Stock Exchange on such date. (This is not deemed an admission that any person whose shares were not included in the computation of the amount set forth in the preceding sentence necessarily is an "affiliate" of the registrant.)

198,568,398 shares of Common Stock, par value $.01 per share, were outstanding on February 15, 2024.

DOCUMENTS INCORPORATED BY REFERENCE

Portions of the registrant's proxy statement for its 2024 annual meeting of stockholders, which proxy statement will be filed with the Securities and Exchange Commission within 120 days after December 31, 2023, are incorporated by reference in Part III.

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PART I

FORWARD-LOOKING STATEMENTS

References herein to HF Sinclair Corporation ("HF Sinclair") include HF Sinclair and its consolidated subsidiaries. In this document, the words "we," "our," "ours" and "us" refer only to HF Sinclair and its consolidated subsidiaries or to HF Sinclair or an individual subsidiary and not to any other person, with certain exceptions. References herein to HF Sinclair, "we," "our," "ours," and "us" with respect to time periods prior to March 14, 2022 refer to HollyFrontier Corporation ("HollyFrontier") and its consolidated subsidiaries and do not include Hippo Holding LLC (now known as Sinclair Holding LLC), the parent company of Sinclair Oil LLC, Sinclair Transportation Company LLC or their respective consolidated subsidiaries (collectively, the "Acquired Sinclair Businesses"). References herein to HF Sinclair "we," "our," "ours," and "us" with respect to time periods from and after March 14, 2022 include the operations of the Acquired Sinclair Businesses. Unless otherwise specified, the financial statements included herein include financial information for HF Sinclair, which for the time period from March 14, 2022 to December 31, 2023 includes the combined business operations of HollyFrontier and the Acquired Sinclair Businesses. References herein to Holly Energy Partners, L.P. ("HEP") with respect to time periods prior to completion of the merger of HEP with a wholly owned subsidiary of HF Sinclair refer to HEP and its consolidated subsidiaries.

This Annual Report on Form 10-K contains certain "forward-looking statements" within the meaning of the federal securities laws. All statements, other than statements of historical fact included in this Annual Report on Form 10-K, including, but not limited to, those under "Business and Properties" in Items 1 and 2, "Risk Factors" in Item 1A, "Legal Proceedings" in Item 3 and "Management's Discussion and Analysis of Financial Condition and Results of Operations" in Item 7, are forward-looking statements. Forward-looking statements use words such as "anticipate," "project," "will," "expect," "plan," "goal," "forecast," "strategy," "intend," "should," "would," "could," "believe," "may," and similar expressions and statements regarding our plans and objectives for future operations. These statements are based on management's beliefs and assumptions using currently available information and expectations as of the date hereof, are not guarantees of future performance and involve certain risks and uncertainties. All statements concerning our expectations for future results of operations are based on forecasts for our existing operations and do not include the potential impact of any future acquisitions. Although we believe that the expectations reflected in these forward-looking statements are reasonable, we cannot assure you that our expectations will prove to be correct. Therefore, actual outcomes and results could materially differ from what is expressed, implied or forecast in these statements. Any differences could be caused by a number of factors including, but not limited to:

- the demand for and supply of feedstocks, crude oil and refined products, including uncertainty regarding the increasing societal expectations that companies address climate change and greenhouse gas emissions;
- risks and uncertainties with respect to the actions of actual or potential competitive suppliers and transporters of refined petroleum products or lubricant and specialty products in our markets;
- the spread between market prices for refined products and market prices for crude oil;
- the possibility of constraints on the transportation of refined products or lubricant and specialty products;
- the possibility of inefficiencies, curtailments or shutdowns in refinery operations or pipelines, whether due to reductions in demand, accidents, unexpected leaks or spills, unscheduled shutdowns, infection in the workforce, weather events, global health events, civil unrest, expropriation of assets, and other economic, diplomatic, legislative, or political events or developments, terrorism, cyberattacks, vandalism or other catastrophes or disruptions affecting our operations, production facilities, machinery, pipelines and other logistics assets, equipment, or information systems, or any of the foregoing of our suppliers, customers, or third-party providers, and any potential asset impairments resulting from, or the failure to have adequate insurance coverage for or receive insurance recoveries from, such actions;
- the effects of current and/or future governmental and environmental regulations and policies, including compliance with existing, new and changing environmental, health and safety laws and regulations, related reporting requirements and pipeline integrity programs;
- the availability and cost of our financing;
- the effectiveness of our capital investments and marketing strategies;
- our efficiency in carrying out and consummating construction projects, including our ability to complete announced capital projects on time and within capital guidance;
- our ability to timely obtain or maintain permits, including those necessary for operations or capital projects;
- our ability to acquire complementary assets or businesses to our existing assets and businesses on acceptable terms and to integrate any existing or future acquired operations and realize the expected synergies of any such transaction on the expected timeline;

- the possibility of vandalism or other disruptive activity, or terrorist or cyberattacks, and the consequences of any such activities or attacks;
- uncertainty regarding the effects and duration of global hostilities, including shipping disruptions in the Red Sea, the Israel-Gaza conflict, the Russia-Ukraine war, and any associated military campaigns which may disrupt crude oil supplies and markets for our refined products and create instability in the financial markets that could restrict our ability to raise capital;
- general economic conditions, including economic slowdowns caused by a local or national recession or other adverse economic condition, such as periods of increased or prolonged inflation;
- limitations on our ability to make future dividend payments or effectuate share repurchases due to market conditions and corporate, tax, regulatory and other considerations; and
- other business, financial, operational and legal risks and uncertainties detailed from time to time in our Securities Exchange Commission filings.

Cautionary statements identifying important factors that could cause actual results to differ materially from our expectations are set forth in this Annual Report on Form 10-K, including, without limitation, the forward-looking statements that are referred to above. You should not put any undue reliance on any forward-looking statements. When considering forward-looking statements, you should keep in mind the risk factors and other cautionary statements set forth in this Annual Report on Form 10-K under “Business and Properties” in Items 1 and 2, “Risk Factors” in Item 1A and in conjunction with the discussion in this Annual Report on Form 10-K in “Management's Discussion and Analysis of Financial Condition and Results of Operations” under the heading “Liquidity and Capital Resources.” All forward-looking statements included in this Annual Report on Form 10-K and all subsequent written or oral forward-looking statements attributable to us or persons acting on our behalf are expressly qualified in their entirety by these cautionary statements. The forward-looking statements speak only as of the date made and, other than as required by law, we undertake no obligation to publicly update or revise any forward-looking statements, whether as a result of new information, future events or otherwise.

DEFINITIONS

Within this report, the following terms have these specific meanings:

"Alkylation" means the reaction of propylene or butylene (olefins) with isobutane to form an iso-paraffinic gasoline (inverse of cracking).

"**BPD**" means the number of barrels per calendar day of crude oil or petroleum products.

"**BPSD**" means the number of barrels per stream day (barrels of capacity in a 24 hour period) of crude oil or petroleum products.

"Base oil" is a lubricant grade oil initially produced from refining crude oil or through chemical synthesis that is used in producing lubricant products such as lubricating greases, motor oil and metal processing fluids.

"Biodiesel" means a clean alternative fuel produced from renewable biological resources.

"**Black wax crude oil**" is a low sulfur, low gravity crude oil produced in the Uintah Basin in Eastern Utah that has certain characteristics that require specific facilities to transport, store and refine into transportation fuels.

"Catalytic reforming" means a refinery process which uses a precious metal (such as platinum) based catalyst to convert low octane naphtha to high octane gasoline blendstock and hydrogen. The hydrogen produced from the reforming process is used to desulfurize other refinery oils and is a primary source of hydrogen for the refinery.

"**Cracking**" means the process of breaking down larger, heavier and more complex hydrocarbon molecules into simpler and lighter molecules.

"**Crude oil distillation**" means the process of distilling vapor from liquid crudes, usually by heating, and condensing the vapor slightly above atmospheric pressure turning it back to liquid in order to purify, fractionate or form the desired products.

"**Ethanol**" means a high octane gasoline blend stock that is used to make various grades of gasoline.

"**FCC**," or fluid catalytic cracking, means a refinery process that breaks down large complex hydrocarbon molecules into smaller more useful ones using a circulating bed of catalyst at relatively high temperatures.

"**Gas oil**" is a group of petroleum distillation products having boiling points between kerosene and lubricating oil and is used as fuel in construction and agricultural machinery.

"**Hydrodesulfurization**" means to remove sulfur and nitrogen compounds from oil or gas in the presence of hydrogen and a catalyst at relatively high temperatures.

"**Hydrogen plant**" means a refinery unit that converts natural gas and steam to high purity hydrogen, which is then used in the hydrodesulfurization, hydrocracking and isomerization processes.

"HF alkylation" or hydrofluoric alkylation, means a refinery process which combines isobutane and C3/C4 olefins using hydrofluoric acid as a catalyst to make high octane gasoline blend stock.

"**Isomerization**" means a refinery process for rearranging the structure of C5/C6 molecules without changing their size or chemical composition and is used to improve the octane of C5/C6 gasoline blendstocks.

"**LPG**" means liquid petroleum gases.

"**Lubricant**" or "**lube**" means a solvent neutral paraffinic product used in commercial heavy duty engine oils, passenger car oils and specialty products for industrial applications such as heat transfer, metalworking, rubber and other general process oil.

"MEK" means a lube process that separates waxy oil from non-waxy oils using methyl ethyl ketone as a solvent.

"MSAT2" means Control of Hazardous Air Pollutants from Mobile Sources, a rule issued by the U.S. Environmental Protection Agency (the "EPA") to reduce hazardous emissions from motor vehicles and motor vehicle fuels.

"**MMBTU**" means one million British thermal units.

"Natural gasoline" means a low octane gasoline blend stock that is purchased and used to blend with other high octane stocks produced to make various grades of gasoline.

"**Refinery gross margin**" means the difference between average net sales price and average cost per barrel sold. This does not include the associated depreciation and amortization costs.

"Reforming" means the process of converting gasoline type molecules into aromatic, higher octane gasoline blend stocks while producing hydrogen in the process.

"**Renewable diesel**" means a diesel fuel derived from renewable feedstock such as vegetable oils or animal fats that is produced through various processes, most commonly through hydrotreating, reacting the feedstock with hydrogen under temperatures and pressure in the presence of a catalyst.

"RINs" means renewable identification numbers and refers to serial numbers assigned to credits generated from renewable fuel production under the EPA's Renewable Fuel Standard ("RFS") regulations, which require blending renewable fuels into the nation's fuel supply. In lieu of blending, refiners may purchase these transferable credits in order to comply with the regulations.

"Roofing flux" is produced from the bottom cut of crude oil and is the base oil used to make roofing shingles for the housing industry.

"ROSE," or **"Solvent deasphalter / residuum oil supercritical extraction,"** means a refinery unit that uses a light hydrocarbon like propane or butane to extract non-asphaltene heavy oils from asphalt or atmospheric reduced crude. These deasphalted oils are then further converted to gasoline and diesel in the FCC process. The remaining asphaltenes are either sold, blended to fuel oil or blended with other asphalt as a hardener.

"Scanfiner" is a refinery unit that removes sulfur from gasoline to produce low sulfur gasoline blendstock.

"**Sour crude oil**" means crude oil containing quantities of sulfur greater than 0.4 percent by weight, while "**sweet crude oil**" means crude oil containing quantities of sulfur equal to or less than 0.4 percent by weight.

"**Vacuum distillation**" means the process of distilling vapor from liquid crudes, usually by heating, and condensing the vapor below atmospheric pressure turning it back to a liquid in order to purify, fractionate or form the desired products.

"White oil" is an extremely pure, highly-refined petroleum product that has a wide variety of applications ranging from pharmaceutical to cosmetic products.

"WTI" means West Texas Intermediate and is a grade of crude oil used as a common benchmark in oil pricing. WTI is a sweet crude oil and has a relatively low density.

Items 1 and 2. Business and Properties

COMPANY OVERVIEW

References herein to HF Sinclair Corporation ("HF Sinclair") include HF Sinclair and its consolidated subsidiaries. In this document, the words "we," "our," "ours" and "us" refer only to HF Sinclair and its consolidated subsidiaries or to HF Sinclair or an individual subsidiary and not to any other person, with certain exceptions. References herein to HF Sinclair "we," "our," "ours" and "us" with respect to time periods prior to March 14, 2022 refer to HollyFrontier Corporation ("HollyFrontier") and its consolidated subsidiaries and do not include Hippo Holding LLC (now known as Sinclair Holding LLC), the parent company of Sinclair Oil LLC, Sinclair Transportation Company LLC or their respective consolidated subsidiaries (collectively, the "Acquired Sinclair Businesses"). References herein to HF Sinclair "we," "our," "ours" and "us" with respect to time periods from and after March 14, 2022 include the operations of the Acquired Sinclair Businesses. Unless otherwise specified, the financial statements included herein include financial information for HF Sinclair, which for the time period from March 14, 2022 to December 31, 2023 includes the combined business operations of HollyFrontier and the Acquired Sinclair Businesses. References herein to Holly Energy Partners, L.P. ("HEP") with respect to time periods prior to the HEP Merger Transaction (as defined below) refer to HEP and its consolidated subsidiaries.

We are an independent energy company that produces and markets high-value light products such as gasoline, diesel fuel, jet fuel, renewable diesel and other specialty products. We were incorporated in Delaware in 1947 and maintain our principal corporate offices at 2828 N. Harwood, Suite 1300, Dallas, Texas 75201-1507. Our telephone number is 214-871-3555, and our internet website address is www.hfsinclair.com. We use our website and social media accounts, including LinkedIn (@HF Sinclair) and Facebook (@HF Sinclair), as a means of disclosing information about us and our services, which information may be deemed material. Except as specifically noted, the information found on our website and social media accounts are not incorporated by reference into, and do not constitute part of, this Annual Report on Form 10-K or any other report filed with or furnished to the Securities and Exchange Commission ("SEC"). A print copy of this Annual Report on Form 10-K will be provided without charge upon written request to the Vice President, Investor Relations at the above address. A direct link to our SEC filings is available on our website under the Investor Relations tab. Also available on our website are copies of our Corporate Governance Guidelines, Audit Committee Charter, Compensation Committee Charter, Nominating, Governance and Social Responsibility Committee Charter, Finance Committee Charter, Environmental, Health, Safety, and Public Policy Committee Charter and Code of Business Conduct and Ethics, all of which will be provided without charge upon written request to the Vice President, Investor Relations at the above address. Our Code of Business Conduct and Ethics applies to all of our officers, employees and directors, including our principal executive officer, principal financial officer and principal accounting officer. Our common stock is traded on the New York Stock Exchange under the trading symbol "DINO."

HEP Merger Transaction

On December 1, 2023, pursuant to that certain Agreement and Plan of Merger, dated as of August 15, 2023 (the "Merger Agreement"), by and among HEP, HF Sinclair, Navajo Pipeline Co., L.P., a Delaware limited partnership and an indirect wholly owned subsidiary of HF Sinclair ("HoldCo"), Holly Apple Holdings LLC, a Delaware limited liability company and a wholly owned subsidiary of HoldCo ("Merger Sub"), HEP Logistics Holdings, L.P., a Delaware limited partnership and the general partner of HEP ("HLH"), and Holly Logistic Services, L.L.C., a Delaware limited liability company and the general partner of HLH, Merger Sub merged with and into HEP, with HEP surviving as an indirect, wholly owned subsidiary of HF Sinclair (the "HEP Merger Transaction").

Under the terms of the Merger Agreement, each outstanding common unit representing a limited partner interest in HEP (an "HEP common unit"), other than the HEP common units already owned by HF Sinclair and its subsidiaries, was converted into the right to receive 0.315 shares of HF Sinclair common stock and $4.00 in cash, without interest. The Merger Agreement consideration totaled $267.6 million in cash and resulted in the issuance of 21,072,326 shares of HF Sinclair common stock from treasury stock.

Sinclair Acquisition

On March 14, 2022 (the "Closing Date"), HollyFrontier and HEP announced the establishment of HF Sinclair as the new parent holding company of HollyFrontier and HEP and their subsidiaries, and the completion of their respective acquisitions of Sinclair Oil Corporation (now known as Sinclair Oil LLC, "Sinclair Oil") and Sinclair Transportation Company LLC ("STC") from The Sinclair Companies (now known as REH Company and referred to herein as "REH Company"). On the Closing Date, HF Sinclair completed its previously announced acquisition of Sinclair Oil by effecting (a) a holding company merger with HollyFrontier surviving such merger as a direct wholly owned subsidiary of HF Sinclair (the "HFC Merger"), and (b) immediately following the HFC Merger, a contribution whereby REH Company contributed all of the equity interests of Hippo Holding LLC (now known as Sinclair Holding LLC), the parent company of Sinclair Oil (the "Target Company") to HF Sinclair in exchange for shares of HF Sinclair, resulting in the Target Company becoming a direct wholly owned subsidiary of HF Sinclair (the "HFC Transactions"). At the effective time of the HFC Merger, all of HollyFrontier's outstanding shares were automatically converted into equivalent corresponding shares of HF Sinclair, and HF Sinclair became the successor issuer to HollyFrontier pursuant to Rule 12g-3(a) under the Securities and Exchange Act of 1934, as amended (the "Exchange Act"), and replaced HollyFrontier as the public company trading on the New York Stock Exchange ("NYSE") under the symbol "DINO."

In connection with the closing of the HFC Transactions, HF Sinclair issued 60,230,036 shares of HF Sinclair common stock, par value $0.01 per share, to REH Company, representing 27% of the pro forma equity of HF Sinclair with a value of approximately $2,149 million based on HollyFrontier's fully diluted shares of common stock outstanding and closing stock price on March 11, 2022. Pursuant to that certain Business Combination Agreement, dated as of August 2, 2021 (as amended on March 14, 2022, the "Business Combination Agreement"), REH Company made a $77.5 million cash payment to HF Sinclair, inclusive of final working capital adjustments, which reduced the aggregate transaction value to approximately $2,072 million. Of the 60,230,036 shares of HF Sinclair common stock, 2,570,000 shares were held in escrow to secure REH Company's RINs credit obligations under Section 6.22 of the Business Combination Agreement. As of December 31, 2023, REH Company had satisfied their RINs credit obligations to HF Sinclair and the corresponding shares were released from escrow in January 2024. Additionally, on the Closing Date, and immediately prior to the consummation of the HFC Transactions, HEP completed its acquisition of STC, REH Company's integrated crude and refined products midstream business, and issued 21,000,000 HEP common units with a value of approximately $349.0 million based on HEP's fully diluted common units outstanding and HEP's closing unit price on March 11, 2022, and paid cash consideration of $329.0 million, inclusive of final working capital adjustments, to REH Company in exchange for all the outstanding equity interests of STC (the "HEP Transaction" and together with the HFC Transactions, the "Sinclair Transactions"). Of these 21,000,000 HEP common units, 5,290,000 units were held in escrow and were released to REH Company in April 2023 upon their satisfaction of the corresponding RINs credit obligations to HF Sinclair under Section 6.22 of the Business Combination Agreement.

HF Sinclair acquired REH Company's refining, branded marketing, renewables, and midstream businesses. At the time of closing, the Sinclair branded marketing business supplied high-quality fuels to more than 1,300 Sinclair branded stations and licensed the use of the Sinclair brand at more than 300 additional locations throughout the United States. The renewables business includes the operation of a renewable diesel unit located in Sinclair, Wyoming. The refining business includes two Rocky Mountains-based refineries located in Casper, Wyoming and Sinclair, Wyoming. Under the terms of that certain Contribution Agreement (as amended on March 14, 2022, the "Contribution Agreement"), HEP acquired STC, REH Company's integrated crude and refined products pipelines and terminal assets, including approximately 1,200 miles of integrated crude and refined product pipeline supporting the Sinclair refineries and third parties, eight product terminals and two crude terminals with approximately 4.5 million barrels of operated storage. In addition, HEP acquired STC's interests in three pipeline joint ventures for crude gathering and product offtake including: Saddle Butte Pipeline III, LLC (at the time of closing, 25.06%, and currently, a 25.12% non-operated interest); Pioneer Investments Corp. (49.995% non-operated interest); and UNEV Pipeline, LLC ("UNEV") (the 25% non-operated interest not already owned by HEP, resulting in UNEV becoming a wholly owned subsidiary of HEP). The addition of Sinclair Oil and STC to the HollyFrontier business created a combined company with increased scale and ability to diversify and is expected to drive growth through the expanded refining and renewables business. In addition, the HFC Transactions added an integrated branded wholesale distribution network to our business.

Puget Sound Refinery Acquisition
On May 4, 2021, HollyFrontier Puget Sound Refining LLC (now known as HF Sinclair Puget Sound Refining LLC), a wholly owned subsidiary of HollyFrontier, entered into a sale and purchase agreement with Equilon Enterprises LLC d/b/a Shell Oil Products US ("Shell") to acquire Shell's Puget Sound refinery and related assets (the "Puget Sound Refinery"). The acquisition closed on November 1, 2021 for aggregate cash consideration of $624.3 million. The Puget Sound Refinery is strategically located on approximately 850 acres in Anacortes, Washington. The 149,000 BPD facility is a large, high quality and complex refinery with catalytic cracking and delayed coking units and is well positioned geographically and logistically to source advantaged Canadian and Alaskan North Slope crudes. In addition to refining assets and an on-site cogeneration facility, the transaction included a deep-water marine dock, a light product loading rack, a rail terminal and storage tanks with approximately 5.8 million barrels of crude, product and other hydrocarbon storage capacity, including the on-site cogeneration facility and related logistics assets.

Sale of Real Property
On April 27, 2021, our wholly owned subsidiary, 7037619 Canada Inc., entered into a contract for sale of real property in Mississauga, Ontario for base consideration of $98.8 million, or CAD 125 million. The transaction closed on September 15, 2021.

Renewable Diesel Construction
In November 2019, we announced our plans to construct a new renewable diesel unit ("RDU") at our Artesia facility. The Artesia RDU was completed and operational in the second quarter of 2022. The Artesia RDU has a production capacity of approximately 135 million gallons a year and allows us to process soybean oil and other renewable feedstocks into renewable diesel.

In the third quarter of 2020, we permanently ceased petroleum refining operations at our facility in Cheyenne, Wyoming (the "Cheyenne Refinery") and subsequently began converting certain assets at the Cheyenne Refinery to renewable diesel production. The Cheyenne RDU was mechanically complete in the fourth quarter of 2021 and operational in the first quarter of 2022. The Cheyenne RDU has a production capacity of approximately 90 million gallons a year.

Additionally, we constructed a pre-treatment unit ("PTU") at our Artesia facility that provides feedstock flexibility for both our Artesia and Cheyenne RDUs. This PTU was completed and operational in the first quarter of 2022.

As of December 31, 2023, we:

- owned and operated a refinery in El Dorado, Kansas (the "El Dorado Refinery"), two refinery facilities located in Tulsa, Oklahoma (collectively, the "Tulsa Refineries"), the Puget Sound Refinery in Anacortes, Washington, a refinery in Artesia, New Mexico that is operated in conjunction with crude oil distillation and vacuum distillation and other facilities situated 65 miles away in Lovington, New Mexico (collectively, the "Navajo Refinery"), a refinery in West Bountiful, Utah (the "Woods Cross Refinery"), a refinery in Sinclair, Wyoming (the "Parco Refinery,") and a refinery in Casper, Wyoming (the "Casper Refinery");
- owned and operated RDUs in Artesia, New Mexico (the "Artesia RDU"), in Cheyenne, Wyoming (the "Cheyenne RDU") and in Sinclair, Wyoming (the "Sinclair RDU") and a PTU in Artesia, New Mexico (the "Artesia PTU");
- owned and operated a manufacturing facility in Mississauga, Ontario, which produces base oils and other specialized lubricant products for our Petro-Canada Lubricants business;
- owned and operated manufacturing facilities in Petrolia, Pennsylvania and the Netherlands, which produce specialty lubricant products for our Sonneborn business, such as white oils, petrolatums and waxes;
- owned and operated Red Giant Oil Company LLC ("Red Giant Oil"), which supplies locomotive engine oil and has storage and distribution facilities in Iowa and Wyoming, along with a blending and packaging facility in Texas;
- owned and operated HF Sinclair Asphalt Company LLC ("Asphalt"), which operates various asphalt terminals in Arizona, New Mexico and Oklahoma; and
- owned and operated logistics and refinery assets consisting of petroleum product and crude oil pipelines, terminals, tankage, loading rack facilities and refinery processing units that principally support our refining operations in the Mid-Continent, Southwest and Rocky Mountains geographic regions of the United States.

As of December 31, 2023, our operations were organized into five reportable segments, Refining, Renewables, Marketing, Lubricants & Specialties and Midstream. The Refining segment includes the operations of our El Dorado, Tulsa, Puget Sound, Navajo, Woods Cross, Parco and Casper Refineries and Asphalt. The Renewables segment includes the operations of the Artesia, Cheyenne and Sinclair RDUs and the Artesia PTU. The Marketing segment includes branded fuel sales. The Lubricants & Specialties segment, which was previously referred to as our Lubricant and Specialty Products segment, includes the operations of our Petro-Canada Lubricants, Red Giant Oil and Sonneborn businesses in addition to specialty lubricant products produced at our Tulsa West refinery. As of December 31, 2023, the Midstream segment, which was previously referred to as our HEP segment, includes petroleum product and crude pipelines, terminal, tankage and loading rack facilities, and refinery processing units that primarily support our refining operations. See Note 20 “Segment Information” in the Notes to Consolidated Financial Statements for additional information on our reportable segments.

REFINERY OPERATIONS

Our refinery operations serve the Mid-Continent, Southwest and Rocky Mountains extending into the Pacific Northwest geographic regions of the United States. We own and operate seven complex refineries having a combined crude oil processing capacity of 678,000 BPSD. Each of our refineries has the complexity to convert discounted, heavy or sour crude oils into a high percentage of gasoline, diesel and other high-value refined products.

Products and Customers

Light products are shipped to customers via product pipelines or are available for loading at our refinery truck facilities and terminals. Light products are also made available to customers at various other locations via exchange with other parties.

Our principal customers for gasoline include other refiners, branded sites, convenience store chains, independent marketers and retailers. Diesel fuel is sold to other refiners, branded sites, truck stop chains, wholesalers and railroads. Jet fuel is sold for commercial airline use. Base oils are intercompany sales to our Lubricants & Specialties segment. LPG's are sold to LPG wholesalers and LPG retailers. We produce and purchase asphalt products that are sold to governmental entities, paving contractors or manufacturers. Asphalt is also blended into fuel oil and is either sold locally or is shipped to the Gulf Coast. For the years ended December 31, 2023, 2022 and 2021, we had one customer, Shell, together with certain of its affiliates, that accounted for 10% or more of our total annual revenues at approximately 12%, 15% and 13%, respectively. See Note 5 “Revenues” in the Notes to Consolidated Financial Statements for additional information on our customers revenues.

Mid-Continent Region

Facilities
The El Dorado Refinery is a high-complexity coking refinery with a 135,000 BPSD processing capacity and the ability to process significant volumes of heavy and sour crudes. The integrated refining processes at the Tulsa West and East refinery facilities provide us with a highly complex refining operation having a combined crude processing rate of approximately 125,000 BPSD.

The El Dorado Refinery is located on 1,100 acres south of El Dorado, Kansas and is a fully integrated refinery. The principal processing units at the El Dorado Refinery consist of crude and vacuum distillation; hydrodesulfurization of naphtha, kerosene, diesel, and gas oil streams; isomerization; catalytic reforming; aromatics recovery; catalytic cracking; alkylation; delayed coking; hydrogen production; and sulfur recovery.

The Tulsa West facility is located on a 750-acre site in Tulsa, Oklahoma. The principal processing units at the Tulsa West facility consist of crude and vacuum distillation (with light ends recovery), naphtha hydrodesulfurization, propane de-asphalting, lubes extraction, MEK dewaxing, delayed coker and butane splitter units.

The Tulsa East facility is located on a 466-acre site also in Tulsa, Oklahoma. The principal processing units at the Tulsa East facility consist of crude and vacuum distillation, naphtha hydrodesulfurization, FCC, isomerization, catalytic reforming, alkylation, scanfiner, diesel hydrodesulfurization and sulfur units.

Crude Oil and Feedstock Supplies
Both of our Mid-Continent refineries are connected via pipeline to Cushing, Oklahoma, a significant crude oil pipeline trading and storage hub. The El Dorado Refinery and the Tulsa Refineries are located approximately 125 miles and 50 miles, respectively, from Cushing, Oklahoma. Local pipelines provide direct access to regional Oklahoma crude production as well as access to United States onshore and Canadian crudes. The proximity of the refineries to the Cushing pipeline and storage hub provides the flexibility to optimize their crude slate with a wide variety of crude oil supply options. Additionally, we have transportation service agreements to transport Canadian crude oil on the Spearhead and Keystone Pipelines, enabling us to transport Canadian crude oil to Cushing for subsequent shipment to either of our Mid-Continent refineries.

We also purchase isobutane, natural gasoline, butane and other feedstocks for processing at our Mid-Continent refineries. The El Dorado Refinery is connected to Conway, Kansas, a major gas liquids trading and storage hub, via the Oneok Pipeline. From time to time, other feedstocks such gas oil, naphtha and light cycle oil are purchased from other refiners for use at our refineries.

Markets and Competition
The El Dorado Refinery primarily serves Colorado and the Plains states, which include the Kansas City metropolitan area. The gasoline, diesel and jet fuel produced by the El Dorado Refinery are primarily shipped via pipeline to terminals for distribution by truck or rail. We ship product via the NuStar Pipeline Operating Partnership L.P. pipeline to the northern Plains states, via the Magellan Pipeline Company, L.P. (“Magellan”) mountain pipeline to Denver, Colorado, and on the Magellan mid-continent pipeline to the Plains states. Additionally, our midstream operations’ on-site truck and rail racks facilitate access to local refined product markets.

The El Dorado Refinery faces competition from other Plains states and Mid-Continent refiners, but the principal competitors for the El Dorado Refinery are Gulf Coast refiners. The Gulf Coast refiners typically have lower production costs due to greater economies of scale; however, they incur higher refined product transportation costs, which allows the El Dorado Refinery to compete effectively in the Plains states and Rocky Mountains region.

The Tulsa Refineries serve the Mid-Continent geographic region of the United States. Distillates and gasolines are primarily delivered from the Tulsa Refineries to market via pipelines owned and operated by Magellan. These pipelines connect the refinery to distribution channels throughout Colorado, Oklahoma, Kansas, Missouri, Illinois, Iowa, Minnesota, Nebraska and Arkansas. Additionally, our midstream operations' on-site truck and rail racks facilitate geographic refined product distribution within the Mid-Continent region.

The Tulsa Refineries’ principal customers for conventional gasoline include other refiners, branded sites, convenience store chains, independent marketers and retailers. Truck stop operators and railroads are the primary diesel customers. Jet fuel is sold primarily for commercial use. The refineries’ asphalt and roofing flux products are sold via truck or railcar directly from the refineries or to customers throughout the Mid-Continent geographic region primarily to paving contractors and manufacturers of roofing products.

West Region

Facilities
The Navajo Refinery has a crude oil processing capacity of 100,000 BPSD and has the ability to process sour crude oils into high-value light products such as gasoline, diesel fuel and jet fuel. The Woods Cross Refinery has a crude oil processing capacity of 45,000 BPSD and processes regional sweet and black wax crude oil into high-value light products. On November 1, 2021, we acquired the Puget Sound Refinery, which is a complex refinery with a 149,000 BPSD processing capacity and has the ability to process a variety of light, medium, heavy sweet and sour crudes. On March 14, 2022, we acquired the Parco Refinery, which has a crude oil processing capacity of 94,000 BPSD and has the ability to process heavy and sweet crudes, and the Casper Refinery, which has a crude oil processing capacity of 30,000 BPSD and processes regional sweet crude into high-value light products.

The Navajo Refinery's Artesia, New Mexico facility is located on a 561-acre site and is a fully integrated refinery with crude oil distillation, vacuum distillation, FCC, ROSE (solvent deasphalter), HF alkylation, catalytic reforming, hydrodesulfurization, mild hydrocracking, isomerization, sulfur recovery and product blending units.

The Artesia facility is operated in conjunction with a refining facility located in Lovington, New Mexico, approximately 65 miles east of Artesia. The principal equipment at the Lovington facility consists of a crude oil distillation unit and associated vacuum distillation units. The Lovington facility processes crude oil into intermediate products that are transported to Artesia by three of our midstream operations' intermediate pipelines. These products are then upgraded into finished products at the Artesia facility. The combined crude oil capacity of the Navajo Refinery facilities is 100,000 BPSD and it typically processes or blends an additional 10,000 BPSD of natural gasoline, butane, gas oil and naphtha.

The Woods Cross Refinery facility is located on a 200-acre site in West Bountiful, Utah and is a fully integrated refinery with crude oil distillation, solvent deasphalter, FCC, HF alkylation, catalytic reforming, hydrodesulfurization, isomerization, sulfur recovery and product blending units. The facility typically processes or blends an additional 2,000 BPSD of natural gasoline, butane and gas oil over its 45,000 BPSD capacity.

The Puget Sound Refinery facility is located on approximately 850 acres in Anacortes, Washington and is a fully integrated refinery. The principal processing units at the Puget Sound Refinery consist of crude and vacuum distillation, FCC, delayed coking, sulfuric alkylation, catalytic reforming, hydrodesulfurization, isomerization, sulfur recovery, cogeneration and product blending. In addition to refining assets and an on-site cogeneration facility, the Puget Sound Refinery also includes a deep-water marine dock, a light product loading rack, a rail terminal and storage tanks with approximately 5.8 million barrels of crude, product and other hydrocarbon storage capacity.

The Parco Refinery facility is located on approximately 420 acres in Sinclair, Wyoming and is a fully integrated refinery. The principal processing units at the Parco Refinery consist of gas oil hydrocracking, gas oil hydrotreating, delayed coking units and associated hydrogen generation.

The Casper Refinery facility is located on approximately 250 acres in Casper, Wyoming and is a fully integrated refinery. The principal processing units at the Casper Refinery consist of crude oil distillation, FCC, catalytic reforming, hydrodesulfurization, sulfur recovery and product blending. In addition to refining assets, the Casper Refinery also includes a light product loading rack, a heavy oil rail terminal and crude and product storage tanks.

Crude Oil and Feedstock Supplies

The Navajo Refinery is situated near the Permian Basin, an area that has historically, and continues to have, abundant supplies of crude oil available both for regional users and for export to other areas. We purchase crude oil from independent producers in southeastern New Mexico and west Texas as well as from major oil companies. The crude oil is gathered through our midstream operations' pipelines and through third-party tank trucks and crude oil pipeline systems for delivery to the Navajo Refinery.

We also purchase volumes of isobutane, natural gasoline and other feedstocks to supply the Navajo Refinery from sources in Texas and the Mid-Continent area that are delivered to this region on a common carrier pipeline owned by Enterprise Products, L.P. Ultimately all volumes of these products are shipped to the Artesia refining facilities on our midstream operations' intermediate pipelines running from Lovington to Artesia. From time to time, we purchase gas oil, naphtha and light cycle oil from other refiners for use as feedstock.

The Woods Cross Refinery currently obtains crude oil from suppliers in Canada, Wyoming and Utah as delivered via common carrier pipelines, including the SLC Pipeline and Frontier Pipeline. Supplies of black wax crude oil are shipped via truck.

The Puget Sound Refinery is well positioned geographically and logistically to source advantaged Canadian and Alaskan North Slope crudes. The Canadian crudes are sourced from Edmonton, Alberta and are supplied directly to the Puget Sound Refinery by the Trans Mountain pipeline system. The Alaskan North Slope crudes are supplied by oil tankers that load the crude from Valdez, Alaska and offload at the Puget Sound Refinery's marine dock. The dock also allows the refinery to receive other crude oil via marine transport.

The Parco Refinery and Casper Refinery purchase much of their crude oil from inland domestic sources, primarily in areas of Wyoming, North Dakota and Colorado, as well as crude oil delivered via pipeline from other regions, including Canada. A long-term agreement with the Express Pipeline allows delivery of up to 40,000 BPD of crude oil from Canada directly to these refineries and affords the optionality between Western Canadian Select and Syncrude.

Markets and Competition

The Navajo Refinery primarily serves the southwestern United States, including the metropolitan areas of El Paso, Texas; Albuquerque, Moriarty and Bloomfield, New Mexico; Phoenix and Tucson, Arizona; and portions of northern Mexico. Our products are shipped through our midstream operations' pipelines from Artesia, New Mexico to El Paso, Texas and from El Paso to Albuquerque and to Mexico via products pipeline systems owned by Magellan and from El Paso to Tucson and Phoenix via a products pipeline system owned by SFPP, L.P. ("SFPP"). In addition, petroleum products from the Navajo Refinery are transported to northwest New Mexico, to Moriarty, New Mexico, near Albuquerque, via our pipelines running from Artesia to San Juan County, New Mexico, and to Bloomfield, New Mexico. We have refined product storage through our midstream operations at terminals in Artesia and Moriarty, New Mexico.

The Woods Cross Refinery primarily serves Utah, which is currently supplied by a number of local refiners and the Pioneer Pipeline. It also supplies a small percentage of the refined products consumed in Idaho, Wyoming, eastern Washington and Nevada. Our Woods Cross Refinery ships refined products over a common carrier pipeline system owned by Tesoro Logistics Northwest Pipelines LLC to numerous terminals, including our terminal at Spokane, Washington and third-party terminals at Pocatello and Boise, Idaho and Pasco, Washington as well as to Cedar City, Utah and Las Vegas, Nevada via the UNEV Pipeline.

The Puget Sound Refinery primarily serves the Pacific Northwest, including Washington, Oregon and British Columbia. It supplies jet fuel for the Seattle-Tacoma, Washington, Portland, Oregon and Vancouver, British Columbia airports. Products are shipped to Seattle, Tacoma, and Portland terminals by the common carrier Olympic Pipeline. Additionally, products are loaded across the Puget Sound Refinery's marine dock to deliver to the same locations in the Pacific Northwest and to expanded locations in California and Alaska. The Puget Sound Refinery can also load products for export sales across its marine dock.

The Parco Refinery's products are primarily distributed by major direct pipelines to Denver, Colorado and to Salt Lake City, Utah connecting to pipelines owned by us, our joint ventures or third parties, serving an extensive network of terminals or by exchange. The refinery's customers are located in Colorado, Utah, Arizona, Idaho, Nebraska, Nevada, Oregon, South Dakota, Washington and Wyoming.

The Casper Refinery's products are primarily distributed by major direct pipelines serving the Rocky Mountain region and western South Dakota through an extensive network of terminals or by exchange. The refinery's customers are located in Colorado, Utah, South Dakota and Wyoming.

HF Sinclair Asphalt Company

We manufacture commodity and modified asphalt products at our manufacturing facilities located in Glendale, Arizona; Albuquerque, New Mexico; Artesia, New Mexico and Catoosa, Oklahoma. Our Albuquerque and Artesia facilities manufacture commodity and modified hot asphalt products as well as commodity and modified asphalt emulsions from base asphalt materials provided by our refineries and third-party suppliers. Our Glendale facility manufactures commodity, modified and specialty modified hot asphalt products from base asphalt materials provided by our refineries and third-party suppliers. Our Catoosa facility manufactures commodity, modified and specialty modified hot asphalt products and commodity asphalt products from base asphalts supplied by our refineries. We primarily sell our finished asphalt products in Arizona, California, Colorado, New Mexico, Oklahoma, Kansas, Missouri, Texas, Arkansas and northern Mexico. Our products are shipped via third-party trucking companies to commercial customers that provide asphalt based materials for private, commercial and government agency projects.

RENEWABLES OPERATIONS

Our renewables business includes the operations ofthe Cheyenne RDU, the Artesia RDU and the Sinclair RDU, which we acquired from REH Company in the Sinclair Transactions. Our Cheyenne RDU was mechanically complete in the fourth quarter of 2021 and operational in the first quarter of 2022 and has a production capacity of approximately 90 million gallons a year. Our Artesia RDU, co-located with the Navajo Refinery, was completed and operational in the second quarter of 2022. The Artesia RDU facility also includes rail infrastructure and storage tanks and has a production capacity of approximately 135 million gallons a year. Our Artesia PTU, co-located with the Navajo Refinery, was completed and operational in the first quarter of 2022. The Artesia PTU provides feedstock flexibility for both our Cheyenne RDU and Artesia RDU. On March 14, 2022, we acquired the Sinclair RDU, co-located with the Parco Refinery. The Sinclair RDU has been operational since 2018 and produces approximately 153 million gallons a year. The Artesia RDU and Sinclair RDU are dependent upon and share certain infrastructure, including a hydrogen plant, with the refineries where they are co-located, respectively.

Our RDUs allow us to process soybean oil and other renewable feedstocks into renewable diesel. Renewable diesel is a cleaner burning fuel with 50% to 80% (results dependent on the feedstock) lower lifecycle greenhouse gas ("GHG") emissions than conventional diesel. The renewable diesel produced by the RDUs is sold to customers in California and Canada where Low Carbon Fuel Standard ("LCFS") credit value can be realized.

MARKETING OPERATIONS

Our marketing operations include branded fuel sales to more than 1,500 branded sites in the United States and licensing fees for the use of the Sinclair brand at more than 300 additional locations throughout the United States. Our marketing operations also include revenues from branded gasoline, diesel and other marketing activities. Our branded sites are located in several states across the United States with the highest concentration of the sites located in our West and Mid-Continent regions.

LUBRICANTS & SPECIALTIES OPERATIONS

Our lubricants and specialties operations consist of our Petro-Canada Lubricants, Red Giant Oil, Sonneborn and the Tulsa lubricants businesses.

Our Petro-Canada Lubricants business produces automotive, industrial and food grade lubricants and greases, base and process oils and specialty fluids. It is one of the largest manufacturers of high margin Group III base oils in North America. Products are marketed in over 80 countries worldwide to a diverse customer base through a global sales force and distributor network.

Our Red Giant Oil business provides high quality lubricants to the railroad industry.

Our Sonneborn business produces specialty products such as white oils, petrolatums and waxes for the personal care, cosmetic, pharmaceutical and food processing industries. Combined with Petro-Canada Lubricants, it is one of the world's largest producers of pharmaceutical white oils.

Our Tulsa Refinery produces high quality base oils, process oils, waxes, horticultural oils and asphalt performance products. Products are marketed worldwide through strategically located terminals in the United States and selected distributors internationally.

Petro-Canada Lubricants Inc. ("PCLI") owns and operates a production facility located in Mississauga, Ontario having lubricant production capacity of 15,600 BPD and has the flexibility to match unique lubricant product formulations. The primary operating units are high-pressure hydrotreating and hydrofinishing, solvent dewaxing and catalytic dewaxing. In addition, the facility operates a hydrogen plant, naphtha hydrotreater and catalytic reformer, along with other utility units to support production. The Mississauga plant also includes packaging facilities and has extensive distribution capabilities with marine, truck and rail access.

Red Giant Oil, headquartered in Council Bluffs, Iowa, owns and operates blending and distribution facilities in Council Bluffs, Iowa, Joshua, Texas and Newcastle, Wyoming.

Sonneborn has manufacturing facilities in Petrolia, Pennsylvania and the Netherlands. The Sonneborn Petrolia site has a production capacity of 6,000 BPD with flexibility to produce a full range of finished specialty products. The primary operating unit is a high-pressure hydrotreater with hydrofinishing. In addition, the facility operates a hydrogen plant along with other utility units to support production. The Petrolia plant also includes packaging facilities with distribution capabilities through rail and trucking. The Sonneborn Netherlands site is located in Amsterdam and has a production capacity of approximately 1,500 BPD. The primary operating units include base oil acid treating, percolation filtration, and bleaching & steaming operations. The Netherlands sites include packaging facilities with distribution capabilities through truck and marine.

MIDSTREAM OPERATIONS

As of December 31, 2023, our midstream operations were comprised of logistic andrefinery assets consisting of petroleum product and crude oil pipelines, terminals, tankage, loading rack facilities and refinery processing units that principally support our refining operations, as well as other third-party refineries, in the Mid-Continent, Southwest and Rocky Mountains geographic regions of the United States. Additionally, we owned a 50% ownership interest in each of Osage Pipe Line Company, LLC, the owner of a pipeline running from Cushing, Oklahoma to El Dorado, Kansas (the "Osage Pipeline"), Cheyenne Pipeline, LLC, the owner of a pipeline running from Fort Laramie, Wyoming to Cheyenne, Wyoming (the "Cheyenne Pipeline") and Cushing Connect Pipeline & Terminal LLC ("Cushing Connect"), the owner of a crude oil storage terminal in Cushing, Oklahoma and a pipeline that runs from Cushing, Oklahoma to our Tulsa Refineries; a 25.12% ownership interest in Saddle Butte Pipeline III, LLC, the owner of a pipeline from the Powder River Basin to Casper, Wyoming (the "Saddle Butte Pipeline"); and a 49.995% ownership interest in Pioneer Investments Corp., the owner of a pipeline from Sinclair, Wyoming to the North Salt Lake City, Utah Terminal (the "Pioneer Pipeline").

As of December 31, 2023, our midstream operations generate revenues by charging tariffs for transporting petroleum products and crude oil through its pipelines, by charging fees for terminalling refined products and other hydrocarbons, by storing and providing other services at its storage tanks and terminals and charging a tolling fee per barrel or thousand standard cubic feet of feedstock throughput in its refinery processing units. Our midstream operations do not take ownership of products that it transports, terminals, stores or refines; therefore, it is not directly exposed to changes in commodity prices.

Investment in Joint Venture

Cushing Connect Joint Venture
In October 2019, HEP Cushing LLC, then a wholly owned subsidiary of HEP and now a wholly owned subsidiary of HF Sinclair, and Plains Marketing, L.P., a wholly owned subsidiary of Plains All American Pipeline, L.P. ("Plains"), formed a 50/50 joint venture, Cushing Connect, for (i) the development, construction, ownership and operation of a new 160,000 barrel per day common carrier crude oil pipeline (the "Cushing Connect Pipeline") that connects the Cushing, Oklahoma crude oil hub to our Tulsa Refineries and (ii) the ownership and operation of 1.5 million barrels of crude oil storage in Cushing, Oklahoma (the "Cushing Connect Terminal"). The Cushing Connect Terminal was fully in service beginning in April 2020, and the Cushing Connect Pipeline was placed in service at the end of the third quarter of 2021. Long-term commercial agreements have been entered into to support the Cushing Connect assets.

Cushing Connect entered into a contract with an affiliate of HEP, now a subsidiary of HF Sinclair, to manage the operation of the Cushing Connect Pipeline and with an affiliate of Plains to manage the operation of the Cushing Connect Terminal. The total investment in Cushing Connect was generally shared proportionately among the partners.

As of December 31, 2023, our midstream assets included:

Pipelines
- approximately 660 miles of refined product pipelines, including 340 miles of leased pipelines, that transport gasoline, diesel and jet fuel principally from our Navajo Refinery in New Mexico to our customers in the metropolitan and rural areas of Texas, New Mexico, Arizona, Colorado, Utah and northern Mexico;
- one 205-mile pipeline that transports refined product from our Parco Refinery in Wyoming to our customers in Colorado;
- one 114-mile bi-directional products pipeline that transports finished and intermediate products between our Parco and Casper Refineries in Wyoming;
- one 220-mile refined products pipeline that transports products between Olathe, Kansas and Montrose, Iowa;
- approximately 510 miles of refined product pipelines that transport refined products from Delek US Holding, Inc.'s ("Delek") Big Spring refinery in Texas to its customers in Texas and Oklahoma;

- two 65-mile pipelines that transport intermediate feedstocks and crude oil from our Navajo Refinery crude oil distillation and vacuum facilities in Lovington, New Mexico to our petroleum refinery facilities in Artesia, New Mexico;
- one 65-mile intermediate pipeline that is used for the shipment of crude oil from the gathering systems in Barnsdall and Beeson, New Mexico to our Navajo Refinery;
- the SLC Pipeline, a 95-mile intrastate crude oil pipeline system that transports crude oil into the Salt Lake City, Utah area from the Utah terminus of the Frontier Pipeline, as well as crude oil flowing from Wyoming and Utah via the Marathon Wamsutter system;
- the Frontier Pipeline, a 289-mile crude oil pipeline running from Casper, Wyoming to Frontier Station, Utah through a connection to the SLC Pipeline;
- approximately 990 miles of crude oil trunk, gathering and connection pipelines located in west Texas, New Mexico and Oklahoma that primarily deliver crude oil to our Navajo Refinery;
- approximately 535 miles of crude oil trunk, gathering and connection pipelines in Wyoming that primarily deliver crude oil to our Parco and Casper Refineries;
- approximately 10 miles of refined product pipelines that support our Woods Cross Refinery located near Salt Lake City, Utah;
- gasoline and diesel connecting pipelines that support our Tulsa East facility;
- five intermediate product and gas pipelines between our Tulsa East and Tulsa West facilities;
- crude receiving assets located at our Cheyenne facility;
- the UNEV Pipeline, which is a 427-mile, 12-inch refined products pipeline running from Woods Cross, Utah to Las Vegas, Nevada and Cedar City, Utah;
- a 50% interest in the Osage Pipeline, a 135-mile pipeline that transports crude oil from Cushing, Oklahoma to our El Dorado Refinery and also has a connection to the Jayhawk pipeline that services the CHS refinery in McPherson, Kansas;
- a 50% interest in the Cheyenne Pipeline, an 87-mile crude oil pipeline running from Fort Laramie, Wyoming to Cheyenne, Wyoming;
- a 50% interest in Cushing Connect Pipeline, a 50-mile crude oil pipeline running from Cushing, Oklahoma to our Tulsa Refineries;
- a 49.995% interest in the Pioneer Pipeline, a 312-mile refined product pipeline running from Sinclair Station in Wyoming to the terminal in North Salt Lake City, Utah. Through connections, this pipeline is also able to deliver refined products to the UNEV refined products pipeline, and
- a 25.12% interest in the Saddle Butte Pipeline, a crude oil pipeline that collects crude oil from the Powder River Basin in Wyoming and primarily delivers into our crude oil pipeline system that supplies our Parco and Casper Refineries.

Refined Product Terminals and Refinery Tankage

- two refined product terminals located in Orla, Texas and Moriarty, New Mexico, with an aggregate capacity of approximately 240,000 barrels, that are integrated with our refined product pipeline system that serves our Navajo Refinery;
- one refined product terminal located in Spokane, Washington, with a capacity of approximately 430,000 barrels, that serves third-party common carrier pipelines;
- one refined product terminal near Mountain Home, Idaho, with a capacity of approximately 120,000 barrels, that serves a nearby United States Air Force Base;
- two refined product terminals, located in Wichita Falls and Abilene, Texas, and one tank farm in Orla, Texas with aggregate capacity of approximately 560,000 barrels, that are integrated with our refined product pipelines that serve Delek's Big Spring, Texas refinery;
- a refined product terminal in Catoosa, Oklahoma with a capacity of approximately 70,000 barrels that stores specialty lubricant products and is utilized by our Tulsa Refineries;
- a refined product loading rack facility at each of our El Dorado, Tulsa, Navajo and Woods Cross Refineries and our Cheyenne facility, heavy product / asphalt loading rack facilities at our Tulsa East facility, Lovington facility and Cheyenne facility, LPG loading rack facilities at our El Dorado Refinery, Tulsa West facility and Cheyenne facility, lube oil loading racks at our Tulsa West facility and crude oil Leased Automatic Custody Transfer units located at our Cheyenne facility;
- refined product storage capacity at our Parco and Casper Refineries as well as at seven refined product terminals in Colorado, Idaho and Wyoming serving our Parco and Casper Refineries with a capacity of approximately 1,960,000 barrels;
- three refined product terminals in Kansas City, Missouri, Carrollton, Missouri and Montrose, Iowa having an aggregate storage capacity of approximately 880,000 barrels;
- on-site crude oil tankage at our Tulsa, Navajo and Woods Cross Refineries and Cheyenne facility having an aggregate storage capacity of approximately 1,440,000 barrels;

- on-site refined and intermediate product tankage at our El Dorado, Tulsa, Navajo and Woods Cross Refineries and Cheyenne facility having an aggregate storage capacity of approximately 8,280,000 barrels;
- eleven crude oil tanks adjacent to our El Dorado Refinery with a capacity of approximately 1,040,000 barrels that primarily serve our El Dorado Refinery;
- crude oil tankage at five crude oil terminals with a capacity of approximately 1,790,000 barrels that primarily serve our Parco and Casper Refineries;
- crude oil tankage with an aggregate storage capacity of approximately 450,000 barrels that primarily serve our Navajo Refinery;
- SLC Pipeline and Frontier Pipeline's tankage with an aggregate capacity of approximately 380,000 barrels;
- the UNEV Pipeline's product terminals near Cedar City, Utah and Las Vegas, Nevada with an aggregate capacity of approximately 650,000 barrels;
- a 50% interest in Cushing Connect Terminal with a capacity of approximately 1,500,000 barrels of crude oil storage in Cushing, Oklahoma;
- a 25.12% interest in Saddle Butte Pipeline III, LLC, which has approximately 160,000 barrels of crude oil storage in Wyoming, and
- a 49.995% ownership interest in Pioneer Investments Corp., which has approximately 655,000 barrels of refined product storage in Utah.

Refinery Processing Units
- a naphtha fractionation tower at our El Dorado Refinery, with a capacity of 50,000 BPD of desulfurized naphtha;
- a hydrogen generation unit at our El Dorado Refinery, with a capacity of 6.1 million standard cubic feet per day of natural gas;
- a crude unit, which is primarily an atmospheric distillation tower, a desalter and heat exchangers, at our Woods Cross Refinery, with a feedstock capacity of 15,000 BPD of crude oil;
- a FCC unit at our Woods Cross Refinery, which converts crude oil to high-value refined products such as gasoline, diesel and liquefied petroleum gases, with a capacity of 8,000 BPD; and
- a polymerization unit at our Woods Cross Refinery, that uses the output of the fluid cracking unit and converts them into gasoline blendstock, with a capacity of 2,500 BPD.

ADDITIONAL OPERATIONS AND OTHER INFORMATION

Corporate Offices
Our principal corporate offices are leased and located in Dallas, Texas. Functions performed in our Dallas office include overall corporate management, refinery and midstream management, planning and strategy, corporate finance, crude acquisition, logistics, contract administration, marketing, investor relations, governmental affairs, accounting, tax, treasury, information technology, legal and human resources support functions.

Human Capital
Our People
Our people differentiate us from our peers. Our “One HF Sinclair Culture” focuses on five key values – safety, integrity, teamwork, ownership and inclusion. These values influence our decisions, shape our behaviors and provide the opportunity for our employees to thrive. Safety is our first priority. We care about our people and have implemented policies and procedures designed to help them return home safely every day. We focus on integrity and doing the right thing. We champion a culture of teamwork and ownership by supporting each other and empowering employees to take action where they see a need or opportunity. Inclusion reflects our desire to foster a work environment in which employees feel valued and included in decisions, opportunities and challenges.

As of December 31, 2023, we had 5,218 employees located in the following geographies: 4,347 employees in the United States, 661 employees in Canada and 210 employees in Europe. As of December 31, 2023, 1,432 employees were covered by collective bargaining agreements. The current collective bargaining agreements have various expiration dates ranging between 2024 and 2026. We have experienced no material interruptions of operations due to disputes with our employees and management attempts to have and believes that we have positive working relationships with our local unions and their members.

Oversight
Our board of directors (the "Board of Directors") and Board of Directors committees provide oversight on our strategies and policies related to human capital management. Our Compensation Committee is responsible for periodically reviewing HF Sinclair's strategies, policies and practices regarding the promotion of employee diversity, equity and inclusion, talent and performance management, pay equity and employee engagement, as well as our executive succession planning. Our Nominating, Governance and Social Responsibility Committee oversees our policies and practices regarding human rights in our operations and supply chain. This process is designed to provide high level oversight of our strategies related to attracting, retaining and developing a workforce that aligns with our values and strategies.

Diversity & Inclusion
Our leadership is committed to attracting, retaining and developing a highly engaged, high-performing, diverse workforce and cultivating an inclusive workplace where all employees feel valued and have a sense of belonging. Of our total employees as of December 31, 2023, approximately 17% identified as female and approximately 83% identified as male. Approximately 22% of our total employees identified as Hispanic or Latino, Black or African American, Asian, American Indian or Alaskan Native, Native Hawaiian or Other Pacific Islander, or as two or more races or ethnicities. Veterans and reservists of the U.S. armed forces also represented approximately 6.5% of our U.S. workforce as of December 31, 2023.

Maintaining diversity and inclusion efforts is an organizational priority, and strategic oversight of our efforts is provided by our Compensation Committee. Our university recruiting team has partnered with various diversity and inclusion groups within universities to engage a diverse population of candidates from the student and alumni base. We held Inclusion and Diversity workshops throughout 2023 for frontline, senior and executive leadership roles. In addition, to help foster a culture of inclusion, we have five voluntary employee resource groups ("ERGs") open to all employees: Women in Energy, Veterans in Energy, Family Caregivers in Energy, Toastmasters in Energy and Cultural Awareness in Energy. Each of the ERGs focuses on developing talent at HF Sinclair by fostering relationships through education, networking and leadership development opportunities.

Health & Safety
The safety of our employees, contractors and communities is an overarching priority and fundamental to our operational success. We are grounded by our "Goal Zero" vision, which reflects our belief that safe production can be achieved each and every day. Our commitment to safety is embedded throughout our organization, from frontline employees and contractors to our executive leadership and Board of Directors. Our Operational Excellence Management System provides the framework through which we identify, monitor and reduce risks. Our Environmental, Health and Safety ("EHS") Leadership Council, comprised of company executives, including our CEO, business unit leaders and corporate safety specialists, sets EHS strategy and reviews performance. The Environmental, Health, Safety and Public Policy Committee of our Board of Directors provides board-level oversight of our strategies and performance in these areas.

In an effort to achieve Goal Zero, our employee and contractor safety education and training programs are conducted on an ongoing basis. We set specific goals for workplace safety and measure attainment of those goals. Over the past five years ended December 31, 2023, our Occupational Safety and Health Administration's ("OSHA") total recordable incident rate declined by 52%.

Total Rewards & Development
We believe that the health of our company is linked to the performance and health of our people. We want to inspire and empower our employees to feel confident in their long-term well-being and are committed to offering a comprehensive and competitive total rewards programs for our employees, as benchmarked against our peers. While our benefit offerings vary depending on each country's market practices, they are designed to support employee health, financial and emotional needs. Our offered benefits include comprehensive coverage for health care, a competitive retirement savings benefit, vacation and holiday time and other income protection and work life benefits. We also provide tools to help recognize and reward employee performance consistent with our One HF Sinclair Culture.

Consistent with our culture of ownership and growth, we offer training, development and engagement programs across every level of our organization to provide employees the opportunity to develop their career by enhancing skills and capabilities consistent with the needs of the business. Our suite of programs include: Accelerate, a curated collection of on-demand e-learning for all employees; Refine, interactive, instructor-led workshops focusing on professional development at any career level; Front Line Leadership Development, a series of leadership training for new and existing supervisors; Catalyst, a guided cohort of new leaders learning about leadership styles and executive presence; and Leading the HF Sinclair Way, a deep-dive for our senior leaders on leading through our cultural values and business objectives. We invested $8.0 million in our employee training and development programs in fiscal 2023.

Community Outreach / Volunteerism
Being a good neighbor and engaged member of the communities in which we operate is important to us. In addition to supporting charitable organizations through financial contributions, throughout 2023, we were equally pleased to support our communities through in-kind giving. From filling backpacks with school essentials to helping prepare meals to planting trees, our employees showed up for their communities and dedicated volunteer efforts to a variety of organizations.

Governmental Regulation
Our operations are subject to international, federal, state, provincial and local laws and regulations regarding, among other things, the generation, storage, handling, use, transportation and distribution of petroleum and hazardous materials by pipeline, truck, rail, ship and barge, the emission and discharge of materials into the environment, waste management, characteristics and composition of gasoline and diesel fuels, and other matters otherwise relating to the protection of human health and the environment. Permits or other authorizations are required under these laws and regulations for the operation of our refineries, pipelines and other facilities, and these permits and authorizations are subject to revocation, modification and renewal or may require operational changes, which may involve significant costs. Our operations are also subject to various international and domestic laws and regulations relating to health and safety, and failure to appropriately manage health and safety risks associated with our business could adversely impact our employees, communities, stakeholders, reputation and results of operations.

A violation of permit conditions or a failure to comply with applicable laws and regulations may result in the assessment of sanctions, including administrative, civil and criminal penalties; the imposition of investigatory, remedial or corrective action obligations or the incurrence of capital expenditures; the occurrence of delays in the permitting, development or expansion of projects or facilities; the issuance of injunctive relief limiting or prohibiting certain operations; and reputational harm. There is also the potential for liability for spill response and remediation, natural resource damage claims, and personal and property damage claims if there were to be an oil spill at a facility located near federal, state or provincial waters or a release of hazardous or other substances into the environment.

Compliance with applicable environmental laws, regulations and permits or other authorizations and health and safety laws and regulations will continue to have an impact on our operations, the results of our operations and our capital expenditures.

Federal Transportation Rate Regulation - Some of our existing pipelines provide interstate transportation services subject to regulation by the Federal Energy Regulatory Commission (the "FERC") pursuant to the Interstate Commerce Act (the "ICA"). The ICA requires that the rates charged by these pipelines (referred to as "interstate liquids pipelines") must be just and reasonable. The ICA also prohibits interstate liquids pipelines from providing services in a manner that unduly discriminates against or confers undue preference upon any shipper. The ICA permits interested persons to challenge newly proposed or changed rates or rules and authorizes the FERC to suspend the effectiveness of such proposed rates or rules for a period of up to seven months, during which the FERC may investigate whether the proposed rate or rules are just and reasonable. Upon completion of an investigation, the FERC may require the interstate liquids pipeline to refund the revenues collected during the pendency of the investigation that are in excess of the amount the FERC determines to be just and reasonable, together with interest. The FERC also may investigate, upon complaint or on its own motion, rates that are already in effect and may order an interstate liquids pipeline to change its rates prospectively. Upon an appropriate showing, a shipper may obtain reparations (including interest) for damages sustained during the two years prior to the filing of a complaint.

As a general matter, interstate liquids pipelines may change their rates within prescribed ceiling levels that are tied to an inflation index that the FERC reviews every five years. Cost-of-service ratemaking, market-based rates and settlement rates are alternatives to the indexing approach and may be used in certain specified circumstances to change rates. When an interstate liquids pipeline adjusts its rates using the index methodology, shippers may challenge rate increases made within the ceiling levels. The FERC's regulations provide that a protest against an index rate increase must allege "reasonable grounds" that the index rate increase is "so substantially in excess of the actual cost increases incurred by the carrier that the rate is unjust and unreasonable."

The current five-year price index is subject to review in the United States Court of Appeals for the District of Columbia (the "D.C. Circuit"). In response to requests for rehearing of its December 2020 price index proposal, on January 20, 2022, the FERC established a revised price index for the five-year period commencing July 1, 2021 and ending June 30, 2026, in which interstate liquids pipelines charging indexed rates were permitted to adjust their index ceilings annually by Producer Price Index for Finished Goods ("PPI-FG") plus 0.21% (a decrease from the PPI-FG plus 0.78% proposed in December 2020). In early 2022, industry participants petitioned the D.C. Circuit to review the FERC's orders setting the price index. Oral argument in the case was held on October 25, 2023. The D.C. Circuit has not yet issued a decision. The D.C. Circuit's decision could result in a further change to the index and impact our rates set using the PPI-FG plus 0.21% price index.

We believe the transportation rates currently charged by our interstate liquids pipelines are in accordance with the ICA and applicable FERC regulations. However, due to the complexity of rate making, the lawfulness of any rate is never assured. Adverse decisions by the FERC related to our rates could adversely affect our revenue, financial position, results of operations, and cash flows. In addition, if any of our pipelines were found to have provided services or otherwise operated in violation of the ICA, this could result in the imposition of administrative and criminal remedies and civil penalties, as well as a requirement to disgorge charges collected for such services in excess of the rate established by the FERC. Any of the foregoing could adversely affect revenues and cash flow related to the affected assets.

State Transportation Rate Regulation - While the FERC regulates the rates for interstate shipments on our interstate liquids pipelines, the New Mexico Public Regulation Commission regulates the rates for intrastate shipments on our pipelines in New Mexico, the Texas Railroad Commission regulates the rates for intrastate shipments on our pipelines in Texas and the Oklahoma Corporation Commission regulates the rates for intrastate shipments on our pipelines in Oklahoma. Generally, these state agencies have not investigated the rates or practices of intrastate pipelines subject to their jurisdiction in the absence of shipper complaints. However, a state regulatory commission could investigate our rates if such a challenge were filed and any adverse decisions could adversely affect our revenue, financial position, results of operations, and cash flows.

Commodity Regulation - The Federal Trade Commission, the FERC, and the Commodity Futures Trading Commission hold statutory authority to monitor certain segments of the physical and futures energy commodities markets. These agencies have imposed broad regulations prohibiting fraud and manipulation of such markets. With regard to our physical sales of oil or other energy commodities, and any related hedging activities that we undertake, we are required to observe the market-related regulations enforced by these agencies, which hold substantial enforcement authority. Failure to comply with such regulations, as interpreted and enforced, could have a material adverse effect on our business, results of operations and financial condition.

Safety and Maintenance - Many of our pipelines are subject to regulation by the Pipeline and Hazardous Materials Safety Administration ("PHMSA") of the Department of Transportation ("DOT"). PHMSA has promulgated regulations governing, among other things, maximum operating pressures, pipeline patrols and leak surveys, control room management, valve spacing and rupture mitigation, spill response and emergency procedures, as well as other matters intended to prevent accidents and failures. Additionally, PHMSA has promulgated regulations requiring pipeline operators to develop and implement integrity management programs for certain pipelines that, in the event of a pipeline leak or rupture, could affect "high consequence areas", which are areas where a release could have the most significant adverse consequences, including certain population areas, certain drinking water sources and unusually sensitive ecological areas.

In addition, many states have adopted regulations similar to, or which go above and beyond, existing PHMSA regulations, for certain intrastate pipelines. For example, Texas has developed regulatory programs that largely parallel the federal regulatory scheme and impose additional requirements for certain pipelines. Furthermore, other related programs, such as the EPA's Risk Management Program and the OSHA's Process Safety Management ("PSM") standard apply to some of our terminals and associated facilities.

We perform preventive and normal maintenance on all of our pipeline and terminal systems and make repairs and replacements when necessary or appropriate. We also conduct routine and required inspections of our pipelines and other assets as required by regulations. Corrosion inhibitors, external coatings and impressed current cathodic protection systems are used to protect against internal and external corrosion. We regularly monitor, test and record the effectiveness of these corrosion-control systems. We monitor the structural integrity of covered segments of our pipeline systems through a program of periodic internal inspections using electronic "smart pigs", hydrostatic testing, or other measures. We follow these inspections with a review of the data, and we make repairs as necessary to maintain the integrity of the pipeline. We have initiated a risk-based approach to prioritizing the pipeline segments for future smart pig runs or other appropriate integrity testing methods. This approach is intended to allow the pipelines that have the greatest risk potential to receive the highest priority in being scheduled for inspections or pressure tests for integrity. Nonetheless, the adoption of new or amended regulations or the reinterpretation of existing laws and regulations by PHMSA or states that result in more stringent or costly pipeline integrity management or safety standards could possibly have a substantial effect on us and similarly situated midstream operators.

Maintenance facilities containing equipment for pipe repairs, spare parts, and trained response personnel are located along the pipelines. Employees participate in simulated spill response exercises on a regular basis. They also participate in actual spill response boom deployment exercises in planned spill scenarios in accordance with Oil Pollution Act of 1990 requirements.

At our terminals, tanks designed for gasoline storage are equipped with internal or external floating roofs that minimize emissions and prevent potentially flammable vapor accumulation between fluid levels and the roof of the tank. Our terminal facilities have facility response plans, spill prevention and control plans, and other plans and programs to respond to emergencies.

Many of our terminal loading racks are protected with water deluge systems activated by either heat sensors or an emergency switch. Several of our terminals are also protected by foam systems that are activated in case of fire. All of our terminals participate in a comprehensive environmental management program to assure compliance with applicable air, solid waste and wastewater regulations.

For further information on pipeline safety and regulatory requirements related to maintenance, see the discussion under our risk factor "*We may incur significant costs and liabilities resulting from performance of pipeline integrity programs and related repairs*" in Item 1A – "Risk Factors."

Air Regulations - Our operations are subject to certain requirements of the federal Clean Air Act ("CAA") as well as related state and local laws and regulations, and similar laws in Canada and the Netherlands. Certain CAA regulatory programs (and similar programs in Canada) applicable to our facilities require capital expenditures for the installation of certain air pollution control devices, operational procedures and expenditures to minimize emissions and to prevent accidental releases of air pollutants, and monitoring and reporting of emissions. For example, implementation of the revised National Ambient Air Quality Standards ("NAAQS") for ozone could result in stricter permitting requirements, a delay in or the inability to obtain such permits, and increased expenditures for pollution control equipment, the costs of which could be significant. Moreover, an EPA rule became effective in January 2018 that requires, among other things, benzene monitoring at the refinery fence line and submittal of fence line monitoring data to the EPA on a quarterly basis; upgraded storage tank controls requirements, including new applicability thresholds; enhanced performance requirements for flares, continuous monitoring of flares and pressure release devices, and analysis and remedy of flare release events; compliance with emissions standards for delayed coking units; and requirements related to air emissions resulting from startup, shutdown and maintenance events. These rules, as well as subsequent rulemaking under the CAA or similar laws in all jurisdictions in which we operate, or new agency interpretations of existing laws and regulations, may necessitate additional expenditures in future years and result in increased costs on our operations.

Fuel Quality Regulations - We are subject to the EPA's regulations governing fuels and fuel additives used in motor vehicles, non-road equipment and airplanes. The EPA has the authority under the CAA to modify the formulation of the refined transportation fuel products we manufacture in order to limit the emissions associated with their final use. In February 2007, the EPA finalized MSAT2 regulations that impose reductions in the benzene content of our produced gasoline. In addition to reducing benzene concentration in our gasoline, our refineries currently purchase benzene credits to meet these requirements. If economically justified or otherwise determined to be beneficial, we may implement additional benzene reduction projects to eliminate or reduce the need to purchase benzene credits. Additionally, in April 2014, the EPA finalized the Tier 3 Motor Vehicle Emission and Fuel Standards, which require a reduction in annual average gasoline sulfur content from 30 parts per million ("ppm") to 10 ppm. These requirements, other CAA requirements, and other presently existing or future environmental regulations may cause us to make substantial capital expenditures and purchase sulfur credits at significant cost to enable our refineries to produce fuels that meet the applicable requirements. Lastly, in December 2020, the EPA streamlined and consolidated its existing fuel quality regulations that apply across all gasoline and diesel fuel programs, also known as the EPA's Fuels Regulatory Streamlining Rule at 40 CFR Part 1090. The streamlining rule is intended to improve overall compliance assurance and reduce compliance costs for the industry and the EPA, while maintaining environmental performance.

Pricing and Profits Reporting and Regulation - Some state governments have considered legislation that would require companies in the petroleum industry to report detailed market information (including supply, pricing and profit information) and would potentially impose penalties for profits deemed to be excessive. For example, in January 2024, Washington, House Bill 2232 ("HB 2232") was introduced in the legislature, and as proposed, would establish extensive reporting requirements for companies, including refiners, seeking information such as volume, price, type, feedstock inputs, origin of petroleum receipts, imports and exports of finished petroleum products, blendstocks and ethanol, refinery outputs and refinery stocks, and finished product supply and distribution, among others. Additionally, HB 2232 would create a new division within the Washington Utilities and Transportation Commission charged with providing independent oversight and analysis of petroleum fuel markets. Although HB 2232 was not adopted by the Washington legislature, whether HB 2232 or similar legislation will be introduced and enacted in the future is unknown at this time. If legislation similar to HB 2232 is passed into law, such legislation could result in increased compliance costs and affect the results of our operations and financial position. Such laws and similar regulations could also increase our litigation risks or may increase risks related to our reputation or goodwill as we cannot predict how additional reporting under this law may be perceived or interpreted by our customers and stakeholders.

Renewable Fuel Standard - Pursuant to the Energy Independence and Security Act of 2007, and the EPA's corresponding RFS regulations, most refiners are required to blend increasing amounts of biofuels with refined products or purchase RINs in lieu of blending. Under the RFS, the percentage of renewable fuels that refineries are obligated to blend into their finished petroleum

products is adjusted annually. In June 2023, the EPA finalized the RFS targets for 2023 through 2025, which set the annual renewable volume obligations ("RVOs") for conventional (i.e., corn ethanol) renewable fuel, biomass-based diesel, advanced biofuels and cellulosic biofuel. The targets increase RVOs in each of the concurrent years.

The EPA's implementation of the RFS has historically been subject to numerous court challenges. Lawsuits have been filed by the renewable fuel industry and refining industry challenging the EPA's decisions on small refinery exemptions. For additional information regarding risks relating to our small refinery exemptions, see the discussion under our risk factor "Compliance with, or developments with respect to, renewable and low carbon fuel blending programs, and other regulations, policies, and standards impacting the demand for low-carbon fuels could have an adverse effect on our financial condition and results of operations" in Item 1A – "Risk Factors." Legal challenges to the EPA's decision are ongoing. We cannot predict the outcome of these matters or whether they may result in increased RFS compliance costs. There also continues to be a shortage of advanced biofuel production resulting in increased difficulties meeting RFS mandates. As a result, we may be unable to blend sufficient quantities of renewable fuel to meet our requirements and, therefore, may have to purchase an increasing number of RINs. It is not possible at this time to predict with certainty what those volumes or costs may be, but given the potential increase in volumes and the volatile price of RINs, increases in RVOs could have an adverse impact on our results of operations.

Finally, while there is no current regulatory standard that authenticates RINs that may be purchased on the open market from third parties, we believe that the RINs we purchase are from reputable sources, are valid and serve to demonstrate compliance with applicable RFS requirements. However, if any of the RINs purchased by us on the open market are subsequently found by the EPA to be invalid, we could incur significant costs, penalties, or other liabilities in connection with replacing any invalid RINs and resolving any enforcement action brought by the EPA.

California Low Carbon Fuel Standard - Under California's Global Warming Solutions Act of 2006, the California Air Resources Board was required to undertake a statewide effort to reduce GHG emissions. One of the programs designed to help achieve those reductions is the LCFS program. The LCFS program is designed to reduce GHG emissions by decreasing the carbon intensity ("CI") of transportation fuels consumed in the state. Under this program, each fuel is assigned a CI value, which is intended to represent the GHG emissions associated with the feedstocks from which the fuel was produced, the fuel production and distribution activities, and the use of the finished fuel. Each producer or importer of fuel must demonstrate that the overall mix of fuels it supplies for use in California meets the CI benchmarks for each compliance period. A producer or importer with a fuel mix that is above the CI benchmark must purchase LCFS credits sufficient to meet the CI benchmark. Fuels produced by our Renewables segment have CI scores that are lower than traditional petroleum-based transportation fuels, and we benefit from the demand from other regulated entities for these low-carbon transportation fuels. California has proposed to amend the LCFS program, including strengthening CI targets from a 20% reduction to a 30% reduction by 2030, and adding a 5% CI benchmark stringency increase for 2025.

Oregon Clean Fuels Program - The Clean Fuels Program ("CFP") became effective in 2016 with the intended goal of reducing the CI of Oregon's transportation fuels by establishing annual standards that decrease over time. The baseline year for the program is 2015, and the standard for that year represents 10% ethanol blended with gasoline and 5% biodiesel blended with diesel. The CFP requires a 10% reduction in average CI from 2015 levels by 2025, followed by a 20% reduction by 2030, and a 37% reduction by 2035. Businesses that create fuels with a lower CI than the annual limit generate credits, while higher CI fuels create deficits. Credits and deficits are measured in metric tons of GHG emissions. CFP credits began to trade at the end of 2016, and the credit market has grown steadily in the subsequent years. We import transportation fuels produced at our Puget Sound Refinery into Oregon, and are thereby subject to the CFP, including, among other things, the requirement to purchase and retire CFP credits to offset the deficit created through our sale of gasoline and diesel fuel into Oregon.

Washington Clean Fuel Standard - The Clean Fuel Standard ("CFS") became effective on January 1, 2023, and similar to the low carbon fuel programs in California and Oregon, it is designed to decrease the CI of Washington's transportation fuels. The CFS requires a 20% reduction in average CI from 2017 levels by 2034. The CFS functions similar to its sister programs in California and Oregon. We sell gasoline and diesel fuel produced at our Puget Sound Refinery in Washington, and are thereby subject to the CFS, including, among other things, the requirement to purchase and retire CFS credits to offset the deficit created through our sale of gasoline and diesel fuel in Washington.

Other Low Carbon Fuel Standards - Many international, federal, state, provincial and local governments, including the state of New Mexico where our Navajo Refinery is located, have issued, or are considering issuing, low carbon fuel regulations, policies, and standards to reduce GHG emissions and increase the percentage of low-carbon fuels in the transportation fuel mix. While these regulations result in additional costs to our refining business, they have created opportunities to develop our renewables business, and should continue to help drive demand for our renewable diesel products. We believe that our ability to supply low-carbon fuels can play an important role in helping achieve GHG emissions reduction targets.

Blender's Tax Credit Incentives - The U.S. federal government has enacted tax incentives to encourage the production of low-carbon fuels and/or reduce GHG emissions. For example, Section 6426 of the Internal Revenue Code of 1986, as amended (the "Code"), provides a tax credit (generally referred to as the blender's tax credit) to blenders of certain renewable fuels to encourage the production and blending of those fuels with traditional petroleum-based transportation fuels. Only blenders that have produced a mixture and either sold or used the fuel mixture as fuel are eligible for the blender's tax credit. The renewable diesel produced by our Renewables segment is a liquid fuel derived from biomass that meets the EPA's fuel registration requirements; therefore, renewable diesel that we produce and blend qualifies for this refundable tax credit of one dollar per gallon. The Inflation Reduction Act of 2022 extended this credit through December 31, 2024, but there is no certainty that this legislation authorizing the credit or the amount of the credit will not be revised or extended.

Oregon and Washington Carbon Cap & Trade Programs - In March 2020, the Governor of Oregon signed Executive Order 20-04 (the "Executive Order"), directing state agencies to take actions to reduce and regulate GHG emissions and consider climate change in agency planning. The Executive Order directed the Oregon Environmental Quality Commission ("OEQC") and the Oregon Department of Environmental Quality ("ODEQ") to take a variety of actions, including the development of a new program to limit GHG emissions from large stationary sources, transportation fuels, and other liquid and gaseous fuels. In August 2021, ODEQ published a notice of proposed rulemaking with the draft emissions cap and reduction rules for the Climate Protection Program (the "CPP"). The OEQC considered and adopted final CPP rules in December 2021, which became effective January 1, 2022. Using a 1990 baseline, the CPP includes GHG reduction goals of 50% by 2035 and 90% by 2050. Our subsidiary, HF Sinclair Refining & Marketing LLC, presumably became subject to the CPP rules in the second quarter of 2022 due to its sale of transportation fuel over the rack in Oregon primarily from the Puget Sound Refinery. As a presumed covered fuel supplier under the CPP rules at that time, HF Sinclair Refining & Marketing LLC submitted a CPP permit application to ODEQ in February 2023 and is purportedly required to acquire, and then surrender, compliance instruments (corresponding to each metric ton of CO2 equivalents imported/sold into Oregon) starting in calendar year 2025 (i.e., at the end of the first three-year compliance period, which runs from 2022 to 2024), and for each three-year compliance period thereafter to demonstrate compliance with the CPP rules. Further, the CPP rules were amended in November 2023 to provide, among other changes, a one-year lookback for purposes of calculating compliance instrument obligations. However, the authority of OEQC to adopt the CPP rules was challenged in litigation, including in a lawsuit filed in March 2022 by the Western States Petroleum Association and others. In December 2023, the Court of Appeals of Oregon issued an order in that lawsuit declaring the 2021 CPP rules "invalid" (based on ODEQ's failure to strictly follow administrative notice requirements at the time of rule adoption). On January 22, 2024, ODEQ announced that it does not intend to appeal the Court of Appeals of Oregon's decision, but instead will proceed with a new rulemaking to re-adopt the CPP rules (under a public process that typically takes about 12 months). It is unclear whether the new rules will address any prior fuel imports or prior distributions of compliance instruments. More legal challenges to the underlying authority of ODEQ to adopt such rules may also be brought. Accordingly, we will closely monitor the Oregon CPP rulemaking process to determine future compliance obligations. In 2021, the State of Washington enacted its own Climate Commitment Act ("CCA"), which establishes a comprehensive cap-and-invest program that provides an overall limit for GHG emissions from major sources in the state that begins on January 1, 2023 and declines yearly to 95% below 1990 levels by 2050. As the operator of the Puget Sound Refinery, our subsidiary HF Sinclair Puget Sound Refining LLC, and as a covered fuel supplier, our subsidiary HF Sinclair Refining & Marketing LLC, must obtain an allowance for each ton of covered CO2e emissions. An initiative to repeal the CCA has been filed in Washington. If the Washington legislature does not choose to adopt the initiative into law, it will be placed on the November 2024 ballot. Because both programs are in a state of flux, it is too early to predict the financial and operational impact of the CPP and CCA on our business.

Climate Change - In recent years, various legislative and regulatory measures to address climate change and GHG emissions (including carbon dioxide, methane and nitrous oxides) have been discussed or implemented. They include proposed and enacted federal regulation and state actions to develop statewide, regional or nationwide programs designed to control and reduce GHG emissions from stationary sources, such as our refineries, as well as power plants, mobile transportation sources and fuels. Measures to date have included but are not limited to cap and trade programs, carbon taxes, vehicle efficiency standards, electric vehicle mandates, combustion engine phaseouts, LCFS and renewable fuel requirements. Although it is not possible to predict the requirements of any GHG legislation that may be enacted, any laws or regulations that may be adopted to restrict or reduce GHG emissions will likely require us to incur increased operating and capital costs.

The EPA has previously issued two rules (the "Clean Power Plan" and the "Affordable Clean Energy Rule") to reduce carbon dioxide emissions from coal-fired power plants, both of which have been invalidated by courts. In May 2023, the EPA proposed a replacement rule, and we do not expect such a rule will directly affect our operations. However, to the extent the EPA fully implements rules that imposes higher costs on electricity generating units, it could result in increased power costs for our refineries in future years. Furthermore, the EPA could propose to apply similar regulations to other facilities, including refineries, in the future.

EPA rules require us to report GHG emissions from our refinery operations and consumer use of fuel products produced at our refineries on an annual basis. While the cost of compliance with the reporting rule is not material, data gathered under these rules may be used in the future to support additional regulation of GHG emissions. Moreover, the EPA directly regulates GHG emissions from refineries and other major sources through the Prevention of Significant Deterioration ("PSD") and Federal Operating Permit programs and may require Best Available Control Technology ("BACT") for GHG emissions above a certain threshold if emissions of other pollutants would otherwise require PSD permitting. While this does not impose any limits or controls on GHG emissions from current operations, future projects or operational changes that increase GHG emissions, such as capacity increases, may be subject to emission limits or technological requirements pertaining to GHG emissions, such as BACT.

In March 2022, the SEC issued proposed rules that, if adopted, would require public companies to include certain climate-related disclosures in their registration statements and periodic reports, including information about climate-related risks, climate-related financial statement metrics, and GHG emissions. Additionally, states are also considering their own climate accounting and risk and disclosure rules, independent of the SEC. For example, in October 2023, California enacted new laws requiring additional disclosure with respect to climate-related risks and GHG emissions reduction claims for certain companies doing business in the state. In November 2022, the Biden Administration issued a proposed rule that would require government contractors to publicly disclose their GHG emissions and set "science based" emissions reduction targets, which generally do not allow for the use of carbon offsets, that align with the goals of the Paris Agreement, which could affect us as a government contractor and/or through our contractual and business arrangements with government contractors.

Stakeholder concerns about climate change could also adversely affect demand for the refined petroleum products that we produce. Recently, certain financial institutions, funds and other sources of capital have made pledges to reduce GHGs in their lending portfolios, leading some to restrict or eliminate their investment in oil and natural gas activities. There is also a risk that future government regulations could require financial institutions to adopt policies that have the effect of reducing the funding provided to or investments in the fossil fuel sector altogether or unless certain climate-related standards are met. Ultimately, this could make it more difficult or costly to secure funding for exploration and production activities and result in decreased production of oil, which indirectly could have an adverse impact on our operations.

Climatic events in the areas in which we operate, whether from climate change or otherwise, can cause disruptions and in some cases delays in our production activities or ability to deliver our products to our customers. These events, including, but not limited to, drought, winter storms, wildfire, extreme temperatures, tornados, extreme precipitation or flooding, may become more intense or more frequent as a result of climate change and could have an adverse effect on our continued operations as well as the operations of our suppliers and customers. Additionally, changing meteorological conditions, particularly temperature, may result in changes to demand for our products. Our customers or suppliers may also be subject to similar risks, any of which may adversely impact our business, financial condition, or operations.

The Biden Administration has adopted an "all of government" approach to climate change in which the federal government would use not only its regulatory and enforcement authority but also its policy and purchasing power to encourage investment and use of renewable energy sources and to otherwise impede and reduce fossil fuel use and GHG emissions. This approach may include elements that could directly or indirectly result in decreased demand for transportation fuel and could have an adverse impact on our operations. For example, in 2021, President Biden issued several executive orders that committed to substantial action on climate change and called for, among other things, the increased use of zero-emission vehicles by the federal government, the elimination of subsidies provided to the fossil fuel industry, and increased emphasis on climate-related risks across governmental agencies and economic sectors. In 2021, the EPA announced its intent to reconsider and revise rules related to the oil and gas sector (primarily oil production and natural gas production, distribution and storage) to further reduce GHG emissions, and, on December 2, 2023, the EPA announced the promulgation of a final rule that expands on the 2021 proposal and significantly limits methane and other emissions from certain oil and natural gas operations. This new rule could increase the cost of domestic crude oil and natural gas. In addition, the EPA, together with the DOT, implemented GHG emission and corporate average fuel economy standards for vehicles manufactured in the United States, which standards have been revised from time-to-time to impose increasingly stringent requirements for emissions reductions. The EPA and DOT proposed regulations again in August and December 2023 to make certain of those requirements more stringent for future model year vehicles. President Biden also reinstated the Interagency Working Group on the Social Cost of Greenhouse Gases in 2021 and directed the group to publish interim estimates of the social cost of carbon dioxide, nitrous oxide, and methane, with a view to using such estimates in federal rulemakings on GHG emissions, which it did. In November 2022, the EPA published a draft report assigning new and higher social cost values to GHG emissions for use in its rulemaking initiatives. In September 2023, the Biden Administration directed federal agencies to begin considering the social cost of GHG emissions in procurement processes and environmental reviews under the National Environmental Policy Act.

Internationally, the United Nations-sponsored Paris Agreement requires member countries to submit non-binding, individually determined emissions reduction goals every five years after 2020. The United States initially joined and then withdrew from such agreement in 2020. In 2021, the United States rejoined the Paris Agreement and issued its corresponding "nationally determined contribution" ("NDC") to reduce economy-wide net GHG emissions to 50-52% below 2005 levels by 2030. While the NDC does not identify specific actions necessary to achieve these reductions, it lists several sectors as pathways for reductions, including the power, transportation, building, industrial, and agricultural sectors. The Biden Administration has acknowledged a combination of regulatory actions and legislation will be necessary to achieve the U.S. NDC. In regards to legislation, in November 2021, the United States enacted a nearly $1 trillion bipartisan infrastructure law, which provided significant funding for electric vehicles and clean energy technologies, and in August 2022 the United States enacted the Inflation Reduction Act of 2022, which allocated $369 billion to climate change and environmental initiatives, including transportation electrification, fees on and greater regulation of methane emissions, and support for green energy manufacturing programs. Certain of these initiatives are subject to ongoing litigation, and the impacts of these laws and orders, and the terms of any legislation or regulation to implement the United States' commitment under the Paris Agreement, remain unclear at this time.

In the Netherlands and Canada, increased attention to climate change has led to changes in laws and to new laws, such as the Climate Act (Netherlands) and the Canadian Net-Zero Emissions Accountability Act, and has also led to increased frequency of climate change lawsuits. Additionally, the EU has promulgated environmental, social and governance ("ESG") reporting requirements under the EU's Corporate Sustainability Reporting Directive ("CSRD"), which requires in-scope companies to disclose extensive sustainability information, including disclosing risks and opportunities arising from environmental and social matters, and the impact of their business on people and the environment. Certain of our subsidiaries are expected to have to begin reporting from 2026 onward for financial years beginning on January 1, 2025. Ultimate parent companies that are not incorporated in the EU, such as HF Sinclair, may be required to report under CSRD from 2029 onward for financial years beginning on or after January 1, 2028. CSRD and similar laws and regulations could result in increased compliance costs and affect the results of our operations and financial position. Such laws and similar regulations could also increase our litigation risks or may increase risks related to our reputation or goodwill as we cannot predict how additional reporting under these laws may be perceived or interpreted by our customers and stakeholders.

Water Discharges - Our operations are also subject to the Federal Clean Water Act ("CWA"), the Federal Safe Drinking Water Act ("SDWA") and comparable state and local requirements, as well as similar laws in Canada and the Netherlands. The CWA, the SDWA and analogous laws prohibit any discharge into surface waters, ground waters, injection wells and publicly-owned treatment works except in conformance with legal authorization, such as pre-treatment permits and National Pollutant Discharge Elimination System ("NPDES") permits, issued by federal, state and local governmental agencies. The EPA commenced a study from 2015-2017 related to the discharges of metals and dioxin from petroleum refining operations and wastewater discharges from refineries in connection with the consideration of new effluent limitation guidelines that would be incorporated into refinery sector NPDES permits. To date, the EPA has not proposed any new effluent limitation guidelines applicable to our operations, but future rulemakings related to this issue could require us to incur increased costs related to the treatment of wastewater resulting from our operations.

The CWA also regulates filling or discharges to wetlands and other "waters of the United States," including certain wetlands. The Biden Administration issued a new rule effective in March 2023 that significantly expanded CWA jurisdiction relative to a prior June 2020 rule that is now largely rescinded. This new rule is being challenged in federal court. Shortly after the Biden Administration issued the new rule, the United States Supreme Court issued a decision in *Sackett v. Environmental Protection Agency* that reduced the scope of waters and wetlands subject to federal jurisdiction. In September 2023, the EPA and the U.S. Army Corps of Engineers revised the regulatory definition of "waters of the United States" to incorporate the Supreme Court's *Sackett* decision. Legal challenges to the January 2023 rulemaking, as revised by the September 2023 rulemaking, are proceeding, creating uncertainty as to the final regulatory definition. Further, some states either have changed or are contemplating changes to their own laws in order to regulate waters and wetlands that were previously subject to federal regulation.

Hazardous Substances and Wastes - We generate wastes that may be subject to the Resource Conservation and Recovery Act and comparable state and local requirements, as well as similar laws in Canada and the Netherlands. The EPA and various state agencies have limited the approved methods of disposal for certain hazardous and non-hazardous wastes. Although the EPA is currently working on several rulemakings that could impact how our refineries manage various waste streams, it does not appear that these rules will significantly impact our refineries.

The Comprehensive Environmental Response, Compensation and Liability Act ("CERCLA"), also known as "Superfund," imposes strict, and under certain circumstances, joint and several liability on certain classes of persons who are considered to be responsible for the cost of cleaning up hazardous substances that have been released into the environment and for damages to natural resources. These persons include current and former owners or operators of property where a release has occurred, and any persons who disposed of, or arranged for the transport or disposal of, hazardous substances at the property. In the course of our historical operations, as well as in our current operations, we have generated waste, some of which falls within the statutory definition of a "hazardous substance" and some of which may have been disposed of at sites that may be subject to cleanup and cost recovery actions under CERCLA in the future. Similarly, locations now owned or operated by us, where third parties have disposed such hazardous substances in the past, may also be subject to cleanup and cost recovery actions under CERCLA. Some states have enacted laws similar to CERCLA which impose similar responsibilities and liabilities on responsible parties. It is also not uncommon for neighboring landowners and other third parties to file claims under state law for personal injury and property damage allegedly caused by hazardous substances or other pollutants released into the environment. Many states also have similar liability regimes that impose strict and potentially joint and several liability for releases into the environment.

Emerging Contaminants - Various regulators are considering regulation of so-called emerging contaminants, including for example a proposal by the EPA in August 2022 to list two per- and polyfluoroalkyl substances ("PFAS") as CERCLA hazardous substances. In 2023, the EPA undertook several other regulatory actions related to PFAS chemicals, including, among others, updates to the EPA's Toxics Release Inventory ("TRI") program to remove an exemption for reporting PFAS and included PFAS exposure as a new enforcement priority for fiscal years 2024-2027. The EPA has also added seven additional PFAS to the list of chemicals covered by TRI reporting. We cannot predict the impact of future regulation of presently unregulated substances, but the prospect of such regulation creates additional uncertainty.

Oil Pollution Liability - The Oil Pollution Act of 1990 ("OPA") and regulations thereunder generally subject owners and operators of facilities to strict, joint and several liability for all containment and cleanup costs, natural resource damages, and potential governmental oversight costs arising from oil spills into the waters of the U.S. The OPA also imposes ongoing requirements on a responsible party, including the preparation of oil spill response plans and proof of financial responsibility to cover environmental cleanup and restoration costs that could be incurred in connection with an oil spill. Likewise, the CWA contains provisions that also impose similar liabilities for oil spills, and regulations under the CWA imposed prevention and response planning requirements applicable to many of our facilities. These liability regimes, as well as the rules under the OPA and CWA, or new agency interpretations of existing laws and regulations, may necessitate additional expenditures in future years and result in increased costs of our operations.

Other Environmental Regulations - Our Canadian assets and operations are also required to comply with various Canadian federal, provincial and municipal regulations. The regulations are in many cases conceptually similar to those described above for our U.S. operations. The principal legislation affecting our Canadian operations is the Canadian Environmental Protection Act, the Fisheries Act, the Greenhouse Gas Pollution Pricing Act and their regulations at a federal level and various provincial statutes and regulations such as the Ontario Environmental Protection Act, the Ontario Occupational Health and Safety Act and the Ontario Water Resources Act. All these laws contain broad prohibitions against causing harm to air, land, water, people or any other living organism and in many cases contain detailed prescriptive rules governing many aspects of our operations. Regulatory trends towards more stringent emission requirements and operating controls are expected to continue at federal, provincial and local levels.

Additionally, our assets and operations in the Netherlands are required to comply with Dutch regulations that are similar to, and in some cases more stringent than, those described above for our U.S. operations. The statutes to which our Dutch assets and operations are subject include the Environmental Protection Act, the Activities Decree, the Environmental Licensing (General Provisions) Act, the Water Act, the Soil Protection Act, the Major Accidents (Risks) Decree, the Climate Act, the European Birds and Habitats Directive implemented in the Nature Conservation Act, and other subordinate decrees and regulations relative to environmental control, permitting and enforcement. A large legislative operation was developed that will lead to the integration of all environmental laws in one, being the Environment and Planning Act, which entered into force on January 1, 2024. Generally, these regulations create a system of environmental permits covering the most significant emissions to water, air and soil, as well as other environmental impacts. The Netherlands also participates in certain broader European legal initiatives, including GHG cap and trade programs. Additionally, in December 2019, the High Council of the Netherlands upheld a court order for the government of the Netherlands to reduce the country's GHG emissions by 25% (compared to 1990) by 2020, and in January 2020, the Climate Act came into force, with (on the basis of the latest amendments following European law) the goal of significantly reducing GHG emissions by 55% (compared to 1990) by 2030, achieving climate neutrality by 2050 and striving for negative GHG emissions after 2050. Furthermore, the target is that 100% of the electricity production will be CO_2 neutral in 2050.

Enforcement and Litigation Proceedings - We are and have been the subject of various local, state, provincial, federal and private proceedings and inquiries relating to compliance with environmental laws and regulations and conditions. These matters include statutory and regulatory programs related to soil and water discharges and contamination, air pollution, pipeline safety and integrity, and GHG emissions, as well as personal injury and property damage allegedly caused by substances that we manufactured, handled, used, released or disposed. We currently have environmental remediation projects that relate to recovery, treatment and monitoring activities resulting from past releases of refined product and crude oil into the environment. As of December 31, 2023, we had an accrual of $195.4 million related to such environmental liabilities.

Compliance with current and future environmental regulations is expected to require additional expenditures, including expenditures for investigation and remediation, which may be significant. To the extent that future expenditures for these purposes are material and can be reasonably determined, these costs are disclosed and accrued, if applicable.

Safety and Accident Prevention - Our operations are subject to various laws and regulations relating to health and safety, including the OSHA, comparable state statutes, Canadian regulations applicable to our operations in Canada and Dutch regulations, including the Health and Safety Act and other subordinate decrees and regulations, applicable to our operations in the Netherlands. We maintain a comprehensive safety program, including mechanical integrity and safety-related maintenance programs and training, to comply with all applicable laws and regulations to protect the safety of our workers and the public. Some of our operations are also subject to OSHA PSM regulations and the EPA's CAA Risk Management Plan ("RMP") regulations, both of which are designed to prevent or minimize chemical accidents and any resulting releases of toxic, reactive, flammable or explosive chemicals. In January 2017, the EPA revised the RMP requirements for incident investigation and accident history reporting, emergency preparedness, and the performance of process hazard analyses and third-party compliance audits. Some of the revised requirements have not yet become effective, and the EPA issued a final rule in December 2019 that rescinded several of the requirements of the 2017 rule. That rescission was challenged in court, but the case has not proceeded because the EPA proposed a further RMP amendment in August 2022 that would reverse much of the 2019 rescission, with some modifications, and add provisions for employee participation, community engagement, and additional accident prevention requirements. Among those revisions is a proposal to require refineries with HF alkylation processes to perform a "safer technology and alternatives analysis" as part of the process hazard analysis to consider and document the practicality of inherently safer technologies and other risk management measures. Also in January 2017, OSHA announced changes to its National Emphasis Program, which specifically identified oil refineries as facilities for increased inspections and instructed inspectors to use data gathered from EPA RMP inspections to identify refiners for additional PSM inspections. Compliance with applicable state and federal health and safety laws and regulations, as well as environmental regulations, has required, and continues to require, substantial expenditures.

Environmental, health and safety legislation, regulations and regulatory programs change frequently. We cannot predict what additional environmental, health and safety legislation or regulations will be enacted or become effective in the future or how existing or future laws or regulations will be administered or interpreted with respect to our operations. Compliance with more stringent laws or regulations or adverse changes in the interpretation of existing laws or regulations by government agencies could have an adverse effect on our financial position and the results of our operations and could require substantial expenditures for the installation and operation of systems and equipment that we do not currently possess.

Insurance

Our operations are subject to hazards of operations, including fire, explosion and weather-related perils. We maintain various insurance coverages, including business interruption insurance, subject to certain deductibles. We are not fully insured against certain risks because such risks are not fully insurable, coverage is unavailable, or premium costs, in our judgment, do not justify such expenditures.

We have a risk management oversight committee consisting of members from our senior management. This committee oversees our risk enterprise program, monitors our risk environment and provides direction for activities to mitigate identified risks that may adversely affect the achievement of our goals.

Item 1A. Risk Factors

Risk Factor Summary

Investing in us involves a degree of risk. You should carefully consider all information in this Annual Report on Form 10-K, including the Management's Discussion & Analysis section and the financial statements and related notes, prior to investing in our common stock. These risks and uncertainties include, but are not limited to, the following:

Risks Related to our Business/Industry:

- The prices of crude oil, renewable feedstocks and refined, finished lubricant and renewable diesel products materially affect our profitability, and are dependent upon many factors that are beyond our control.
- Our operations are subject to catastrophic losses, operational hazards and unforeseen interruptions and other disruptive risks for which we may not be adequately insured.
- A disruption to or proration of the product distribution systems or manufacturing facilities we utilize could negatively impact our profitability.
- A material decrease in the supply, or a material increase in the price, of crude oil, renewable feedstocks or other raw materials or equipment available to our refineries and other facilities could significantly reduce our production levels and negatively affect our operations.
- To successfully operate our facilities, we are required to expend significant amounts for capital outlays and operating expenditures. If we are unable to complete capital projects at their expected costs or in a timely manner, our financial condition, results of operations, or cash flows could be materially and adversely affected.
- The refining and marketing industry and the lubricants and specialties industry are highly competitive, and an increase in competition could adversely affect our earnings and profitability.
- Our business is subject to the risks of international operations.
- Negative publicity or an erosion of our business reputation could have a material adverse effect on our earnings, cash flows and financial condition.
- Potential product, service or other related liability claims and litigation could adversely affect our business, reputation and results of operations.
- Terrorist attacks, and the threat of terrorist attacks or vandalism, have resulted in increased costs to our business. Continued global hostilities or other sustained military campaigns may adversely impact our results of operations.
- Our business may suffer due to a change in the composition of our Board of Directors, or the departure of any of our key executives or other key employees. A shortage of skilled labor may make it difficult for us to maintain labor productivity.
- A portion of our workforce is unionized, and any disruptions in our labor force or adverse employee relations could adversely affect our business.
- The widespread outbreak of an illness or pandemic or other public health crisis, and actions taken in response thereto, may have a material adverse effect on our business.
- Acquisitions involve numerous risks, any of which could adversely affect us.
- Certain of our facilities, pipelines and assets are located on or adjacent to Native American tribal lands or on other lands which we do not own. Our operations are subject to potentially disruptive activity by those concerned with our industry.
- General economic conditions may adversely affect our business, operating results and financial condition.
- An impairment of our goodwill or long-lived assets could reduce our earnings or negatively impact our financial condition and results of operations.
- We sell many of our lubricants and specialties products through distributors, which presents risks that could adversely affect our operating results.
- The market price of our common stock may fluctuate significantly, and the value of a stockholder's investment could be impacted.
- REH Company became a significant holder of our common stock following the completion of the Sinclair Transactions.

Risks Related to Government Regulation

- We are subject to significant regulation and oversight by governmental agencies.
- We incur significant costs and liabilities, and expect to incur additional costs and liabilities in the future, resulting from compliance with existing, new and changing environmental, health and safety laws and regulations, and face potential exposure for environmental matters.
- We may incur significant costs and liabilities resulting from performance of pipeline integrity programs and related repairs.

- There are various risks associated with GHGs and climate change that could result in increased operating costs and litigation and reduced demand for the refined products we produce and investment in our industry.
- Increasing attention to ESG matters may adversely impact our business, financial results, stock price or price of debt securities.
- Compliance with, or developments with respect to, renewable and low carbon fuel blending programs, and other regulations, policies, and standards impacting the demand for low-carbon fuels could have an adverse effect on our financial condition and results of operations.
- Increases in required fuel economy and regulation of GHG emissions from motor vehicles may reduce demand for petroleum-based transportation fuels.
- State regulation of petroleum product markets and reporting requirements could adversely impact our business, costs of operation and financial results.
- Physical impacts of climate change could have an adverse effect on our financial condition and results of operations.
- Compliance with and changes in tax laws could materially and adversely impact our financial condition, results of operations and cash flows.

Risks Related to Cybersecurity, Data Security and Privacy, Information Technology and Intellectual Property

- Our information technology systems, operational systems, security systems, infrastructure, communications networks, software integrated in our manufacturing and administrative processes, and customer data processed by us, third-party vendors or suppliers are subject to risks presented by cyber events, including incidents or breaches of security, any of which could prevent us or third parties we rely on from effectively operating our business, and could harm our reputation or materially adversely affect our company's assets, growth efforts, operations, facilities, business reputation or financial condition.
- We may be subject to information technology system failures, communications network disruptions and data breaches that are generally beyond our control.
- Our business is subject to complex and evolving global laws, regulations and security standards regarding data privacy, cybersecurity and data protection, which could result in claims or increased cost of operations, or other harm to our business.
- We may be unable to adequately maintain, enforce and protect our intellectual property and may not be able to prevent third parties from unauthorized access or use of our intellectual property, which may increase our cost of doing business or otherwise hurt our ability to compete in the market.
- If we fail to comply with our obligations under license or technology agreements with third parties or are unable to license rights to use technologies on reasonable terms or at all, we may be required to pay damages or could potentially lose license rights that are critical to our business.

Risks Related to Liquidity, Financial Instruments and Credit

- Changes in our credit profile, or a significant increase in the price of crude oil, may affect our relationship with our suppliers, which could have a material adverse effect on our liquidity and limit our ability to purchase sufficient quantities of crude oil to operate our refineries at desired capacity.
- We may not be able to obtain funding on acceptable terms or at all because of volatility and uncertainty in the credit and capital markets. This may hinder or prevent us from meeting our future capital needs.
- We are exposed to the credit risks, and certain other risks, of our key customers and vendors.
- Our credit facilities contain certain covenants and restrictions that may constrain our business and financing activities.
- Our hedging transactions may limit our gains and expose us to other risks.
- We may be unable to pay future dividends.

Additional risks and uncertainties not currently known to us or that we currently deem to be immaterial may also materially and adversely affect our business operations. If any of the following risks were to actually occur, our business, financial condition, results of operations could be materially and adversely affected. The headings provided in this Item 1A. are for convenience and reference purposes only and shall not affect or limit the extent or interpretation of the risk factors.

RISKS RELATED TO OUR BUSINESS/INDUSTRY

The prices of crude oil, renewable feedstocks and refined, finished lubricant and renewable diesel products materially affect our profitability, and are dependent upon many factors that are beyond our control, including general market demand and economic conditions, seasonal and weather-related factors, regional and grade differentials and governmental regulations and policies.

Among these factors is the demand for crude oil, renewable feedstocks (such as soybean oil), refined, finished lubricant and renewable diesel products, which can vary by type and class or product and is largely driven by the conditions of local and worldwide economies, as well as by weather patterns, changes in consumer preferences and the taxation of these products relative to other energy sources. Governmental regulations and policies, particularly in the areas of taxation, energy and the environment, also have a significant impact on our activities. Operating results can be affected by these industry factors, product and crude pipeline capacities, crude oil differentials (including regional and grade differentials), the price and availability of renewable feedstocks, changes in transportation costs, accidents or interruptions in transportation, competition in the particular geographic areas that we serve, global market conditions, actions by foreign nations and factors that are specific to us, such as the success of particular marketing programs and the efficiency of our refinery and facility operations. Developments in the global oil markets, such as actual or potential hostilities or other conflicts in oil producing areas, including shipping disruptions in the Red Sea, the Israel-Gaza conflict and the Russia-Ukraine war, and worldwide demand for crude oil, particularly in developing countries, can affect the prices of crude oil and result in inflated energy prices. The demand for crude oil and refined and finished lubricant products can also be reduced due to a local or national recession or other adverse economic condition, which results in lower spending by businesses and consumers on gasoline and diesel fuel, higher gasoline prices due to higher crude oil prices, a shift by consumers to more fuel-efficient vehicles or alternative fuel vehicles (such as ethanol or wider adoption of electric, gas/electric hybrid or hydrogen-powered vehicles), or an increase in vehicle fuel economy, whether as a result of technological advances by manufacturers, legislation mandating or encouraging higher fuel economy or the use of alternative fuel.

We do not produce crude oil or our renewable feedstocks and must purchase nearly all of the feedstocks we process, the price of which fluctuates based upon worldwide and local market conditions. The profitability of our Refining, Lubricants & Specialties and Marketing segments depends largely on the spread between market prices for refined petroleum products and crude oil prices. The profitability of our Renewables segment depends largely on the spread between market prices for renewable diesel plus state and federal low carbon fuel incentives and renewable feedstocks, such as soybean oil. This margin is continually changing and may fluctuate significantly from time to time. Crude oil and refined and renewable products are commodities whose price levels are determined by market forces beyond our control. For example, the reversal of certain existing pipelines or the construction of certain new pipelines transporting additional crude oil or refined products to markets that serve competing refineries could affect the market dynamic that has allowed us to take advantage of favorable pricing. In addition, the volume of renewable diesel produced by our competitors is expected to increase going forward, and as the market becomes more competitive, or if there are changes in the regulations, policies, and standards affecting the demand for low-carbon fuels or our ability to obtain approved fuel pathways, our Renewables segment may experience increased volatility in product margins. A deterioration of crack spreads or price differentials between domestic and foreign crude oils or renewable diesel product margins could have a material adverse effect on our business, financial condition, results of operations and cash flows.

Additionally, due to the seasonality of refined and renewable products markets and refinery maintenance schedules, results of operations for any particular quarter of a fiscal year are not necessarily indicative of results for the full year and can vary year to year in the event of unseasonably cool weather in the summer months and/or unseasonably warm weather in the winter months in the markets in which we sell our products. Commodity prices have, and in the future may, fluctuate due to adverse weather events, regulatory interventions and geo-political conditions. In general, prices for refined products are influenced by the price of crude oil and prices for renewable diesel are influenced by the price of renewable feedstocks. Although an increase or decrease in the price for crude oil or renewable feedstocks may result in a similar increase or decrease in prices for refined products or renewable diesel, there may be a time lag in the realization of the similar increase or decrease in prices for refined products or renewable diesel. The effect of changes in crude oil or renewable feedstock prices on operating results, therefore, depends in part on how quickly refined product or renewable diesel prices adjust to reflect these changes. A substantial or prolonged increase in crude oil or renewable feedstock prices without a corresponding increase in refined product or renewable diesel prices, a substantial or prolonged decrease in refined product or renewable diesel prices without a corresponding decrease in crude oil or renewable feedstock prices, or a substantial or prolonged decrease in demand for refined products or renewable diesel could have a significant negative effect on our earnings and cash flow. Also, our crude oil and refined and renewable diesel product inventories are valued at the lower of cost or market under the last-in, first-out ("LIFO") inventory valuation methodology. If the market value of our inventory were to decline to an amount less than our LIFO cost, we would record a write-down of inventory and a non-cash charge to cost of products sold even when there is no underlying economic impact at

that point in time. Continued volatility in crude oil and refined product or renewable diesel prices could result in lower of cost or market inventory charges in the future, or in reversals reducing cost of products sold in subsequent periods should prices recover.

Our operations are subject to catastrophic losses, operational hazards and unforeseen interruptions and other disruptive risks for which we may not be adequately insured.

Our operations are subject to catastrophic losses, operational hazards, unforeseen interruptions and other disruptive risks such as natural disasters, adverse weather, accidents, maritime disasters or casualties (including those involving marine vessels/terminals), fires, explosions, hazardous materials releases or spills (such as the release of crude oil on the Osage Pipeline in July 2022), terror or cyberattacks, vandalism, power failures, mechanical failures and other events beyond our control, and we have experienced certain of these events in the past. These events could result in an injury or loss of life, and have in the past and could in the future result in, property damage or destruction or curtailment or an interruption in our operations and may affect our ability to meet customer commitments. For example, historic spills along our existing pipelines and terminals as a result of past operations have resulted in contamination of the environment, including soils and groundwater. Additionally, third-party damage, mechanical malfunctions, undetected leaks in pipelines, faulty measurement or other errors may result in significant costs or lost revenues. Further, the consequences of any operational incident (including as a result of a maritime disaster or casualty) at our marine terminal facilities may be even more significant as a result of the complexities involved in addressing releases or spills occurring in U.S. federal and/or state waters (or in waters of other jurisdictions in which we operate) and/or the repair of marine terminal facilities.

We may not be able to maintain or obtain insurance of the type and amount we desire at commercially reasonable rates and exclusions from coverage may limit our ability to recover the amount of the full loss in all situations. As a result of market conditions, premiums and deductibles for certain of our insurance policies and insurance policies for our joint ventures are increasing. In some instances, certain insurance has become unavailable or has become available only for reduced amounts of coverage or at a significantly increased cost.

There can be no assurance that insurance will cover all or any damages and losses resulting from these types of hazards. We are not fully insured against all risks to our business and therefore, we self-insure certain risks. If any of our facilities were to experience an interruption in operations, our earnings could be materially adversely affected (to the extent not recoverable through insurance) because of lost production and repair costs.

The energy industry is highly capital intensive, and the entire or partial loss of individual facilities can result in significant costs to both industry companies, such as us, and their insurance carriers. In recent years, several large energy industry claims have resulted in significant increases in the level of premium costs and deductible periods for participants in the energy industry. As a result of large energy industry claims, insurance companies that have historically participated in underwriting energy-related facilities may discontinue that practice or demand significantly higher premiums or deductible periods to cover these facilities. If significant changes in the number or financial solvency of insurance underwriters for the energy industry occur, or if other adverse conditions over which we have no control prevail in the insurance market, we may be unable to obtain and maintain adequate insurance at reasonable cost. In addition, we cannot assure you that our insurers will renew our insurance coverage on acceptable terms, if at all, or that we will be able to arrange for adequate alternative coverage in the event of non-renewal. Further, our underwriters could have credit issues that affect their ability to pay claims. If a significant accident or event occurs that is self-insured or not fully insured, it could have a material adverse effect on our business, financial condition and results of operations.

A disruption to or proration of the product distribution systems or manufacturing facilities we utilize could negatively impact our profitability.

We utilize various third party pipeline systems to deliver our products from our refineries to market. The key third party pipeline systems utilized by the Casper, El Dorado, Navajo, Parco, Puget Sound, Woods Cross, and Tulsa Refineries are Magellan (RMPS), NuStar Energy Magellan (Mid-Con), SFPP, Pioneer, Olympic, MPLX, and Magellan (Mid-Con), respectively. Our refineries also utilize systems owned by our Midstream segment. If these key pipelines or their associated tanks and terminals become inoperative or decrease the capacity available to us due to testing, line repair, reduced operating pressures, catastrophic events, terror or cyberattacks, vandalism or other causes, we may not be able to sell our product, or we may be required to hold our product in inventory or supply products to our customers through an alternative pipeline or by rail or additional tanker trucks from the refinery, all of which could increase our costs and result in a decline in profitability.

Additionally, our Refining, Lubricants & Specialties and Renewables segments depend on rail and marine transportation for the delivery of feedstocks used in the production of our products and to deliver products to market, the availability of which is

subject to various risks, including those associated with rail or marine strikes, equipment shortages, operating hazards and transportation regulations. If rail or marine transportation is disrupted, we may be unable to produce and/or deliver our products in a competitive or profitable manner and, if such disruption were to occur over an extended period of time, it could have a material adverse effect on our business, financial condition and results of operations.

We have manufacturing facilities in foreign countries that support the Lubricants & Specialties segment. If one of our facilities is damaged or disrupted, resulting in production being halted for an extended period, we may not be able to timely supply our customers. We take steps to mitigate this risk, including business continuity and contingency planning and procuring property insurance (including resulting business interruption) and casualty insurance. Nevertheless, the loss of sales in any one region over an extended period of time could have a material adverse effect on our business, financial condition and results of operations.

A material decrease in the supply, or a material increase in the price, of crude oil, renewable feedstocks or other raw materials or equipment available to our refineries and other facilities could significantly reduce our production levels and negatively affect our operations.

To maintain or increase production levels at our refineries and facilities, we must continually contract for crude oil and renewable feedstock supplies from third parties. There are a limited number of crude oil and renewable feedstock suppliers in certain geographic regions, and in such cases, we may be required to source from a single third party supplier. If we are unable to maintain or extend our existing contracts with any such crude oil or renewable feedstock suppliers, or enter into new agreements on similar terms, the supply of crude oil or renewable feedstocks could be adversely impacted, or we may incur a higher cost. A material decrease in crude oil production from the fields that supply our refineries, as a result of depressed commodity prices, decreased demand, lack of drilling activity, natural production declines, governmental regulations, including travel bans and restrictions, quarantines, shelter in place orders, and shutdowns, catastrophic events or otherwise, could result in a decline in the volume of crude oil available to our refineries. As the volume of renewable diesel produced increases, competition for renewable feedstocks may also increase and result in an increase in feedstock costs and a decrease in renewable diesel margins. In addition, any prolonged disruption of a significant pipeline that is used in supplying crude oil to our refineries or the potential operation of a new, converted or expanded crude oil pipeline that transports crude oil to other markets could result in a decline in the volume of crude oil available to our refineries. Such an event could result in an overall decline in volumes of refined products processed at our refineries and therefore a corresponding reduction in our cash flow. In addition, the future growth of our operations will depend in part upon whether we can contract for additional supplies of crude oil or renewable feedstocks at a greater rate than the rate of natural decline in our currently connected supplies. If we are unable to secure additional crude oil supplies or renewable feedstocks of sufficient quality or crude pipeline expansion to our refineries, we will be unable to take full advantage of current and future expansion of our refineries' and renewable facilities' production capacities.

For certain raw materials and utilities used by our refineries and other facilities, there are a limited number of suppliers and, in some cases, we source from a single supplier and/or suppliers in economies that have experienced instability or the supplies are specific to the particular geographic region in which a facility is located. Any significant disruption in supply could affect our ability to obtain raw materials, or increase the cost of such raw materials, which could significantly reduce our production levels or have a material adverse effect on our business, financial condition and results of operations. In addition, certain raw materials that we use are subject to various regulatory laws, and a change in the ability to legally use such raw materials may impact our liquidity, financial position and results of operations.

It is also common in the refining industry for a facility to have a sole, dedicated source for its utilities, such as steam, electricity, hydrogen, water and gas. Having a sole or limited number of suppliers may limit our negotiating power, particularly in the case of rising raw material costs. Any new supply agreements we enter into may not have terms as favorable as those contained in our current supply agreements. Additionally, there is growing concern over the reliability of water sources. The decreased availability or less favorable pricing for water as a result of population growth, drought or regulation could negatively impact our operations.

Our Artesia RDU and Sinclair RDU are co-located with the Navajo Refinery and Parco Refinery, respectively, and their operations are dependent upon certain shared infrastructure at the co-located facilities. For example, the hydrogen plants at the Navajo Refinery and Parco Refinery support both refinery and renewable diesel operations. As a result, any disruption due to weather events, mechanical failure or other interruption that negatively impacts, or causes a shut down of, shared infrastructure at the co-located facilities could result in lost production and have a material adverse effect on earnings for both refinery and renewable diesel operations at the co-located facility. In addition, in the event equipment or raw materials at the co-located facilities are constrained, we may not have adequate inputs to support both refinery and renewable diesel operations and have in the past made, and may in the future have to make, commercial decisions that prioritize the continuing operation of one segment over the other in order to maximize earnings of our consolidated business.

In addition, from time to time, periods of disruption in the global supply chain, including as a result of COVID-19, shipping disruptions in the Red Sea, the Israel-Gaza conflict and the Russia-Ukraine war, have caused delays or shortages in the equipment and parts necessary to operate our facilities and complete our capital projects. Certain suppliers have experienced, and may continue to experience, delays related to a variety of factors, including logistical delays and component shortages from vendors. We continue to monitor the situation and work closely with our suppliers to minimize disruption to our operations as a result of supply chain interruptions.

If our raw material, utility, gas, hydrogen or water supplies or access to the equipment necessary to operate our facilities were disrupted, our businesses may incur increased costs to procure alternative supplies or equipment or incur excessive downtime, which would have a direct negative impact on our operations.

To successfully operate our facilities, we are required to expend significant amounts for capital outlays and operating expenditures. If we are unable to complete capital projects at their expected costs or in a timely manner, or if the market conditions assumed in our project economics deteriorate, our financial condition, results of operations, or cash flows could be materially and adversely affected.

Our facilities consist of many processing units, a number of which have been in operation for many years. One or more of the units may require unscheduled downtime for unanticipated maintenance or repairs that are more frequent than our scheduled turnaround for such units. Scheduled and unscheduled maintenance could reduce our revenues during the period of time that the units are not operating. We have taken significant measures to expand and upgrade units in our facilities by installing new equipment and redesigning older equipment to improve refinery capacity or to address changes in consumer preferences, such as the growing demand for renewable diesel and other lower carbon fuels. The installation and redesign of key equipment at our facilities involves significant uncertainties, including the following: our upgraded equipment may not perform at expected levels; operating costs of the upgraded equipment may be higher than expected; and the yield and product quality of new equipment may differ from design and/or specifications and redesign, modification or replacement of the equipment may be required to correct equipment that does not perform as expected, which could require facility shutdowns until the equipment has been redesigned or modified. Any of these risks associated with new equipment, redesigned older equipment, or repaired equipment could lead to lower revenues or higher costs or otherwise have a negative impact on our future financial condition and results of operations. For example, in the third quarter of 2020, we ceased refining operations at our Cheyenne Refinery due, in part, to uncompetitive operating and maintenance costs for the refinery.

One of the ways we may grow our business is through the construction of new refinery processing units (or the purchase and refurbishment of used units from another refinery), pipelines and terminals and the conversion or expansion of existing ones, such as the conversion of the Cheyenne Refinery to renewable diesel production and the connection of a new renewable diesel and a pre-treatment unit in Artesia, New Mexico. Projects are generally initiated to increase the yields of higher-value products, increase the amount of lower cost crude oils that can be processed, increase refinery production capacity, meet new governmental requirements or take advantage of new government incentive programs, or maintain the operations of our existing assets. Additionally, our growth strategy includes projects that permit access to new and/or more profitable markets, including the growing demand for renewable diesel and other lower carbon fuels. The construction process involves numerous regulatory, environmental, political, and legal uncertainties, most of which are not fully within our control, including:

- third party challenges to, denials, or delays with respect to the issuance of requisite regulatory approvals and/or obtaining or renewing permits, licenses, registrations and other authorizations;
- societal and political pressures and other forms of opposition;
- compliance with or liability under environmental or pipeline safety regulations;
- unplanned increases in the cost of construction materials or labor;
- disruptions in transportation of modular components and/or construction materials;
- severe adverse weather conditions, natural disasters, terror or cyberattacks, vandalism or other events (such as equipment malfunctions, explosions, fires or spills) affecting our facilities, or those of vendors and suppliers;

- shortages of sufficiently skilled labor, or labor disagreements resulting in unplanned work stoppages;
- market-related increases in a project's debt or equity financing costs; and/or
- nonperformance or force majeure by, or disputes with, vendors, suppliers, contractors, or sub-contractors involved with a project.

If we are unable to complete capital projects at their expected costs or in a timely manner, our financial condition, results of operations, or cash flows could be materially and adversely affected. Delays in making required changes or upgrades to our facilities could subject us to fines or penalties as well as affect our ability to supply certain products we make. Further, delays or cost increases related to capital spending programs involving construction of new pipeline facilities (or improvements and increased maintenance or repair expenditures on our existing pipeline facilities) could adversely affect our ability to achieve forecasted operating results. In addition, our revenues may not increase immediately upon the expenditure of funds on a particular project. Moreover, we may construct facilities to capture anticipated future growth in demand for refined products or renewable diesel in a region in which such growth does not materialize. As a result, new capital investments may not achieve our expected investment return, which could adversely affect our financial condition or results of operations.

In addition, we expect to execute turnarounds at a number of our refineries in 2024, which involve numerous risks and uncertainties, including delays and incurrence of additional and unforeseen costs. The turnarounds allow us to perform maintenance, upgrades, overhaul and repair of process equipment and materials, during which time all or a portion of the refinery will be under scheduled downtime.

Our forecasted internal rates of return are also based upon our projections of future market fundamentals which are not within our control, including changes in general economic conditions, available alternative supply, global market conditions, actions by foreign nations and customer demand.

Competition in the refining and marketing industry is intense, and an increase in competition in the markets in which we sell our products could adversely affect our earnings and profitability.

We compete with a broad range of refining and marketing companies, including certain multinational oil companies. Because of their geographic diversity, larger and more complex refineries, integrated operations and greater resources, some of our competitors may be better able to withstand volatile market conditions, to obtain crude oil in times of shortage and to bear the economic risks inherent in all areas of the refining industry.

We are not engaged in petroleum exploration and production activities and do not produce any of the crude oil feedstocks used at our refineries. We do not have a retail business and therefore are dependent upon others for outlets for our refined products. Certain of our competitors, however, obtain a portion of their feedstocks from company-owned production and have retail outlets. Competitors that have their own production or extensive retail outlets, with brand-name recognition, are at times able to offset losses from refining operations with profits from producing or retailing operations, and may be better positioned to withstand periods of depressed refining margins or feedstock shortages.

In recent years there have been several refining and marketing consolidations or acquisitions between entities competing in the geographies where we operate. These transactions could increase the future competitive pressures on us.

We may be impacted by competitors' plans for expansion projects and refinery improvements that could increase the production of refined products in our areas of operation and significantly affect our profitability.

Also, the potential operation of new or expanded refined product transportation pipelines, or the conversion of existing pipelines into refined product transportation pipelines, could impact the supply of refined products to the geographies we supply and negatively affect our profitability.

We compete with other industries that provide alternative means to satisfy the energy and fuel requirements of our industrial, commercial and individual consumers. The more successful these alternatives become as a result of governmental regulations, technological advances, consumer demand, improved pricing or otherwise, the greater the impact on pricing and demand for our products and our profitability. There are presently significant governmental and consumer pressures to increase the use of alternative fuels in the United States.

The market for our Lubricants & Specialties segment is highly competitive and requires us to continuously develop and introduce new products and product enhancements.

Our ability to grow our Lubricants & Specialties segment depends, in part, on our ability to continuously develop, manufacture and introduce new products and product enhancements on a timely and cost-effective basis, in response to customers' demands for higher performance process lubricants, coatings, greases and other product offerings. Our competitors may develop new products or enhancements to their products that offer performance, features and lower prices that may render our products less competitive or obsolete, and, as a consequence, we may lose business and/or significant market share. Our efforts to respond to changes in consumer demand in a timely and cost-efficient manner to drive growth could be adversely affected by unfavorable margins or difficulties or delays in product development and service innovation, including the inability to identify viable new products, successfully complete research and development, obtain regulatory approvals, obtain intellectual property protection or gain market acceptance of new products or service techniques. The development and commercialization of new products require significant expenditures over an extended period of time, and some products that we seek to develop may never become profitable, and we could be required to write-off our investments related to a new product that does not reach commercial viability.

Our business is subject to the risks of international operations.

We derive a portion of our revenue and earnings from international operations. Our acquisitions of the Petro-Canada Lubricants and Sonneborn businesses expanded our operations and sales to over 80 countries and increased our exposure to foreign exchange risks. Any significant change in the value of the currencies of the countries in which we do business against the U.S. dollar could affect our revenue, competitiveness and cost of doing business, which could have a material adverse effect on our business, financial condition and results of operations.

In addition, compliance with applicable U.S. and foreign laws and regulations, such as import and export requirements, economic or trade sanctions, anti-corruption laws, data privacy regulations and foreign exchange controls and cash repatriation restrictions, environmental laws, labor laws and anti-competition regulations, increase the cost of doing business in foreign jurisdictions. Although we have implemented policies and procedures to comply with these laws and regulations, a violation by any of our employees, contractors, distributors or agents could nevertheless occur. In some cases, compliance with the laws and regulations of one country could violate the laws and regulations of another country. Violations of these laws and regulations could materially adversely affect our company's brand, reputation, international growth efforts and business.

In addition, global market risks, actions by foreign nations and other international conditions, particularly in a time of increasing political, economic and global instability, may have a material adverse effect on our results and operations. The consequences of such uncertainty cannot be anticipated or quantified.

Negative publicity or an erosion of our business reputation could have a material adverse effect on our earnings, cash flows and financial condition.

Our reputation and our brands, including, without limitation, our existing Sinclair, HollyFrontier Specialty Products, Petro-Canada Lubricants, Red Giant Oil and Sonneborn brands, and any brands we may acquire or establish in the future, are an important corporate asset. Factors that could have a negative impact on our reputation and our brands include, by way of example and not limitation, an operating incident or significant cybersecurity disruption; changes in consumer views concerning our products; a perception by investors or others that we are making insufficient progress with respect to our carbon emission reduction goals, or that pursuit of this ambition may result in allocation of capital to investments with reduced returns; and other adverse events such as those described in this Item 1A. Negative impacts on our reputation and our brands could in turn make it more difficult for us to compete successfully for new opportunities, obtain necessary regulatory approvals, obtain financing, attract talent, or could reduce consumer demand for our branded products. Our reputation may also be harmed by events which negatively affect the image of our industry as a whole. The materialization of risks discussed in this section could negatively affect our reputation and could have a material adverse effect on our earnings, cash flows and financial condition.

Potential product, service or other related liability claims and litigation could adversely affect our business, reputation and results of operations.

A significant portion of our operating responsibility on refined product pipelines is to maintain the quality and purity of the products loaded at our loading racks. If our quality control measures were to fail, we may have contaminated or off-specification commingled pipelines and storage tanks or off-specification product could be sent to public gasoline stations. The development, manufacture and sale of renewable diesel and specialty lubricant products also involves an inherent risk of exposure to potential product liability claims. These types of incidents could result in product liability claims from our

customers. Our products could also be subject to false advertising claims, product recalls, workplace exposure, product seizures and related adverse publicity.

Any of these incidents is a significant commercial risk. Substantial damage awards have been made in certain jurisdictions against manufacturers and resellers based upon claims for injuries caused by the use of or exposure to various products. While we have received and resolved immaterial product liability claims in the past, there can be no assurance that future product liability claims against us would not have a material adverse effect on our business, reputation or results of operations or our ability to maintain existing customers or retain new customers. Although we maintain product and other general liability insurance, there can be no assurance that the types or levels of coverage maintained are adequate to cover these potential risks, or that we will be able to continue to maintain existing insurance or obtain comparable insurance at a reasonable cost, if at all.

Terrorist attacks, and the threat of terrorist attacks or vandalism, have resulted in increased costs to our business. Continued global hostilities or other sustained military campaigns may adversely impact our results of operations.

The long-term impacts of terrorist attacks and the threat of future terrorist attacks and vandalism on the energy transportation industry in general, and on us in particular, are unknown. Any attack on our facilities, those of our customers or suppliers and, in some cases, those of other pipelines, could have a material adverse effect on our business. Increased security measures taken by us as a precaution against possible terrorist attacks or domestic vandalism have resulted in increased costs to our business.

The U.S. government has issued public warnings that indicate that pipelines and other assets could be specific targets of terrorist organizations. These potential targets may include our pipeline systems or operating systems and might affect our ability to operate or control our pipeline assets or our operations could be disrupted. The occurrence of one of these events could cause a substantial decrease in revenues, increased costs to respond or other financial loss, damage to reputation, increased regulation or litigation and or inaccurate information reported from our operations. These developments may subject our operations to increased risks, as well as increased costs, and, depending on their ultimate magnitude, could have a material adverse effect on our business, financial condition and results of operations.

Uncertainty surrounding continued global hostilities or other sustained military campaigns, and the possibility that infrastructure facilities could be direct targets of, or indirect casualties of, an act of terror, may affect our operations in unpredictable ways, including disruptions of crude oil supplies and markets for refined products. In addition, disruption or significant increases in energy prices could result in government-imposed price controls. Any one of, or a combination of, these occurrences could have a material adverse effect on our business, financial condition and results of operations.

Changes in the insurance markets attributable to terrorist attacks, vandalism, or cyberattacks or extortion could make certain types of insurance more difficult for us to obtain. Moreover, the insurance that may be available to us may be significantly more expensive than our existing insurance coverage. Instability in the financial markets as a result of terrorism, cyberattacks, vandalism or war could also affect our ability to raise capital including our ability to repay or refinance debt.

Our business may suffer due to a change in the composition of our Board of Directors, or the departure of any of our key executives or other key employees. A shortage of skilled labor may make it difficult for us to maintain labor productivity.

Our future performance depends to a significant degree upon the continued contributions of our Board of Directors, our senior management team and key technical personnel. We do not currently maintain key person life insurance, or employment agreements with respect to any member of our senior management team. The loss or unavailability to us of any member of our senior management team or a key technical employee could significantly harm us. We face competition for these professionals from our competitors, our customers and other companies operating in our industry. To the extent that the services of members of our senior management team and key technical personnel would be unavailable to us for any reason, we may be required to hire other personnel to manage and operate our company. We may not be able to locate or employ such qualified personnel on acceptable terms, or at all.

Furthermore, our operations require skilled and experienced laborers with proficiency in multiple tasks. A shortage of trained workers due to retirements, an increase in labor costs as a result of inflation or otherwise could have an adverse impact on productivity and costs and our ability to expand production in the event there is an increase in the demand for our products and services, which could adversely affect our operations.

A portion of our workforce is unionized, and any disruptions in our labor force or adverse employee relations could adversely affect our business.

We depend on unionized labor for the operation of many of our facilities. As of December 31, 2023, approximately 31% of our employees were represented by a labor union or works council under collective bargaining or labor agreements with various expiration dates. In addition, employees who are not currently represented by a labor union or works council may seek representation in the future. We may not be able to renegotiate our collective bargaining or labor agreements when they expire on satisfactory terms or at all. If we are unable to renegotiate our collective bargaining or labor agreements when they expire, any work stoppages or other labor disturbances at these facilities could have an adverse effect on our business, impact our ability to pay dividends to our stockholders and make payments on our debt obligations, and increase our costs. In addition, our existing labor agreements may not prevent a strike or work stoppage or other adverse employee relations event at any of our facilities in the future, and any work stoppage could negatively affect our results of operations and financial condition.

The widespread outbreak of an illness or pandemic or other public health crisis, and actions taken in response thereto, may have a material adverse effect on our operations, business, financial condition and results of operations and cash flows.

The spread of an illness or pandemic across the globe, like the COVID-19 pandemic, and any governmental actions in response thereto could negatively affect worldwide economic and commercial activity, significantly impact global demand for oil, gas and refined products, and create significant volatility and disruption of financial and commodity markets.

The widespread outbreaks of illness or pandemics could cause us to modify our business practices from time to time as needed (including limiting employee and contractor presence at our work locations, restricting travel unless approved by senior leadership, quarantining employees when necessary and reducing utilization at our refineries), and could significantly disrupt our operations and ability to perform critical functions in the future.

The effects of such pandemics are difficult to predict and the duration of any potential business disruption or the extent to which such illnesses may negatively affect our operating results is uncertain. The effects of an illness or pandemic, such as COVID-19, volatility in global oil markets as well as the potential for a recession, while uncertain, have and may continue to, materially adversely affect our business, financial condition, results of operations and/or cash flows, as well as our ability to pay dividends to our stockholders.

Acquisitions involve numerous risks, any of which could adversely affect us.

An additional component of our growth strategy is to selectively acquire complementary assets or businesses for our existing assets and businesses in order to increase earnings and cash flow. Recent acquisitions include our acquisition of all of the remaining outstanding HEP common units, the Sinclair refining, renewables, midstream and marketing assets and the Puget Sound Refinery. Our ability to do so will be dependent upon a number of factors, including our ability to identify attractive acquisition candidates, consummate acquisitions on favorable terms, successfully integrate acquired assets and obtain financing to fund acquisitions and to support our growth, and other factors beyond our control. Risks associated with acquisitions include those relating to:

- diversion of significant management time and attention from our existing business;
- challenges in managing the increased scope, geographic diversity and complexity of operations and inefficiencies that may result therefrom;
- difficulties in integrating the financial, technological and management standards, processes, procedures and controls of an acquired business with those of our existing operations;
- the inability to integrate multiple acquisitions simultaneously or within a short timeframe of each other;
- difficulties integrating personnel from the acquired business while maintaining focus on providing consistent, high-quality products and services or the loss of key employees;
- difficulties integrating relationships with customers, vendors and business partners;
- liability for known or unknown environmental conditions or other contingent liabilities not covered by indemnification or insurance or potential unknown and unforeseen expenses, delays or regulatory conditions associated with such acquisitions;
- greater than anticipated expenditures required for compliance with environmental or other regulatory standards or for investments to improve operating results;
- difficulties or delays in achieving anticipated operational improvements or benefits or inaccurate assumptions about future synergies or revenues;
- incurrence of additional indebtedness to finance acquisitions or capital expenditures relating to acquired assets;
- issuance of additional equity, which could result in further dilution of the ownership interest of existing stockholders;

- disruption of, or the loss of momentum in, ongoing business or inconsistencies in standards, controls, procedures and policies;
- incurrence of substantial, nonrecurring transaction-related expenses; and
- potential securities class action and derivative lawsuits, which could result in substantial costs.

Any acquisitions that we do consummate may have adverse effects on our business and operating results.

Delays or difficulties in the integration process could adversely affect our business, financial results, financial condition and stock price. Even if we are able to integrate our business operations successfully, there can be no assurance that this integration will result in the realization of the full benefits of synergies, cost savings, innovation and operational efficiencies that we currently expect or have communicated from this integration or that these benefits will be achieved within the anticipated time frame.

Certain of our facilities, pipelines and assets are located on or adjacent to Native American tribal lands or on other lands which we do not own. Our operations are subject to potentially disruptive activity by those concerned with our industry.

Certain of our facilities, pipelines and other assets are located on or adjacent to Native American tribal lands. Various federal agencies, along with each Native American tribe, promulgate and enforce regulations, including environmental standards, regarding operations on Native American tribal lands. In addition, each Native American tribe is a sovereign nation having the right to enforce laws and regulations (including various taxes, fees, and other requirements and conditions) and to grant approvals independent from federal, state and local statutes and regulations. In 2020, the Supreme Court ruled in *McGirt v. Oklahoma* that the Muscogee (Creek) Nation reservation in Eastern Oklahoma has not been disestablished, and therefore retains jurisdiction over criminal matters, and a subsequent ruling in July 2022 in *Oklahoma v. Castro-Huerta* narrowed *McGirt*'s holding to find concurrent tribal and state jurisdiction with respect to crimes committed by non-Native Americans against Native Americans on tribal lands, creating substantial uncertainty with respect to matters over which tribes may have exclusive or concurrent jurisdiction. Although the ruling in *McGirt* indicates that it is limited to criminal law, the ruling has significant potential implications for civil law. At this time, we cannot predict how these jurisdictional issues may ultimately be resolved. Furthermore, our operations may be disrupted by restrictions on our access to railways and waterways on or adjacent to tribal lands, including, for example, through restrictions on the number of trains permitted to cross certain reservations. These factors may increase our cost of doing business on Native American tribal lands.

We do not own all of the land on which our pipeline systems and other assets are located, and we are, therefore, subject to the risk of increased costs or more burdensome terms to maintain necessary land use. We obtain the right to construct and operate pipelines and other assets on land owned by third parties and government agencies for specified periods. If we were to lose these rights through an inability to renew leases, right-of-way contracts or similar agreements, we may be required to relocate our pipelines or other assets and our business could be adversely affected. Additionally, it may become more expensive for us to obtain new rights-of-way or leases or to renew existing rights-of-way or leases. If the cost of obtaining or renewing such agreements increases, it may adversely affect our operations and cash flows.

The adoption or amendment of laws and regulations that limit or eliminate a state's ability to exercise eminent domain over private property in a state in which we operate could make it more difficult or costly for us to secure rights-of-way for future pipeline construction and other projects.

In addition, our industry is subject to potentially disruptive activities by those concerned with the possible environmental impacts of crude oil and refined products. Activists, non-governmental organizations and others may seek to restrict our operations or the transportation of crude oil and refined products by exerting social or political pressure. This interference could have a material adverse effect on our business, financial condition and results of operations

General economic conditions may adversely affect our business, operating results and financial condition.

Economic slowdowns may have serious negative consequences for our business and operating results because our performance is subject to domestic economic conditions and their impact on levels of consumer spending. Some of the factors affecting consumer spending include general economic conditions, unemployment, consumer debt, inflation, reductions in net worth based on declines in equity markets and residential real estate values, adverse developments in mortgage markets, taxation, energy prices, interest rates, consumer confidence and other macroeconomic factors. Political instability and global health crises, such as the COVID-19 pandemic, can also impact the global economy and decrease worldwide demand for oil and refined products. Increased volatility in the global oil markets, including the prices our customers or our joint ventures' customers pay for crude oil and other raw materials, has, and may continue to, materially adversely affect our business, financial condition, results of operations and/or cash flows.

Adverse developments in the global economy or in regional economies could also negatively impact our customers and suppliers, and therefore have a negative impact on our business or financial condition. In the event of adverse developments or stagnation in the economy or financial markets, our customers and suppliers may experience deterioration of their businesses, reduced demand for their products, cash flow shortages and difficulty obtaining financing. As a result, existing or potential customers might delay or cancel plans to use our services and may not be able to fulfill their obligations to us in a timely fashion. Further, suppliers may experience similar conditions, which could impact their ability to fulfill their obligations to us. Moreover, a financial market crisis may have a material adverse impact on financial institutions and limit access to capital and credit. This could, among other things, make it more difficult for us to obtain (or increase our cost of obtaining) capital and financing for our operations. Our access to additional capital may not be available on terms acceptable to us or at all.

Inflation has the potential to adversely affect our liquidity, business, financial condition and results of operations by increasing our overall cost structure, particularly if we are unable to achieve commensurate increases in the prices we charge our customers. Current and future inflationary pressures may be driven by, among other things, supply chain disruptions and governmental stimulus or fiscal policies. Continuing increases in inflation could affect the U.S. economy generally, the overall demand for our products and services, our costs for labor, material and services and our operating margins, all of which could have an adverse effect on our business, financial position, results of operations and cash flows. As a result of inflation, we have experienced and may continue to experience, cost increases. Although we may take measures to mitigate the impact of this inflation, if these measures are not effective our business, financial condition, results of operations and liquidity could be materially adversely affected. Even if such measures are effective, there could be a difference between the timing of when these beneficial actions impact our results of operations and when the cost inflation is incurred.

An impairment of our goodwill or long-lived assets could reduce our earnings or negatively impact our financial condition and results of operations.

An impairment of our goodwill or long-lived assets could reduce our earnings or negatively impact our results of operations and financial condition. We continually monitor our business, the business environment and the performance of our operations to determine if an event has occurred that indicates that a goodwill or long-lived asset may be impaired. If a triggering event occurs, which is a determination that involves judgment, we may be required to utilize cash flow projections to assess our ability to recover the carrying value based on the ability to generate future cash flows. We may also conduct impairment testing based on both the guideline public company and guideline transaction methods. Our goodwill and long-lived assets impairment analyses are sensitive to changes in key assumptions used in our analysis, estimates of future crack spreads, forecasted production levels, operating costs and capital expenditures. If the assumptions used in our analysis are not realized, it is possible a material impairment charge may need to be recorded in the future. We cannot accurately predict the amount and timing of any additional impairments of goodwill or long-lived assets in the future.

As market prices for refined products and market prices for crude oil continue to fluctuate, we will need to continue to evaluate the carrying value of our refinery reporting units. During the year ended December 31, 2020, we recorded long-lived asset impairment charges of $232.2 million that related to our Cheyenne Refinery, $26.5 million for construction-in-progress consisting primarily of engineering work for potential upgrades to certain processing units at our Tulsa and El Dorado Refineries and $204.7 million related to PCLI. Also, during the year ended December 31, 2020, we recorded a goodwill impairment charge of $81.9 million that related to Sonneborn. Although no impairment charges have been taken since 2020, a reasonable expectation exists that a deterioration in our operating results or overall economic conditions could result in an impairment of goodwill and/or additional long-lived asset impairments at some point in the future. Future impairment charges could be material to our results of operations and financial condition.

We sell many of our lubricants and specialties products through distributors, which presents risks that could adversely affect our operating results.

A large portion of our lubricants and specialties product sales, both in domestic and international markets, occur through distributors. As a result, we are dependent on these distributors to promote and create demand for our products. We cannot assure you that we will be successful in maintaining and strengthening our relationships with our distributors or establishing relationships with new distributors who have the ability to market, sell and support our products effectively. We may rely on one or more key distributors for a product or a region, and the loss of these distributors could reduce our revenue. The sales, business practices and reputation of our distributors may affect our business and our reputation. The consolidation of distributors, loss of a relationship with a distributor, significant disagreement with a distributor, or significant deterioration in the financial condition of a distributor could also have an adverse effect on our operating results and may also result in increased competition in the applicable jurisdiction.

The market price of our common stock may fluctuate significantly, and the value of a stockholder's investment could be impacted.

The market price of our common stock may be influenced by many factors, some of which are beyond our control, including:

- our quarterly or annual earnings or those of other companies in our industry;
- changes in accounting standards, policies, guidance, interpretations or principles;
- general economic, industry global and stock market conditions;
- the failure of securities analysts to cover our common stock or changes in financial estimates by analysts;
- future sales of our common stock;
- announcements by us or our competitors of significant contracts or acquisitions;
- sales of common stock by us, our senior officers, our affiliates or REH Company; and/or
- the other factors described in these Risk Factors, specifically, the Risk Factor titled "REH Company became a significant holder of our common stock following the completion of the Sinclair Transactions."

In recent years, the stock market has experienced extreme price and volume fluctuations. This volatility has had a significant impact on the market price of securities issued by many companies, including companies in our industry. The price of our common stock could fluctuate based upon factors that have little or nothing to do with our company, and these fluctuations could materially reduce our stock price.

REH Company became a significant holder of our common stock following the completion of the Sinclair Transactions.

As of February 15, 2024, REH Company owned approximately 13.3% of our common stock. In connection with the Sinclair Transactions, the REH Company stockholders agreed to certain customary registration rights. Subject to the filing of a resale registration statement or satisfaction of the requirements of Rule 144, REH Company and its stockholders may seek to sell their shares of common stock. These sales (or the perception that sales may occur) may affect the market for, and the market price of, our common stock in an adverse manner.

Additionally, pursuant to the stockholders agreement ("Stockholders Agreement") by and among HF Sinclair, REH Company and the stockholders of REH Company (together with REH Company and each of their permitted transferees, the "REH Parties"), the REH Parties are currently entitled to nominate one person to our Board of Directors for so long as the REH Parties beneficially own less than 15% but more than or equal to 5% of all outstanding common stock. As a result, REH Company (and the REH Parties) has the ability to influence our management and affairs. Further, the existence of a significant stockholder may have the effect of deterring hostile takeovers, delaying or preventing changes in control or changes in management, or limiting the ability of our other stockholders to approve transactions that they may deem to be in the best interests of our company.

So long as REH Company continues to control a significant amount of our common stock, it will continue to be able to influence all matters requiring stockholder approval, subject to the voting agreements of the REH Parties set forth in the Stockholders Agreement. Moreover, this concentration of stock ownership may also adversely affect the trading price of our common stock to the extent investors perceive a disadvantage in owning stock of a company with a significant stockholder.

RISKS RELATED TO GOVERNMENT REGULATION

We are subject to significant regulation and oversight by governmental agencies.

Regulation affects almost every part of our business. For instance, we are subject to laws and regulations related to working conditions, health and safety, equal employment opportunity, employee benefit and other labor and employment matters, and competition and antitrust matters. Our facilities, pipelines, and other operations are subject to regulation and oversight by international, federal, state, provincial and local regulatory authorities, including the FERC, Commodities Futures Trading Commission, EPA, PHMSA, OSHA, the SEC and the United States Department of Justice, and similar authorities in Canada and the Netherlands, each of which may impose significant civil and criminal penalties to enforce compliance with its requirements. Any such regulatory violations could have a material adverse effect on our financial operating results including earnings, cash flow and liquidity. Further, our financial results may be materially affected by the adoption of new or amended financial accounting standards, and regulatory or outside auditor guidance or interpretations.

New laws, policies, regulations, rulemaking and oversight, unexpected policy changes or interpretations of existing laws or regulations as well as changes to those currently in effect, could adversely impact our earnings, cash flows and operations. Legislative changes, as well as regulatory actions taken by these agencies, have the potential to adversely affect our profitability. Furthermore, we could incur additional costs to comply with such statutes, rules, regulations and orders. Should we fail to comply with any applicable statutes, rules, regulations, and orders of regulatory authorities, we could be subject to substantial penalties and fines and withdrawal or denial of permits to operate.

We incur significant costs and liabilities, and expect to incur additional costs and liabilities in the future, resulting from compliance with existing, new and changing environmental, health and safety laws and regulations, and face potential exposure for environmental matters.

Operations of our facilities, pipelines and distribution operations are subject to international, federal, state, provincial and local laws and regulations regarding, among other things, the manufacture, storage, handling, use, transportation and distribution of petroleum and hazardous substances by pipeline, truck, rail, ship and barge, the emission and discharge of materials into the environment, waste management, and characteristics and composition of gasoline and diesel fuels, and other matters otherwise relating to the protection of human health and the environment, including climate change. Permits or other authorizations are required under these laws and regulations for the operation of our facilities, pipelines and other operations, and these permits and authorizations are subject to revocation, modification and renewal or may require operational changes, which may involve significant costs. A violation of permit conditions or a failure to comply with applicable laws and regulations may result in the assessment of sanctions, including administrative, civil and criminal penalties; the imposition of investigatory, remedial or corrective action obligations and the incurrence of capital expenditures; the occurrence of delays in the permitting or the denial of permits, development or expansion of projects; the issuance of injunctive relief limiting or prohibiting certain operations; and reputational harm. There is also the potential for liability for spill response and remediation, natural resource damage claims, and personal and property damage claims in the event of an oil or other refined product spill at a facility located near federal, state or provincial waters or a release of hazardous or other substances into the environment.

Changes in laws or regulations could require major modifications of our operations, including expensive upgrades to our existing pollution control equipment, which could have a material adverse effect on our business, financial condition, or results of operations. For example, implementation of the revised NAAQS for ozone could result in stricter permitting requirements, a delay in or inability to obtain required permits, and increased expenditures for pollution control equipment, the costs of which could be significant. Also, an EPA rule became effective in January 2018 that requires, among other things, benzene monitoring at refinery fence lines and submittal of fence line monitoring data to the EPA on a quarterly basis; upgraded storage tank controls requirements, including new applicability thresholds; enhanced performance requirements for flares, continuous monitoring of flares and pressure release devices, and analysis and remedy of flare release events; compliance with emissions standards for delayed coking units; and requirements related to air emissions resulting from startup, shutdown and maintenance events. These rules, as well as subsequent rulemaking under the CAA or similar laws, or new agency interpretations of existing laws and regulations, may necessitate additional expenditures in future years and result in increased costs on our operations. For example, in 2022, the EPA proposed an amendment to the risk management program rules that would, among other provisions, require refineries with HF Alkylation process (including three of our refineries) to perform a “safer technology and alternatives analysis” as part of the process hazard analysis to consider and document the practicality of inherently safer technologies and other risk management measures. If finalized, the analysis of our HF Alkylation processes under such a rule may lead to capital expenditures in future years or otherwise constrain our operations. Updated or new determinations under the Endangered Species Act and comparable international, federal, state, provincial and local laws and regulations could also impact our operations or those of our suppliers. Our operations and those of our suppliers could also be impacted by new or revised federal restrictions or laws pertaining to oil and gas operations on federal lands, which could include pauses on leasing, enhanced

environmental reviews, and emissions regulations. Compliance with new international and domestic environmental laws, regulations and interpretations will continue to have an adverse impact on our operations, results of our operations and capital requirements.

As is the case with all companies engaged in industries similar to ours, we face potential exposure to future claims and lawsuits involving environmental matters. The matters include, but are not limited to, soil, groundwater and water discharges and contamination, air pollution, accident prevention and personal injury and property damage allegedly caused by substances which we processed, manufactured, handled, used, transported, stored, released or disposed and climate change.

We are, and have been, the subject of various local, state, provincial, federal, international and private proceedings relating to environmental regulations, conditions and inquiries. Instances of non-compliance with environmental regulations could require additional expenditures, including expenditures for investigation and remediation, which could be significant.

Our operations are also subject to various international and domestic laws and regulations relating to health and safety, including, for example, chemical accident prevention. We maintain safety, training and maintenance programs as part of our ongoing efforts to comply with applicable laws and regulations but cannot guarantee that these efforts will always be successful. Compliance with applicable health and safety laws and regulations has required and continues to require substantial expenditures. Failure to appropriately manage health and safety risks associated with our business could also adversely impact our employees, communities, stakeholders, reputation and results of operations.

The costs of environmental, health and safety regulations are already significant and compliance with more stringent laws or regulations or adverse changes in the interpretation of existing regulations by government agencies or courts could have an adverse effect on the financial position and the results of our operations and could require substantial expenditures for the installation and operation of systems and equipment that we do not currently possess.

We are also subject to existing, and may in the future be subject to new or changing, domestic and international energy policy legislation. For example, in the United States, the Energy Independence and Security Act mandates annually increasing levels for the use of renewable fuels such as ethanol and increasing energy efficiency goals, among other steps. Dutch policy also aims to increase the share of renewable energy used in transportation and reduce GHG emissions from transportation fuels. In Canada, fuel content legislation exists at the federal and provincial level. These statutory mandates may have the impact over time of offsetting projected increases in the demand for refined petroleum products, particularly gasoline, in certain markets. In the near term, the increasing RFS obligations, as discussed in our Risk Factor titled “Compliance with, or developments with respect to, renewable and low carbon fuel blending programs, and other regulations, policies and standards impacting the demand for low-carbon fuels could have an adverse effect on our financial condition and results of operations” below, present ethanol production and logistics challenges for both the ethanol and refining industries and may require additional capital expenditures or expenses by us to accommodate increased ethanol use. Other legislative changes may similarly alter the expected demand and supply projections for refined petroleum products in ways that cannot be predicted.

For additional information on regulations and related liabilities or potential liabilities affecting our business, see “Regulation” under Items 1 and 2, “Business and Properties,” and Item 3, “Legal Proceedings.”

We may incur significant costs and liabilities resulting from performance of pipeline integrity programs and related repairs.

We are regulated under federal pipeline safety statutes by DOT through PHMSA. PHMSA sets and enforces pipeline safety regulations. Failure to comply with PHMSA or comparable state pipeline safety regulations could result in a number of consequences which may have a materially adverse effect on our operations. PHMSA’s enforcement authority includes the ability to assess civil penalties for violations of pipeline safety regulations, issue orders directing compliance, and issue orders directing corrective action to abate hazardous conditions. Among other things, pipeline safety laws and regulations require pipeline operators to develop integrity management programs, including more frequent inspections and other measures for pipelines located in “high consequence areas,” which are areas where a release could have the most significant adverse consequences, including certain population areas, certain drinking water sources and unusually sensitive ecological areas. These regulations require operators of covered pipelines to perform a variety of heightened assessments, analysis, prevention, mitigation and repair activities on the segments of pipe located within high consequence areas. Routine assessments under the integrity management program may result in findings that require repairs or other actions.

Moreover, changes to pipeline safety laws by Congress and regulations by PHMSA or states that result in more stringent or costly pipeline integrity management or safety standards could possibly have a substantial effect on us and similarly situated midstream operators. Congress is currently considering potential amendments to the Pipeline Safety Act, as it does approximately every four years. Should amendments be introduced and later become law, this could affect our operations and

result in additional compliance, maintenance and other obligations. Additionally, PHMSA adopted new regulations related to Valve Installation and Minimum Rupture Detection Standards, which became effective on October 5, 2022. These regulations expand PHMSA's regulation of the safety of hazardous liquid pipelines by establishing certain new procedural and notification requirements for managing rupture events and requiring the installation of rupture-mitigation valves on new or certain replaced pipelines. This final rule may result in additional capital and operations and maintenance costs in the coming years. Furthermore, where PHMSA has not issued any legal requirements, state agencies, to the extent authorized, could enact regulatory standards for certain pipelines.

We perform preventive and normal maintenance on all of our pipeline and terminal systems and make repairs and replacements when necessary or appropriate. We also conduct routine and required inspections of our pipelines and other assets as required by regulations. Corrosion inhibitors, external coatings and impressed current cathodic protection systems are used to protect against internal and external corrosion. We regularly monitor, test and record the effectiveness of these corrosion-control systems. We monitor the structural integrity of covered segments of our pipeline systems through a program of periodic internal inspections using electronic "smart pigs," hydrostatic testing, and other measures. We follow these inspections with a review of the data, and we make repairs as necessary to maintain the integrity of the pipeline. We have initiated a risk-based approach to prioritizing the pipeline segments for future smart pig runs or other appropriate integrity testing methods. This approach is intended to allow the pipelines that have the greatest risk potential to receive the highest priority in being scheduled for inspections or pressure tests for integrity. Nonetheless, the adoption of new or amended regulations or the reinterpretation of existing laws and regulations by PHMSA or states that result in more stringent or costly pipeline integrity management or safety standards could possibly have a substantial effect on us and similarly situated midstream operators.

There are various risks associated with GHGs and climate change that could result in increased operating costs and litigation and reduced demand for the refined products we produce and investment in our industry.

Climate change continues to attract considerable attention in the United States, Canada, Europe, and other regions. Numerous proposals have been made and could continue to be made at the international, national, regional and state levels of government to monitor and limit existing emissions of GHGs, to limit or eliminate future emissions, and to require or incentivize the use of lower carbon or renewable alternatives, including through accounting and risk disclosures. As a result, our operations, and those of our customers, are subject to a series of regulatory, political, litigation, and financial risks associated with the refining, transportation and use of petroleum products and emission of GHGs.

The EPA has adopted rules that, among other things, establish construction and operating permit reviews for GHG emissions from certain large stationary sources, require the monitoring and annual reporting of GHG emissions from certain petroleum and natural gas sources in the United States or require control or reduction of emissions of GHGs, including methane, from such sources.

In 2021, President Biden issued several executive orders that committed to substantial action on climate change, released "The Long-Term Strategy of the United States: Pathways to Net-Zero Greenhouse Gas Emissions by 2050", which established a roadmap to net zero emissions in the United States by 2050 and called for, among other things, improving energy efficiency, decarbonizing energy sources, the increased use of zero-emission vehicles by the federal government, the elimination of subsidies provided to the fossil fuel industry, and increased emphasis on climate-related risks across governmental agencies and economic sectors. In 2023, the EPA finalized a rule to further reduce GHGs and other emissions primarily from oil production and natural gas production, transmission and storage that could increase the cost of domestic crude oil and natural gas. The EPA also proposed rules in 2023 to regulate GHG emissions from power plants that could increase the cost of electricity use by our facilities. President Biden also reinstated the Interagency Working Group on the Social Cost of Greenhouse Gases in 2021 and directed the group to publish interim estimates of the social cost of GHGs, which it did. In November 2022, the EPA published a draft report assigning new and higher social cost values to GHG emissions for use in its rulemaking initiatives. In September 2023, the Biden Administration directed federal agencies to begin considering the social cost of GHG emissions in procurement processes and environmental reviews under the National Environmental Policy Act. Additionally, various states and groups of states have adopted or are considering adopting legislation, regulations or other regulatory initiatives that are focused on such areas as GHG cap and trade programs, carbon taxes, reporting and tracking programs, restriction of emissions, electric vehicle mandates and combustion engine phaseouts. See also "*Compliance with, or developments with respect to, renewable and low carbon fuel blending programs, and other regulations, policies, and standards impacting the demand for low carbon fuels could have an adverse effect on our financial condition and results of operations*" below for more information on how these programs may impact us.

In March 2022, the SEC issued proposed rules that, if adopted, would require public companies to include certain climate-related disclosures in their registration statements and periodic reports, including information about climate-related risks, climate-related financial statement metrics, and greenhouse gas emissions. In November 2022, the Biden Administration issued

a proposed rule that would require government contractors to publicly disclose their greenhouse gas emissions and set emissions reduction targets, which could affect us if we enter into contractual and business arrangements with government contractors. At the state level, in October 2023, the Governor of California signed the Climate Corporate Data Accountability Act ("CCDAA") and Climate-Related Financial Risk Act ("CRFRA") into law. The CCDAA requires both public and private U.S. companies that are "doing business in California" and that have a total annual revenue of $1 billion to publicly disclose and verify, on an annual basis, Scope 1, 2, and 3 GHG emissions. The CRFRA requires the disclosure of a climate-related financial risk report (in line with the Task Force on the Climate-Related Financial Disclosures recommendations or equivalent disclosure requirements under the International Sustainability Standards Board's climate-related disclosure standards) every other year for public and private companies "doing business in California" with a total annual revenue of $500 million. Reporting under both laws would begin in 2026. Non-compliance with these new laws may result in the imposition of substantial fines or penalties. Additionally, other states are considering similar laws. Any new laws or regulations imposing more stringent requirements on our business related to the disclosure of climate-related risks may result in (i) reputational harms among certain stakeholders if they disapprove of our reported GHG emissions data or disagree with our approach to mitigating climate-related risks, (ii) increased compliance costs resulting from the development of any such disclosures, and (iii) increased costs of and restrictions on access to capital to the extent that we do not meet any climate-related expectations or requirements of financial institutions. Additionally, similar climate reporting legislation is being considered in other states, and if any such new laws come into effect, may pose similar risks and costs to our business.

Similar regulations exist at the provincial and federal levels in Canada, including a nation-wide GHG pricing initiative and regulations related to the control of GHGs from vehicles as well as industrial facilities and either cap and trade programs or carbon taxes at both federal and provincial levels. Canada's 2030 Emissions Reduction Plan issued pursuant to the Canadian Net-Zero Emissions Accountability Act, includes a projected contribution from the oil and gas sector of emissions reductions to 31% below 2005 levels in 2030 (or to 42% below 2019 levels) with an ultimate goal of achieving net zero emissions by 2050. At this time, the applicability to, and impact on, if any, our Canadian operations is not clear.

The Netherlands also participates in European legal initiatives, including GHG cap and trade programs. Furthermore, the Climate Act has come into effect with the goal of significantly reducing GHG emissions by 55% (compared to 1990) by 2030 and to be climate neutral by 2050. The Climate Act also establishes that the government must prepare a Climate Plan. This plan contains, among others, the principles by which the Dutch government intends to achieve the goals set out in the Climate Act.

At the international level, the United Nations-sponsored "Paris Agreement" requires member nations to limit their GHG emissions through nationally-determined reduction goals reevaluated every five years after 2020. The United States initially joined and then withdrew from such agreement in 2020. The United States rejoined the Paris Agreement in 2021 and issued its corresponding NDC to reduce economy-wide net GHG emissions to 50-52% below 2005 levels by 2030. While the NDC does not identify specific actions necessary to achieve these reductions, it lists several sectors as pathways for reductions, including the power, transportation, building, industrial, and agricultural sectors. The administration has acknowledged that a combination of regulatory actions and legislation will be necessary to achieve the U.S. NDC. European Union ("EU") member states have agreed to reduce GHG emissions by at least 55% reduction by 2030. They have also agreed to be climate neutral and climate sustainable by 2050 (net zero). Moreover, following the United States' reentry into Paris Agreement, the United States, the EU and other partner countries have announced further pledges and agreements at several United Nations Climate Change Conference of Parties, including a pledge to reduce global methane emissions by 30% from 2020 to 2030 and a commitment to develop international standards for monitoring and reporting on methane emissions. Most recently, at the 28th Conference of the Parties (COP28), participants agreed to transition "away from fossil fuels in energy systems in a just, orderly, and equitable manner" and increase renewable energy capacity so as to achieve net zero by 2050, although no timeline for doing so was set. Such international pledges and treaties may cause signatory nations to increase regulations, taxes or restrictions, or subsidize competitive industries, which could have a detrimental impact on our industry in general and our business in particular.

The adoption of legislation or regulatory programs to reduce emissions of GHGs could require us to incur increased operating costs, such as costs to purchase and operate emissions control systems and electricity, to acquire emissions allowances or comply with new regulatory or reporting and disclosure requirements or otherwise result in decreased demand for the petroleum products we refine and produce. For example, in November 2021, the United States enacted a nearly $1 trillion bipartisan infrastructure law, which provided significant funding for electric vehicles and clean energy technologies, and in August 2022 the United States enacted the Inflation Reduction Act of 2022, which allocated $369 billion to climate change and environmental initiatives, including transportation electrification, fees on and greater regulation of methane emissions, and financial incentives for low or zero-carbon forms of energy, products, or processes, which could result in changes in consumer preferences or otherwise increase competition within our industry. In addition, several states have also taken steps to incentivize the production of electric vehicles or otherwise limit the sale of gasoline or diesel-powered vehicles. These and any future legislation or regulatory programs could increase the cost of consuming or otherwise reduce demand for, the refined petroleum

products that we produce and transport. Additionally, political, litigation and financial risks may result in curtailed refinery activity, increased liability, or other adverse effects on our business, financial condition and results of operations.

There are also increasing risks of litigation related to climate change effects. Governments and third-parties have brought suit against some fossil fuel companies alleging, among other things, that such companies created public and private nuisances by producing fuels that contributed to climate change, such as rising sea levels, and therefore are responsible for resulting roadway and infrastructure damages, or alleging that the companies have been aware of the adverse effects of climate change for some time but defrauded their investors or customers by failing to adequately disclose those impacts. In 2022, the EPA announced an ongoing enforcement initiative targeting GHG compliance. In the Netherlands, The Hague District Court has ordered Royal Dutch Shell ("RDS") to reduce the CO2 emissions of the RDS group by net 45% by 2030, compared to 2019 levels, through the RDS group's corporate policy. Although this judgment is still subject to appeal (which is expected to be heard by the Courts of Appeal in the spring of 2024), other environmental organizations may bring similar cases against other parties. In Canada, in March 2023, Greenpeace filed a complaint with the Competition Bureau of Canada alleging that Pathways Alliance, a coalition of six of Canada's largest oil sand producers, made false and misleading claims that they are actively reducing emissions and helping Canada achieve its climate targets. The remedies being sought include a fine in the amount of the greater of CAN 10 million or 3% of worldwide gross revenues. While we are not party to such suits at this time, we may become subject to similar litigation in the future. Such cases could also adversely impact public perception and the demand for fossil fuels and petroleum products, which could subsequently result in decreased demand for our services and refined products and a drop in our stock price.

In addition, increasing societal expectations for companies to address environmental issues, including climate change, and increased consumer use of substitutes to energy commodities may result in increased costs, reduced demand for our products and our services, reduced profits, increased investigations and litigation, and negative impacts on our stock price and costs of and access to capital markets. To the extent that societal pressures or political or other factors are involved, it is possible that such liability could be imposed on us without regard to our causation of or contribution to the asserted damage, or to other mitigating factors.

Furthermore, our stock price could be adversely impacted if existing shareholders, including institutional investors, elect in the future to shift some or all of their investments into renewable energy or non-energy related sectors. Additionally, there is the possibility that financial institutions may be pressured or required to adopt policies that limit funding for fossil fuel energy companies. The Glasgow Financial Alliance for Net Zero ("GFANZ") announced commitments from over 550 firms around the world, resulting in over $130 trillion in capital committed to net zero goals. The various sub-alliances of GFANZ generally require participants to set short-term, sector-specific targets to transition their financing, investing, and/or underwriting activities to net zero emissions by 2050. More recently, in October 2023, the Federal Reserve, the Office of the Comptroller of the Currency and the Federal Deposit Insurance Corp. released a finalized set of principles guiding financial institutions with $100 billion or more in assets on the management of physical and transition risks associated with climate change. Moreover, institutional lenders have been lobbied intensively, and often publicly, by environmental activists, advocates of international climate change treaties and agreements, proponents of the international Paris Agreement, and foreign citizenry concerned about climate change not to provide funding for fossil fuel energy companies. Accordingly, limitation of investments in and financings for fossil fuel energy companies or subsidies or other governmental support for competitive technologies or industries could result in the direct or indirect restriction, delay or cancellation of drilling programs or development or production activities, could result in a reduction of available capital funding for potential development projects and could also adversely affect demand for our services and refined petroleum products, all of which could adversely impact our future financial results.

Increasing attention to ESG matters may adversely impact our business, financial results, stock price or price of debt securities.

In recent years the investment community, including investment advisors, sovereign wealth funds, pension funds, universities, financial institutions, including institutional banks, lenders, and insurance companies, and other groups have become more attentive to ESG and sustainability related practices and have been lobbied intensively, and often publicly, by environmental activists concerned about climate change to limit or curtail activities with fossil fuel energy companies. There has also been an increase in third-party providers of company ESG ratings, and more ESG-focused voting policies among proxy advisory firms, portfolio managers, and institutional investors. As a result, some investors, funds, financial institutions and other capital markets participants may screen companies such as ours for ESG performance before investing in our common stock or debt securities, or lending to us or have imposed restrictions upon or otherwise limited lending to, investing in, or providing insurance coverage for, companies that operate in industries with higher perceived environmental exposure, such as the energy industry. If we are unable to meet the ESG standards or investment, lending, ratings, or voting criteria and policies set by these parties, we may lose investors, investors may allocate a portion of their capital away from us, we may become a target for ESG-

focused activism, we may face increased costs of or limitations on access to capital or insurance necessary to sustain or grow our business, the price of our common stock or debt securities may be adversely impacted, demand for our services and refined petroleum products may be adversely impacted, and our reputation may be adversely affected, all of which could adversely impact our future financial results. Furthermore, there has recently been backlash from certain governments and investors against ESG funds and investment practices, which has resulted in increased scrutiny and withdrawals from such funds. Such backlash has also resulted in "anti-ESG" focused activism and investment funds. If we are unable to meet the often conflicting ESG standards or investment, lending, ratings, or voting criteria and policies set by these parties, we may lose investors, investors may allocate a portion of their capital away from us, we may face increased ESG- or anti-ESG-focused activism, our cost of capital may increase, and our reputation may also be negatively affected.

In recent years, members of the investment community have also increased their focus on ESG practices and disclosures, including those related to climate change, GHG emissions targets, business resilience under the assumptions of demand-constrained scenarios, and net-zero ambitions in the energy industry in particular, as well as diversity, equity, and inclusion initiatives, political activities, and governance standards among companies more generally. As a result, we may face increasing pressure or negative publicity regarding our ESG practices and disclosures and demands for ESG-focused engagement from investors, stakeholders, and other interested parties. This could result in higher costs, disruption and diversion of management attention, an increased strain on our resources, and the implementation of certain ESG practices or disclosures that may present a heightened level of legal and regulatory risk, or that threaten our credibility with other investors and stakeholders.

Investors, stakeholders and other interested parties are also increasingly focusing on issues related to environmental justice. This may result in increased scrutiny, protests, and negative publicity with respect to our business and operations, and those of our counterparties, which could in turn result in the cancellation or delay of projects, the revocation or delay of permits, termination of contracts, lawsuits, regulatory action, and policy change that may adversely affect our business strategy, increase our costs, adversely affect our reputation and financial performance, and impact our ability to attract and retain employees.

Furthermore, public statements with respect to ESG matters, such as emissions reduction goals or progress, other environmental targets or other commitments addressing certain social issues, are becoming increasingly subject to heightened scrutiny from public and governmental authorities related to the risk of potential "greenwashing," i.e., misleading information or false claims overstating potential ESG benefits. For example, in March 2021, the SEC established the Climate and ESG Task Force in the Division of Enforcement to identify and address potential ESG-related misconduct, including greenwashing. Certain non-governmental organizations and other private actors have also filed lawsuits under various securities and consumer protection laws alleging that certain ESG-statements, goals or standards were misleading, false or otherwise deceptive. As a result, we may face increased litigation risks from private parties and regulatory enforcement from governmental authorities related to our ESG efforts. Additionally, we could face increasing costs as we attempt to comply with and navigate further regulatory focus and scrutiny. Furthermore, we could be criticized by various anti-ESG stakeholders for the scope of our climate or ESG-related goals or policies, our strategic choices regarding ESG matters as they may impact our operations now or in the future, or for any revisions to the same, as well as initiatives we may pursue or any public statements we may make. We could be subjected to negative responses by governmental actors (such as anti-ESG legislation or retaliatory legislative or administrative treatment) or consumers (such as boycotts or negative publicity campaigns), which could adversely affect our reputation, business, financial performance, market access and growth.

For financial years beginning on or after January 1, 2024, certain undertakings that are incorporated in the EU (including EU-incorporated subsidiaries of non-EU incorporated parent entities) will be subject to ESG reporting requirements under the EU's CSRD, which requires in-scope companies to disclose extensive sustainability information, including disclosing risks and opportunities arising from environmental and social matters, and the impact of their business on people and the environment. Reporting requirements under CSRD will be implemented in stages from financial years beginning in 2024 through 2028 for different categories of companies, and certain of our subsidiaries are expected to have to begin reporting from 2026 onward for financial years beginning on January 1, 2025. Ultimate parent companies that are not incorporated in the EU, such as HF Sinclair, may be required to report under CSRD from 2029 onward for financial years beginning on or after January 1, 2028. CSRD and similar laws and regulations could result in increased compliance costs and affect the results of our operations and financial position. Such laws and similar regulations could also increase our litigation risks or may increase risks related to our reputation or goodwill as we cannot predict how additional reporting under these laws may be perceived or interpreted by our customers and stakeholders.

Compliance with, or developments with respect to, renewable and low carbon fuel blending programs, and other regulations, policies, and standards impacting the demand for low-carbon fuels could have an adverse effect on our financial condition and results of operations.

As described under Items 1 and 2. "Business and Properties – Additional Operations and Other Information – Governmental Regulation," many international, federal, state, provincial and local governments have issued, or are considering issuing, low carbon fuel regulations, policies, and standards to help reduce GHG emissions and increase the percentage of low-carbon fuels in the transportation fuel mix.

Pursuant to the 2007 Energy Independence and Security Act, the EPA promulgated the RFS regulations reflecting the increased volume of renewable fuels mandated to be blended into the nation's fuel supply. The regulations, in part, require refiners to add annually increasing amounts of "renewable fuels" to their petroleum products or purchase credits, known as RINs, in lieu of such blending. We currently purchase RINs for some fuel categories on the open market in order to comply with the quantity of renewable fuels we are required to blend under the RFS regulations. Since the EPA first began mandating biofuels in excess of the "blend wall" (the 10% ethanol limit prescribed by most automobile warranties), the price of RINs has been extremely volatile. While we cannot predict the future prices of RINs, the costs to obtain the necessary number of RINs could be material. If we are unable to pass the costs of compliance with the RFS regulations on to our customers, if sufficient RINs are unavailable for purchase, if we have to pay a significantly higher price for RINs or if we are otherwise unable to meet the RFS mandates, our financial condition and results of operations could be adversely affected.

In the past, we have received small refinery exemptions under the RFS program for certain of our refineries. However, there is no assurance that such an exemption will be obtained for any of our refineries in future years. For example, the EPA has recently denied all pending small refinery exemption petitions on the belief that small refineries are able to pass through compliance costs to customers. Based on this new approach to evaluating exemption petitions, the EPA has indicated that it is unlikely to grant such exemptions in future years. The failure to obtain such exemptions for certain of our refineries could result in the need to purchase more RINs than we currently have estimated and accrued for in our consolidated financial statements.

In addition, the RFS regulations are highly complex and evolving, requiring us to periodically update our compliance systems. In July 2023, the EPA published RFS regulations that establish annual RVOs and percentage standards for 2023 through 2025 for cellulosic biofuel, biomass-based diesel, advanced biofuel, and total renewable fuel. The biofuel blending percentages increase in each of the three years and are higher than previous years. Higher blending percentages may increase the cost of compliance because the future cost of RINs is difficult to estimate. Moreover, in addition to increased price volatility in the RINs market, there have been multiple instances of RINs fraud occurring in the marketplace over the past several years. The EPA has initiated several enforcement actions against refiners who purchase fraudulent RINs, resulting in substantial costs to the refiner. We cannot predict with certainty our exposure to increased RINs costs in the future, nor can we predict the extent by which costs associated with RFS regulations will impact our future results of operations.

We strategically market our low-carbon fuels based on regional policies, feedstock preferences, CI scores, and our ability to obtain fuel pathways. A significant portion of our low-carbon fuels are sold in California and Canada. We are exposed to the volatility in the market price of LCFS program credits. We cannot predict the future prices of LCFS program credits. Prices for LCFS program credits are dependent upon a variety of factors, including, as applicable, changes in regulations, the availability of LCFS program credits for purchase, transportation fuel production levels, which can vary significantly each quarter, approved CI pathways, and CI scores. If an insufficient number of LCFS program credits are available for purchase, if we have to pay significantly higher prices for them, or if we are otherwise unable to meet other obligations under the LCFS programs, our business, financial condition, results of operations, and liquidity could be adversely affected.

In addition to state LCFS (e.g., California LCFS, Oregon CFP and Washington CFS), and certain carbon cap and trade programs (e.g., Washington CAA and Oregon CPP, which, we note, is in current legal limbo due to a recent court decision purportedly striking down the CPP rules – and OEDQ's recent notice that it intends to re-adopt the CPP rules under a projected 12-month administrative rulemaking process), we do business in multiple jurisdictions that have issued, or are considering issuing, similar low-carbon fuel regulations, policies, and standards. The LCFS, carbon cap and trade programs and similar U.S. state and international low carbon fuel regulations, policies, and standards are extremely complex, often have different or conflicting requirements or methodologies, and are frequently evolving, requiring us to periodically update our systems and controls to maintain compliance, which could require significant expenditures, and presents an increased risk of administrative error. Our Refining segment could be materially and adversely affected if (i) we are unable to comply with these programs in the states where we sell our petroleum products or we incur a significant cost to comply or (ii) we are unable to continue to sell our products in markets where we currently sell our products. While these regulations, policies and standards may materially and adversely impact our Refining segment, they do create opportunity for our Renewables segment. As a result, our Renewables segment could be materially and adversely affected if (i) these regulations, policies, and standards are adversely

changed, not enforced, or discontinued, (ii) the benefits therefrom are reduced (such as the blender's tax credit and other incentives), (iii) any of the products we produce are deemed not to qualify for compliance therewith, or (iv) we are unable to satisfy or maintain any approved pathways. Such changes could also negatively impact the economic assumptions and projections with respect to many of our Renewables segment investments and could have a material adverse impact on the returns achieved from those investments.

Increases in required fuel economy and regulation of GHG emissions from motor vehicles may reduce demand for petroleum-based transportation fuels.

The EPA and the National Highway Traffic Safety Administration ("NHTSA") are required to promulgate requirements regarding the Corporate Average Fuel Economy ("CAFE") of the nation's passenger fleet. The EPA and NHTSA previously adopted such standards, which were most recently revised in 2022 to impose more stringent requirements for emissions reductions. Additionally, the agencies proposed regulations in August and December 2023 to make certain of those requirements more stringent for future model year vehicles. These rulemakings will likely be subject to challenge by a variety of parties seeking even stricter GHG and CAFE standards to increase fuel economy. Furthermore, California adopted Advanced Clean Cars I and Advanced Clean Cars II ("ACC II") in 2012 and 2022, respectively, which are intended to scale down the use of internal combustion engines and instead require an increased number of zero emission vehicles. ACC II mandates that, by 2035, all new passenger cars, trucks and SUVs sold in California will be zero emission vehicles. Although states besides California are not permitted to develop their own emissions standards, Section 177 of the Clean Air Act authorizes other states to choose to adopt California's standards in lieu of federal requirements. Several states, including, for example, New Mexico and Colorado, are seeking to adopt the California standards and promote zero emission vehicles and mandate the transition away from internal combustion engines. Any further increases in fuel economy standards, along with mandated increases in use of renewable fuels discussed above, as well as electric vehicle mandates or internal combustion engine bans, could result in decreasing demand for petroleum-based transportation fuels. Decreasing demand for petroleum-based transportation fuels could have a material adverse effect on our financial condition and results of operation.

State regulation of petroleum product markets and reporting requirements could adversely impact our business, costs of operation, and financial results.

In recent years, high gasoline prices have raised questions about the operation of petroleum product markets and company profits. As a result, we may become subject to state regulation of petroleum product markets and reporting requirements relating to the petroleum industry. For example, California recently enacted legislation (SB 1322 and SBx 1-2), which greatly expanded reporting requirements for the petroleum industry and authorized the California Energy Commission to establish a maximum gross gasoline refining margin, impose financial penalties for profits exceeding the established maximum margin, and regulate refinery turnaround and maintenance activities that may affect fuel supply and pricing. Although we are not subject to these California statutes, other states could adopt similar regulations or reporting requirements. For example, the Washington legislature recently introduced HB 2232, which although it was not ultimately adopted, proposed to establish extensive petroleum industry reporting requirements (including the reporting of supply, pricing and profit information) and create a new division within the Washington Utilities and Transportation Commission charged with providing independent oversight and analysis of petroleum fuel markets. If HB 2232 or similar legislation is enacted, it could result in increased compliance costs and affect the results of our operations and financial position. Such laws and similar regulations could also increase our litigation risks or may increase risks related to our reputation or goodwill as we cannot predict how additional reporting under this law may be perceived or interpreted by our customers and stakeholders.

Physical impacts of climate change could have an adverse effect on our financial condition and results of operations.

Scientists have noted that changes occurring in the environment may produce changes to the climate, which have significant effects on the weather. These events, including, but not limited to, drought, winter storms, wildfire, tornados, extreme temperatures, extreme precipitation or flooding, may become more intense or more frequent as a result of climate change and could have an adverse effect on our continued operations as well as the operations of our suppliers and customers. Additionally, changing meteorological conditions, particularly temperature, may result in changes to demand for our products. Our suppliers or customers may also be subject to similar risks, any of which may adversely impact our business, financial condition, or operations.

Compliance with and changes in tax laws could materially and adversely impact our financial condition, results of operations and cash flows.

We are subject to extensive tax liabilities, including federal and state income taxes and transactional taxes such as excise, sales and use, payroll, franchise, withholding and property taxes. In addition, many tax liabilities are subject to periodic audits by

taxing authorities, and such audits could subject us to interest and penalties. New tax laws and regulations and changes in existing tax laws and regulations could result in increased expenditures by us for tax liabilities in the future and could materially and adversely impact our financial condition, results of operations and cash flows.

Additionally, U.S. federal tax law provides various tax credits to producers of qualifying renewable fuels. These tax credits have the effect of making the production of qualifying renewable fuels more competitive by offsetting a portion of the production costs or by supplementing a portion of the revenue, but may be repealed by governmental action at any time. Alternatively, the types of renewable fuels to which these tax credits apply, as well as the applicable value, duration, and requirements, may be modified or amended by governmental action in a form where the types of fuel mixtures we produce are not eligible for the tax credits or are eligible for relatively less benefits than we anticipated. As a result, a loss of, or reduction in, U.S. federal tax credits for producers of qualifying renewable fuels could increase our production costs or decrease our revenues which, in turn, could adversely impact the financial condition, results of operations and cash flows of our renewables business.

RISKS RELATED TO CYBERSECURITY, DATA SECURITY AND PRIVACY, INFORMATION TECHNOLOGY AND INTELLECTUAL PROPERTY

Our information technology systems, operational systems, security systems, infrastructure, communications networks, software integrated in our manufacturing and administrative processes, and customer data processed by us, third-party vendors or suppliers are subject to risks presented by cyber events, including incidents or breaches of security, any of which could prevent us or third parties we rely on from effectively operating our business, and could harm our reputation or materially adversely affect our company's assets, growth efforts, operations, facilities, business reputation or financial condition.

Our business is dependent upon increasingly complex information technology systems and other digital technologies for controlling our plants and pipelines, processing transactions and summarizing and reporting results of operations. The secure collection, processing, maintenance, storage, and transmission of information is critical to our operations. We are at risk for interruptions, outages and breaches of operational systems, including business, financial, accounting, product development, data processing or manufacturing processes, owned by us or our third-party vendors or suppliers; or third-party data that we process or our third-party partners process on our behalf. Such cyber incidents could materially disrupt or shut down operational systems; result in loss of, unauthorized access to, or copying or transfer of intellectual property assets, trade secrets or other proprietary or competitively sensitive information; compromise certain information of customers, employees, suppliers or others; and jeopardize the security of our facilities. We and vendors on our behalf monitor our information technology systems on a 24/7 basis in an effort to detect cyberattacks, security breaches or unauthorized access. Preventative and detective measures we utilize include independent cybersecurity audits and penetration tests. We implemented these efforts along with other risk mitigation procedures designed to detect and address unauthorized and damaging activity on our network, stay abreast of the increasing cybersecurity threat landscape and improve our cybersecurity posture, but there is no guarantee that such measures will be adequate to detect, prevent or mitigate cyber incidents. Any implementation, maintenance, segregation and improvement of our systems may require significant management time, support and cost and may not be effective or adequate. Such costs may include making organizational changes, deploying additional personnel and protection technologies, training employees, and engaging third-party experts and consultants. These efforts may come at the potential cost of revenues and human resources that could be utilized to continue to enhance our product offerings, and such increased costs and diversion of resources may adversely affect operating margins.

A cyber incident could be caused by disasters, insiders (through inadvertence or with malicious intent) or malicious third parties (including nation-states or nation-state supported actors) using sophisticated, targeted methods to circumvent firewalls, encryption and other security defenses, including hacking, fraud, trickery or other forms of deception that are generally beyond our control despite our implementation of protective measures. Emerging artificial intelligence technologies may improve or expand the capabilities of malicious third parties in a way we cannot predict at this time, including being used to develop new hacking tools, exploit vulnerabilities, obscure malicious activities, and increase the difficulty of detecting threats. Strategic targets, such as energy-related assets, may be at greater risk of future attacks from hackers, nation-sponsored groups or other nefarious actors than other targets in the United States, especially during times of war with nations with a history of funding cyberattacks. While there have been immaterial incidents of unauthorized access to our information technology systems, we have not experienced any material impact on our business or operations from these attacks; however, there is no assurance that other such incidents have not already occurred and we are unaware of it, and that we will not suffer such a loss in the future. In addition, information technology system failures, communications network disruptions, and security breaches could still impact equipment and software used to control plants and pipelines, resulting in improper operation of our assets or delays in the delivery or availability of our customers' products, contamination or degradation of the products we transport, store or distribute, or releases of hydrocarbon products and other damage to our facilities for which we could be held liable. These

information technology system failures, communications network disruptions, and security breaches could also cause us to breach our contractual arrangements with other parties, subject us to regulatory actions or litigation and harm our brand and business relationships as well as our financial condition.

Furthermore, we collect and store sensitive data in the ordinary course of our business, including personally identifiable information of our employees as well as our proprietary business information and that of our customers, suppliers, contractors, investors and other stakeholders. Additionally, new laws and regulations governing cybersecurity, data privacy and unauthorized disclosure of confidential information, including international comprehensive data privacy regulations and recent U.S. state legislation in Texas and Utah, among others, pose increasingly complex compliance challenges and could potentially elevate our costs over time. We also work with third-party partners that may in the course of their business relationship with us collect, store, process and transmit such data on our behalf and in connection with our products and services offerings. Despite our cybersecurity measures, our or our third-party partners' information technology systems may become the target of cyberattacks or security breaches (including employee error, malfeasance or other intentional or unintentional breaches) which are generally beyond our control, which could result in the theft or loss of the stored information, misappropriation of assets, disruption of transactions and reporting functions, harm our ability to protect customer or company information and impact our financial reporting. Our efforts to improve security and protect data may result in increased distraction to management and capital and operating costs to modify, upgrade or enhance our security measures that are designed to protect against such cyberattacks and we may face difficulties in fully anticipating or implementing adequate security measures or mitigating potential harm. Moreover, as technologies evolve and cyberattacks become increasingly sophisticated, we may not be able to anticipate, detect or prevent cyberattacks or security breaches, particularly because the methodologies used by attackers change frequently or may not be recognized until after such attack is launched, and because attackers are increasingly using technologies specifically designed to circumvent cybersecurity measures and avoid detection. Even with insurance coverage for cyberattacks, data breaches or unauthorized access of our or a third-party partner's information technology systems, a claim could be denied or coverage delayed. Moreover, it is increasingly difficult to buy sufficient cyber insurance coverages as the insurance market has been limiting both liability under cyber policies and the issuance of said policies, generally. A cyberattack or security breach could result in liability under data privacy laws, regulatory penalties, damage to our reputation or a loss of confidence in our products and services, or additional costs for remediation and modification or enhancement of our information systems to prevent future occurrences, all of which could have a material and adverse effect on our business, financial condition or results of operations.

We may be subject to information technology system failures, communications network disruptions and data breaches that are generally beyond our control.

We depend on the efficient and uninterrupted operation of third-party hardware and software systems and infrastructure, including our operating, communications and financial reporting systems. These systems are critical in meeting customer expectations, effectively tracking, maintaining and operating our equipment, directing and compensating our employees, and interfacing with our financial reporting system. We have implemented safeguards and other preventative measures designed to protect our systems and confidential and proprietary data, including sophisticated network security and internal control measures; however, our information technology systems and communications network, and those of our information technology and communication service providers, remain vulnerable to interruption by natural disasters, power loss, telecommunications failure, terrorist attacks, internet failures, computer malware, ransomware, cyberattacks, data breaches and other events unforeseen or generally beyond our control. Additionally, the implementation of social distancing measures and other limitations on our employees, service providers and other third parties in response to the COVID-19 pandemic have necessitated in certain cases to switching to remote work arrangements on less secure systems and environments. The increase in companies and individuals working remotely has increased the risk of cyberattacks and potential cybersecurity incidents, both deliberate attacks and unintentional events. Any of these events could cause system interruptions, delays, and loss of critical data, and could prevent us from developing or manufacturing products or providing services on a timely basis, which could make our business and services less attractive and subject us to liability. Any of these events could damage our reputation and be expensive to remedy.

Our business is subject to complex and evolving global laws, regulations and security standards regarding data privacy, cybersecurity and data protection ("data protection obligations"). Many of these data protection obligations are subject to change and uncertain interpretation, and any real or perceived failure to comply with such obligations could result in claims, increased cost of operations, or other harm to our business.

The constantly evolving global regulatory and legislative environment surrounding data privacy and protection poses increasingly complex compliance challenges, and complying with such data protection obligations could increase the costs and complexity of compliance and enforcement risks. These laws also are not uniform, as certain laws may be more stringent or broader in scope, or offer greater individual rights, with respect to sensitive and personal information, and such laws may differ

from each other, which may complicate compliance efforts. We, and our third-party service providers, collect some personal information from consumers, including credit card data, and we have personal information from our employees, job applicants and some business partners, such as customers, contractors and distributors. Any failure, whether real or perceived, by us to comply with applicable data protection obligations could result in proceedings or actions against us by governmental entities or others, subject us to significant fines, penalties, judgments, and negative publicity, require us to change our business practices, increase the costs and complexity of compliance, and adversely affect our business. Our compliance with current laws such as the General Data Protection Regulation and other similar current and upcoming data privacy/security laws, as well as any associated inquiries or investigations or any other government actions related to these laws, may increase our operating costs or subject us to legal and reputational risks, including significant fines, civil or criminal penalties or judgments, proceedings or litigation by governmental agencies or customers, class action privacy litigation in certain jurisdictions and negative publicity.

Our pipeline operations are subject to the Department of Homeland Security's ("DHS") Transportation Security Administration ("TSA") security directives. These directives require us to comply with mandatory reporting measures, including, among other things, to appoint personnel, report confirmed and potential cybersecurity incidents to the DHS Cybersecurity and Infrastructure Security Agency ("CISA") and provide vulnerability assessments. As legislation continues to develop and cyber incidents continue to evolve, we may be required to expend significant additional resources to respond to cyberattacks, to continue to modify or enhance our protective measures, or to detect, assess, investigate and remediate any security vulnerabilities and report any cyber incidents to the applicable regulatory authorities. Any failure to maintain compliance with these evolving government regulations may result in enforcement actions which may then result in significant time, support and cost and have a material adverse effect on our business and operations.

We may be unable to adequately maintain, enforce and protect our intellectual property and may not be able to prevent third parties from unauthorized access or use of our intellectual property, which may increase our cost of doing business or otherwise hurt our ability to compete in the market.

We develop and use intellectual property in the ordinary course of our business, including trademarks, trade secrets, copyrighted work and innovations, some of which are material to our business. We take measures designed to identify and protect our intellectual property through practices appropriate for securing and protecting exclusive rights in and to our intellectual property, including applying for registrations in the United States and in various foreign jurisdictions. Despite our efforts to protect such intellectual property, it is possible that competitors or other unauthorized third parties may illegally obtain, copy, use or disclose our trademarks (or other marks likely to cause confusion among our consumers), technologies, products and processes or seek court declarations that they do not infringe, misappropriate, dilute, or otherwise violate our intellectual property. In addition, the laws and/or judicial systems and enforcement mechanisms of foreign jurisdictions in which we create, market and sell our products may afford little or no effective protection of our intellectual property. We may also be subject to infringement, misappropriation, dilution, or other violation complaints from others challenging our use of a valid technology or intellectual property right. We cannot guarantee that our efforts to enforce our intellectual property rights against unauthorized use and appropriation, or our efforts to defend against third party claims of infringement would be successful. These potential risks to our intellectual property and any failure to adequately maintain, enforce and protect our intellectual property could subject us to increased competition and potentially result in the loss of our competitive advantage, key technology and resources, and customer trust, and negatively impact our brand. These risks could also lead to a decrease in our revenue which would negatively impact our business, prospects, liquidity, financial position and results of operations.

If we fail to comply with our obligations under license or technology agreements with third parties or are unable to license rights to use technologies on reasonable terms or at all, we may be required to pay damages or could potentially lose license rights that are critical to our business.

We obtain rights and licenses to certain intellectual property, including technologies, data, content and software from third parties, that are important to our business, and in the future we may enter into additional agreements that provide us with rights or licenses to valuable intellectual property or technology. If a licensor alleges that we failed to comply with any of the obligations under our license agreements, we may be required to pay damages, enter costly litigation, and the licensor may have the right to terminate part of or all of the license. Termination by the licensor would cause us to lose valuable rights that may be difficult to acquire elsewhere, and could prevent us from using, selling, marketing, manufacturing, importing or producing our products and services, or inhibit our ability to commercialize future products and services. Our business could suffer if any current or future licenses are limited or terminated, if the licensors fail to abide by the terms of the license, if the licensed intellectual property rights are found to be invalid or unenforceable, if the licensed technology is found to infringe, misappropriate, dilute, or otherwise violate the rights of a third party, or if we are unable to enter into necessary licenses on acceptable terms or at all. Moreover, our licensors may own or control intellectual property that has not been licensed to us and, as a result, we may be subject to claims, regardless of their merit, that we are infringing, misappropriating, diluting, or otherwise violating the licensor's or another party's rights.

In the future, we may identify additional third-party intellectual property that we believe is useful or necessary to engage in our business. However, such licenses may not be available on acceptable terms or at all. The licensing or acquisition of third-party intellectual property rights is a competitive area, and several companies may pursue strategies to license or acquire third-party intellectual property rights that we may consider attractive or necessary. In addition, companies that perceive us to be a competitor may be unwilling to assign or license rights to us. Even if such licenses are available, we may be required to pay the licensor substantial license fees or royalties based on sales of our products and services. Such fees and royalties are a component of the cost of our products and services and may affect the attractiveness of and the margins on our products and services. In addition, such licenses may be non-exclusive, which could give our competitors access to the same intellectual property licensed to us, and sometimes at a cheaper cost. Any of the foregoing could have a material adverse effect on our competitive position, business, financial condition, results of operations and on our ability to prevent others from interfering with our goodwill and commercialization of our products and services.

RISKS RELATED TO LIQUIDITY, FINANCIAL INSTRUMENTS AND CREDIT

Changes in our credit profile, or a significant increase in the price of crude oil, may affect our relationship with our suppliers, which could have a material adverse effect on our liquidity and limit our ability to purchase sufficient quantities of crude oil to operate our refineries at desired capacity.

An unfavorable credit profile, or a significant increase in the price of crude oil, could affect the way crude oil suppliers view our ability to make payments and induce them to shorten the payment terms of their invoices with us or require credit enhancement. Due to the large dollar amounts and volume of our crude oil and other feedstock purchases, any imposition by our suppliers of more burdensome payment terms or credit enhancement requirements on us may have a material adverse effect on our liquidity and our ability to make payments to our suppliers. This in turn could cause us to be unable to operate our refineries at desired capacity. A failure to operate our refineries at desired capacity could adversely affect our profitability and cash flow.

We may not be able to obtain funding on acceptable terms or at all because of volatility and uncertainty in the credit and capital markets. This may hinder or prevent us from meeting our future capital needs.

The domestic and global financial markets and economic conditions are disrupted and volatile from time to time due to a variety of factors, including low consumer confidence, high unemployment, geoeconomic and geopolitical issues, weak economic conditions and uncertainty in the financial services sector. In addition, the fixed-income markets have experienced periods of extreme volatility, which negatively impacted market liquidity conditions. As a result, the cost of raising money in the debt and equity capital markets has increased substantially at times while the availability of funds from these markets diminished significantly. In particular, as a result of concerns about the stability of financial markets generally and the solvency of lending counterparties specifically, the cost of obtaining money from the credit markets may increase as many lenders and institutional investors increase interest rates, enact tighter lending standards, refuse to refinance existing debt on similar terms or at all and reduce, or in some cases cease to provide, funding to borrowers. We use both fixed and variable rate debt, and we are exposed to market risk due to the floating interest rates on our credit facilities. In addition, lending counterparties under any existing revolving credit facility and other debt instruments may be unwilling or unable to meet their funding obligations, or we may experience a decrease in our capacity to issue debt or obtain commercial credit or a deterioration in our credit profile, including a rating agency lowering or withdrawing our credit ratings if, in its judgment, the circumstances warrant. Due to these factors, we cannot be certain that new debt or equity financing will be available on acceptable terms. If funding is not available when needed, or is available only on unfavorable terms, we may be unable to meet our obligations as they come due or we may be required to sell assets. Moreover, without adequate funding, we may be unable to execute our growth strategy, complete future acquisitions or construction projects, take advantage of other business opportunities or respond to competitive pressures, comply with regulatory requirements, or meet our short-term or long-term working capital requirements, any of which could have a material adverse effect on our revenues and results of operations. Failure to comply with regulatory requirements in a timely manner or meet our short-term or long-term working capital requirements could subject us to regulatory action.

We are exposed to the credit risks, and certain other risks, of our key customers and vendors.

We are subject to risks of loss resulting from nonpayment or nonperformance by our customers. We derive a significant portion of our revenues from contracts with key customers. Additionally, certain key customers of our Puget Sound Refinery contribute significantly to the cash flows and profitability of that facility. Any default or delay in payment by, or disputes with, those customers, such delays or disputes which have occurred in the past and may occur in the future, could have an adverse impact on our financial results.

If any of our key customers default on their obligations to us, our financial results could be adversely affected. Furthermore, some of our customers may be highly leveraged and subject to their own operating and regulatory risks. For example, in a low commodity price environment, certain of our customers have been or could be negatively impacted, causing them significant economic stress and resulting in an increase in nonpayment and/or nonperformance by them. Any such default or delay in payment by our customers could negatively affect our operating results in the periods in which they occur, and, if significant, could materially or adversely affect our business, financial condition, results of operations, and cash flows. In addition, nonperformance by vendors who have committed to provide us with products or services could result in higher costs or interfere with our ability to successfully conduct our business.

Any substantial increase in the nonpayment and/or nonperformance by our customers or vendors could have a material adverse effect on our results of operations and cash flows.

Our credit facilities contain certain covenants and restrictions that may constrain our business and financing activities.

The operating and financial restrictions and covenants in our credit facility, HEP's credit facility and any future financing agreements could adversely affect our ability to finance future operations or capital needs or to engage, expand or pursue our business activities. For example, each of our revolving credit facility and HEP's revolving credit facility imposes usual and customary requirements for this type of credit facility, including: (A) for us and HEP, (i) limitations on liens and indebtedness, (ii) a prohibition on changes in control, and (iii) restrictions on engaging in mergers and consolidations, and (B) for HEP, (i) limitations on asset sales, (ii) limitations on investments, and (iii) limitations on restricted payments, including certain debt prepayments. If we fail to satisfy the covenants set forth in the credit facilities or another event of default occurs under the credit facilities, the maturity of the loans could be accelerated or we could be prohibited from borrowing for our future working capital needs and issuing letters of credit. We might not have, or be able to obtain, sufficient funds to make these immediate payments. If we desire to undertake a transaction that is prohibited by the covenants in our credit facility or HEP's credit facility, as applicable, we will need to obtain consent under our credit facility or HEP's credit facility, as applicable. Such refinancing may not be possible or may not be available on commercially acceptable terms.

Our leverage may adversely affect our ability to fund future working capital, capital expenditures and other corporate purposes, future acquisitions, construction or development activities, or to otherwise realize fully the value of our assets and opportunities because of the need to dedicate a significant portion of our cash flow from operations to payments on our indebtedness or to comply with any restrictive terms of our indebtedness. Our leverage also may make our results of operations more susceptible to adverse economic and industry conditions by limiting our flexibility in planning for, or reacting to, changes in our business and the industry in which we operate and may place us at a competitive disadvantage as compared to our competitors that have less debt.

Our hedging transactions may limit our gains and expose us to other risks.

We periodically enter into derivative transactions as it relates to inventory levels and/or future production to manage the risks from changes in the prices of crude oil, refined products and other feedstocks. These transactions limit our potential gains if commodity prices move above or below the certain price levels established by our hedging instruments. We hedge price risk on inventories above our target levels to minimize the impact these price fluctuations have on our earnings and cash flows. Consequently, our hedging results may fluctuate significantly from one reporting period to the next depending on commodity price fluctuations and our relative physical inventory positions. These transactions may also expose us to risks of financial losses; for example, if our production is less than we anticipated at the time we entered into a hedge agreement or if a counterparty to our hedge agreements fails to perform its obligations under the agreements.

We may be unable to pay future dividends.

We will only be able to pay dividends from our available cash on hand, cash from operations or borrowings under our credit agreement. The declaration of future dividends on our common stock is evaluated quarterly and will be at the discretion of our Board of Directors and will depend upon many factors, including our results of operations, financial condition, earnings, capital requirements, and restrictions in our debt agreements and legal requirements. We cannot assure you that any dividends will be paid or the frequency or amounts of such payments.

Item 1B. Unresolved Staff Comments

We do not have any unresolved staff comments.

Item 1C. Cybersecurity

Description of Processes for Assessing, Identifying, and Managing Cybersecurity Risks

We are focusing on cybersecurity risk, particularly as our operations become increasingly dependent on digital technologies for controlling our plants and pipelines, processing transactions and summarizing and reporting results of operations. Globally, as cybersecurity incidents are occurring more often and using increasingly sophisticated methods, we are at risk for interruptions, outages and breaches of operational systems, including business, financial, accounting, product development, data processing or manufacturing processes, owned by us or our third-party vendors or suppliers, or data that we process or that our third-party service providers process on our behalf. Any such cyber incidents have the potential to materially disrupt or shut down operational systems; result in loss of, unauthorized access to, or copying or transfer of intellectual property assets, trade secrets or other proprietary or competitively sensitive information; compromise certain information of customers, employees, suppliers or others; and/or jeopardize the security of our facilities. We collect and store sensitive data in the ordinary course of our business, including certain personally identifiable information and proprietary business information for our business and our customers, suppliers, contractors, investors and other stakeholders. We also work with third-party service providers that may in the course of their business relationship with us collect, store, process and transmit such data on our behalf.

As further described in Item 1A. “Risk Factors – Risks Related to Cybersecurity, Data Security, and Privacy, Information Technology and Intellectual Property,” the Department of Homeland Security’s Transportation Safety Administration has issued a series of security directives that require us to take a number of actions, including among other things, to appoint personnel, report confirmed and potential cybersecurity incidents to the DHS Cybersecurity and Infrastructure Security Agency and provide vulnerability assessments. We have adopted a cybersecurity program, which uses technology and processes designed to help mitigate cybersecurity risks, with our information technology (“IT”) and operational technology (“OT”) teams working together to protect, identify, detect, mitigate and respond to potential cybersecurity incidents that threaten our Company.

We have made efforts to implement the National Institute of Standards and Technology (NIST) Cybersecurity Framework as well as supplemental guidance for information and operational technologies. We seek to follow federal and state statutory and regulatory guidance and have adopted internal policies and standards designed to align with these requirements.

We regularly engage independent third-party security consultants to help assess and monitor our IT and OT environments for vulnerabilities, to conduct penetration testing and to recommend mitigation strategies. In addition, we use third-party tools for vulnerability scans to identify external and internal risks.

Our cybersecurity program includes a process for overseeing and identifying cybersecurity risks associated with our third-party service providers.

Each employee’s and contractor’s ability to recognize and report cybersecurity threats is an important component of our cybersecurity program. On an annual basis, all Company employees are required to complete cybersecurity training. In addition, we regularly utilize employee exercises and communications designed to reinforce key cybersecurity training messages.

The above cybersecurity risk management processes are integrated into our overall risk management program. In addition to our efforts to continually evaluate our cybersecurity program and cybersecurity risks based upon emerging threats as a part of our risk management processes, cybersecurity risks to the Company are evaluated periodically through internal audits and annually by independent consultants, and we seek to incorporate learnings into our overall risk matrices.

We continue to make investments in new cybersecurity technologies to protect our facilities, users, and stakeholders, and to protect the personally identifiable information we maintain.

Board of Directors’ Oversight of Risks from Cybersecurity Risks

Cybersecurity risks are overseen by our full Board of Directors with input from the Audit Committee, which reviews the results of internal audit assessments and tests related to cybersecurity. As part of this oversight, the Board of Directors and the Audit Committee meet regularly to discuss the progress of ongoing initiatives and to seek coordination between enterprise stakeholders. At these meetings, our Chief Information Officer (“CIO”), who oversees the Company’s cybersecurity program, along with key subject matter experts, as necessary, review current and emerging cybersecurity-related threats as well as key performance indicators for cybersecurity process maturity, operational performance, and enterprise performance in countering

these threats. Based on the information provided through these various processes, our Board of Directors evaluates the risks facing us and provides guidance to management on our risk management strategy.

Management's Role in Assessing and Managing Cybersecurity Risks

The CIO, in collaboration with our Corporate Instruments and Control Systems Lead, and General Manager, Control Center Operations, are primarily responsible for assessing and managing our material risks from cybersecurity threats, monitoring the effectiveness of our cybersecurity detection and response processes in countering current threats and providing updates to our executive team.

The CIO has over 30 years of combined accounting, financial consulting, IT and Committee of Sponsoring Organizations (referred to as COSO) risk-based operational, financial and IT auditing experience, and has had oversight over the Company's cybersecurity activities as CIO for more than five years. The Corporate Control Systems, Advanced Process Controls and Cybersecurity Lead has over 30 years of experience overseeing the Company's refining operational technology systems and serves as the Company's refining operational technology cybersecurity leader. The General Manager, Control Center Operations, has over 10 years of pipeline control systems experience and serves as the Company's midstream operational technology cybersecurity leader.

The CIO serves as Chair of the Company's management level Cyber Risk Committee, which provides oversight over the Company's strategy and controls to identify, manage and mitigate risks related to cybersecurity and incident response and resiliency associated with the Company's IT and OT environments, and is comprised of representatives from compliance, IT and OT cybersecurity, internal audit, legal and risk. The Chair of the Cyber Risk Committee reports to the Company's management level Risk Management Oversight Committee and the Board of Directors on a regular basis.

The Company has adopted multiple incident response plans that establish guidelines for responding to incidents that may compromise the confidentiality, integrity and availability of Company information and systems, including referring matters to the Company's Incident Response Team and, as appropriate, to the Chief Executive Officer and the Board of Directors for additional evaluation and oversight.

As of the date of this Report, we are not aware of any previous cybersecurity threats that have materially affected or are reasonably likely to materially affect our Company. However, we acknowledge that cybersecurity threats are continually evolving, and the possibility of future cybersecurity incidents remains. See Item 1A. "Risk Factors – Risks Related to Cybersecurity, Data Security, and Privacy, Information Technology and Intellectual Property" for additional information about the risks to our business associated with a breach or compromise to our IT systems.

Item 3. Legal Proceedings

Commitment and Contingency Reserves

In the ordinary course of business, we may become party to legal, regulatory or administrative proceedings or governmental investigations, including environmental and other matters. Damages or penalties may be sought from us in some matters and certain matters may require years to resolve. While the outcome and impact of these proceedings and investigations on us cannot be predicted with certainty, based on advice of counsel and information currently available to us, management believes that the resolution of these proceedings and investigations through settlement or adverse judgment will not either individually or in the aggregate have a material adverse effect on our financial condition, results of operations or cash flows.

The environmental proceedings are reported to comply with SEC regulations which require us to disclose proceedings arising under provisions regulating the discharge of materials into the environment or protecting the environment when a governmental authority is party to the proceedings and such proceedings involve potential monetary sanctions that we reasonably believe could exceed $300,000 or more.

Environmental Matters

Navajo

HF Sinclair Navajo Refining LLC ("HFS Navajo") has been engaged in discussions with, and has responded to document requests from, the EPA, the United States Department of Justice (the "DOJ") and the New Mexico Environment Department (the "NMED") (collectively, the "Agencies") regarding HFS Navajo's compliance with the Clean Air Act ("CAA") and underlying regulations, and similar New Mexico laws and regulations, at its Artesia and Lovington, New Mexico refineries. The discussions have included the following topics: (a) alleged noncompliance with CAA's National Emission Standards for Hazardous Air Pollutants ("NESHAP") and New Source Performance Standards ("NSPS") at the Artesia refinery, which were set forth in a Notice of Violation ("May 2020 NOV") issued by the EPA in May 2020; (b) a Post Inspection Notice issued in June 2020 by the NMED, alleging noncompliance issues similar to those alleged by the EPA in its May 2020 NOV as well as alleged noncompliance with the State Implementation Plan ("SIP") and the Title V permit operating programs; (c) an information request issued in September 2020 by the EPA, pursuant to CAA Section 114, related to benzene fenceline monitoring, flare fuel gas, leak detection and repair, storage vessels and tanks, and other information regarding the Artesia refinery; (d) an information request issued by the EPA in May 2021, pursuant to CAA Section 114, requesting additional information and testing related to certain tanks at the Artesia refinery; and (e) informal information requests related to, among other things, the Artesia refinery's wastewater treatment plant, oil water separators and heat exchangers. In each of April 2022, June 2023 and August 2023, the EPA alleged additional CAA noncompliance at the Artesia refinery beyond the allegations in the May 2020 NOV, including alleged noncompliance with NESHAP, NSPS, SIP, Title V and other requirements.

Beginning in the spring of 2021, HFS Navajo and the Agencies began monthly meetings to discuss potential injunctive relief measures to address the alleged noncompliance at the Artesia refinery. In September 2021 and August 2023, the EPA presented to HFS Navajo potential claims for stipulated penalties for alleged noncompliance with a 2002 consent decree.

HFS Navajo continues to work with the Agencies to resolve these issues. At this time, no penalties have been demanded, and it is too early to predict the outcome of this matter.

Renewable Fuel Standard

On April 7, 2022, the EPA issued a decision reversing the grant of small refinery exemptions for our Woods Cross and Cheyenne refineries for the 2018 compliance year. On June 3, 2022, the EPA issued a decision reversing the grant of small refinery exemptions for our Woods Cross and Cheyenne refineries for the 2016 compliance year and denying small refinery exemption petitions for our Woods Cross and Cheyenne refineries for the 2019 and 2020 compliance years.

Various subsidiaries of HollyFrontier are currently pursuing legal challenges to the EPA's decisions to reverse its grant of small refinery exemptions for the 2016 and 2018 compliance years. The first lawsuit, filed against the EPA on May 6, 2022 and currently pending before the U.S. Court of Appeals for the DC Circuit, seeks to have the EPA's reversal of our 2018 small refinery exemption petitions overturned. The second lawsuit, filed against the EPA on August 5, 2022 and currently pending before the U.S. Court of Appeals for the DC Circuit, seeks to have the EPA's reversal of our 2016 small refinery exemption petitions overturned and to have the EPA's denial of our 2019 and 2020 small refinery exemption petitions reversed.

In addition, for both the 2016 and 2018 compliance years, pursuant to the June 2022 and April 2022 decisions, respectively, the EPA established an alternative compliance demonstration for small refineries pursuant to which the EPA is not imposing any obligations for the small refineries whose exemptions were reversed. On June 24, 2022, Growth Energy filed two lawsuits in the U.S. Court of Appeals for the DC Circuit against the EPA challenging the alternative compliance demonstration for the 2016 and 2018 compliance years. On July 25, 2022, various subsidiaries of HollyFrontier intervened on behalf of the EPA to aid the defense of the EPA's alternative compliance demonstration decision.

It is too early to predict the outcome of these matters.

Osage Pipeline

On July 8, 2022, the Osage Pipeline, which is owned by Osage Pipe Line Company, LLC ("Osage"), a joint venture between El Dorado Osage Company LLC and CHS McPherson Refinery Inc., experienced a release of crude oil at a location approximately 5 miles north of Cushing, Oklahoma.

Osage and Holly Energy Partners – Operating, L.P. (“HEP Operating”), the operator of the Osage Pipeline, are working with federal, state, tribal, and local governmental agencies, as well as the affected landowners. Discussions with the U.S. Department of Transportation Pipeline and Hazardous Materials Safety Administration, the EPA and the DOJ regarding resolution of their potential claims relating to the incident are ongoing. On September 13, 2023, Osage and HEP Operating received an offer for settlement from the EPA and the DOJ. After several months of negotiations, the EPA, the DOJ, Osage, and HEP Operating reached an agreement that resolves the civil claims of the United States under the CWA, subject to certain reservations of rights by the United States, in exchange for the payment of a $7.4 million civil penalty and performance of certain items of injunctive relief by Osage and HEP Operating. The agreement is set forth in a Consent Decree lodged in federal court on January 30, 2024. The Consent Decree is subject to a review and comment period of 30 days.

Other

We are a party to various other litigation and proceedings that we believe, based on advice of counsel, will not either individually or in the aggregate have a materially adverse impact on our financial condition, results of operations or cash flows.

Item 4. Mine Safety Disclosures

Not Applicable.

PART II

Item 5. Market for the Registrant's Common Equity, Related Stockholder Matters and Issuer Purchases of Equity Securities

Our common stock is traded on the New York Stock Exchange under the trading symbol “DINO.”

In August 2023, our Board of Directors approved a $1.0 billion share repurchase program, which replaced all existing share repurchase programs. This share repurchase program authorizes us to repurchase common stock in the open market or through privately negotiated transactions. Privately negotiated repurchases from REH Company (formerly known as The Sinclair Companies) are also authorized under this share repurchase program, subject to REH Company’s interest in selling its shares and other limitations. The timing and amount of share repurchases, including those from REH Company, will depend on market conditions and corporate, tax, regulatory and other relevant considerations. This share repurchase program may be discontinued at any time by our Board of Directors. In addition, we are authorized by our Board of Directors to repurchase shares in an amount sufficient to offset shares issued under our compensation programs. The following table includes repurchases made under this program during the fourth quarter of 2023.

Period	Total Number of Shares Purchased	Average Price Paid Per Share	Total Number of Shares Purchased as Part of Publicly Announced Plans or Programs	Maximum Dollar Value of Shares that May Yet Be Purchased under the Plans or Programs
October 2023	1,376,652	$ 54.48	1,376,652	$ 750,000,036
November 2023	—	$ —	—	$ 750,000,036
December 2023	1,334,454	$ 55.14	1,334,454	$ 676,422,860
Total for October to December 2023	2,711,106		2,711,106	

On January 3, 2024, we repurchased 454,380 shares of our outstanding common stock from REH Company in a privately negotiated transaction under our share repurchase program and pursuant to the Stock Purchase Agreement, dated January 3, 2024 (the “January Stock Purchase Agreement”), between us and REH Company. The price paid by us under the January Stock Purchase Agreement was $55.02 per share resulting in an aggregate purchase price of $25.0 million. The purchase price was funded with cash on hand.

On February 8, 2024, we repurchased 1,061,946 shares of our outstanding common stock from REH Company in a privately negotiated transaction under our new share repurchase program and pursuant to the Stock Purchase Agreement, dated February 8, 2024 (the “February Stock Purchase Agreement”), between us and REH Company. The price paid by us under the February Stock Purchase Agreement was $56.50 per share resulting in an aggregate purchase price of $60.0 million. The purchase price was funded with cash on hand. As of February 15, 2024, we had remaining authorization to repurchase up to $591.4 million under our share repurchase program.

As of February 15, 2024, we had approximately 1,594 registered holders of our common stock. A substantially greater number of holders of our common stock are “street name” or beneficial holders, whose shares of record are held by banks, brokers and other financial institutions.

We intend to consider the declaration of a dividend on a quarterly basis, although there is no assurance as to future dividends since they are dependent upon future earnings, capital requirements, our financial condition and other factors.

Item 6. [Reserved]

Item 7. Management's Discussion and Analysis of Financial Condition and Results of Operations

This Item 7 contains "forward-looking" statements. See "Forward-Looking Statements" at the beginning of this Annual Report on Form 10-K. In this document, the words "we," "our," "ours" and "us" refer only to HF Sinclair and its consolidated subsidiaries or to HF Sinclair or an individual subsidiary and not to any other person with certain exceptions. References herein to HF Sinclair "we," "our," "ours" and "us" with respect to time periods prior to March 14, 2022 refer to HollyFrontier and its consolidated subsidiaries and do not include the Acquired Sinclair Businesses. References herein to HF Sinclair "we," "our," "ours," and "us" with respect to time periods from and after March 14, 2022 include the operations of the Acquired Sinclair Businesses. Unless otherwise specified, the financial statements included herein include financial information for HF Sinclair, which for the time period from March 14, 2022 to December 31, 2023 includes the combined business operations of HollyFrontier and the Acquired Sinclair Businesses. References herein to HEP with respect to time periods prior to the closing of the HEP Merger Transaction on December 1, 2023 refers to HEP and its consolidated subsidiaries.

OVERVIEW

We are an independent energy company that produces and markets high-value light products such as gasoline, diesel fuel, jet fuel, renewable diesel and other specialty products. We own and operate refineries located in Kansas, Oklahoma, New Mexico, Wyoming, Washington and Utah. We provide petroleum product and crude oil transportation, terminalling, storage and throughput services to our refineries and the petroleum industry. We market our refined products principally in the Southwest United States, the Rocky Mountains extending into the Pacific Northwest and in other neighboring Plains states, and we supply high-quality fuels to more than 1,500 branded stations and license the use of the Sinclair brand at more than 300 additional locations throughout the country. We produce renewable diesel at two of our facilities in Wyoming and our facility in New Mexico. In addition, our subsidiaries produce and market base oils and other specialized lubricants in the United States, Canada and the Netherlands, and export products to more than 80 countries.

Market Developments

For the year ended December 31, 2023, net income attributable to HF Sinclair stockholders was $1,589.7 million compared to $2,922.7 million and $558.3 million for the years ended December 31, 2022, and 2021, respectively. Gross refining margin per produced barrel sold in our Refining segment for 2023 decreased 20% over the year ended December 31, 2022.

Our results for the year ended December 31, 2023 were favorably impacted by healthy demand for transportation fuels, lubricants and transportation and terminal services and constrained refined product supply. We continue to adjust our operational plans to evolving market conditions. The extent to which our future results are affected by volatile regional and global economic conditions will depend on various factors and consequences beyond our control.

In the Refining segment, we saw healthy refining margins in both the West and Mid-Continent regions in 2023 as a result of steady demand and tight supply during the period. We completed a significant number of planned turnarounds during the year ended December 31, 2023. These turnarounds have provided us with the opportunity to execute maintenance strategies focused on improving operational reliability. For the first quarter of 2024, we expect to run between 585,000 – 615,000 barrels per day of crude oil. This guidance reflects planned maintenance activities at our Puget Sound Refinery during the first quarter. Refined product margins are expected to be impacted by typical seasonal weakness in gasoline.

In the Renewables segment, we continued to optimize the operation of our assets during 2023. For the first quarter of 2024, we expect continued weakening in RINs and LCFS prices to impact margins, and we will continue to focus on the economic optimization of our assets.

In the Marketing segment, we continued to see strong value in the Sinclair brand during 2023 as the marketing business continued to provide a consistent sales channel with margin uplift for our produced fuels. We continue to target 5% or more annual growth in the number of sites.

In the Lubricants & Specialties segment, effective the first quarter of 2023, management views the segment as an integrated business of processing feedstocks into base oils and processing base oils into finished lubricant products along with the packaging, distribution and sales to customers. During 2023, despite weakening base oil prices during the period, we continued to see strong performance (excluding FIFO) driven by sales mix optimization across our finished products portfolio.

In the Midstream segment, 2023 was favorably impacted by our refining activity.

In August 2023, our Board of Directors authorized a new $1.0 billion share repurchase program, and we expect to repurchase shares in the first quarter of 2024. On February 14, 2024, our Board of Directors announced that it declared a regular quarterly dividend in the amount of $0.50 per share, an increase of $0.05 over our previous dividend of $0.45 per share. The dividend is payable on March 5, 2024 to holders of record of common stock on February 26, 2024.

HEP Merger Transaction
On December 1, 2023, pursuant to the Agreement and Plan of Merger, dated as of August 15, 2023 (the "Merger Agreement"), by and among HEP, HF Sinclair, Navajo Pipeline Co., L.P., a Delaware limited partnership and an indirect wholly owned subsidiary of HF Sinclair ("HoldCo"), Holly Apple Holdings LLC, a Delaware limited liability company and a wholly owned subsidiary of HoldCo ("Merger Sub"), HEP Logistics Holdings, L.P., a Delaware limited partnership and the general partner of HEP ("HLH"), and Holly Logistic Services, L.L.C., a Delaware limited liability company and the general partner of HLH, Merger Sub merged with and into HEP, with HEP surviving as an indirect, wholly owned subsidiary of HF Sinclair (the "HEP Merger Transaction").

Under the terms of the Merger Agreement, each outstanding common unit representing a limited partner interest in HEP (an "HEP common unit"), other than the HEP common units already owned by HF Sinclair and its subsidiaries, was converted into the right to receive 0.315 shares of HF Sinclair common stock and $4.00 in cash, without interest. The Merger Agreement consideration totaled $267.6 million in cash and resulted in the issuance of 21,072,326 shares of HF Sinclair common stock from treasury stock.

For a description of our existing indebtedness, as well as the changes thereto associated with the HEP Merger Transaction, see Note 13 "Debt" in the Notes to Consolidated Financial Statements.

Sinclair Acquisition
On March 14, 2022, HollyFrontier and HEP announced the establishment of HF Sinclair as the new parent holding company of HollyFrontier and HEP and their subsidiaries, and the completion of their respective acquisitions (the "Sinclair Transactions") of Sinclair Oil Corporation (now known as Sinclair Oil LLC, "Sinclair Oil") and Sinclair Transportation Company LLC ("STC") from The Sinclair Companies (now known as REH Company and referred to herein as "REH Company").

HF Sinclair acquired REH Company's refining, branded marketing, renewables, and midstream businesses. The branded marketing business supplies high-quality fuels to Sinclair branded stations and licenses the use of the Sinclair brand to additional locations throughout the United States. The renewables business includes the operation of a renewable diesel unit located in Sinclair, Wyoming. The refining business includes two Rocky Mountains-based refineries located in Casper, Wyoming and Sinclair, Wyoming. Under the terms of that certain Contribution Agreement as amended on March 14, 2022 (the "Contribution Agreement"), HEP acquired STC, REH Company's integrated crude and refined products pipelines and terminal assets, including approximately 1,200 miles of integrated crude and refined product pipeline supporting the Sinclair refineries and third parties, eight product terminals and two crude terminals with approximately 4.5 million barrels of operated storage. In addition, HEP acquired STC's interests in three pipeline joint ventures for crude gathering and product offtake including: Saddle Butte Pipeline III, LLC (at the time of closing, 25.06% and currently, a 25.12% non-operated interest); Pioneer Investments Corp. (49.995% non-operated interest); and UNEV Pipeline, LLC ("UNEV") (the 25% non-operated interest not already owned by HEP, resulting in UNEV becoming a wholly owned subsidiary of HEP).

See Note 2 "Acquisitions" in the Notes to Consolidated Financial Statements for additional information.

Puget Sound Refinery Acquisition
On May 4, 2021, HollyFrontier Puget Sound Refining LLC (now known as HF Sinclair Puget Sound Refining LLC), a wholly owned subsidiary of HollyFrontier, entered into a sale and purchase agreement with Equilon Enterprises LLC d/b/a Shell Oil Products US ("Shell") to acquire the Puget Sound Refinery. The acquisition closed on November 1, 2021.

Renewable Fuel Standard Regulations
Pursuant to the 2007 Energy Independence and Security Act, the EPA promulgated the RFS regulations, which increased the volume of renewable fuels mandated to be blended into the nation's fuel supply. The regulations, in part, require refiners to add annually increasing amounts of "renewable fuels" to their petroleum products or purchase credits, known as RINs, in lieu of such blending. Compliance with RFS regulations significantly increases our cost of products sold, with RINs costs totaling $790.8 million for the year ended December 31, 2023.

Under the RFS regulations, the EPA is required to set annual volume targets of renewable fuels that obligated parties, such as us, must blend into petroleum-based transportation fuels consumed in the United States. These volume requirements are used to determine an obligated party's renewable volume obligation ("RVO"). The EPA released a final rule on June 3, 2022 that, among other things, reduced the volume targets for 2020 and established targets for 2021 and 2022. In 2020, we recognized the cost of the RVO using the 2020 volume targets set by the EPA at that time, and in 2021 and the three months ended March 31, 2022, we recognized the cost of the RVO using our estimates. As a result of the final rule released by the EPA on June 3, 2022 as noted above, we recognized a benefit of $72.0 million in the year ended December 31, 2022 related to the modification of the 2020 and 2021 volume targets. In June 2023, the EPA established the targets for 2023 through 2025, which increase RVOs in each of the concurrent years.

A more detailed discussion of our financial and operating results for the years ended December 31, 2023 and 2022 is presented in the following sections. Discussions of year-over-year comparisons for 2022 and 2021 can be found in "Management's Discussion and Analysis of Financial Condition and Results of Operations" in Part II, Item 7 of our Annual Report on Form 10-K for the year ended December 31, 2022.

RESULTS OF OPERATIONS

Financial Data

	Years Ended December 31,		
	2023	2022	2021
	(In thousands, except per share data)		
Sales and other revenues	$ 31,964,395	$ 38,204,839	$ 18,389,142
Operating costs and expenses:			
Cost of products sold (exclusive of depreciation and amortization):			
Cost of products sold (exclusive of lower of cost or market inventory valuation adjustment)	25,784,449	30,680,013	15,567,052
Lower of cost or market inventory valuation adjustment	270,419	52,412	(310,123)
	26,054,868	30,732,425	15,256,929
Operating expenses (exclusive of depreciation and amortization)	2,438,148	2,334,893	1,517,478
Selling, general and administrative expenses (exclusive of depreciation and amortization)	498,240	426,485	362,010
Depreciation and amortization	770,573	656,787	503,539
Total operating costs and expenses	29,761,829	34,150,590	17,639,956
Income from operations	2,202,566	4,054,249	749,186
Other income (expense):			
Earnings (loss) of equity method investments	17,369	(260)	12,432
Interest income	93,468	30,179	4,019
Interest expense	(190,796)	(175,628)	(125,175)
Gain on business interruption insurance settlement	—	15,202	—
Gain on tariff settlement	—	—	51,500
Gain on early extinguishment of debt	—	604	—
Gain (loss) on foreign currency transactions	2,530	(1,637)	(2,938)
Gain on sale of assets and other	27,370	13,337	98,128
	(50,059)	(118,203)	37,966
Income before income taxes	2,152,507	3,936,046	787,152
Income tax expense	441,612	894,872	123,898
Net income	1,710,895	3,041,174	663,254
Less net income attributable to noncontrolling interest	121,229	118,506	104,930
Net income attributable to HF Sinclair stockholders	$ 1,589,666	$ 2,922,668	$ 558,324
Earnings per share:			
Basic	$ 8.29	$ 14.28	$ 3.39
Diluted	$ 8.29	$ 14.28	$ 3.39
Cash dividends declared per common share	$ 1.80	$ 1.20	$ 0.35
Average number of common shares outstanding:			
Basic	190,035	202,566	162,569
Diluted	190,035	202,566	162,569

Other Financial Data

	Years Ended December 31,		
	2023	2022	2021
	(In thousands)		
Net cash provided by operating activities	$ 2,297,235	$ 3,777,159	$ 406,682
Net cash used for investing activities	$ (371,323)	$ (774,488)	$ (1,327,219)
Net cash used for financing activities	$ (2,243,882)	$ (1,560,759)	$ (211,803)
Capital expenditures	$ 385,413	$ 524,007	$ 813,409
EBITDA [(1)]	$ 2,899,179	$ 4,619,776	$ 1,306,917

(1) Earnings before interest, taxes, depreciation and amortization, which we refer to as "EBITDA," is calculated as net income attributable to HF Sinclair stockholders plus (i) income tax provision, (ii) interest expense, net of interest income and (iii) depreciation and amortization. EBITDA is not a calculation provided for under GAAP; however, the amounts included in the EBITDA calculation are derived from amounts included on our consolidated financial statements. EBITDA should not be considered as an alternative to net income or operating income as an indication of our operating performance or as an alternative to operating cash flow as a measure of liquidity. EBITDA is not necessarily comparable to similarly titled measures of other companies. EBITDA is presented here because it is a widely used financial indicator used by investors and analysts to measure performance. EBITDA is also used by our management for internal analysis and as a basis for financial covenants. EBITDA presented above is reconciled to net income under "Reconciliations to Amounts Reported Under Generally Accepted Accounting Principles" following Item 7A of Part II of this Annual Report on Form 10-K.

Supplemental Segment Operating Data

Our operations are organized into five reportable segments, Refining, Renewables, Marketing, Lubricants & Specialties and Midstream. See Note 20 "Segment Information" in the Notes to Consolidated Financial Statements for additional information on our reportable segments.

Refining Segment Operating Data

The disaggregation of our refining geographic operating data is presented in two regions, Mid-Continent and West, to best reflect the economic drivers of our refining operations. The Mid-Continent region is comprised of the El Dorado and Tulsa Refineries. The West region is comprised of the Puget Sound, Navajo, Woods Cross, Parco and Casper Refineries. The Puget Sound Refinery was acquired November 1, 2021, and thus is included for the period November 1, 2021 through December 31, 2023. In addition, the refinery operations of the Parco and Casper Refineries are included for the period March 14, 2022 (date of acquisition) through December 31, 2023. The following tables set forth information, including non-GAAP performance measures, about our consolidated refinery operations. The refinery gross and net operating margins do not include lower of cost or market inventory valuation adjustments and depreciation and amortization. Reconciliations to amounts reported under GAAP are provided under "Reconciliations to Amounts Reported Under Generally Accepted Accounting Principles" following Item 7A of Part II of this Annual Report on Form 10-K.

	Years Ended December 31,		
	2023	**2022 (8)**	**2021 (9)**
Mid-Continent Region			
Crude charge (BPD) (1)	237,510	283,160	260,350
Refinery throughput (BPD) (2)	256,810	299,380	276,430
Sales of produced refined products (BPD) (3)	248,330	280,800	265,470
Refinery utilization (4)	91.4 %	108.9 %	100.1 %
Average per produced barrel sold (5)			
Refinery gross margin	$ 17.49	$ 22.01	$ 9.44
Refinery operating expenses (6)	7.02	6.19	6.42
Net operating margin	$ 10.47	$ 15.82	$ 3.02
Refinery operating expenses per throughput barrel (7)	$ 6.79	$ 5.81	$ 6.17
Feedstocks:			
Sweet crude oil	56 %	58 %	61 %
Sour crude oil	20 %	20 %	15 %
Heavy sour crude oil	16 %	16 %	18 %
Other feedstocks and blends	8 %	6 %	6 %
Total	100 %	100 %	100 %

	Years Ended December 31,		
	2023	2022 [8]	2021 [9]
Mid-Continent Region			
Sales of refined products:			
Gasolines	51 %	51 %	52 %
Diesel fuels	30 %	33 %	33 %
Jet fuels	6 %	6 %	5 %
Fuel oil	1 %	1 %	1 %
Asphalt	4 %	3 %	3 %
Base oils	4 %	4 %	4 %
LPG and other	4 %	2 %	2 %
Total	100 %	100 %	100 %
West Region			
Crude charge (BPD) [1]	330,030	323,820	140,370
Refinery throughput (BPD) [2]	360,200	347,590	155,440
Sales of produced refined products (BPD) [3]	353,950	347,540	158,630
Refinery utilization [4]	79.0 %	81.4 %	82.7 %
Average per produced barrel sold [5]			
Refinery gross margin	$ 24.13	$ 30.64	$ 13.32
Refinery operating expenses [6]	10.14	9.31	8.09
Net operating margin	$ 13.99	$ 21.33	$ 5.23
Refinery operating expenses per throughput barrel [7]	$ 9.97	$ 9.31	$ 9.27
Feedstocks:			
Sweet crude oil	30 %	28 %	22 %
Sour crude oil	45 %	50 %	58 %
Heavy sour crude oil	11 %	10 %	1 %
Black wax crude oil	6 %	5 %	10 %
Other feedstocks and blends	8 %	7 %	9 %
Total	100 %	100 %	100 %
Sales of refined products:			
Gasolines	54 %	53 %	54 %
Diesel fuels	31 %	32 %	35 %
Jet fuels	6 %	5 %	1 %
Fuel oil	2 %	3 %	3 %
Asphalt	2 %	3 %	4 %
LPG and other	5 %	4 %	3 %
Total	100 %	100 %	100 %
Consolidated			
Crude charge (BPD) [1]	567,540	606,980	400,720
Refinery throughput (BPD) [2]	617,010	646,970	431,870
Sales of produced refined products (BPD) [3]	602,280	628,340	424,100
Refinery utilization [4]	83.7 %	92.3 %	93.1 %
Average per produced barrel [5]			
Refinery gross margin	$ 21.39	$ 26.78	$ 10.89
Refinery operating expenses [6]	8.86	7.92	7.04
Net operating margin	$ 12.53	$ 18.86	$ 3.85
Refinery operating expenses per throughput barrel [7]	$ 8.65	$ 7.69	$ 6.92

	Years Ended December 31,		
	2023	2022 [8]	2021 [9]
Consolidated			
Feedstocks:			
Sweet crude oil	42 %	42 %	47 %
Sour crude oil	34 %	36 %	31 %
Heavy sour crude oil	13 %	13 %	12 %
Black wax crude oil	3 %	3 %	4 %
Other feedstocks and blends	8 %	6 %	6 %
Total	100 %	100 %	100 %
Sales of refined products:			
Gasolines	53 %	52 %	53 %
Diesel fuels	30 %	32 %	34 %
Jet fuels	6 %	6 %	4 %
Fuel oil	1 %	2 %	1 %
Asphalt	3 %	3 %	3 %
Base oils	2 %	2 %	2 %
LPG and other	5 %	3 %	3 %
Total	100 %	100 %	100 %

(1) Crude charge represents the barrels per day of crude oil processed at our refineries.
(2) Refinery throughput represents the barrels per day of crude and other refinery feedstocks input to the crude units and other conversion units at our refineries.
(3) Represents barrels sold of refined products produced at our refineries (including Asphalt and inter-segment sales) and does not include volumes of refined products purchased for resale or volumes of excess crude oil sold.
(4) Represents crude charge divided by total crude capacity (BPSD). As a result of our acquisition of the Puget Sound Refinery on November 1, 2021, our consolidated crude capacity increased from 405,000 BPSD at December 31, 2020 to 554,000 BPSD at December 31, 2021, and further increased to 678,000 BPSD at December 31, 2022 as a result of our acquisition of the Parco and Casper Refineries on March 14, 2022.
(5) Represents average amount per produced barrel sold, which is a non-GAAP measure. Reconciliations to amounts reported under GAAP are provided under "Reconciliations to Amounts Reported Under Generally Accepted Accounting Principles" following Item 7A of Part II of this Annual Report on Form 10-K.
(6) Represents total Mid-Continent and West regions operating expenses, exclusive of depreciation and amortization, divided by sales volumes of refined products produced at our refineries.
(7) Represents total Mid-Continent and West regions operating expenses, exclusive of depreciation and amortization, divided by refinery throughput.
(8) We acquired the Parco and Casper Refineries on March 14, 2022. Refining operating data for the year ended December 31, 2022 includes crude oil and feedstocks processed and refined products sold at our Parco and Casper Refineries for the period March 14, 2022 through December 31, 2022 only, averaged over the 365 days in the year ended December 31, 2022.
(9) We acquired the Puget Sound Refinery on November 1, 2021. Refining operating data for the year ended December 31, 2021 includes crude oil and feedstocks processed and refined products sold at our Puget Sound Refinery for the period November 1, 2021 through December 31, 2021 only, averaged over the 365 days in the year ended December 31, 2021.

Renewables Operating Data

The following table sets forth information, including non-GAAP performance measures, about our renewables operations and includes our Wyoming renewable diesel unit acquired as part of the Sinclair Transactions for the period March 14, 2022 (the date of acquisition) through December 31, 2023. The renewables gross and net operating margins do not include the non-cash effects of lower of cost or market inventory valuation adjustments and depreciation and amortization. Reconciliations to amounts reported under GAAP are provided under "Reconciliations to Amounts Reported Under Generally Accepted Accounting Principles" following Item 7A of Part II of this Annual Report on Form 10-K.

	Years Ended December 31,	
	2023	2022
Renewables		
Sales volumes (in thousand gallons)	215,510	136,204
Average per produced gallon [1]		
Renewables gross margin	$ 0.50	$ 0.30
Renewables operating expenses [2]	0.51	0.82
Net operating margin	$ (0.01)	$ (0.52)

(1) Represents average amount per produced gallon sold, which is a non-GAAP measure. Reconciliations to amounts reported under GAAP are provided under "Reconciliations to Amounts Reported Under Generally Accepted Accounting Principles" following Item 7A of Part II of this Annual Report on Form 10-K.
(2) Represents total Renewables segment operating expenses, exclusive of depreciation and amortization, divided by sales volumes of renewable diesel produced at our renewable diesel units.

Marketing Operating Data

The following table sets forth information, including non-GAAP performance measures, about our marketing operations and includes our Sinclair branded fuel business for the period March 14, 2022 (the date of acquisition) through December 31, 2023. The marketing gross margin does not include the non-cash effects of depreciation and amortization. Reconciliations to amounts reported under GAAP are provided under "Reconciliations to Amounts Reported Under Generally Accepted Accounting Principles" following Item 7A of Part II of this Annual Report on Form 10-K.

	Years Ended December 31,	
	2023	2022
Marketing		
Number of branded sites at period end [1]	1,540	1,513
Sales volumes (in thousand gallons)	1,441,607	1,118,444
Margin per gallon of sales [2]	$ 0.07	$ 0.06

(1) Includes non-Sinclair branded sites from legacy HollyFrontier agreements.
(2) Represents average amount per gallon sold, which is a non-GAAP measure. Reconciliations to amounts reported under GAAP are provided under "Reconciliations to Amounts Reported Under Generally Accepted Accounting Principles" following Item 7A of Part II of this Annual Report on Form 10-K.

Lubricants & Specialties Segment Operating Data

The following table sets forth information about our lubricants and specialties operations.

	Years Ended December 31,		
	2023	2022	2021
Lubricants & Specialties			
Sales of produced barrels sold (BPD)	30,210	32,530	34,016
Sales of produced refined products:			
Finished products	50 %	51 %	51 %
Base oils	27 %	28 %	27 %
Other	23 %	21 %	22 %
Total	100 %	100 %	100 %

Effective the first quarter of 2023, management views the Lubricants & Specialties segment as an integrated business of processing feedstocks into base oils and processing base oils into finished lubricant products along with the packaging, distribution and sales to customers.

Midstream Segment Operating Data

The following table sets forth information about our midstream operations.

	Years Ended December 31,		
	2023	2022	2021
Midstream			
Volumes (BPD)			
Pipelines:			
Affiliates—refined product pipelines	152,462	143,303	108,767
Affiliates—intermediate pipelines	110,720	129,295	125,225
Affiliates—crude pipelines	437,586	456,797	279,514
	700,768	729,395	513,506
Third parties—refined product pipelines	38,834	38,000	49,356
Third parties—crude pipelines	197,659	144,478	129,084
	937,261	911,873	691,946
Terminals and loading racks:			
Affiliates	728,128	560,038	391,698
Third parties	42,567	38,211	51,184
	770,695	598,249	442,882
Affiliates—refinery processing units	62,057	70,222	69,628
Total for pipelines, terminals and refinery processing unit assets (BPD)	1,770,013	1,580,344	1,204,456

Results of Operations - Year Ended December 31, 2023 Compared to Year Ended December 31, 2022

Summary

Net income attributable to HF Sinclair stockholders for the year ended December 31, 2023 was $1,589.7 million ($8.29 per basic and diluted share), a $1,333.0 million decrease compared to net income of $2,922.7 million ($14.28 per basic and diluted share) for the year ended December 31, 2022. The decrease in net income was principally driven by lower refinery gross margins and lower refined product sales volumes. Lower of cost or market inventory reserve adjustments decreased pre-tax earnings by $270.4 million and $52.4 million for the years ended December 31, 2023 and 2022, respectively. Refinery gross margins for the year ended December 31, 2023 decreased to $21.39 per produced barrel sold from $26.78 for the year ended December 31, 2022.

Sales and Other Revenues

Sales and other revenues decreased 16% from $38,204.8 million for the year ended December 31, 2022 to $31,964.4 million for the year ended December 31, 2023, principally due to decreased refined product sales prices and lower refined product sales volumes.Sales and other revenues included $4,146.3 million, $2,762.8 million and $781.3 million in unaffiliated revenues related to our Marketing, Lubricants & Specialties and Renewables segments, respectively, for the year ended December 31, 2023. Sales and other revenues included $3,911.9 million, $3,149.1 million and $654.9 million in unaffiliated revenues related to our Marketing, Lubricants & Specialties and Renewables segments, respectively, for the year ended December 31, 2022.

Cost of Products Sold

Total cost of products sold decreased 15% from $30,732.4 million for the year ended December 31, 2022 to $26,054.9 million for the year ended December 31, 2023, principally due to lower crude oil costs and lower refined product sales volumes. During the years ended December 31, 2023 and 2022, we recognized a lower of cost or market inventory valuation adjustment charge of $270.4 million and $52.4 million, respectively. Within our Lubricants & Specialties segment, FIFO impact was a charge of $13.4 million for the year ended December 31, 2023 and a benefit of $77.6 million for the year ended December 31, 2022.

Gross Refinery Margins

Gross refinery margin per barrel sold decreased 20% from $26.78 for the year ended December 31, 2022 to $21.39 for the year ended December 31, 2023The decrease was due to lower average per barrel sold sales prices, partially offset by lower crude oil and feedstock prices. Gross refinery margin per barrel does not include the non-cash effects of lower of cost or market inventory valuation adjustments or depreciation and amortization. See “Reconciliations to Amounts Reported Under Generally Accepted Accounting Principles” following Item 7A of Part II of this Annual Report on Form 10-K for a reconciliation to the income statement of sale prices of products sold and cost of products purchased.

Operating Expenses

Operating expenses, exclusive of depreciation and amortization, increased 4% from $2,334.9 million for the year ended December 31, 2022 to $2,438.1 million for the year ended December 31, 2023, primarily due to increased maintenance activities and our acquisition of the Acquired Sinclair Businesses, partially offset by lower natural gas costs.

Selling, General and Administrative Expenses

Selling, general and administrative expenses increased 17% from $426.5 million for the year ended December 31, 2022 to $498.2 million for the year ended December 31, 2023, primarily due to higher costs related to information technology, other professional services and employee costs as compared to the prior period and our acquisition of the Acquired Sinclair Businesses, partially offset by a decrease in acquisition integration and regulatory costs. We incurred $39.4 million and $52.9 million in acquisition integration and regulatory costs during the years ended December 31, 2023 and 2022, respectively.

Depreciation and Amortization Expenses

Depreciation and amortization increased 17% from $656.8 million for the year ended December 31, 2022 to $770.6 million for the year ended December 31, 2023. This increase was principally due to depreciation and amortization attributable to capitalized turnaround costs, capitalized improvement projects and the Acquired Sinclair Businesses.

Earnings (Loss) of Equity Method Investments

For the year ended December 31, 2023, we recorded net earnings of $17.4 million of equity method investments as compared to a net loss of $0.3 million for the year ended December 31, 2022. Net loss during the year ended December 31, 2022 was primarily due to HEP’s 50% share of incurred and estimated environmental remediation and recovery expenses, net of insurance proceeds received to date, for the Osage Pipeline. In July 2022, the Osage Pipeline, which carries crude oil from Cushing, Oklahoma to El Dorado, Kansas, suffered a release of crude oil. The pipeline resumed operations during the third quarter of 2022.

Interest Income
Interest income was $93.5 million for the year ended December 31, 2023 compared to $30.2 million for the year ended December 31, 2022. The increase in interest income was primarily due to the increase in the average cash balance and higher interest rates on cash investments.

Interest Expense
Interest expense was $190.8 million for the year ended December 31, 2023 compared to $175.6 million for the year ended December 31, 2022. This increase was primarily due to the April 2022 issuance of $400 million in aggregate principal amount of 6.375% senior notes maturing in April 2027 and higher market interest rates on HEP's revolving credit facility during the year ended December 31, 2023.

Gain on Business Interruption Insurance Settlement
During the year ended December 31, 2022, we recorded a gain of $15.2 million from a settlement of our business interruption claim related to winter storm Uri that occurred in the first quarter of 2021.

Gain on Early Extinguishment of Debt
For the year ended December 31, 2022, we recorded a $0.6 million gain on the extinguishment of debt related to our open market repurchase of $42.2 million in principal of our $350 million aggregate principal amount of our 2.625% senior notes maturing October 2023 at a cost of $41.4 million.

Gain (loss) on Foreign Currency Transactions
Remeasurement adjustments resulting from the foreign currency conversion of the intercompany financing notes payable by PCLI net of mark-to-market valuations on foreign exchange forward contracts with banks which hedge the foreign currency exposure on these intercompany notes was a gain of $2.5 million for the year ended December 31, 2023 compared to a loss of $1.6 million for the year ended December 31, 2022. For the years ended December 31, 2023 and 2022, gain (loss) on foreign currency transactions included a loss of $7.4 million and a gain of $27.8 million, respectively, on foreign exchange forward contracts (utilized as an economic hedge).

Gain on Sale of Assets and Other
For the year ended December 31, 2023, we recorded a $15.0 million gain from the settlement of a preservation of property claim related to winter storm Uri that occurred in the first quarter of 2021.

Income Taxes
For the year ended December 31, 2023, we recorded an income tax expense of $441.6 million compared to $894.9 million for the year ended December 31, 2022. This decrease was principally due to lower pre-tax income during the year ended December 31, 2023 compared to the year ended December 31, 2022. Our effective tax rates were 20.5% and 22.7% for the years ended December 31, 2023 and 2022, respectively. The year-over-year decrease in the effective tax rate is principally due to the relationship between the pre-tax results and the earnings attributable to the noncontrolling interest that is not included in income for tax purposes and other non-taxable permanent differences.

LIQUIDITY AND CAPITAL RESOURCES

HF Sinclair Credit Agreement
We have a $1.65 billion senior unsecured revolving credit facility maturing in April 2026 (the “HF Sinclair Credit Agreement”). The HF Sinclair Credit Agreement may be used for revolving credit loans and letters of credit from time to time and is available to fund general corporate purposes. At December 31, 2023, we were in compliance with all covenants, had no outstanding borrowings and had outstanding letters of credit totaling $0.3 million under the HF Sinclair Credit Agreement.

HEP Credit Agreement
Through our wholly owned subsidiary, HEP, we have a $1.2 billion senior secured revolving credit facility maturing in July 2025 (the “HEP Credit Agreement”). The HEP Credit Agreement is available to fund capital expenditures, investments, acquisitions, distribution payments, working capital and for general partnership purposes. It is also available to fund letters of credit up to a $50 million sub-limit and has an accordion feature that allows us to increase the commitments under the HEP Credit Agreement up to a maximum amount of $1.7 billion.

In connection with the consummation of the HEP Merger Transaction, we amended the HEP Credit Agreement to, among other things, (a) provide a guaranty from us and terminated all guaranties from subsidiaries of HEP, (b) amend the definition of "Investment Grade Rating" (as defined in the HEP Credit Agreement) to reference the credit rating of our senior unsecured indebtedness, (c) eliminate the requirement to deliver separate audited and unaudited financial statements for HEP and its subsidiaries and only provide certain segment-level reporting for HEP with any compliance certificate delivered in accordance with the HEP Credit Agreement and (d) amend certain covenants to eliminate certain restrictions on (i) amendments to intercompany contracts, (ii) transactions with us and our subsidiaries and (iii) investments in and contributions, dividends, transfers and distributions to us and our subsidiaries.

During the year ended December 31, 2023, HEP had net repayments of $212.5 million under the HEP Credit Agreement. At December 31, 2023, we were in compliance with all of its covenants, had outstanding borrowings of $455.5 million and no outstanding letters of credit under the HEP Credit Agreement.

HF Sinclair Senior Notes and HEP Senior Notes Exchange

In October 2023, we repaid at maturity our $59.637 million aggregate principal amount HollyFrontier 2.625% senior notes maturing October 2023 (the "HollyFrontier 2.625% Senior Notes") and $248.190 million aggregate principal amount HF Sinclair 2.625% senior notes maturing October 2023 (the "HF Sinclair 2.625% Senior Notes").

On December 4, 2023, we completed our offers to exchange any and all outstanding HEP 5.000% senior notes maturing February 2028 (the "HEP 5.000% Senior Notes") and HEP 6.375% senior notes maturing April 2027 (the "HEP 6.375% Senior Notes") (and, collectively, the "HEP Senior Notes") for HF Sinclair 5.000% senior notes maturing February 2028 (the "HF Sinclair 5.000% Senior Notes") and HF Sinclair 6.375% senior notes maturing April 2027 (the "HF Sinclair 6.375% Senior Notes") (and, collectively, the "New HF Sinclair Senior Notes") to be issued by HF Sinclair with registration rights and cash. In connection with the exchange offers, HEP amended the indenture governing the HEP Senior Notes to eliminate (i) substantially all of the restrictive covenants, (ii) certain of the events which may lead to an "Event of Default," (iii) the SEC reporting covenant and (iv) the requirement of HEP to offer to purchase the HEP Senior Notes upon a change of control.

The New HF Sinclair Senior Notes are unsecured and unsubordinated obligations of ours and rank equally with all our other existing and future unsecured and unsubordinated indebtedness. Each series of the New HF Sinclair Senior Notes has the same interest rate, interest payment dates, maturity date and redemption terms as the corresponding series of HEP Senior Notes. The New HF Sinclair Senior Notes were issued in exchange for the HEP Senior Notes pursuant to a private exchange offer exempt from registration under the Securities Act of 1933, as amended. This exchange was part of a broader corporate strategy, including the HEP Merger Transaction, which closed on December 1, 2023.

HF Sinclair Financing Arrangements

Certain of our wholly owned subsidiaries entered into financing arrangements whereby such subsidiaries sold a portion of their precious metals catalyst to a financial institution and then leased back the precious metals catalyst in exchange for cash. The volume of the precious metals catalyst and the lease rate are fixed over the term of each lease, and the lease payments are recorded as interest expense. The current leases mature in one year or less. Upon maturity, we must either satisfy the obligation at fair market value or refinance to extend the maturity.

HF Sinclair may, from time to time, issue letters of credit pursuant to uncommitted letters of credit facilities with its lenders. At December 31, 2023, there were no letters of credit outstanding under such facilities.

See Note 13 "Debt" in the Notes to Consolidated Financial Statements for additional information on our debt instruments.

Liquidity

We believe our current cash and cash equivalents, along with future internally generated cash flow and funds available under our credit facilities, will provide sufficient resources to fund currently planned capital projects and our liquidity needs for the foreseeable future. We expect that, to the extent necessary, we can raise additional funds from time to time through equity or debt financings in the public and private capital markets. Further, we may from time to time seek to retire some or all of our outstanding debt or debt agreements through cash purchases, and/or exchanges, open market purchases, privately negotiated transactions, tender offers or otherwise. Such transactions, if any, may be material and will depend on prevailing market conditions, our liquidity requirements and other factors. In addition, components of our long-term growth strategy include the optimization of existing units at our facilities and selective acquisition of complementary assets for our operations intended to increase earnings and cash flow. We also expect to use cash for payment of cash dividends, which are at the discretion of our Board of Directors, and for the repurchase of common stock under our share repurchase program.

Our liquidity was approximately $3.75 billion at December 31, 2023, consisting of cash and cash equivalents of $1.35 billion, n undrawn $1.65 billion credit facility under the HF Sinclair Credit Agreement and $744.5 million remaining availability under the HEP Credit Agreement.

We consider all highly liquid instruments with a maturity of three months or less at the time of purchase to be cash equivalents. These primarily consist of investments in conservative, highly rated instruments issued by financial institutions, government and corporate entities with strong credit standings and money market funds. Cash equivalents are stated at cost, which approximates market value.

In September 2022, our Board of Directors approved a $1.0 billion share repurchase program (the"September 2022 Share Repurchase Program"), which replaced all existing share repurchase programs at that time, authorizing us to repurchase common stock in the open market or through privately negotiated transactions. Privately negotiated repurchases from REH Company were also authorized under the September 2022 Share Repurchase Program, subject to REH Company's interest in selling its shares and other limitations. As of August 15, 2023, we had repurchased $995.0 million under the September 2022 Share Repurchase Program.

On August 15, 2023, our Board of Directors approved a new $1.0 billion share repurchase program (the"August 2023 Share Repurchase Program"), which replaced all existing share repurchase programs, including the $5.0 million remaining authorization under the September 2022 Share Repurchase Program. The August 2023 Share Repurchase Program authorizes us to repurchase common stock in the open market or through privately negotiated transactions. Privately negotiated repurchases from REH Company are also authorized under the August 2023 Share Repurchase Program, subject to REH Company's interest in selling its shares and other limitations. The timing and amount of share repurchases, including those from REH Company, will depend on market conditions and corporate, tax, regulatory and other relevant considerations. In addition, we are authorized by our Board of Directors to repurchase shares in an amount sufficient to offset shares issued under our compensation programs. The August 2023 Share Repurchase Program may be discontinued at any time by our Board of Directors.

During the year ended December 31, 2023, we made open market and privately negotiated purchases of 18,779,880 shares for $974.5 million under our share repurchase programs, of which 15,515,302 shares were repurchased for $810.6 million pursuant to privately negotiated repurchases from REH Company. As of December 31, 2023, we had remaining authorization to repurchase up to $676.4 million under the August 2023 Share Repurchase Program.

On January 3, 2024, we repurchased 454,380 shares of our outstanding common stock from REH Company in a privately negotiated transaction under the August 2023 Share Repurchase Program and pursuant to the Stock Purchase Agreement, dated January 3, 2024 (the "January Stock Purchase Agreement"), between us and REH Company. The price paid by us under the January Stock Purchase Agreement was $55.02 per share resulting in an aggregate purchase price of $25.0 million. The purchase price was funded with cash on hand.

On February 8, 2024, we repurchased 1,061,946 shares of our outstanding common stock from REH Company in a privately negotiated transaction under the August 2023 Share Repurchase Program and pursuant to the Stock Purchase Agreement, dated February 8, 2024 (the "February Stock Purchase Agreement"), between us and REH Company. The price paid by us under the February Stock Purchase Agreement was $56.50 per share resulting in an aggregate purchase price of $60.0 million. The purchase price was funded with cash on hand. As of February 15, 2024, we had remaining authorization to repurchase up to $591.4 million under the August 2023 Share Repurchase Program.

Cash Flows – Operating Activities

Year Ended December 31, 2023 Compared to Year Ended December 31, 2022

Net cash flows provided by operating activities were $2,297.2 million for the year ended December 31, 2023 compared to $3,777.2 million for the year ended December 31, 2022, a decrease of $1,479.9 million primarily driven by lower income from operations combined with higher turnaround spend during the year ended December 31, 2023. Changes in working capital decreased operating cash flows by $119.1 million and increased operating cash flows by $28.7 million for the years ended December 31, 2023 and 2022, respectively. Additionally, for the year ended December 31, 2023, turnaround expenditures were $555.7 million compared to $144.8 million for the year ended December 31, 2022.

Cash Flows – Investing Activities and Planned Capital Expenditures

Year Ended December 31, 2023 Compared to Year Ended December 31, 2022

For the year ended December 31, 2023, our net cash flows used for investing activities were $371.3 million. Cash expenditures for properties, plants and equipment for the year ended December 31, 2023 were $385.4 million for the year ended December 31, 2023.

For the year ended December 31, 2022, our net cash flows used for investing activities were $774.5 million. On March 14, 2022, we closed the Sinclair Transactions for cash consideration of $251.4 million. The remainder of the purchase consideration was funded with the issuance of HF Sinclair common stock and HEP common units. See Note 2 “Acquisitions” in the Notes to Consolidated Financial Statements for additional information on the Sinclair Transactions. Cash expenditures for properties, plants and equipment for the year ended December 31, 2022 were $524.0 million, which included HEP capital expenditures of $39.0 million for the year ended December 31, 2022.

Each year our Board of Directors approves our annual capital budget which includes specific projects that management is authorized to undertake. When conditions warrant or as new opportunities arise, additional projects may be approved. The funds appropriated for a particular capital project may be expended over a period of several years, depending on the time required to complete the project. Therefore, our planned capital expenditures for a given year consist of expenditures appropriated in that year’s capital budget plus expenditures for projects appropriated in prior years which have not yet been completed. Refinery turnaround spending is amortized over the useful life of the turnaround.

The refining industry is capital intensive and requires on-going investments to sustain our refining operations. This includes replacement of, or rebuilding, refinery units and components that extend the useful life. We also invest in projects that improve operational reliability and profitability via enhancements that improve refinery processing capabilities as well as production yield and flexibility. Our capital expenditures also include projects related to renewable diesel, environmental, health and safety compliance and include initiatives as a result of federal and state mandates.

Our refinery operations and related emissions are highly regulated at both federal and state levels, and we invest in our facilities as needed to remain in compliance with these standards. Additionally, when faced with new emissions or fuels standards, we seek to execute projects that facilitate compliance and also improve the operating costs and / or yields of associated refining processes.

Expected capital and turnaround cash spending for 2024 is as follows:

	Expected Cash Spending
	(In millions)
HF Sinclair	
Refining	$ 235.0
Renewables	5.0
Lubricants & Specialties	40.0
Marketing	10.0
Midstream	30.0
Corporate	65.0
Turnarounds and catalyst	415.0
Total sustaining	800.0
Growth capital	75.0
Total capital	$ 875.0

Cash Flows – Financing Activities

Year Ended December 31, 2023 Compared to Year Ended December 31, 2022

For the year ended December 31, 2023, our net cash flows used for financing activities were $2,243.9 million. During the year ended December 31, 2023, we purchased $999.3 million of treasury stock, paid $340.7 million in dividends, paid $307.8 million upon the maturity of our HF Sinclair 2.625% Senior Notes and HollyFrontier 2.625% Senior Notes, paid $267.6 million as cash consideration in connection with the HEP Merger Transaction and had net repayments of $212.5 million under the HEP Credit Agreement.

For the year ended December 31, 2022, our net cash flows used for financing activities were $1,560.8 million. During the year ended December 31, 2022, we purchased $1,371.7 million of treasury stock, paid $255.9 million in dividends and paid $41.4 million to extinguish $42.2 million in principal of the HF Sinclair 2.625% Senior Notes and HollyFrontier 2.625% Senior Notes. During the year ended December 31, 2022, HEP received $400.0 million in proceeds from the issuance of the HEP 6.375% Senior Notes, had net repayments of $172.0 million under the HEP Credit Agreement and paid distributions of $96.2 million to noncontrolling interests.

Contractual Obligations and Commitments

The following table presents our long-term contractual obligations as of December 31, 2023 in total and by period due beginning in 2024.

		Payments Due by Period			
Contractual Obligations and Commitments	**Total**	**2024**	**2025 & 2026**	**2027 & 2028**	**Thereafter**
		(In thousands)			
HF Sinclair Corporation					
Long-term debt - principal [1]	$ 2,755,500	$ —	$ 1,455,500	$ 900,000	$ 400,000
Long-term debt - interest [1]	490,770	159,500	229,250	70,520	31,500
Financing arrangements [2]	37,043	37,043	—	—	—
Supply agreements [3]	987,658	541,729	435,072	9,038	1,819
Transportation and storage agreements [4]	2,173,435	237,534	444,719	411,226	1,079,956
Operating and finance leases [5]	563,919	133,381	141,768	78,250	210,520
Other long-term obligations	179,368	45,412	49,179	22,148	62,629
Total	$ 7,187,693	$ 1,154,599	$ 2,755,488	$ 1,491,182	$ 1,786,424

(1) See Note 13 “Debt” in the Notes to Consolidated Financial Statements for a description of our outstanding debt.
(2) We have financing arrangements related to the sale and subsequent lease-back of certain of our precious metals.
(3) We have long-term supply agreements to secure certain quantities of crude oil, feedstock and other resources used in the production process at market prices. We have estimated future payments under these fixed-quantity agreements expiring between 2024 and 2028 using current market rates.
(4) Consists of contractual obligations under agreements with third parties for the transportation of crude oil, natural gas and feedstocks to our refineries and for terminal and storage services under contracts expiring between 2024 and 2038.
(5) Operating and finance lease obligations include options to extend terms that are reasonably certain of being exercised.

CRITICAL ACCOUNTING POLICIES AND ESTIMATES

Our discussion and analysis of our financial condition and results of operations are based upon our consolidated financial statements, which have been prepared in accordance with accounting principles generally accepted in the United States. The preparation of these financial statements requires us to make estimates and judgments that affect the reported amounts of assets, liabilities, revenues and expenses, and related disclosure of contingent assets and liabilities as of the date of the financial statements. Actual results may differ from these estimates under different assumptions or conditions. We consider the following policies to be the most critical to understanding the judgments that are involved and the uncertainties that could impact our results of operations, financial condition and cash flows. For additional information, see Note 1 “Description of Business and Summary of Significant Accounting Policies” in the Notes to Consolidated Financial Statements.

Goodwill and Long-lived Assets
As of December 31, 2023, our goodwill balance was $3.0 billion, with goodwill assigned to our Refining, Renewables, Marketing, Lubricants & Specialties and Midstream segments. Goodwill represents the excess of the cost of an acquired entity over the fair value of the assets acquired and liabilities assumed. Goodwill is not subject to amortization and is tested annually or more frequently if an event occurs or circumstances change that would more likely than not reduce the fair value of a reporting unit below its carrying amount. Our goodwill impairment testing first entails either a quantitative assessment or an optional qualitative assessment to determine whether it is more likely than not that the fair value of a reporting unit is less than its carrying amount. If we determine that based on the qualitative factors that it is more likely than not that the carrying value of the reporting unit is greater than its fair value, a quantitative test is performed in which we estimate the fair value of the related reporting unit. If the carrying amount of a reporting unit exceeds its fair value, the goodwill of that reporting unit is impaired, and we measure goodwill impairment as the excess of the carrying amount of reporting unit over the related fair value.

We performed our annual goodwill impairment testing quantitatively as of July 1, 2023 and determined there was no impairment of goodwill attributable to our reporting units. The estimated fair values of our reporting units were derived using a combination of income and market approaches. The income approach reflects expected future cash flows based on estimated forecasted production levels, selling prices, gross margins, operating costs and capital expenditures. Our market approaches include both the guideline public company and guideline transaction methods. Both methods utilize pricing multiples derived from historical market transactions of other like-kind assets. The excess of the fair values of the reporting units over their respective carrying values ranged from 23% to 91%. Increasing the discount rate by 1.0% or reducing the terminal cash flow growth rate by 1.0% would not have changed the results of our annual goodwill testing.

In performing our impairment test of goodwill, we developed cash flow forecasts for each of our reporting units. Significant judgment is involved in performing these fair value estimates since the results are based on forecasted financial information. The cash flow forecasts include significant assumptions such as planned utilization, end-user demand, selling prices, gross margins, operating costs and capital expenditures. Another key assumption applied to these forecasts to determine the fair value of a reporting unit is the discount rate. The discount rate is intended to reflect the weighted average cost of capital for a market participant and the risks associated with the realization of the estimated future cash flows. Our fair value estimates are based on
projected cash flows, which we believe to be reasonable.

We continually monitor and evaluate various factors for potential indicators of goodwill and long-lived asset impairment. A reasonable expectation exists that further deterioration in our operating results or overall economic conditions could result in an impairment of goodwill and / or long-lived asset impairments at some point in the future. Future impairment charges could be material to our results of operations and financial condition.

Contingencies
We are subject to proceedings, lawsuits and other claims related to environmental, labor, product and other matters. We are required to assess the likelihood of any adverse judgments or outcomes to these matters as well as potential ranges of probable losses. A determination of the amount of reserves required, if any, for these contingencies is made after careful analysis of each individual issue. The required reserves may change in the future due to new developments in each matter or changes in approach such as a change in settlement strategy in dealing with these matters.

RISK MANAGEMENT

We use certain strategies to reduce some commodity price and operational risks. We do not attempt to eliminate all market risk exposures when we believe that the exposure relating to such risk would not be significant to our future earnings, financial position, capital resources or liquidity or that the cost of eliminating the exposure would outweigh the benefit.

Commodity Price Risk Management
Our primary market risk is commodity price risk. We are exposed to market risks related to the volatility in crude oil and refined products, as well as volatility in the price of natural gas used in our refining operations. We periodically enter into derivative contracts in the form of commodity price swaps, collar contracts, forward purchase and sales and futures contracts to mitigate price exposure with respect to our inventory positions, natural gas purchases, sales prices of refined products and crude oil costs.

Foreign Currency Risk Management
We are exposed to market risk related to the volatility in foreign currency exchange rates. We periodically enter into derivative contracts in the form of foreign exchange forward contracts to mitigate the exposure associated with fluctuations on intercompany notes with our foreign subsidiaries that are not denominated in the U.S. dollar.

As of December 31, 2023, we have the following notional contract volumes related to all outstanding derivative contracts used to mitigate commodity price and foreign currency risk (all maturing in 2024):

Contract Description	Total Outstanding Notional	Unit of Measure
NYMEX futures (WTI) - short	640,000	Barrels
Forward gasoline contracts - long	800,000	Barrels
Foreign currency forward contracts	387,613,367	U.S. dollar
Forward commodity contracts (platinum) [1]	36,969	Troy ounces
Natural gas price swaps (basis spread) - long	6,667,000	MMBTU

(1) Represents an embedded derivative within our catalyst financing arrangements, which may be refinanced or require repayment under certain conditions. See Note 13 “Debt” in the Notes to Consolidated Financial Statements for additional information on these financing arrangements.

The following sensitivity analysis provides the hypothetical effects of market price fluctuations in commodity prices for our open commodity derivative contracts at December 31, 2023 and 2022:

	Derivative Fair Value Gain (Loss) at December 31,	
	2023	2022
	(In thousands)	
10% increase in underlying commodity prices	$ (4,682)	$ (3,502)
10% decrease in underlying commodity prices	$ 4,682	$ 3,298

Interest Rate Risk Management

The market risk inherent in our fixed-rate debt is the potential change arising from increases or decreases in interest rates as discussed below.

For the fixed rate HF Sinclair Senior Notes, HollyFrontier Senior Notes and HEP Senior Notes (each as defined in Note 13 “Debt” in the Notes to Consolidated Financial Statements), changes in interest rates will generally affect fair value of the debt, but not earnings or cash flows. The outstanding principal, estimated fair value and estimated change in fair value (assuming a hypothetical 10% change in the yield-to-maturity rates) for this debt as of December 31, 2023 is presented below:

	Outstanding Principal	Estimated Fair Value	Estimated Change in Fair Value
		(In thousands)	
HF Sinclair, HollyFrontier and HEP Senior Notes	$ 2,300,000	$2,271,856	$ 41,358

For the variable rate HEP Credit Agreement, changes in interest rates would affect cash flows, but not the fair value. At December 31, 2023, outstanding borrowings under the HEP Credit Agreement were $455.5 million. A hypothetical 10% change in interest rates applicable to the HEP Credit Agreement would not materially affect cash flows.

Our operations are subject to catastrophic losses, operational hazards and unforeseen interruptions, including but not limited to fire, explosion, releases or spills, cyberattacks, weather-related perils, vandalism, power failures, mechanical failures and other events beyond our control. We maintain various insurance coverages, including general liability, property damage, business interruption and cyber insurance, subject to certain deductibles and insurance policy terms and conditions. We are not fully insured against certain risks because such risks are not fully insurable, coverage is unavailable, or premium costs, in our judgment, do not justify such expenditures.

Financial information is reviewed on the counterparties in order to review and monitor their financial stability and assess their ongoing ability to honor their commitments under the derivative contracts. We have not experienced, nor do we expect to experience, any difficulty in the counterparties honoring their commitments.

We have a risk management oversight committee consisting of members from our senior management. This committee oversees our risk enterprise program, monitors our risk environment and provides direction for activities to mitigate identified risks that may adversely affect the achievement of our goals.

Item 7A. Quantitative and Qualitative Disclosures About Market Risk

See "Risk Management" under "Management's Discussion and Analysis of Financial Condition and Results of Operations."

Reconciliations to Amounts Reported Under Generally Accepted Accounting Principles

Reconciliations of earnings before interest, taxes, depreciation and amortization ("EBITDA") to amounts reported under generally accepted accounting principles in financial statements.

Earnings before interest, taxes, depreciation and amortization, which we refer to as EBITDA, is calculated as net income attributable to HF Sinclair stockholders plus (i) income tax provision, (ii) interest expense, net of interest income and (iii) depreciation and amortization. EBITDA is not a calculation provided for under GAAP; however, the amounts included in the EBITDA calculation are derived from amounts included on our consolidated financial statements. EBITDA should not be considered as an alternative to net income or operating income as an indication of our operating performance or as an alternative to operating cash flow as a measure of liquidity. EBITDA is not necessarily comparable to similarly titled measures of other companies. EBITDA is presented here because it is a widely used financial indicator used by investors and analysts to measure performance. EBITDA is also used by our management for internal analysis and as a basis for financial covenants.

Set forth below is our calculation of EBITDA.

	Years Ended December 31,		
	2023	2022	2021
	(In thousands)		
Net income attributable to HF Sinclair stockholders	$ 1,589,666	$ 2,922,668	$ 558,324
Add interest expense	190,796	175,628	125,175
Subtract interest income	(93,468)	(30,179)	(4,019)
Add income tax expense	441,612	894,872	123,898
Add depreciation and amortization	770,573	656,787	503,539
EBITDA	$ 2,899,179	$ 4,619,776	$ 1,306,917

Reconciliations of refinery operating information (non-GAAP performance measures) to amounts reported under generally accepted accounting principles in financial statements.

Refinery gross margin and net operating margin are non-GAAP performance measures that are used by our management and others to compare our refining performance to that of other companies in our industry. We believe these margin measures are helpful to investors in evaluating our refining performance on a relative and absolute basis. Refinery gross margin per produced barrel sold is total Refining segment revenues less total Refining segment cost of products sold, exclusive of lower of cost or market inventory valuation adjustments, divided by sales volumes of produced refined products sold. Net operating margin per barrel sold is the difference between refinery gross margin and refinery operating expenses per produced barrel sold. These two margins do not include the non-cash effects of long-lived asset impairment charges, lower of cost or market inventory valuation adjustments or depreciation and amortization. Each of these component performance measures can be reconciled directly to our consolidated statements of income. Other companies in our industry may not calculate these performance measures in the same manner.

Reconciliation of average refining net operating margin per produced barrel sold to refinery gross margin to refining sales and other revenues

	Years Ended December 31,		
	2023	2022	2021
	(Dollars in thousands, except per barrel amounts)		
Refining segment sales and other revenues	$ 28,672,604	$ 34,412,909	$ 16,358,558
Refining segment cost of products sold (exclusive of lower of cost or market inventory valuation adjustment)	23,969,557	28,270,195	14,673,062
Lower of cost or market inventory valuation adjustment	220,558	—	(318,353)
	4,482,489	6,142,714	2,003,849
Add (subtract) lower of cost or market inventory valuation adjustment	220,558	—	(318,353)
Refinery gross margin	$ 4,703,047	$ 6,142,714	$ 1,685,496
Refining segment operating expenses	$ 1,946,958	$ 1,815,931	$ 1,090,424
Produced barrels sold (BPD)	602,280	628,340	424,100
Refinery gross margin per produced barrel sold	$ 21.39	$ 26.78	$ 10.89
Less average refinery operating expenses per produced barrel sold	8.86	7.92	7.04
Net operating margin per produced barrel sold	$ 12.53	$ 18.86	$ 3.85

Reconciliation of renewables operating information (non-GAAP performance measures) to amounts reported under generally accepted accounting principles in financial statements.

Renewables gross margin and net operating margin are non-GAAP performance measures that are used by our management and others to compare our renewables performance to that of other companies in our industry. We believe these margin measures are helpful to investors in evaluating our renewables performance on a relative and absolute basis. Renewables gross margin per produced gallon sold is total Renewables segment revenues less total Renewables segment cost of products sold, exclusive of lower of cost or market inventory valuation adjustments, divided by sales volumes of produced renewables products sold. Net operating margin per produced gallon sold is the difference between renewables gross margin and renewables operating expenses per produced gallon sold. These two margins do not include the non-cash effects of lower of cost or market inventory valuation adjustments and depreciation and amortization. Each of these component performance measures can be reconciled directly to our consolidated statements of income. Other companies in our industry may not calculate these performance measures in the same manner.

Reconciliation of renewables gross margin and operating expenses to gross margin per produced gallon sold and net operating margin per produced gallon sold

	Years Ended December 31,	
	2023	2022
	(In thousands, except for per gallon amounts)	
Renewables segment sales and other revenues	$ 1,188,990	$ 1,015,499
Renewables segment cost of products sold	1,080,919	974,167
Lower of cost or market inventory valuation adjustment	49,861	52,412
	58,210	(11,080)
Add lower of cost or market inventory valuation adjustment	49,861	52,412
Renewables gross margin	$ 108,071	$ 41,332
Renewables operating expenses	$ 109,056	$ 111,974
Produced gallons sold (in thousand gallons)	215,510	136,204
Renewables gross margin per produced gallon sold	$ 0.50	$ 0.30
Less operating expenses per produced gallon sold	0.51	0.82
Net operating margin per produced gallon sold	$ (0.01)	$ (0.52)

Reconciliation of marketing operating information (non-GAAP performance measures) to amounts reported under generally accepted accounting principles in financial statements.

Marketing gross margin is a non-GAAP performance measure that is used by our management and others to compare our marketing performance to that of other companies in our industry. We believe this margin measure is helpful to investors in evaluating our marketing performance on a relative and absolute basis. Marketing gross margin per gallon sold is total Marketing segment revenues less total Marketing segment cost of products sold divided by sales volumes of marketing products sold. This margin does not include the non-cash effects of depreciation and amortization. This component performance measure can be reconciled directly to our consolidated statements of income. Other companies in our industry may not calculate these performance measures in the same manner.

Reconciliation of marketing gross margin to gross margin per gallon sold

	Years Ended December 31,	
	2023	2022
Marketing segment sales and other revenues	$ 4,146,292	$ 3,911,922
Marketing segment cost of products sold	4,050,759	3,845,625
Marketing gross margin	$ 95,533	$ 66,297
Sales volumes (in thousand gallons)	1,441,607	1,118,444
Marketing segment gross margin per gallon sold	$ 0.07	$ 0.06

Item 8. Financial Statements and Supplementary Data

MANAGEMENT'S REPORT ON ITS ASSESSMENT OF THE COMPANY'S INTERNAL CONTROL OVER FINANCIAL REPORTING

Management of HF Sinclair Corporation (the “Company”) is responsible for establishing and maintaining adequate internal control over financial reporting.

All internal control systems, no matter how well designed, have inherent limitations. Therefore, even those systems determined to be effective can provide only reasonable assurance with respect to financial statement preparation and presentation.

Management assessed the Company's internal control over financial reporting as of December 31, 2023 using the criteria for effective control over financial reporting established in “Internal Control - Integrated Framework” issued by the Committee of Sponsoring Organizations of the Treadway Commission (2013 framework). Based on this assessment, management concludes that, as of December 31, 2023, the Company maintained effective internal control over financial reporting.

The Company's independent registered public accounting firm has issued an attestation report on the effectiveness of the Company's internal control over financial reporting as of December 31, 2023. That report is included herein.

REPORT OF INDEPENDENT REGISTERED PUBLIC ACCOUNTING FIRM

To the Stockholders and the Board of Directors of HF Sinclair Corporation

Opinion on Internal Control over Financial Reporting

We have audited HF Sinclair Corporation's internal control over financial reporting as of December 31, 2023, based on criteria established in Internal Control—Integrated Framework issued by the Committee of Sponsoring Organizations of the Treadway Commission (2013 framework) (the COSO criteria). In our opinion, HF Sinclair Corporation (the Company) maintained, in all material respects, effective internal control over financial reporting as of December 31, 2023, based on the COSO criteria.

We also have audited, in accordance with the standards of the Public Company Accounting Oversight Board (United States) (PCAOB), the consolidated balance sheets of the Company as of December 31, 2023 and 2022, the related consolidated statements of income, comprehensive income, equity and cash flows for each of the three years in the period ended December 31, 2023, and the related notes and our report dated February 21, 2024 expressed an unqualified opinion thereon.

Basis for Opinion

The Company's management is responsible for maintaining effective internal control over financial reporting and for its assessment of the effectiveness of internal control over financial reporting included in the accompanying Management's Report on its Assessment of the Company's Internal Control Over Financial Reporting. Our responsibility is to express an opinion on the Company's internal control over financial reporting based on our audit. We are a public accounting firm registered with the PCAOB and are required to be independent with respect to the Company in accordance with the U.S. federal securities laws and the applicable rules and regulations of the Securities and Exchange Commission and the PCAOB.

We conducted our audit in accordance with the standards of the PCAOB. Those standards require that we plan and perform the audit to obtain reasonable assurance about whether effective internal control over financial reporting was maintained in all material respects.

Our audit included obtaining an understanding of internal control over financial reporting, assessing the risk that a material weakness exists, testing and evaluating the design and operating effectiveness of internal control based on the assessed risk, and performing such other procedures as we considered necessary in the circumstances. We believe that our audit provides a reasonable basis for our opinion.

Definition and Limitations of Internal Control Over Financial Reporting

A company's internal control over financial reporting is a process designed to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles. A company's internal control over financial reporting includes those policies and procedures that (1) pertain to the maintenance of records that, in reasonable detail, accurately and fairly reflect the transactions and dispositions of the assets of the company; (2) provide reasonable assurance that transactions are recorded as necessary to permit preparation of financial statements in accordance with generally accepted accounting principles, and that receipts and expenditures of the company are being made only in accordance with authorizations of management and directors of the company; and (3) provide reasonable assurance regarding prevention or timely detection of unauthorized acquisition, use, or disposition of the company's assets that could have a material effect on the financial statements.

Because of its inherent limitations, internal control over financial reporting may not prevent or detect misstatements. Also, projections of any evaluation of effectiveness to future periods are subject to the risk that controls may become inadequate because of changes in conditions, or that the degree of compliance with the policies or procedures may deteriorate.

/s/ Ernst & Young LLP

Dallas, Texas
February 21, 2024

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REPORT OF INDEPENDENT REGISTERED PUBLIC ACCOUNTING FIRM

To the Stockholders and the Board of Directors of HF Sinclair Corporation

Opinion on the Financial Statements

We have audited the accompanying consolidated balance sheets of HF Sinclair Corporation (the Company) as of December 31, 2023 and 2022, the related consolidated statements of income, comprehensive income, equity and cash flows for each of the three years in the period ended December 31, 2023, and the related notes (collectively referred to as the "consolidated financial statements"). In our opinion, the consolidated financial statements present fairly, in all material respects, the financial position of the Company at December 31, 2023 and 2022, and the results of its operations and its cash flows for each of the three years in the period ended December 31, 2023, in conformity with U.S. generally accepted accounting principles.

We also have audited, in accordance with the standards of the Public Company Accounting Oversight Board (United States) (PCAOB), the Company's internal control over financial reporting as of December 31, 2023, based on criteria established in Internal Control—Integrated Framework issued by the Committee of Sponsoring Organizations of the Treadway Commission (2013 framework), and our report dated February 21, 2024 expressed an unqualified opinion thereon.

Basis for Opinion

These financial statements are the responsibility of the Company's management. Our responsibility is to express an opinion on the Company's financial statements based on our audits. We are a public accounting firm registered with the PCAOB and are required to be independent with respect to the Company in accordance with the U.S. federal securities laws and the applicable rules and regulations of the Securities and Exchange Commission and the PCAOB.

We conducted our audits in accordance with the standards of the PCAOB. Those standards require that we plan and perform the audit to obtain reasonable assurance about whether the financial statements are free of material misstatement, whether due to error or fraud. Our audits included performing procedures to assess the risks of material misstatement of the financial statements, whether due to error or fraud, and performing procedures that respond to those risks. Such procedures included examining, on a test basis, evidence regarding the amounts and disclosures in the financial statements. Our audits also included evaluating the accounting principles used and significant estimates made by management, as well as evaluating the overall presentation of the financial statements. We believe that our audits provide a reasonable basis for our opinion.

Critical Audit Matter

The critical audit matter communicated below is a matter arising from the current period audit of the financial statements that was communicated or required to be communicated to the audit committee and that: (1) relates to accounts or disclosures that are material to the financial statements and (2) involved our especially challenging, subjective or complex judgments. The communication of the critical audit matter does not alter in any way our opinion on the consolidated financial statements, taken as a whole, and we are not, by communicating the critical audit matter below, providing a separate opinion on the critical audit matter or on the account or disclosure to which it relates.

Valuation of Goodwill

Description of the Matter

At December 31, 2023, the Company's goodwill was $2,978 million, including goodwill assigned to the Refining, Renewables, Marketing, Lubricants & Specialties, and Midstream segments of $1,977 million, $159 million, $164 million, $246 million, and $432 million, respectively. As described in Note 1 and Note 11 of the consolidated financial statements, goodwill is tested for impairment at least annually on July 1 at the reporting unit level or more frequently if events or changes in circumstances indicate the asset might be impaired.

Auditing management's goodwill impairment testing was complex and highly judgmental for the Company's El Dorado Refinery reporting unit due to the significant estimation required to determine the fair value of this reporting unit. In particular, the fair value estimates were sensitive to significant assumptions, such as revenue, gross margins, and EBITDA, and discount rates which are affected by expectations about future market or economic conditions. These assumptions have a significant effect on the fair value estimates.

How We Addressed the Matter in Our Audit

We obtained an understanding, evaluated the design and tested the operating effectiveness of controls over the Company's goodwill impairment testing process. For example, we tested controls over management's review of the significant inputs and assumptions used in determining the reporting unit fair value.

To test the estimated fair value of the Company's El Dorado Refinery reporting unit, we performed audit procedures with the support of a valuation specialist that included, among others, assessing the methodologies used and testing the significant assumptions discussed above and the underlying data used by the Company in its analysis. We compared the significant assumptions used by management to relevant industry and economic trends, published forward prices, historical operating results and other relevant factors. We performed sensitivity analyses on significant assumptions to evaluate the changes in the fair value of the reporting unit that would result from changes in the assumptions. We also tested management's reconciliation of the fair value of the reporting units to the market capitalization of the Company.

/s/ Ernst & Young LLP

We have served as the Company's auditor since 1977.
Dallas, Texas
February 21, 2024

HF SINCLAIR CORPORATION
CONSOLIDATED BALANCE SHEETS
(In thousands, except share data)

	December 31, 2023	December 31, 2022
ASSETS		
Current assets:		
Cash and cash equivalents	$ 1,353,747	$ 1,665,066
Accounts receivable: Product and transportation	1,527,950	1,626,199
Crude oil resales	197,169	76,950
	1,725,119	1,703,149
Inventories: Crude oil and refined products	2,645,724	2,853,425
Materials, supplies and other	276,107	361,103
	2,921,831	3,214,528
Income taxes receivable	56,528	53,563
Prepayments and other	89,229	112,013
Total current assets	6,146,454	6,748,319
Properties, plants and equipment, at cost	10,533,432	10,146,652
Less accumulated depreciation	(3,906,600)	(3,457,747)
	6,626,832	6,688,905
Operating lease right-of-use assets	348,006	351,068
Other assets: Turnaround costs	644,957	376,158
Goodwill	2,977,744	2,978,315
Intangibles and other	972,272	982,718
	4,594,973	4,337,191
Total assets	$ 17,716,265	$ 18,125,483
LIABILITIES AND EQUITY		
Current liabilities:		
Accounts payable	$ 2,205,759	$ 2,334,107
Income taxes payable	8,772	7,818
Operating lease liabilities	106,973	109,926
Current debt	—	306,959
Accrued liabilities	453,045	486,719
Total current liabilities	2,774,549	3,245,529
Long-term debt	2,739,083	2,948,513
Noncurrent operating lease liabilities	249,479	254,215
Deferred income taxes	1,297,130	1,262,165
Other long-term liabilities	418,726	397,489
Commitments and contingencies (Note 19)		
Equity:		
HF Sinclair stockholders' equity:		
Preferred stock, $1.00 par value – 5,000,000 shares authorized; none issued	—	—
Common stock $0.01 par value – 320,000,000 shares authorized; 223,231,546 shares issued as of December 31, 2023 and December 31, 2022	2,232	2,232
Additional capital	5,993,661	6,468,775
Retained earnings	5,379,182	4,130,252
Accumulated other comprehensive loss	(11,784)	(22,013)
Common stock held in treasury, at cost - 23,235,599 and 26,152,344 shares as of December 31, 2023 and December 31, 2022, respectively	(1,194,201)	(1,335,431)
Total HF Sinclair stockholders' equity	10,169,090	9,243,815
Noncontrolling interest	68,208	773,757
Total equity	10,237,298	10,017,572
Total liabilities and equity	$ 17,716,265	$ 18,125,483

See accompanying notes.

HF SINCLAIR CORPORATION
CONSOLIDATED STATEMENTS OF INCOME
(In thousands, except per share data)

	Years Ended December 31,		
	2023	**2022**	**2021**
Sales and other revenues	$ 31,964,395	$ 38,204,839	$ 18,389,142
Operating costs and expenses:			
Cost of products sold (exclusive of depreciation and amortization):			
Cost of products sold (exclusive of lower of cost or market inventory valuation adjustment)	25,784,449	30,680,013	15,567,052
Lower of cost or market inventory valuation adjustment	270,419	52,412	(310,123)
	26,054,868	30,732,425	15,256,929
Operating expenses (exclusive of depreciation and amortization)	2,438,148	2,334,893	1,517,478
Selling, general and administrative expenses (exclusive of depreciation and amortization)	498,240	426,485	362,010
Depreciation and amortization	770,573	656,787	503,539
Total operating costs and expenses	29,761,829	34,150,590	17,639,956
Income from operations	2,202,566	4,054,249	749,186
Other income (expense):			
Earnings (loss) of equity method investments	17,369	(260)	12,432
Interest income	93,468	30,179	4,019
Interest expense	(190,796)	(175,628)	(125,175)
Gain on business interruption insurance settlement	—	15,202	—
Gain on tariff settlement	—	—	51,500
Gain on early extinguishment of debt	—	604	—
Gain (loss) on foreign currency transactions	2,530	(1,637)	(2,938)
Gain on sale of assets and other	27,370	13,337	98,128
	(50,059)	(118,203)	37,966
Income before income taxes	2,152,507	3,936,046	787,152
Income tax expense (benefit):			
Current	249,062	841,704	(4,672)
Deferred	192,550	53,168	128,570
	441,612	894,872	123,898
Net income	1,710,895	3,041,174	663,254
Less net income attributable to noncontrolling interest	121,229	118,506	104,930
Net income attributable to HF Sinclair stockholders	$ 1,589,666	$ 2,922,668	$ 558,324
Earnings per share:			
Basic	$ 8.29	$ 14.28	$ 3.39
Diluted	$ 8.29	$ 14.28	$ 3.39
Average number of common shares outstanding:			
Basic	190,035	202,566	162,569
Diluted	190,035	202,566	162,569

See accompanying notes.

HF SINCLAIR CORPORATION
CONSOLIDATED STATEMENTS OF COMPREHENSIVE INCOME
(In thousands)

	Years Ended December 31,		
	2023	**2022**	**2021**
Net income	$ 1,710,895	$ 3,041,174	$ 663,254
Other comprehensive income:			
Foreign currency translation adjustment	13,161	(32,383)	(13,336)
Hedging instruments:			
Change in fair value of cash flow hedging instruments	(3,236)	(4,962)	(17,548)
Reclassification adjustments to net income on settlement of cash flow hedging instruments	3,236	5,288	17,579
Net unrealized gain on hedging instruments	—	326	31
Pension and other post-retirement benefit obligations:			
Actuarial gain (loss) on pension plans	1,909	(3,836)	2,104
Pension plans (gain) loss reclassified to net income	1,378	(208)	(407)
Actuarial gain on post-retirement healthcare plans	761	7,885	1,133
Post-retirement healthcare plans gain reclassified to net income	(3,859)	(3,440)	(3,328)
Actuarial gain (loss) on retirement restoration plan	(99)	349	2
Retirement restoration plan loss reclassified to net income	11	39	39
Net change in pension and other post-retirement benefit obligations	101	789	(457)
Other comprehensive income (loss) before income taxes	13,262	(31,268)	(13,762)
Income tax expense (benefit)	3,033	(6,584)	(2,971)
Other comprehensive income (loss)	10,229	(24,684)	(10,791)
Total comprehensive income	1,721,124	3,016,490	652,463
Less noncontrolling interest in comprehensive income	121,229	118,506	104,930
Comprehensive income attributable to HF Sinclair stockholders	$ 1,599,895	$ 2,897,984	$ 547,533

See accompanying notes.

HF SINCLAIR CORPORATION
CONSOLIDATED STATEMENTS OF CASH FLOWS
(In thousands)

	Years Ended December 31,		
	2023	**2022**	**2021**
Cash flows from operating activities:			
Net income	$ 1,710,895	$ 3,041,174	$ 663,254
Adjustments to reconcile net income to net cash provided by operating activities:			
Depreciation and amortization	770,573	656,787	503,539
Lower of cost or market inventory valuation adjustment	270,419	52,412	(310,123)
Earnings of equity method investments, inclusive of distributions	8,093	19,769	—
Gain on early extinguishment of debt	—	(604)	—
Gain on sale of assets	(6,879)	(2,118)	(89,765)
Deferred income taxes	192,550	53,168	128,570
Equity-based compensation expense	41,135	30,318	39,273
Change in fair value – derivative instruments	16,555	9,989	(34,096)
(Increase) decrease in current assets:			
Accounts receivable	(17,131)	(4,282)	(614,407)
Inventories	30,246	(224,421)	(344,559)
Income taxes receivable	(2,691)	42,641	(6,415)
Prepayments and other	8,062	(40,810)	(18,672)
Increase (decrease) in current liabilities:			
Accounts payable	(108,859)	194,424	612,410
Income taxes payable	925	(17,169)	23,158
Accrued liabilities	(29,619)	78,349	83,602
Turnaround expenditures	(555,697)	(144,759)	(214,431)
Other, net	(31,342)	32,291	(14,656)
Net cash provided by operating activities	2,297,235	3,777,159	406,682
Cash flows from investing activities:			
Additions to properties, plants and equipment	(353,451)	(485,043)	(725,073)
Additions to properties, plants and equipment – HEP	(31,962)	(38,964)	(88,336)
Acquisitions, net of cash acquired	—	(251,448)	(624,332)
Proceeds from sale of assets	17,361	3,344	106,357
HEP investment in Osage Pipe Line Company LLC	(7,482)	(13,000)	—
Distributions in excess of equity in earnings of equity investments	4,211	10,623	4,165
Net cash used for investing activities	(371,323)	(774,488)	(1,327,219)
Cash flows from financing activities:			
Borrowings under credit agreements	60,000	510,000	555,500
Repayments under credit agreements	(272,500)	(682,000)	(629,000)
Proceeds from issuance of senior notes – HEP	—	400,000	—
Redemption of senior notes	(307,827)	(41,420)	—
Purchase of treasury stock	(999,282)	(1,371,700)	(7,058)
Dividends	(340,736)	(255,928)	(57,663)
Distributions to noncontrolling interest	(102,523)	(96,192)	(75,395)
Contribution from noncontrolling interests	—	—	23,194
Payments on finance leases	(11,923)	(11,713)	(3,990)
HEP Merger Transaction consideration	(267,592)	—	—
Deferred financing costs	(899)	(9,273)	(14,500)
Other, net	(600)	(2,533)	(2,891)
Net cash provided by (used for) financing activities	(2,243,882)	(1,560,759)	(211,803)
Effect of exchange rate on cash flow	6,651	(11,290)	(1,534)
Cash and cash equivalents:			
Increase (decrease) for the period	(311,319)	1,430,622	(1,133,874)
Beginning of period	1,665,066	234,444	1,368,318
End of period	$ 1,353,747	$ 1,665,066	$ 234,444
Supplemental disclosure of cash flow information:			
Cash (paid) received during the period for:			
Interest	$ (203,063)	$ (160,409)	$ (136,429)
Income taxes, net	$ (250,815)	$ (816,379)	$ 19,760
Decrease in accrued and unpaid capital expenditures	$ (5,924)	$ (31,714)	$ (15,319)

See accompanying notes.

HF SINCLAIR CORPORATION
CONSOLIDATED STATEMENTS OF EQUITY
(In thousands)

	HF Sinclair Stockholders' Equity								
	Common Stock					Treasury Stock			
	Shares	Amount	Additional Capital	Retained Earnings	Accumulated Other Comprehensive Income (Loss)	Shares	Amount	Non-controlling Interest	Total Equity
Balance at December 31, 2020	256,046	$ 2,560	$ 4,207,672	$ 3,913,179	$ 13,462	93,632	$ (2,968,512)	$ 553,842	$ 5,722,203
Net income	—	—	—	558,324	—	—	—	104,930	663,254
Dividends ($0.35 declared per common share)	—	—	—	(57,663)	—	—	—	—	(57,663)
Other comprehensive loss, net of tax	—	—	—	—	(10,791)	—	—	—	(10,791)
Issuance of common shares under incentive compensation plans	—	—	(24,313)	—	—	(804)	24,313	—	—
Equity-based compensation	—	—	36,716	—	—	—	—	2,557	39,273
Purchase of treasury stock	—	—	—	—	—	217	(7,058)	—	(7,058)
Distributions to noncontrolling interests	—	—	—	—	—	—	—	(75,395)	(75,395)
Contributions from noncontrolling interests	—	—	—	—	—	—	—	23,194	23,194
Purchase of HEP units for equity grants	—	—	—	—	—	—	—	(2,548)	(2,548)
Other	—	—	—	(4)	—	—	—	—	(4)
Balance at December 31, 2021	256,046	$ 2,560	$ 4,220,075	$ 4,413,836	$ 2,671	93,045	$ (2,951,257)	$ 606,580	$ 6,294,465
Net income	—	—	—	2,922,668	—	—	—	118,506	3,041,174
Dividends ($1.20 declared per common share)	—	—	—	(255,928)	—	—	—	—	(255,928)
Other comprehensive loss, net of tax	—	—	—	—	(24,684)	—	—	—	(24,684)
Issuance of common shares for HFC Transactions	60,230	602	2,148,406	—	—	—	—	—	2,149,008
Issuance of common shares under incentive compensation plans	—	—	(42,962)	—	—	(849)	42,962	—	—
Equity-based compensation	—	—	28,474	—	—	—	—	1,844	30,318
Purchase of treasury stock	—	—	—	—	—	27,001	(1,378,390)	—	(1,378,390)
Retirement of treasury stock	(93,045)	(930)	—	(2,950,324)	—	(93,045)	2,951,254	—	—
Distributions to noncontrolling interests	—	—	—	—	—	—	—	(96,192)	(96,192)
Purchase of HEP units for equity grants	—	—	—	—	—	—	—	(2,363)	(2,363)
Equity attributable to HEP common unit issuance, net of tax	—	—	95,047	—	—	—	—	223,392	318,439
Acquisition of remaining UNEV interests	—	—	19,735	—	—	—	—	(78,010)	(58,275)
Balance at December 31, 2022	223,231	$ 2,232	$ 6,468,775	$ 4,130,252	$ (22,013)	26,152	$ (1,335,431)	$ 773,757	$ 10,017,572
Net income	—	—	—	1,589,666	—	—	—	121,229	1,710,895
Dividends ($1.80 declared per common share)	—	—	—	(340,736)	—	—	—	—	(340,736)
Other comprehensive income, net of tax	—	—	—	—	10,229	—	—	—	10,229
HEP Merger Transaction	—	—	(465,596)	—	—	(21,072)	1,084,593	(725,079)	(106,082)
Issuance of common shares under incentive compensation plans	—	—	(49,229)	—	—	(957)	49,229	—	—
Equity-based compensation	—	—	39,711	—	—	—	—	1,424	41,135
Purchase of treasury stock, inclusive of excise tax	—	—	—	—	—	19,113	(992,592)	—	(992,592)
Distributions to noncontrolling interest holders	—	—	—	—	—	—	—	(102,523)	(102,523)
Purchase of HEP units for equity grants	—	—	—	—	—	—	—	(600)	(600)
Balance at December 31, 2023	223,231	$ 2,232	$ 5,993,661	$ 5,379,182	$ (11,784)	23,236	$ (1,194,201)	$ 68,208	$ 10,237,298

See accompanying notes.

NOTE 1: Description of Business and Summary of Significant Accounting Policies

Description of Business: References herein to HF Sinclair Corporation ("HF Sinclair") include HF Sinclair and its consolidated subsidiaries. In these financial statements, the words "we," "our," "ours" and "us" refer only to HF Sinclair and its consolidated subsidiaries or to HF Sinclair or an individual subsidiary and not to any other person, with certain exceptions. References herein to HF Sinclair "we," "our," "ours" and "us" with respect to time periods prior to March 14, 2022 refer to HollyFrontier Corporation ("HollyFrontier") and its consolidated subsidiaries and do not include Hippo Holding LLC (now known as Sinclair Holding LLC), the parent company of Sinclair Oil LLC, Sinclair Transportation Company LLC or their respective consolidated subsidiaries (collectively, the "Acquired Sinclair Businesses"). References herein to HF Sinclair "we," "our," "ours" and "us" with respect to time periods from and after March 14, 2022 include the operations of the Acquired Sinclair Businesses. Unless otherwise specified, the financial statements included herein include financial information for HF Sinclair, which for the time period from March 14, 2022 to December 31, 2023 includes the combined business operations of HollyFrontier and the Acquired Sinclair Businesses. References herein to Holly Energy Partners, L.P. ("HEP") with respect to time periods prior to the closing of the HEP Merger Transaction (as defined below) on December 1, 2023 refer to HEP and its consolidated subsidiaries.

We are an independent energy company that produces and markets high-value light products such as gasoline, diesel fuel, jet fuel, renewable diesel and other specialty products. We own and operate refineries located in Kansas, Oklahoma, New Mexico, Wyoming, Washington and Utah. We provide petroleum product and crude oil transportation, terminalling, storage and throughput services to our refineries and the petroleum industry. We market our refined products principally in the Southwest United States, the Rocky Mountains extending into the Pacific Northwest and in other neighboring Plains states and we supply high-quality fuels to more than 1,500 branded stations and license the use of the Sinclair brand at more than 300 additional locations throughout the country. We produce renewable diesel at two of our facilities in Wyoming and our facility in New Mexico. In addition, our subsidiaries produce and market base oils and other specialized lubricants in the United States, Canada and the Netherlands, and export products to more than 80 countries.

On December 1, 2023, pursuant to the Agreement and Plan of Merger dated as of August 15, 2023 (the "Merger Agreement") by and among HEP, HF Sinclair, Navajo Pipeline Co., L.P., a Delaware limited partnership and an indirect wholly owned subsidiary of HF Sinclair ("HoldCo"), Holly Apple Holdings LLC, a Delaware limited liability company and a wholly owned subsidiary of HoldCo ("Merger Sub"), HEP Logistics Holdings, L.P., a Delaware limited partnership and the general partner of HEP ("HLH"), and Holly Logistic Services, L.L.C., a Delaware limited liability company and the general partner of HLH, Merger Sub merged with and into HEP, with HEP surviving as an indirect, wholly owned subsidiary of HF Sinclair (the "HEP Merger Transaction").

Under the terms of the Merger Agreement, each outstanding common unit representing a limited partner interest in HEP (an "HEP common unit"), other than the HEP common units already owned by HF Sinclair and its subsidiaries, was converted into the right to receive 0.315 shares of HF Sinclair common stock and $4.00 in cash, without interest. The Merger Agreement consideration totaled $267.6 million in cash and resulted in the issuance of 21,072,326 shares of HF Sinclair common stock from treasury stock.

The HEP Merger Transaction was accounted for in accordance with Financial Accounting Standards Board ("FASB") Accounting Standards Codification ("ASC") 810, "Consolidation." Since we controlled HEP both before and after the HEP Merger Transaction, the changes in our ownership interest in HEP resulting from the HEP Merger Transaction were accounted for as an equity transaction, and no gain or loss was recognized in our Consolidated Statements of Income. The tax effects of the HEP Merger Transaction were recorded as adjustments to deferred income taxes and additional capital consistent with ASC 740, "Income Taxes."

For a description of our existing indebtedness, as well as changes thereto associated with the HEP Merger Transaction, see Note 13.

In connection with the HEP Merger Transaction, for the year ended December 31, 2023, we incurred $23.5 million in incremental direct acquisition and integration costs that principally relate to legal, advisory and other professional fees and are presented as selling, general and administrative expenses in our statements of income.

On May 4, 2021, HollyFrontier Puget Sound Refining LLC (now known as HF Sinclair Puget Sound Refining LLC), a wholly owned subsidiary of HollyFrontier, entered into a sale and purchase agreement with Equilon Enterprises LLC d/b/a Shell Oil Products US ("Shell") to acquire Shell's Puget Sound refinery and related assets, including the on-site cogeneration facility and related logistics assets (the "Puget Sound Refinery"). The acquisition closed on November 1, 2021. See Note 2 for additional information.

On April 27, 2021, our wholly owned subsidiary, 7037619 Canada Inc., entered into a contract for sale of real property in Mississauga, Ontario for base consideration of $98.8 million, or CAD 125 million. The transaction closed on September 15, 2021, and we recorded a gain on sale of assets totaling $86.0 million for the year ended December 31, 2021, which was recognized in "Gain on sale of assets and other" on our consolidated statements of income.

During the first quarter of 2021, we initiated a restructuring within our Lubricants & Specialties segment. As a result of this restructuring, we recorded $7.8 million in employee severance costs for the year ended December 31, 2021, which were recognized primarily as selling, general and administrative expenses in our Lubricants & Specialties segment.

In the third quarter of 2020, we permanently ceased petroleum refining operations at our Cheyenne, Wyoming refinery (the "Cheyenne Refinery") and subsequently began converting certain assets at our Cheyenne Refinery to renewable diesel production. In connection with the cessation of petroleum refining operations at our Cheyenne Refinery, we recognized $1.7 million and $25.8 million in decommissioning expense for the years ended December 31, 2022 and 2021, respectively. We also recognized $1.0 million in employee severance costs for the year ended December 31, 2021. These charges were all recognized in operating expenses in our Corporate and Other segment.

Principles of Consolidation: Our consolidated financial statements include our accounts and the accounts of partnerships and joint ventures that we control through an ownership interest greater than 50% or through a controlling financial interest with respect to variable interest entities. All significant intercompany transactions and balances have been eliminated.

Variable Interest Entities: A variable interest entity ("VIE") is a legal entity whose equity owners do not have sufficient equity at risk for the entity to finance its activities without additional subordinated financial support or, as a group, the equity holders lack the power, through voting rights, to direct the activities that most significantly impact the entity's financial performance, the obligation to absorb the entity's expected losses or rights to expected residual returns. See Note 4 for additional information.

Use of Estimates: The preparation of financial statements in accordance with U.S. generally accepted accounting principles ("GAAP") requires management to make estimates and assumptions that affect the amounts reported in the financial statements and accompanying notes. Actual results could differ from those estimates.

Cash Equivalents: We consider all highly liquid instruments with a maturity of three months or less at the date of purchase to be cash equivalents. Cash equivalents are stated at cost, which approximates market value and are primarily invested in highly-rated instruments issued by government or municipal entities with strong credit standings.

Balance Sheet Offsetting: We purchase and sell inventories of crude oil with certain same-parties that are net settled in accordance with contractual net settlement provisions. Our policy is to present such balances on a net basis since it presents our accounts receivables and payables consistent with our contractual settlement provisions.

Accounts Receivable: Our accounts receivable primarily consist of amounts due from customers that are primarily from sales of refined products and renewable diesel. Credit is extended based on our evaluation of the customer's financial condition, and in certain circumstances collateral, such as letters of credit or guarantees, is required. We reserve for expected credit losses based on our historical loss experience as well as expected credit losses from current economic conditions and management's expectations of future economic conditions. Credit losses are charged to the allowance for expected credit losses when an account is deemed uncollectible. Our allowance for expected credit losses was $3.2 million at December 31, 2023 and $7.7 million at December 31, 2022.

Accounts receivable attributable to crude oil resales generally represent the sale of excess crude oil to other purchasers and / or users in cases when our crude oil supplies are in excess of our immediate needs as well as certain reciprocal buy / sell exchanges of crude oil. At times we enter into such buy / sell exchanges to facilitate the delivery of quantities to certain locations. In many cases, we enter into net settlement agreements relating to the buy / sell arrangements, which may mitigate credit risk.

Inventories: Inventories related to our refining operations are stated at the lower of cost, using the last-in, first-out ("LIFO") method for crude oil and unfinished and finished refined products, or market. Inventories related to our renewable business are stated at the lower of cost, using the LIFO method for feedstock and unfinished and finished renewable products, or market. Cost, consisting of raw material, transportation and conversion costs, is determined using the LIFO inventory valuation methodology and market is determined using current replacement costs. Under the LIFO method, the most recently incurred costs are charged to cost of sales and inventories are valued at the earliest acquisition costs. In periods of rapidly declining prices, LIFO inventories may have to be written down to market value due to the higher costs assigned to LIFO layers in prior periods. In addition, the use of the LIFO inventory method may result in increases or decreases to cost of sales in years that inventory volumes decline as the result of charging cost of sales with LIFO inventory costs generated in prior periods. An actual valuation of inventory under the LIFO method is made at the end of each year based on the inventory levels at that time. Accordingly, interim LIFO calculations are based on management's estimates of expected year-end inventory levels and are subject to the final year-end LIFO inventory valuation.

Inventories of our Petro-Canada Lubricants and Sonneborn businesses are stated at the lower of cost, using the first-in, first-out ("FIFO") method, or net realizable value.

Inventories consisting of process chemicals, materials and maintenance supplies and RINs are stated at the lower of weighted-average cost or net realizable value.

Leases: At inception, we determine if an arrangement is or contains a lease. Right-of-use ("ROU") assets represent our right to use an underlying asset for the lease term and lease liabilities represent our payment obligation under the leasing arrangement. ROU assets and lease liabilities are recognized at the commencement date based on the present value of lease payments over the lease term. We use our estimated incremental borrowing rate ("IBR") to determine the present value of lease payments as most of our leases do not contain an implicit rate. Our IBR represents the interest rate which we would pay to borrow, on a collateralized basis, an amount equal to the lease payments over a similar term in a similar economic environment. We use the implicit rate when readily determinable.

Operating leases are recorded in "Operating lease right-of-use assets" and current and noncurrent "Operating lease liabilities" on our consolidated balance sheets. Finance leases are included in "Properties, plants and equipment, at cost" and "Accrued liabilities" and "Other long-term liabilities" on our consolidated balance sheets.

Our lease term includes an option to extend the lease when it is reasonably certain that we will exercise that option. Leases with a term of 12 months or less are not recorded on our balance sheets. For certain equipment leases, we apply a portfolio approach for the operating lease ROU assets and liabilities. Also, as a lessee, we separate non-lease components that are identifiable and exclude them from the determination of net present value of lease payment obligations. In addition, as a lessor, we do not separate the non-lease (service) component in contracts in which the lease component is the dominant component. We treat these combined components as an operating lease. We bifurcate the consideration received for sales-type lease contracts between lease and service revenue, with the service component accounted for within the scope of ASC 606, "Revenue from Contracts with Customers".

Derivative Instruments: All derivative instruments are recognized as either assets or liabilities on our consolidated balance sheets and are measured at fair value. Changes in the derivative instrument's fair value are recognized in earnings unless specific hedge accounting criteria are met. Cash flows from all our derivative activity are reported in the operating section on our consolidated statements of cash flows. See Note 14 for additional information.

Properties, Plants and Equipment: Properties, plants and equipment are stated at cost. Depreciation is provided by the straight-line method over the estimated useful lives of the assets, primarily 15 to 32 years for refining, pipeline and terminal facilities, 10 to 40 years for buildings and improvements, 5 to 30 years for other fixed assets and 5 years for vehicles.

Asset Retirement Obligations: We record legal obligations associated with the retirement of long-lived assets that result from the acquisition, construction, development and / or the normal operation of long-lived assets. The fair value of the estimated cost to retire a tangible long-lived asset is recorded as a liability with the associated retirement costs capitalized as part of the asset's carrying amount in the period in which it is incurred and when a reasonable estimate of the fair value of the liability can be made. If a reasonable estimate cannot be made at the time the liability is incurred, we record the liability when sufficient information is available to estimate the liability's fair value. Certain of our refining assets have no recorded liability for asset retirement obligations since the timing of any retirement and related costs are currently indeterminable.

Our asset retirement obligations were $64.6 million and $61.8 million at December 31, 2023 and 2022, respectively, which are included in “Other long-term liabilities” on our consolidated balance sheets. Accretion expense was insignificant for the years ended December 31, 2023,2022 and 2021. Asset retirement obligations assumed in the Sinclair Transactions, as defined in Note 2, were $6.2 million.

Intangibles, Goodwill and Long-lived Assets: Intangible assets are assets (other than financial assets) that lack physical substance, and goodwill represents the excess of the cost of an acquired entity over the fair value of the assets acquired and liabilities assumed. Goodwill acquired in a business combination and intangibles with indefinite useful lives are not amortized, whereas intangible assets with finite useful lives are amortized on a straight-line basis. Goodwill and intangible assets that are not subject to amortization are tested for impairment annually or more frequently if an event occurs or circumstances change that would more likely than not reduce the fair value of a reporting unit below its carrying amount. Our goodwill impairment testing first entails either a quantitative assessment or an optional qualitative assessment to determine whether it is more likely than not that the fair value of a reporting unit is less than its carrying amount. If we determine that based on the qualitative factors that it is more likely than not that the carrying amount of the reporting unit is greater than its fair value, a quantitative test is performed in which we estimate the fair value of the related reporting unit. If the carrying amount of a reporting unit exceeds its fair value, the goodwill of that reporting unit is impaired, and we measure goodwill impairment as the excess of the carrying amount of the reporting unit over the related fair value. The carrying amount of our intangible assets and goodwill may fluctuate from period to period due to the effects of foreign currency translation adjustments on goodwill and intangible assets assigned to our Lubricants & Specialties segment.

For purposes of long-lived asset impairment evaluation, we group our long-lived assets as follows: (i) our refinery asset groups, which include certain logistics assets, (ii) our renewables products asset groups, (iii) our Lubricants & Specialties asset groups, (iv) our Marketing assets and (v) our Midstream asset groups, which is comprised of logistics assets not included in our refinery asset groups. These asset groups represent the lowest level for which independent cash flows can be identified. Our long-lived assets are evaluated for impairment by identifying whether indicators of impairment exist and, if so, assessing whether such long-lived assets are recoverable from estimated future undiscounted cash flows. The actual amount of impairment loss measured, if any, is equal to the amount by which the asset group’s carrying value exceeds its fair value.

See Note 11 for additional information regarding goodwill and intangible assets.

Equity Method Investments: We account for investments in which we have a noncontrolling interest, yet have significant influence over the entity, using the equity method of accounting, whereby we record our pro-rata share of earnings of these companies and contributions to and distributions from the joint ventures as adjustments to our investment balance.

The following table summarizes our recorded investment compared to its share of underlying equity for each of its investee. The differences are being amortized as adjustments to our pro-rata share of earnings in the joint ventures.

	Balance at December 31, 2023		
	Underlying Equity	Recorded Investment Balance	Difference
		(In thousands)	
Equity Method Investments			
Osage Pipe Line Company, LLC	$ 1,144	$ 27,135	$ (25,991)
Cheyenne Pipeline, LLC	30,508	41,985	(11,477)
Cushing Connect Terminal Holdings LLC	48,135	32,474	15,661
Pioneer Investments Corp.	24,188	131,244	(107,056)
Saddle Butte Pipeline III, LLC	66,436	33,107	33,329
Total	$ 170,411	$ 265,945	$ (95,534)

	Balance at December 31, 2022		
	Underlying Equity	Recorded Investment Balance	Difference
		(In thousands)	
Equity Method Investments			
Osage Pipe Line Company, LLC	$ 2,901	$ 29,773	$ (26,872)
Cheyenne Pipeline, LLC	27,655	40,019	(12,364)
Cushing Connect Terminal Holdings LLC	49,915	34,746	15,169
Pioneer Investments Corp.	23,835	133,182	(109,347)
Saddle Butte Pipeline III, LLC	67,349	32,884	34,465
Total	$ 171,655	$ 270,604	$ (98,949)

Revenue Recognition: Revenues on refined product, branded fuel sales, renewable diesel and excess crude oil sales are recognized when delivered (via pipeline, in-tank or rack) and the customer obtains control of such inventory, which is typically when title passes and the customer is billed. All revenues are reported inclusive of shipping and handling costs billed and exclusive of any taxes billed to customers. Shipping and handling costs incurred are reported as cost of products sold.

Our lubricants and specialties business has sales agreements with marketers and distributors that provide certain rights of return or provisions for the repurchase of products previously sold to them. Under these agreements, revenues and cost of revenues are deferred until the products have been sold to end customers. Our lubricants and specialties business also has agreements that create an obligation to deliver products at a future date for which consideration has already been received and recorded as deferred revenue. This revenue is recognized when the products are delivered to the customer.

Our midstream business recognizes revenues as products are shipped through its pipelines and terminals and as other services are rendered. Additionally, we have certain throughput agreements that specify minimum volume requirements, whereby we bill a customer for a minimum level of shipments in the event a customer ships below their contractual requirements. If there are no future performance obligations, we recognize these deficiency payments as revenue. In certain of these throughput agreements, a customer may later utilize such shortfall billings as credit towards future volume shipments in excess of its minimum levels within its respective contractual shortfall make-up period. Such amounts represent an obligation to perform future services, which may be initially deferred and later recognized as revenue based on estimated future shipping levels, including the likelihood of a customer's ability to utilize such amounts prior to the end of the contractual shortfall make-up period. We recognize the service portion of these deficiency payments as revenue when we do not expect it will be required to satisfy these performance obligations in the future based on the pattern of rights exercised by the customer. Payment terms under our contracts with customers are consistent with industry norms and are typically payable within 30 days of the date of invoice.

Cost Classifications: Costs of products sold include the cost of crude oil, other feedstocks, blendstocks and purchased finished products, inclusive of transportation costs. We purchase crude oil that at times exceeds the supply needs of our refineries. Quantities in excess of our needs are sold at market prices to purchasers of crude oil that are recorded on a gross basis with the sales price recorded as revenues and the corresponding acquisition cost as cost of products sold. Additionally, we enter into buy / sell exchanges of crude oil with certain parties to facilitate the delivery of quantities to certain locations that are netted at cost. Operating expenses include direct costs of labor, maintenance materials and services, utilities and other direct operating costs. Selling, general and administrative expenses include compensation, professional services and other support costs.

Deferred Maintenance Costs: Our refinery units require regular major maintenance and repairs which are commonly referred to as "turnarounds." Catalysts used in certain refinery processes also require regular "change-outs." The required frequency of the maintenance varies by unit and by catalyst, but generally occurs no less than once every five years. Turnaround costs are deferred and amortized over the period until the next scheduled turnaround. Other repairs and maintenance costs are expensed when incurred. Deferred turnaround and catalyst amortization expense was $238.7 million, $159.3 million and $136.9 million for the years ended December 31, 2023, 2022 and 2021, respectively.

Environmental Costs: Environmental costs are charged to operating expenses if they relate to an existing condition caused by past operations and do not contribute to current or future revenue generation. We have ongoing investigations of environmental matters at various locations and routinely assess our recorded environmental obligations, if any, with respect to such matters. Liabilities are recorded when site restoration and environmental remediation, cleanup and other obligations are either known or considered probable and can be reasonably estimated. Such estimates are undiscounted and require judgment with respect to costs, time frame and extent of required remedial and clean-up activities and are subject to periodic adjustments based on currently available information. Recoveries of environmental costs through insurance, indemnification arrangements or other sources are included in other assets to the extent such recoveries are considered probable.

Contingencies: We are subject to proceedings, lawsuits and other claims related to environmental, labor, product and other matters. We are required to assess the likelihood of any adverse judgments or outcomes to these matters as well as potential ranges of probable losses. We accrue for contingencies when it is probable that a loss has occurred and when the amount of that loss is reasonably estimable. A determination of the amount of reserves required, if any, for these contingencies is made after careful analysis of each individual issue. The required reserves may change in the future due to new developments in each matter or changes in approach such as a change in settlement strategy in dealing with these matters.

Foreign Currency Translation: Assets and liabilities recorded in foreign currencies are translated into U.S. dollars using exchange rates in effect as of the balance sheet date. Revenue and expense accounts are translated using the weighted-average exchange rates during the period presented. Foreign currency translation adjustments are recorded as a component of accumulated other comprehensive income.

We have intercompany notes that were issued to fund certain of our foreign businesses. Remeasurement adjustments resulting from the conversion of such intercompany financing amounts to functional currencies are recorded as gains or losses as a component of other income (expense) on our consolidated statements of income. Such adjustments are not recorded to the Lubricants & Specialties segment operations, but to Corporate and Other. See Note 20 for additional information on our segments.

Income Taxes: Provisions for income taxes include deferred taxes resulting from temporary differences in income for financial and tax purposes, using the liability method of accounting for income taxes. The liability method requires the effect of tax rate changes on deferred income taxes to be reflected in the period in which the rate change was enacted. The liability method also requires that deferred tax assets be reduced by a valuation allowance unless it is more likely than not that the assets will be realized. We account for U.S. tax on global intangible low-taxed income in the period in which it is incurred.

Potential interest and penalties related to income tax matters are recognized in income tax expense. We believe we have appropriate support for the income tax positions taken and to be taken on our income tax returns and that our accruals for tax liabilities are adequate for all open years based on an assessment of many factors, including past experience and interpretations of tax law applied to the facts of each matter.

Inventory Repurchase Obligations: We periodically enter into same-party sell / buy transactions, whereby we sell certain refined product inventory and subsequently repurchase the inventory in order to facilitate delivery to certain locations. Such sell / buy transactions are accounted for as inventory repurchase obligations under which proceeds received under the initial sell is recognized as inventory repurchase obligations that are subsequently reversed when the inventories are repurchased. For the years ended December 31, 2023, 2022 and 2021, we received proceeds of $25.7 million, $42.1 million and $43.5 million, respectively, and subsequently repaid $27.4 million, $42.8 million and $45.4 million, respectively, under these sell / buy transactions.

Accounting Pronouncements - Recently Adopted

In October 2021, Accounting Standards Update ("ASU") 2021-08, "Accounting for Contract Assets and Contract Liabilities from Contracts with Customers," was issued requiring that an acquiring entity recognize and measure contract assets and contract liabilities acquired in a business combination in accordance with ASC 606, "Revenue from Contracts with Customers." We adopted this standard effective January 1, 2023, but did not have a business combination under the scope of ASC 805, "Business Combinations" for the year ended December 31, 2023.

Accounting Pronouncements - Not Yet Adopted

In November 2023, ASU 2023-07, "Improvements to Reportable Segment Disclosures" was issued. ASU 2023-07 requires, among other updates, enhanced disclosures about significant segment expenses that are regularly provided to the chief operating decision maker, as well as the aggregate amount of other segment items included in the reported measure of segment profit or loss. This aims to provide more decision-useful information to stakeholders by giving a clearer picture of the costs incurred by each reportable segment. ASU 2023-07 is effective for fiscal years beginning after December 15, 2023, and interim periods within fiscal years beginning after December 15, 2024, and requires retrospective adoption. Early adoption is permitted. We are assessing the impact of this guidance on our disclosures.

In December 2023, ASU 2023-09, "Improvements to Income Tax Disclosures" was issued. ASU 2023-09 requires enhanced annual disclosures regarding the rate reconciliation and income taxes paid information. ASU 2023-09 is effective for fiscal years beginning after December 15, 2024 and may be adopted on a prospective or retrospective basis. Early adoption is permitted. We are assessing the impact of this guidance on our disclosures.

NOTE 2: Acquisitions

On March 14, 2022 (the "Closing Date"), HollyFrontier and HEP announced the establishment of HF Sinclair as the new parent holding company of HollyFrontier and HEP and their subsidiaries, and the completion of their respective acquisitions of Sinclair Oil Corporation (now known as Sinclair Oil LLC) and Sinclair Transportation Company LLC ("STC") from The Sinclair Companies (now known as REH Company and referred to herein as "REH Company"). On the Closing Date, pursuant to that certain Business Combination Agreement, dated as of August 2, 2021 (as amended on March 14, 2022, the "Business Combination Agreement"), by and among HollyFrontier, HF Sinclair, Hippo Merger Sub, Inc., a wholly owned subsidiary of HF Sinclair ("Parent Merger Sub"), REH Company, and Hippo Holding LLC (now known as Sinclair Holding LLC), a wholly owned subsidiary of REH Company (the "Target Company"), HF Sinclair completed its previously announced acquisition of the Target Company by effecting (a) a holding company merger in accordance with Section 251(g) of the Delaware General Corporation Law whereby HollyFrontier merged with and into Parent Merger Sub, with HollyFrontier surviving such merger as a direct wholly owned subsidiary of HF Sinclair (the "HFC Merger") and (b) immediately following the HFC Merger, a contribution whereby REH Company contributed all of the equity interests of the Target Company to HF Sinclair in exchange for shares of HF Sinclair, resulting in the Target Company becoming a direct wholly owned subsidiary of HF Sinclair (the "HFC Transactions").

In connection with the closing of the HFC Transactions, HF Sinclair issued 60,230,036 shares of HF Sinclair common stock, par value $0.01 per share, to REH Company, representing 27% of the pro forma equity of HF Sinclair with a value of approximately $2,149 million based on HollyFrontier's fully diluted shares of common stock outstanding and closing stock price on March 11, 2022. Pursuant to the Business Combination Agreement, REH Company made a $77.5 million cash payment to HF Sinclair, inclusive of final working capital adjustments, which reduced the aggregate transaction value to approximately $2,072 million. Of the 60,230,036 shares of HF Sinclair common stock, 2,570,000 shares were held in escrow to secure REH Company's RINs credit obligations under Section 6.22 of the Business Combination Agreement. As of December 31, 2023, REH Company had satisfied their RINs credit obligations to HF Sinclair and the corresponding shares were released from escrow in January 2024. Additionally, on the Closing Date, and immediately prior to the consummation of the HFC Transactions, HEP completed its acquisition of STC, REH Company's integrated crude and refined products midstream business, and issued 21,000,000 HEP common units and paid cash consideration of $329.0 million, inclusive of final working capital adjustments, to REH Company in exchange for all the outstanding equity interests of STC (the "HEP Transaction" and together with the HFC Transactions, the "Sinclair Transactions"). Of these 21,000,000 HEP common units, 5,290,000 units were held in escrow and were released to REH Company in April 2023 upon their satisfaction of the corresponding RINs credit obligations to HF Sinclair under Section 6.22 of the Business Combination Agreement.

HollyFrontier's (now HF Sinclair's) senior management team continues to operate the combined company. Pursuant to that certain stockholders agreement (the "Stockholders Agreement") by and among HF Sinclair, REH Company and the stockholders of REH Company (together with REH Company and each of their permitted transferees, the "REH Parties"), REH Company was granted the right to nominate, and has nominated, two directors to our Board of Directors at the Closing Date who continued to serve on our Board of Directors as of December 31, 2023. The REH Company stockholders also agreed to certain customary lock up (which expired in June 2023), voting and standstill restrictions, as well as customary registration rights, for the HF Sinclair common stock issued to the stockholders of REH Company. HF Sinclair is headquartered in Dallas, Texas, with combined business offices in Salt Lake City, Utah.

Under the terms of the Business Combination Agreement, HF Sinclair acquired REH Company's refining, branded marketing, renewables, and midstream businesses. At the time of closing, the branded marketing business supplied high-quality fuels to more than 1,300 Sinclair branded stations and licensed the use of the Sinclair brand at more than 300 additional locations throughout the United States. The renewables business includes the operation of a renewable diesel unit located in Sinclair, Wyoming. The refining business includes two Rocky Mountains-based refineries located in Casper, Wyoming and Sinclair, Wyoming. Under the terms of the Contribution Agreement, HEP acquired STC, REH Company's integrated crude and refined products pipelines and terminal assets, including approximately 1,200 miles of integrated crude and refined product pipeline supporting the Sinclair refineries and third parties, eight product terminals and two crude terminals with approximately 4.5 million barrels of operated storage. In addition, HEP acquired STC's interests in three pipeline joint ventures for crude gathering and product offtake including: Saddle Butte Pipeline III, LLC (at the time of closing, a 25.06%, and currently, a 25.12% non-operated interest); Pioneer Investments Corp. (49.995% non-operated interest); and UNEV Pipeline (the 25% non-operated interest not already owned by HEP, resulting in UNEV Pipeline, LLC ("UNEV") becoming a wholly owned subsidiary of HEP). The addition of the Acquired Sinclair Businesses to the HollyFrontier business created a combined company with increased scale and ability to diversify and is expected to drive growth through the expanded refining and renewables business. In addition, the HFC Transactions added an integrated branded wholesale distribution network to our business.

The Sinclair Transactions were accounted for as a business combination using the acquisition method of accounting, with the assets acquired and liabilities assumed at their respective acquisition date fair values at the effective date, with the excess consideration recorded as goodwill.

The following tables present the purchase consideration and final purchase price allocation of the assets acquired and liabilities assumed on March 14, 2022:

Purchase Consideration (in thousands except for per share amounts)		
Shares of HF Sinclair common stock issued		60,230
Closing price per share of HFC common stock[(1)]	$	35.68
Purchase consideration paid in HF Sinclair common stock		2,149,008
Shares of HEP common units issued to REH Company		21,000
Closing price per share of HEP common units[(2)]	$	16.62
Purchase consideration paid in HEP common units		349,020
Total equity consideration		2,498,028
Cash consideration paid by HEP		328,955
Cash consideration received by HF Sinclair		(77,507)
Total cash consideration		251,448
Total purchase consideration	$	2,749,476

(1) Based on the HollyFrontier closing stock price on March 11, 2022.
(2) Based on the HEP closing unit price on March 11, 2022.

		(In thousands)
Assets Acquired		
Accounts receivable	$	467,530
Inventories: Crude oil and refined products		906,461
Inventories: Materials, supplies and other		39,350
Properties, plants and equipment		1,242,549
Operating lease right-of-use assets		4,585
Other assets: Intangibles and other		495,621
Total assets acquired	$	3,156,096
Liabilities Assumed		
Accounts payable	$	564,385
Operating lease liabilities		1,030
Accrued liabilities		84,298
Noncurrent operating lease liabilities		3,554
Deferred income taxes		351,189
Other long-term liabilities		88,098
Total liabilities assumed	$	1,092,554
Net assets acquired	$	2,063,542
Goodwill	$	685,934

The final purchase price allocation resulted in the recognition of $685.9 million in goodwill. Our Refining, Renewables, Marketing and Midstream segments recognized $244.0 million, $159.0 million, $163.8 million and $119.1 million of goodwill, respectively. The goodwill recognized was primarily attributable to operating and administrative synergies and net deferred tax liabilities arising from the differences between the estimated fair values of assets and liabilities and the tax basis of these assets and liabilities. There are qualitative assumptions of long-term factors that this acquisition creates for our stockholders, including increased scale and diversification that is expected to drive growth through the expanded refining and renewables businesses and the addition of an integrated branded wholesale distribution network. This goodwill was not deductible for income tax purposes.

The fair value measurements for properties, plants and equipment were based on significant inputs that are not observable in the market and, therefore, represent Level 3 measurements.

The fair value of properties, plants and equipment was based on the combination of the cost and market approaches. Key assumptions in the cost approach include determining the replacement cost by evaluating recent published data and adjusting replacement cost for physical deterioration, functional, and economic obsolescence. We used the market approach to measure the value of certain assets through an analysis of recent sales or offerings of comparable properties. The fair value of crude oil and refined products inventory was based on market prices as of the acquisition date.

Intangibles include the Sinclair trade name, fuel agreements and customer relationships totaling $221.4 million that are being amortized on a straight-line basis over a range of four to twenty-year period. The intangible assets were valued using the income approach.

The fair value of equity method investments totaled $234.3 million and was based on a combination of valuation methods including discounted cash flows and the guideline public company method.

Accrued liabilities included $70.6 million of RINs credit obligations, including 2022 obligations through the Closing Date, which were valued based on market prices for RINs at the effective date, a Level 2 input. REH Company is financially responsible for satisfaction of RINs credit obligations for all periods prior to the closing. This receivable totaled $68.4 million and was valued based on market prices for RINs at the effective date.

During the year ended December 31, 2023, we purchased RINs for an aggregate amount o $36.0 million, on behalf of REH Company from third parties at applicable market prices in connection with our provision of services to REH Company under the transition services agreement that we and REH Company entered into at the closing of the Sinclair Transactions. We acted as an agent in these RINs transactions and did not recognize sales or cost of products sold as a result. During the year ended December 31, 2023, we recognized sales of $21.2 million related to the sale of RINs to REH Company based on applicable market prices.

All other fair values discussed above were based on significant inputs that are not observable in the market and, therefore, represent Level 3 measurements.

The fair values of all other current receivable and payables were equivalent to their carrying values due to their short-term nature.

Our consolidated financial and operating results reflect the Acquired Sinclair Businesses operations beginning March 14, 2022. Our results of operations included revenue and income from operations of $9,835.0 million and $865.1 million, respectively, for the period from March 14, 2022 through December 31, 2022 related to the Acquired Sinclair Businesses operations.

During the years ended December 31, 2023 and 2022, we incurred $15.8 million and $52.9 million, respectively, in incremental direct acquisition and integration costs that principally relate to legal, advisory and other professional fees and are presented as selling, general and administrative expenses in our consolidated statements of income.

Puget Sound Refinery

On May 4, 2021, HollyFrontier Puget Sound Refining LLC (now known as HF Sinclair Puget Sound Refining LLC), a wholly owned subsidiary of HollyFrontier, entered into a sale and purchase agreement with Shell to acquire the Puget Sound Refinery. The acquisition closed on November 1, 2021 for aggregate cash consideration of $624.3 million, which consists of a base cash purchase price of $350.0 million, hydrocarbon inventory of $277.9 million and other closing adjustments and accrued liabilities of $3.6 million (the "Puget Sound Acquisition").

This transaction was accounted for as a business combination, using the acquisition method, with the aggregate cash consideration allocated to the acquisition date fair value of assets and liabilities acquired.

In connection with the Puget Sound Acquisition, we incurred $12.2 million of acquisition and integration costs during the year ended December 31, 2021, which are included in selling, general and administrative expenses on the consolidated statements of income.

Our consolidated financial and operating results reflect the Puget Sound Refinery operations beginning November 1, 2021. Our results of operations include revenue and loss from operations of $603.1 million and $8.3 million, respectively, for the period from November 1, 2021 through December 31, 2021 related to these operations.

NOTE 3: Leases

Lessee

We have operating and finance leases for land, buildings, pipelines, storage tanks, transportation and other equipment for our operations. Our leases have remaining terms of one to 56 years, some of which include options to extend the leases for up to10 years. Certain of our leases for pipeline assets include provisions for variable payments which are based on a measure of throughput and also contain a provision for the lessor to adjust the rate per barrel periodically over the life of the lease. These variable costs are not included in the initial measurement of ROU assets and lease liabilities.

The following table presents the amounts and balance sheet locations of our operating and financing leases recorded on our consolidated balance sheets.

	December 31,	
	2023	2022
	(In thousands)	
Operating leases:		
Operating lease right-of-use assets	$ 348,006	$ 351,068
Operating lease liabilities	106,973	109,926
Noncurrent operating lease liabilities	249,479	254,215
Total operating lease liabilities	$ 356,452	$ 364,141
Finance leases:		
Properties, plants and equipment, at cost	$ 108,746	$ 81,454
Accumulated amortization	(25,271)	(21,434)
Properties, plants and equipment, net	$ 83,475	$ 60,020
Accrued liabilities	$ 10,842	$ 10,722
Other long-term liabilities	74,860	50,361
Total finance lease liabilities	$ 85,702	$ 61,083

Supplemental balance sheet information related to our leases was as follows:

	December 31,	
	2023	2022
Weighted average remaining lease term (in years)		
Operating leases	7.9	7.2
Finance leases	8.7	7.8
Weighted average discount rate		
Operating leases	5.0 %	4.2 %
Finance leases	5.8 %	4.2 %

The components of lease expense were as follows:

	Years Ended December 31,		
	2023	2022	2021
	(In thousands)		
Operating lease expense	$ 120,552	$ 116,769	$ 117,292
Finance lease expense:			
Amortization of right-of-use assets	13,007	13,003	4,295
Interest on lease liabilities	3,156	2,593	733
Variable lease cost	12,968	4,448	3,645
Total lease expense	$ 149,683	$ 136,813	$ 125,965

Supplemental cash flow information related to leases was as follows:

	Years Ended December 31,		
	2023	2022	2021
	(In thousands)		
Cash paid for amounts included in the measurement of lease liabilities:			
Operating cash flows from operating leases	$ 127,923	$ 126,048	$ 129,577
Operating cash flows from finance leases	$ 3,156	$ 2,593	$ 733
Financing cash flows from finance leases	$ 11,923	$ 11,713	$ 3,990
Right-of-use assets obtained in exchange for lease obligations:			
Operating leases	$ 103,352	$ 61,403	$ 147,718
Finance leases	$ 38,061	$ 6,149	$ 64,334

As of December 31, 2023, minimum future lease payments of our operating and finance lease obligations were as follows:

	Operating	Finance
	(In thousands)	
2024	$ 118,006	$ 15,375
2025	65,436	13,650
2026	50,145	12,537
2027	31,644	11,442
2028	24,398	10,766
Thereafter	163,651	46,869
Future minimum lease payments	453,280	110,639
Less: imputed interest	96,828	24,937
Total lease obligations	356,452	85,702
Less: current obligations	106,973	10,842
Long-term lease obligations	$ 249,479	$ 74,860

As of December 31, 2023, we entered into certain leases that have not yet commenced. Such leases include a15-year lease for a manufacturing and distribution facility, with estimated future undiscounted lease payments of $62.8 million, expected to commence in the first quarter of 2024.

Lessor

Our consolidated statements of income reflect lease revenue recognized by our midstream operations for contracts with third parties in which we are the lessor.

Substantially all of the assets supporting contracts meeting the definition of a lease have long useful lives, and we believe these assets will continue to have value when the current agreements expire due to our risk management strategy for protecting the residual fair value of the underlying assets by performing ongoing maintenance during the lease term.

Lease income recognized was as follows:

	Years Ended December 31,		
	2023	2022	2021
	(In thousands)		
Operating lease revenues	$ 16,879	$ 14,346	$ 15,281
Sales-type lease interest income	$ 1,634	$ 2,515	$ 2,545
Lease revenues relating to variable lease payments not included in measurement of the sales-type lease receivable	$ 1,325	$ 1,782	$ 2,162

For our sales-type leases, we included customer obligations related to minimum volume requirements in guaranteed minimum lease payments. Portions of our minimum guaranteed pipeline tariffs for assets subject to sales-type lease accounting are recorded as interest income with the remaining amounts recorded as a reduction in net investment in leases. We recognized any billings for throughput volumes in excess of minimum volume requirements as variable lease payments, and these variable lease payments were recorded in lease revenues.

Annual minimum undiscounted lease payments in which we are a lessor to third-party contracts as of December 31, 2023 were as follows:

	Operating	Sales-type
	(In thousands)	
2024	$ 12,994	$ 2,170
2025	3,441	2,170
2026	—	2,170
2027	—	2,170
2028	—	2,170
Thereafter	—	13,560
Total lease payment receipts	$ 16,435	24,410
Less: imputed interest		(15,945)
		8,465
Unguaranteed residual assets at end of leases		25,180
Net investment in leases		$ 33,645

Net investment in sales-type leases recorded on our consolidated balance sheets was composed of the following:

	December 31, 2023	December 31, 2022
	(In thousands)	
Lease receivables	$ 18,830	$ 23,797
Unguaranteed residual assets	14,815	10,383
Net investment in leases	$ 33,645	$ 34,180

NOTE 4: Cushing Connect Joint Venture

In 2019, HEP Cushing LLC, then a wholly owned subsidiary of HEP and now a wholly owned subsidiary of HF Sinclair, and Plains Marketing, L.P., a wholly owned subsidiary of Plains All American Pipeline, L.P. ("Plains") formed a 50/50 joint venture, Cushing Connect Pipeline & Terminal LLC ("Cushing Connect"), for (i) the development, construction, ownership and operation of a new 160,000 barrel per day common carrier crude oil pipeline (the "Cushing Connect Pipeline") that connects the Cushing, Oklahoma crude oil hub to our Tulsa refineries and (ii) the ownership and operation of 1.5 million barrels of crude oil storage in Cushing, Oklahoma (the "Cushing Connect Terminal" and together with Cushing Connect and the Cushing Connect Pipeline, the "Cushing Connect Joint Venture"). The Cushing Connect Terminal was fully in-service beginning in April 2020, and the Cushing Connect Pipeline was placed in service during the third quarter of 2021. Long-term commercial agreements have been entered into to support the Cushing Connect assets. Cushing Connect entered into a contract with an affiliate of HEP, now a subsidiary of HF Sinclair, to manage the operation of the Cushing Connect Pipeline and with an affiliate of Plains to manage the operation of the Cushing Connect Terminal. The total investment in Cushing Connect was generally shared proportionately among the partners. However, HEP was solely responsible for any Cushing Connect Pipeline construction costs that exceeded the budget by more than 10%. HEP's share of the cost of the Cushing Connect Terminal contributed by Plains and Cushing Connect Pipeline construction costs was approximately $74.0 million.

Cushing Connect and its two subsidiaries, Cushing Connect Pipeline and Cushing Connect Terminal are each VIEs as defined under GAAP. A VIE is a legal entity whose equity owners do not have sufficient equity at risk for the entity to finance its activities without additional subordinated financial support or, as a group, the equity holders lack the power, through voting rights, to direct the activities that most significantly impact the entity's financial performance, the obligation to absorb the entity's expected losses or rights to expected residual returns. Cushing Connect and its two subsidiaries are each VIE's because they did not originally have sufficient equity at risk to finance their activities without additional financial support. We are the primary beneficiary of two of these entities as HEP constructed and operates the Cushing Connect Pipeline, and we have more ability to direct the activities that most significantly impact the financial performance of Cushing Connect and Cushing Connect Pipeline. Therefore, we consolidate these two entities. We are not the primary beneficiary of Cushing Connect Terminal, which we account for using the equity method of accounting. Our maximum exposure to loss as a result of our involvement with Cushing Connect Terminal is not expected to be material due to the long-term terminalling agreements in place to support operations.

With the exception of the assets of HEP Cushing, creditors of the Cushing Connect Joint Venture legal entities have no recourse to our assets. Any recourse to HEP Cushing would be limited to the extent of HEP Cushing's assets, which other than its investment in Cushing Connect Joint Venture, are not significant. Furthermore, our creditors have no recourse to the assets of the Cushing Connect Joint Venture legal entities. The most significant assets of Cushing Connect and Cushing Connect Pipeline that are available to settle only their obligations, along with their most significant liabilities for which their creditors do not have recourse to our general credit, were:

	Years Ended December 31,	
	2023	2022
	(In thousands)	
Cash and cash equivalents	1,536	2,147
Properties, plants and equipment, at cost	102,936	102,635
Less accumulated depreciation	(8,022)	(4,484)
Intangibles and other	32,473	34,746

NOTE 5: Revenues

Substantially all revenue-generating activities relate to sales of refined product, branded fuel, renewable diesel and excess crude oil inventories sold at market prices (variable consideration) under contracts with customers. Additionally, we have revenues attributable to logistics services provided under petroleum product and crude oil pipeline transportation, processing, storage and terminalling agreements with third parties.

Disaggregated revenues were as follows:

	Years Ended December 31,		
	2023	2022	2021
	(In thousands)		
Revenues by type			
Refined product revenues			
Transportation fuels [1]	$ 19,867,388	$ 25,895,867	$ 13,414,543
Specialty lubricant products [2]	2,521,025	2,826,206	2,322,242
Asphalt, fuel oil and other products [3]	2,167,049	2,147,710	948,581
Total refined product revenues	24,555,462	30,869,783	16,685,366
Excess crude oil revenues [4]	2,147,466	2,342,288	1,547,696
Renewable diesel revenues [5]	781,309	654,893	—
Transportation and logistic services	117,749	109,200	103,646
Marketing revenues [6]	4,146,292	3,911,922	—
Other revenues [7]	216,117	316,753	52,434
Total sales and other revenues	$ 31,964,395	$ 38,204,839	$ 18,389,142

	Years Ended December 31,		
	2023	2022	2021
	(In thousands)		
Refined product revenues by market			
United States			
Mid-Continent	$ 9,362,688	$ 13,924,566	$ 9,094,885
Southwest	3,814,142	4,965,298	3,477,562
Rocky Mountains	9,131,345	9,533,476	2,118,619
Northeast	959,239	1,037,771	824,900
Canada	1,022,174	1,063,961	836,317
Europe, Asia and Latin America	265,874	344,711	333,083
Total refined product revenues	$ 24,555,462	$ 30,869,783	$ 16,685,366

(1) Transportation fuels revenues are attributable to our Refining segment wholesale marketing of gasoline, diesel and jet fuel.
(2) Specialty lubricant products consist of base oil, waxes, finished lubricants and other specialty fluids.
(3) Asphalt, fuel oil and other products revenue include revenues attributable to our Refining and Lubricants & Specialties segments of $1,928.6 million and $238.4 million, respectively, for the year ended December 31, 2023. For the year ended December 31, 2022 such revenues attributable to our Refining and Lubricants & Specialties were $1,827.3 million and $314.8 million, respectively. For the year ended December 31, 2021 such revenue attributable to our Refining and Lubricants & Specialties segments were $724.3 million and $224.3 million, respectively.
(4) Excess crude oil revenues represent sales of purchased crude oil inventory that at times exceeds the supply needs of our refineries.
(5) Renewable diesel revenues are attributable to our Renewables segment.
(6) Marketing segment revenues consist primarily of branded gasoline and diesel fuel.
(7) Other revenues are principally attributable to our Refining segment.

Our consolidated balance sheets reflect contract liabilities related to unearned revenues attributable to future service obligations under our third-party transportation agreements and production agreements from our Sonneborn operations. The following table presents changes to contract liabilities:

	Years Ended December 31,		
	2023	2022	2021
	(In thousands)		
Balance at January 1	$ 10,722	$ 9,278	$ 6,738
Increase	21,381	32,040	32,301
Recognized as revenue	(24,570)	(30,596)	(29,761)
Balance at December 31	$ 7,533	$ 10,722	$ 9,278

As of December 31, 2023, we have long-term contracts with customers that specify minimum volumes of gasoline, diesel, lubricants and specialties to be sold ratably at market prices through 2032. Future prices are subject to market fluctuations and therefore, we have elected the exemption to exclude variable consideration under these contracts under ASC 606-10-50-14A. Aggregate minimum volumes expected to be sold (future performance obligations) under our long-term product sales contracts with customers are as follows, which include branded sales volumes assumed upon our acquisition of the Acquired Sinclair Businesses:

	2024	2025	2026	Thereafter	Total
			(In thousands)		
Refined product sales volumes (barrels)	35,563	25,506	17,891	47,240	126,200

Additionally, we have long-term contracts with third-party customers that specify minimum volumes of product to be transported through our pipelines and terminals that result in fixed-minimum annual revenues through 2033. Annual minimum revenues attributable to our third-party contracts as of December 31, 2023 are presented below:

	2024	2025	2026	Thereafter	Total
			(In thousands)		
Midstream operations contractual minimum revenues	$ 20,656	$ 11,097	$ 7,656	$ 42,592	$ 82,001

For the years ended December 31, 2023, 2022 and 2021, we had one customer, Shell, together with certain of its affiliates, that accounted for 10% or more of our total annual revenues at approximately 12%, 15% and 13%, respectively, which were primarily generated through our Refining segment operations.

NOTE 6: Fair Value Measurements

Fair value measurements are derived using inputs (assumptions that market participants would use in pricing an asset or liability, including assumptions about risk). GAAP categorizes inputs used in fair value measurements into three broad levels as follows:

- (Level 1) Quoted prices in active markets for identical assets or liabilities.
- (Level 2) Observable inputs other than quoted prices included in Level 1, such as quoted prices for similar assets and liabilities in active markets, similar assets and liabilities in markets that are not active or can be corroborated by observable market data.
- (Level 3) Unobservable inputs that are supported by little or no market activity and that are significant to the fair value of the assets or liabilities. This includes valuation techniques that involve significant unobservable inputs.

The carrying amounts of derivative instruments and RINs receivable and credit obligations at December 31, 2023 and 2022 were as follows:

		Fair Value by Input Level		
	Carrying Amount	Level 1	Level 2	Level 3
		(In thousands)		
December 31, 2023				
Assets:				
NYMEX futures contracts	$ 836	$ 836	$ —	$ —
Commodity forward contracts	2,908	—	2,908	—
Total assets	$ 3,744	$ 836	$ 2,908	$ —
Liabilities:				
Commodity price swaps	$ 7,808	$ —	$ 7,808	$ —
Commodity forward contracts	1,848	—	1,848	—
Foreign currency forward contracts	7,893	—	7,893	—
Total liabilities	$ 17,549	$ —	$ 17,549	$ —

		Fair Value by Input Level		
	Carrying Amount	Level 1	Level 2	Level 3
		(In thousands)		
December 31, 2022				
Assets:				
Commodity price swaps	$ 342	$ —	$ 342	$ —
Commodity forward contracts	2,949	—	2,949	—
RINs receivable [(1)]	81,232	—	81,232	—
Foreign currency forward contracts	15,359	—	15,359	—
Total assets	$ 99,882	$ —	$ 99,882	$ —
Liabilities:				
NYMEX futures contracts	$ 2,750	$ 2,750	$ —	$ —
Commodity collar contracts	6,275	—	6,275	—
Commodity forward contracts	2,987	—	2,987	—
RINs credit obligations [(1)]	81,232	—	81,232	—
Total liabilities	$ 93,244	$ 2,750	$ 90,494	$ —

(1) REH Company was financially responsible for satisfaction of RINs credit obligations for all periods prior to the closing of the Sinclair Transactions. See Note 2 for additional information on RINs credit obligations assumed in the Sinclair Transactions.

Level 1 Fair Value Measurements

Our New York Mercantile Exchange ("NYMEX") futures contracts are exchange traded and are measured and recorded at fair value using quoted market prices, a Level 1 input.

Level 2 Fair Value Measurements

Derivative instruments consisting of foreign currency forward contracts, commodity price swaps, commodity collar contracts and forward sales and purchase contracts are measured and recorded at fair value using Level 2 inputs. The fair value of the commodity price swap contracts is based on the net present value of expected future cash flows related to both variable and fixed rate legs of the respective swap agreements. The measurements are computed using market-based observable input and quoted forward commodity prices with respect to our commodity price swaps. The fair value of the commodity collar contracts is based on forward natural gas prices. The fair value of the forward sales and purchase contracts are computed using quoted forward commodity prices. The fair value of foreign currency forward contracts are based on values provided by a third party, which were derived using market quotes for similar type instruments, a Level 2 input.

Nonrecurring Fair Value Measurements

During the years ended December 31, 2022 and 2021, we recognized assets and liabilities based on fair value measurements for the Sinclair Transactions and the acquisition of Puget Sound Refinery (see Note 2). The fair value measurements were based on a combination of valuation methods including discounted cash flows, the guideline public company and guideline transaction methods and obsolescence adjusted replacement costs, all of which are Level 3 inputs.

NOTE 7: Earnings Per Share

Basic earnings per share is calculated as net income attributable to HF Sinclair stockholders, adjusted for participating securities' share in earnings divided by the average number of shares of common stock outstanding. Diluted earnings per share includes the incremental shares resulting from certain share-based awards. The following is a reconciliation of the denominators of the basic and diluted per share computations for net income attributable to HF Sinclair stockholders:

	Years Ended December 31,		
	2023	2022	2021
	(In thousands, except per share data)		
Net income attributable to HF Sinclair stockholders	$ 1,589,666	$ 2,922,668	$ 558,324
Participating securities' share in earnings[(1)]	14,045	29,465	7,465
Net income attributable to common shares	$ 1,575,621	$ 2,893,203	$ 550,859
Average number of shares of common stock outstanding	190,035	202,566	162,569
Average number of shares of common stock outstanding assuming dilution	190,035	202,566	162,569
Basic earnings per share	$ 8.29	$ 14.28	$ 3.39
Diluted earnings per share	$ 8.29	$ 14.28	$ 3.39

(1) Unvested restricted stock unit awards and unvested performance share units that settle in HF Sinclair common stock represent participating securities because they participate in nonforfeitable dividends or distributions with the common stockholders of HF Sinclair. Participating earnings represent the distributed and undistributed earnings of HF Sinclair attributable to the participating securities. Unvested restricted stock unit awards and performance share units do not participate in undistributed net losses as they are not contractually obligated to do so.

NOTE 8: Stock-Based Compensation

We have a principal share-based compensation plan (the HF Sinclair Corporation Amended and Restated 2020 Long Term Incentive Plan, the "2020 Plan"). The 2020 Plan provides for the grant of unrestricted and restricted stock, restricted stock units, other stock-based awards, stock options, performance awards, substitute awards, cash awards and stock appreciation rights. Subject to adjustment for certain events, an aggregate of6,368,930 of these awards may be issued pursuant to awards granted under the 2020 Plan. We also have a stock compensation deferral plan which allows non-employee directors to defer settlement of vested stock granted under our share-based compensation plan. Our accounting policy for the recognition of compensation expense for awards with pro-rata vesting is to expense the costs ratably over the vesting periods. Share-based awards paid in cash upon vesting are accounted for as liability awards and recorded at fair value at the end of each reporting period with a mark-to-mark adjustment recognized in earnings.

The stock-based compensation expense and associated tax benefit were as follows:

	Years Ended December 31,		
	2023	2022	2021
	(In thousands)		
Compensation expense:			
Restricted stock units	$ 30,067	$ 27,264	$ 29,453
Performance stock units	12,145	8,683	12,591
Total compensation expense	$ 42,212	$ 35,947	$ 42,044
Tax benefit recognized on compensation expense	$ 10,203	$ 8,918	$ 10,545

Additionally, prior to the HEP Merger Transaction, HEP maintained an equity-based compensation plan for Holly Logistic Services, L.L.C.'s non-employee directors and certain executives and employees (the "HEP LTIP"). Compensation cost attributable to HEP's equity-based compensation plan was $ 1.5 million, $1.9 million and $2.6 million for the years ended December 31, 2023, 2022 and 2021, respectively. In connection with the HEP Merger Transaction, on December 4, 2023, HF Sinclair registered additional shares of HF Sinclair common stock under the 2020 Plan pursuant to General Instruction E of Form S-8, which authorized the 2020 Plan's assumption of authorized but unissued HEP common units remaining under the HEP LTIP at the time of the HEP Merger Transaction, adjusted to reflect the applicable exchange rate pursuant to the HEP Merger Transaction.

Restricted Stock Units

Under the 2020 Plan, we grant certain officers and other key employees restricted stock unit awards, which are payable in stock or cash and generally vest over a period of one to three years. Restricted stock unit award recipients have the right to receive dividends, however, restricted stock units do not have any other rights of absolute ownership. Upon vesting, restrictions on the restricted stock units lapse at which time they convert to common shares or cash. In addition, we grant non-employee directors restricted stock unit awards, which typically vest over a period of one year and are payable in stock. The fair value of each restricted stock unit award is measured based on the grant date market price of our common shares and is amortized over the respective vesting period. We account for forfeitures on an estimated basis.

A summary of restricted stock unit activity during the year ended December 31, 2023 is presented below:

Restricted Stock Units	Grants	Weighted Average Grant Date Fair Value
Outstanding at January 1, 2023	1,055,875	$ 39.46
Granted	529,559	$ 52.59
Vested	(641,944)	$ 33.17
Forfeited	(65,952)	$ 38.55
Converted from performance share units	225,217	$ 45.46
Outstanding at December 31, 2023	1,102,755	$ 50.71

For the years ended December 31, 2023, 2022 and 2021, restricted stock units vested having a grant date fair value of $31.3 million, $26.5 million and $28.4 million, respectively. For the years ended December 31, 2022 and 2021, we granted restricted stock units having a weighted average grant date fair value of $59.41 and $33.95, respectively. As of December 31, 2023, there was $28.9 million of total unrecognized compensation cost related to non-vested restricted stock unit grants. That cost is expected to be recognized over a weighted-average period of 1.5 years. For the years ended December 31, 2023, 2022 and 2021, we paid $3.9 million, $5.8 million and $3.4 million, respectively, in cash equal to the value of the stock award on the vest date to certain employees to settle71,589, 96,005 and 105,459 restricted stock units, respectively.

Performance Share Units

Under the 2020 Plan, we grant certain officers and other key employees performance share units, which are payable in stock or cash upon meeting certain criteria over the service period, and generally vest over a period of three years. Under the terms of our performance share unit grants, awards are subject to "financial performance" and "market performance" criteria. Financial performance is based on our financial performance compared to a peer group of independent refining companies, while market performance is based on the relative standing of total shareholder return achieved by HF Sinclair compared to peer group companies. The number of shares ultimately issued or cash paid under these awards can range from zero to 200% of target award amounts. Holders of performance share units have the right to receive dividend equivalents and other distributions with respect to such performance share units based on the target level of payout.

A summary of performance share unit activity and changes during the year ended December 31, 2023 is presented below:

Performance Share Units	Grants	Weighted Average Grant Date Fair Value
Outstanding at January 1, 2023	771,197	$ 41.78
Granted	263,881	$ 67.73
Vested	(298,978)	$ 24.47
Forfeited	(25,352)	$ 40.76
Converted to restricted stock units	(225,217)	$ 52.44
Outstanding at December 31, 2023	485,531	$ 61.66

For the year ended December 31, 2023, we issued375,376 shares of common stock, representing a payout of up to125% on vested performance share units having a grant date fair value of $7.3 million. For the years ended December 31, 2022 and 2021, we issued common stock upon the vesting of the performance share units having a grant date fair value of $6.2 million and $4.5 million, respectively. As of December 31, 2023, there was $20.1 million of total unrecognized compensation cost related to non-vested performance share units. That cost is expected to be recognized over a weighted-average period of 2.2 years. For the years ended December 31, 2023 and 2022, we paid $1.2 million and $0.7 million, respectively, in cash equal to the value of the stock award on the vest date to certain employees to settle23,587 and 12,108 performance share units, respectively.

NOTE 9: Inventories

Inventories consist of the following components:

	December 31,	
	2023	2022
	(In thousands)	
Crude oil	$ 858,411	$ 818,737
Other raw materials and unfinished products [1]	683,066	842,855
Finished products [2]	1,435,817	1,252,984
Lower of cost or market reserve	(331,570)	(61,151)
Process chemicals [3]	50,917	53,900
Repairs and maintenance supplies and other [4]	225,190	307,203
Total inventory	$ 2,921,831	$ 3,214,528

(1) Other raw materials and unfinished products include feedstocks and blendstocks, other than crude.
(2) Finished products include gasolines, jet fuels, diesels, renewable diesels, lubricants, asphalts, LPG's and residual fuels.
(3) Process chemicals include additives and other chemicals.
(4) Includes RINs.

Our Refining segment inventories that are valued at the lower of LIFO cost or market reflect a new market reserve of $220.6 million that was established as of December 31, 2023 based on market conditions and prices at that time. The effect of the change in the lower of cost or market reserve was an increase to cost of products sold totaling $220.6 million for the year ended December 31, 2023. The excess replacement cost over the LIFO value of our Refining segment inventories was $39.0 million at December 31, 2022. For the year ended December 31, 2021, we recorded a decrease to cost of products sold of $18.9 million due to the effect of the change in the lower of cost or market reserve recorded on our Refining segment inventories at that time.

Our Renewables segment inventories that are valued at the lower of LIFO cost or market reflect a valuation reserve of $111.0 million and $61.2 million at December 31, 2023 and 2022, respectively. A new market reserve of $ 111.0 million as of December 31, 2023 was based on market conditions and prices at that time. The effect of the change in the lower of cost or market reserve was an increase of cost of products sold totaling $49.9 million and $52.4 million for the years ended December 31, 2023 and 2022, respectively.

NOTE 10: Properties, Plants and Equipment

The components of properties, plants and equipment are as follows:

	December 31,	
	2023	2022
	(In thousands)	
Land, buildings and improvements	$ 765,826	$ 741,874
Refining facilities	6,593,768	6,346,422
Pipelines and terminals	2,300,781	2,267,052
Transportation vehicles	37,870	43,801
Other fixed assets	543,966	422,583
Construction in progress	291,221	324,920
	10,533,432	10,146,652
Accumulated depreciation	(3,906,600)	(3,457,747)
	$ 6,626,832	$ 6,688,905

We capitalized interest attributable to construction projects of $4.3 million, $6.2 million and $15.2 million for the years ended December 31, 2023, 2022 and 2021, respectively.

Depreciation expense was $474.3 million, $442.2 million and $329.4 million for the years ended December 31, 2023, 2022 and 2021, respectively.

NOTE 11: Goodwill and Intangibles

Goodwill

As of December 31, 2023, our goodwill balance was $3.0 billion. The carrying amount of our goodwill may fluctuate from period to period due to the effects of foreign currency translation adjustments on goodwill assigned to our Lubricants & Specialties segment.

The following is a summary of our goodwill by segment:

	Refining	Renewables	Marketing	Lubricants & Specialties	Midstream	Total
			(In thousands)			
Balance at December 31, 2022	$ 1,977,435	$ 159,020	$ 163,839	$ 246,036	$ 431,985	$ 2,978,315
Goodwill disposal and other changes	—	—	—	(943)	—	(943)
Foreign currency translation adjustment	—	—	—	372	—	372
Balance at December 31, 2023	$ 1,977,435	$ 159,020	$ 163,839	$ 245,465	$ 431,985	$ 2,977,744
Balance at December 31, 2023						
Goodwill	$ 2,286,753	$ 159,020	$ 163,839	$ 480,044	$ 431,985	$ 3,521,641
Accumulated impairment losses	(309,318)	—	—	(234,579)	—	(543,897)
	$ 1,977,435	$ 159,020	$ 163,839	$ 245,465	$ 431,985	$ 2,977,744

We performed our annual goodwill impairment testing quantitatively as of July 1, 2023 and determined there wasno impairment of goodwill attributable to our reporting units. Furthermore, there was no impairment of goodwill during the years ended December 31, 2022 and 2021.

Intangibles

The carrying amounts of our intangible assets presented in “Intangibles and other” on our consolidated balance sheets are as follows:

		December 31	
	Useful Life	2023	2022
		(In thousands)	
Customer relationships	10 - 20 years	$ 347,567	$ 346,354
Transportation agreements	30 years	59,933	59,933
Trademarks, patents and other	10 - 20 years	263,855	261,678
		671,355	667,965
Accumulated amortization		(260,723)	(204,239)
Total intangibles, net		$ 410,632	$ 463,726

Amortization expense was $55.1 million, $51.0 million and $35.6 million for the years ended December 31, 2023, 2022 and 2021, respectively.Estimated future amortization expense related to the intangible assets at December 31, 2023 is as follows:

		(In thousands)
2024	$	55,261
2025	$	55,261
2026	$	46,499
2027	$	40,753
2028	$	34,253

NOTE 12: Environmental

We expensed $26.5 million, $13.4 million and $7.8 million for the years ended December 31, 2023, 2022 and 2021, respectively, for environmental remediation obligations. The accrued environmental liability reflected on our consolidated balance sheets was $195.4 million and $192.3 million at December 31, 2023 and 2022, respectively, of which $161.4 million and $170.0 million, respectively, were classified as other long-term liabilities. These accruals include remediation and monitoring costs expected to be incurred over an extended period of time. Accrued environmental liabilities assumed in the Sinclair Transactions were $72.2 million at the acquisition date and an associated receivable from third parties of $21.5 million. Estimated liabilities could increase in the future when the results of ongoing investigations become known, are considered probable and can be reasonably estimated.

NOTE 13: Debt

HF Sinclair Credit Agreement
We have a $1.65 billion senior unsecured revolving credit facility maturing in April 2026 (the "HF Sinclair Credit Agreement"). The HF Sinclair Credit Agreement may be used for revolving credit loans and letters of credit from time to time and is available to fund general corporate purposes. At December 31, 2023, we were in compliance with all covenants, had no outstanding borrowings and had outstanding letters of credit totaling $0.3 million under the HF Sinclair Credit Agreement.

Indebtedness under the HF Sinclair Credit Agreement bears interest, at our option based on the currency of such indebtedness at either (a) a base rate equal to the highest of the Federal Funds Effective Rate (as defined in the HF Sinclair Credit Agreement) plus half of 1%, Spread Adjusted Term SOFR (as defined in the HF Sinclair Credit Agreement) for a one-month interest period plus 1% and the prime rate (as publicly announced from time to time by the administrative agent), as applicable, plus an applicable margin (ranging from 0.25% - 1.125%), (b) the CDOR Rate (as defined in the HF Sinclair Credit Agreement) plus an applicable margin (ranging from 1.25% to 2.125%), (c) the Spread Adjusted Term SOFR (as defined in the HF Sinclair Credit Agreement) plus an applicable margin (ranging from1.25% to 2.125%) or (d) the Daily Simple RFR (as defined in the HF Sinclair Credit Agreement) plus an applicable margin (ranging from1.25% to 2.125%). In each case, the applicable margin is based on HF Sinclair's debt rating assigned by Standard & Poor's Financial Services LLC and Moody's Investors Service, Inc.

HEP Credit Agreement
Our wholly owned subsidiary, HEP, has a $1.2 billion senior secured revolving credit facility maturing in July 2025 (the "HEP Credit Agreement"). In connection with the consummation of the HEP Merger Transaction, we amended the HEP Credit Agreement to, among other things, (a) provide a guaranty from us and terminated all guaranties from subsidiaries of HEP, (b) amended the definition of "Investment Grade Rating" (as defined in the HEP Credit Agreement) to reference the credit rating of our senior unsecured indebtedness, (c) eliminated the requirement to deliver separate audited and unaudited financial statements for HEP and its subsidiaries and only provide certain segment-level reporting for HEP with any compliance certificate delivered in accordance with the HEP Credit Agreement and (d) amended certain covenants to eliminate certain restrictions on (i) amendments to intercompany contracts, (ii) transactions with us and our subsidiaries and (iii) investments in and contributions, dividends, transfers and distributions to us and our subsidiaries.

The HEP Credit Agreement is available to fund capital expenditures, investments, acquisitions, distribution payments, working capital and for general corporate purposes. It is also available to fund letters of credit up to a $50 million sub-limit and has an accordion feature that allows us to increase the commitments under the HEP Credit Agreement up to a maximum amount of $1.7 billion. At December 31, 2023, we were in compliance with all of its covenants, had outstanding borrowings of $455.5 million and no outstanding letters of credit under the HEP Credit Agreement.

Prior to the Investment Grade Date (as defined in the HEP Credit Agreement), indebtedness under the HEP Credit Agreement bears interest, at our option, at either (a) the Alternate Base Rate (as defined in the HEP Credit Agreement) plus an applicable margin (ranging from 0.75% - 1.75%) or (b) Adjusted Term SOFR (as defined in the HEP Credit Agreement) plus an applicable margin (ranging from 1.75% - 2.75%). In each case, the applicable margin is based upon the Total Leverage Ratio (as defined in the HEP Credit Agreement). The weighted average interest rate in effect under the HEP Credit Agreement on our borrowings was 7.08% and 6.32% as of December 31, 2023 and 2022, respectively.

HEP Senior Notes Exchange
On December 4, 2023, we completed our offers to exchange any and all outstanding HEP 5.000% senior notes maturing February 2028 (the "HEP 5.000% Senior Notes") and HEP 6.375% senior notes maturing April 2027 (the "HEP 6.375% Senior Notes" and, collectively with the HEP 5.000% Senior Notes, the "HEP Senior Notes") for the HF Sinclair 5.000% senior notes maturing February 2028 (the "HF Sinclair 5.000% Senior Notes") and the HF Sinclair 6.375% senior notes maturing April 2027 (the "HF Sinclair 6.375% Senior Notes", and, collectively with the HF Sinclair 5.000% Senior Notes, the "New HF Sinclair Senior Notes") to be issued by HF Sinclair with registration rights and cash. In connection with the exchange offers, HEP amended the indenture governing the HEP Senior Notes to eliminate (i) substantially all of the restrictive covenants, (ii) certain of the events which may lead to an "Event of Default", (iii) the SEC reporting covenant and (iv) the requirement of HEP to offer to purchase the HEP Senior Notes upon a change of control.

The New HF Sinclair Senior Notes are unsecured and unsubordinated obligations of ours and rank equally with all our other existing and future unsecured and unsubordinated indebtedness. Each series of the New HF Sinclair Senior Notes has the same interest rate, interest payment dates, maturity date and redemption terms as the corresponding series of HEP Senior Notes. The New HF Sinclair Senior Notes were issued in exchange for the HEP Senior Notes pursuant to a private exchange offer exempt from registration under the Securities Act of 1933, as amended.

Senior Notes
At December 31, 2023, our senior notes consisted of the following:

- $202.900 million in aggregate principal amount of 5.875% senior notes maturing April 2026 (the "HollyFrontier 5.875% Senior Notes"),
- $74.966 million in aggregate principal amount of 4.500% senior notes maturing October 2030 (the "HollyFrontier 4.500% Senior Notes" and, collectively with the HollyFrontier 5.875% Senior Notes, the "HollyFrontier Senior Notes"),
- $797.100 million in aggregate principal amount of 5.875% senior notes maturing April 2026 (the "HF Sinclair 5.875% Senior Notes"),
- $325.034 million in aggregate principal amount of 4.500% senior notes maturing October 2030 (the "HF Sinclair 4.500% Senior Notes"),
- $498.879 million in aggregate principal amount of HF Sinclair 5.000% Senior Notes,
- $399.875 million in aggregate principal amount of HF Sinclair 6.375% Senior Notes (collectively with the HF Sinclair 5.875% Senior Notes, HF Sinclair 4.500% Senior Notes and HF Sinclair 5.000% Senior Notes, the "HF Sinclair Senior Notes"),
- $1.121 million in aggregate principal amount of HEP 5.000% Senior Notes and
- $0.125 million in aggregate principal amount of HEP 6.375% Senior Notes.

Our senior notes are unsecured and unsubordinated obligations of ours and rank equally with all future unsecured and unsubordinated indebtedness.

In October 2023, we repaid at maturity our $59.6 million aggregate principal amount HollyFrontier 2.625% senior notes maturing October 2023 (the "HollyFrontier 2.625% Senior Notes") and $248.2 million aggregate principal amount HF Sinclair 2.625% senior notes maturing October 2023 (the "HF Sinclair 2.625% Senior Notes").

During the fourth quarter of 2022, we made open market repurchases of HF Sinclair 2.625% Senior Notes and HollyFrontier 2.625% Senior Notes that resulted in the extinguishment of $42.2 million in principal of the HF Sinclair 2.625% Senior Notes and fifteen thousand dollars in principal of the HollyFrontier 2.625% Senior Notes. Total cash consideration paid to repurchase the principal amount outstanding, excluding accrued interest, totaled $41.4 million, and we recognized a $0.6 million gain on the extinguishment of debt during the year ended December 31, 2022.

Further, we may from time to time seek to retire some or all of our outstanding debt or debt agreements through cash purchases, and/or exchanges, open market purchases, privately negotiated transactions, tender offers or otherwise. Such transactions, if any, may be material and will depend on prevailing market conditions, our liquidity requirements and other factors.

HF Sinclair Financing Arrangements

Certain of our wholly owned subsidiaries entered into financing arrangements whereby such subsidiaries sold a portion of their precious metals catalyst to a financial institution and then leased back the precious metals catalyst in exchange for cash. The volume of the precious metals catalyst and the lease rate are fixed over the term of each lease, and the lease payments are recorded as interest expense. The current leases mature in one year or less. Upon maturity, we must either satisfy the obligation at fair market value or refinance to extend the maturity. These financing arrangements are recorded at a Level 2 fair value totaling $37.0 million and $39.8 million at December 31, 2023 and 2022, respectively, and are included in "Accrued liabilities" on our consolidated balance sheets. See Note 6 for additional information on Level 2 inputs.

HF Sinclair may, from time to time, issue letters of credit pursuant to uncommitted letters of credit facilities with its lenders. At December 31, 2023, there wereno letters of credit outstanding under such credit facilities.

The carrying amounts of outstanding debt are as follows:

	December 31,	
	2023	2022
	(In thousands)	
HollyFrontier		
2.625% Senior Notes	$ —	$ 59,637
5.875% Senior Notes	202,900	202,900
4.500% Senior Notes	74,966	74,966
	277,866	337,503
HF Sinclair		
2.625% Senior Notes	$ —	$ 248,190
5.875% Senior Notes	797,100	797,100
4.500% Senior Notes	325,034	325,034
5.000% Senior Notes	498,879	—
6.375% Senior Notes	399,875	—
	2,020,888	1,370,324
HEP		
5.000% Senior Notes	1,121	500,000
6.375% Senior Notes	125	400,000
	1,246	900,000
HEP Credit Agreement	455,500	668,000
Less current debt [1]	—	(306,959)
Unamortized discount and debt issuance costs [1]	(16,417)	(20,355)
Total long-term debt [2]	$ 2,739,083	$ 2,948,513

(1) The 2.625% HollyFrontier Senior Notes and HF Sinclair 2.625% Senior Notes, inclusive of unamortized discount and debt issuance costs of $0.9 million at December 31, 2022 were due October 2023 and were classified as current debt on our consolidated balance sheets.
(2) At December 31, 2022, total HF Sinclair standalone long-term debt, which excluded HEP long-term debt, was $1.4 billion.

The fair values of the senior notes are as follows:

	December 31,	
	2023	2022
	(In thousands)	
HollyFrontier and HF Sinclair Senior Notes	$ 2,270,668	$ 1,655,726
HEP Senior Notes	$ 1,188	$ 852,658

These fair values are based on a Level 2 input. See Note 6 for additional information on Level 2 inputs.

Principal maturities of outstanding debt as of December 31, 2023 are as follows:

Years Ending December 31,	(In thousands)
2024	$ —
2025	455,500
2026	1,000,000
2027	400,000
2028	500,000
Thereafter	400,000
Total	$ 2,755,500

NOTE 14: Derivative Instruments and Hedging Activities

Commodity Price Risk Management
Our primary market risk is commodity price risk. We are exposed to market risks related to the volatility in crude oil and refined products, as well as volatility in the price of natural gas used in our refining operations. We periodically enter into derivative contracts in the form of commodity price swaps, collar contracts, forward purchase and sales and futures contracts to mitigate price exposure with respect to our inventory positions, natural gas purchases, sales prices of refined products and crude oil costs.

Foreign Currency Risk Management
We are exposed to market risk related to the volatility in foreign currency exchange rates. We periodically enter into derivative contracts in the form of foreign exchange forward contracts to mitigate the exposure associated with fluctuations on intercompany notes with our foreign subsidiaries that are not denominated in the U.S. dollar.

Accounting Hedges
We had swap contracts serving as cash flow hedges against price risk on forecasted purchases of natural gas that matured as of December 31, 2021. We also periodically have swap contracts to lock in basis spread differentials on forecasted purchases of crude oil and forward sales contracts that lock in the prices of future sales of crude oil and refined product. These contracts have been designated as accounting hedges and are measured at fair value with offsetting adjustments (gains/losses) recorded directly to other comprehensive income (loss). These fair value adjustments are later reclassified to earnings as the hedging instruments mature.

The following table presents the pre-tax effect on other comprehensive income ("OCI") and earnings due to fair value adjustments and maturities of hedging instruments under hedge accounting:

Derivatives Designated as Cash Flow Hedging Instruments	Net Unrealized Gain Recognized in OCI			Gain (Loss) Reclassified into Earnings			
	Years Ended December 31,				Years Ended December 31,		
	2023	2022	2021	Income Statement Location	2023	2022	2021
				(In thousands)			
Commodity contracts	$ —	$ 326	$ 31	Sales and other revenues	$ (3,236)	$ (5,288)	$ (19,239)
				Operating expenses	—	—	1,660
Total	$ —	$ 326	$ 31		$ (3,236)	$ (5,288)	$ (17,579)

Economic Hedges

We have commodity contracts including NYMEX futures contracts to lock in prices on forecasted purchases and sales of inventory and basis swap contracts to mitigate exposure to natural gas price volatility and forward purchase and sell contracts of refined products, as well as periodically have contracts to lock in basis spread differentials on forecasted purchases of crude oil and collar contracts to mitigate exposure to natural gas price volatility, that serve as economic hedges (derivatives used for risk management, but not designated as accounting hedges). We also have forward currency contracts to fix the rate of foreign currency. In addition, our catalyst financing arrangements discussed in Note 13 could require repayment under certain conditions based on the future pricing of platinum, which is an embedded derivative. These contracts are measured at fair value with offsetting adjustments (gains/losses) recorded directly to earnings.

The following table presents the pre-tax effect on earnings due to maturities and fair value adjustments of our economic hedges:

Derivatives Not Designated as Hedging Instruments	Income Statement Location	Gain (Loss) Recognized in Earnings		
		Years Ended December 31,		
		2023	2022	2021
			(In thousands)	
Commodity contracts	Cost of products sold	$ 10,162	$ (17,189)	$ (22,909)
	Operating expenses	(20,783)	(13,780)	—
	Interest expense	2,050	(4,420)	11,816
Foreign currency contracts	Gain (loss) on foreign currency transactions	(7,414)	27,826	(4,013)
	Total	$ (15,985)	$ (7,563)	$ (15,106)

As of December 31, 2023, we have the following notional contract volumes related to outstanding derivative instruments (all maturing in 2024):

	Total Outstanding Notional	Unit of Measure
Derivatives not designated as hedging instruments:		
NYMEX futures (WTI) - short	640,000	Barrels
Forward gasoline contracts - long	800,000	Barrels
Foreign currency forward contracts	387,613,367	U. S. dollar
Forward commodity contracts (platinum)	36,969	Troy ounces
Natural gas price swaps (basis spread) - long	6,667,000	MMBTU

The following table presents the fair value and balance sheet locations of our outstanding derivative instruments. These amounts are presented on a gross basis with offsetting balances that reconcile to a net asset or liability position on our consolidated balance sheets. We present on a net basis to reflect the net settlement of these positions in accordance with provisions of our master netting arrangements.

	Derivatives in Net Asset Position			Derivatives in Net Liability Position		
	Gross Assets	Gross Liabilities Offset in Balance Sheet	Net Assets Recognized in Balance Sheet	Gross Liabilities	Gross Assets Offset in Balance Sheet	Net Liabilities Recognized in Balance Sheet
			(In thousands)			
December 31, 2023						
Derivatives not designated as cash flow hedging instruments:						
NYMEX futures contracts	$ 836	$ —	$ 836	$ —	$ —	$ —
Commodity price swap contracts	—	—	—	7,808	—	7,808
Commodity forward contracts	2,908	—	2,908	1,848	—	1,848
Foreign currency forward contracts	—	—	—	7,893	—	7,893
	$ 3,744	$ —	$ 3,744	$ 17,549	$ —	$ 17,549
Total net balance			$ 3,744			$ 17,549
Balance sheet classification:	Prepayment and other		$ 3,744	Accrued liabilities		$ 17,549

	Derivatives in Net Asset Position			Derivatives in Net Liability Position		
	Gross Assets	Gross Liabilities Offset in Balance Sheet	Net Assets Recognized in Balance Sheet	Gross Liabilities	Gross Assets Offset in Balance Sheet	Net Liabilities Recognized in Balance Sheet
			(In thousands)			
December 31, 2022						
Derivatives not designated as cash flow hedging instruments:						
NYMEX futures contracts	$ —	$ —	$ —	$ 2,750	$ —	$ 2,750
Commodity price swap contracts	342	—	342	—	—	—
Commodity collar contracts	—	—	—	6,275	—	6,275
Commodity forward contracts	2,949	—	2,949	2,987	—	2,987
Foreign currency forward contracts	15,359	—	15,359	—	—	—
	$ 18,650	$ —	$ 18,650	$ 12,012	$ —	$ 12,012
Total net balance			$ 18,650			$ 12,012
Balance sheet classification:	Prepayments and other		$ 18,650	Accrued liabilities		$ 12,012

NOTE 15: Income Taxes

The provision for income taxes is comprised of the following:

	Years Ended December 31,		
	2023	2022	2021
	(In thousands)		
Current			
Federal	$ 179,677	$ 674,977	$ (33,206)
State	24,419	108,993	(1,802)
Foreign	44,966	57,734	30,336
Deferred			
Federal	154,996	38,535	94,353
State	31,285	21,121	1,386
Foreign	6,269	(6,488)	32,831
	$ 441,612	$ 894,872	$ 123,898

The statutory federal income tax rate applied to pre-tax book income reconciles to income tax expense as follows:

	Years Ended December 31,		
	2023	2022	2021
	(In thousands)		
Tax computed at statutory rate	$ 452,027	$ 826,570	$ 165,302
State income taxes, net of federal tax benefit	55,734	123,442	13,588
Noncontrolling interest in net income	(29,386)	(28,726)	(25,931)
Effect of change in state rate	—	(15,800)	(13,342)
Nontaxable permanent differences	(49,420)	—	—
CARES Act benefits	—	—	(10,384)
Foreign rate differential	5,753	6,608	331
Federal tax credits	(5,344)	(23,853)	(29,777)
US tax on non-US operations	7,239	12,920	18,547
Other	5,009	(6,289)	5,564
	$ 441,612	$ 894,872	$ 123,898

Deferred income taxes reflect the net tax effects of temporary differences between the carrying amounts of assets and liabilities for financial reporting purposes and the amounts used for income tax purposes. Our deferred income tax assets and liabilities as of December 31, 2023 and 2022 are as follows:

	December 31, 2023		
	Assets	Liabilities	Total
	(In thousands)		
Deferred income taxes			
Properties, plants, equipment and intangibles (due primarily to tax in excess of book depreciation)	$ —	$ (1,171,633)	$ (1,171,633)
Lease obligation	106,161	—	106,161
Accrued employee benefits	20,260	—	20,260
Accrued post-retirement benefits	9,985	—	9,985
Accrued environmental costs	42,916	—	42,916
Hedging instruments	1,791	—	1,791
Inventory differences	—	(164,178)	(164,178)
Deferred turnaround costs	—	(155,833)	(155,833)
Net operating loss and tax credit carryforwards	35,294	—	35,294
Valuation allowance	—	(10,614)	(10,614)
Other	—	(11,279)	(11,279)
Total	$ 216,407	$ (1,513,537)	$ (1,297,130)

	December 31, 2022		
	Assets	Liabilities	Total
	(In thousands)		
Deferred income taxes			
Properties, plants, equipment and intangibles (due primarily to tax in excess of book depreciation)	$ —	$ (1,032,048)	$ (1,032,048)
Lease obligation	129,727	—	129,727
Accrued employee benefits	17,665	—	17,665
Accrued post-retirement benefits	9,951	—	9,951
Accrued environmental costs	37,868	—	37,868
Hedging instruments	3,260	—	3,260
Inventory differences	—	(230,112)	(230,112)
Deferred turnaround costs	—	(88,574)	(88,574)
Net operating loss and tax credit carryforwards	27,963	—	27,963
Investment in HEP	—	(134,160)	(134,160)
Valuation allowance	—	(3,691)	(3,691)
Other	—	(14)	(14)
Total	$ 226,434	$ (1,488,599)	$ (1,262,165)

We have tax benefits attributable to net operating losses of $17.6 million in Luxembourg that can be carried forward 16 years which will begin expiring in 2034. We also have tax benefits attributable to net operating losses of $11.2 million in the Netherlands that can be carried forward indefinitely, and tax benefits attributable to net operating losses in China of $3.5 million which can be carried forward five years. We have reflected a valuation allowance of $10.6 million in 2023 and $3.7 million in 2022, with respect to net operating carryforwards that primarily relate to losses in Luxembourg and China.

A reconciliation of the beginning and ending amount of unrecognized tax benefits is as follows:

	Years Ended December 31,		
	2023	2022	2021
		(In thousands)	
Balance at January 1	$ 1,354	$ 54,605	$ 54,899
Reductions for tax positions of prior years	—	(53,023)	(49)
Settlements	—	—	(125)
Lapse of statute of limitations	(129)	(228)	(120)
Balance at December 31	$ 1,225	$ 1,354	$ 54,605

At December 31, 2023, 2022 and 2021, there were$1.2 million, $1.4 million, and $54.6 million, respectively, of unrecognized tax benefits that, if recognized, would affect our effective tax rate. Unrecognized tax benefits are adjusted in the period in which new information about a tax position becomes available or the final outcome differs from the amount recorded.

Approximately $0.7 million of the unrecognized tax benefits relates to claims filed with the U.S. Internal Revenue Service (“IRS”) on the federal income tax treatment of refundable biodiesel/ethanol blending tax credits for prior years. We filed suit related to these claims in the Federal District Court of Dallas in March of 2022; the suit was stayed pending the outcome of controlling cases in the U.S. Court of Appeals for the Fifth Circuit, which were decided in favor of the IRS and were not appealed. As such precedent is controlling for us, we intend to file a motion to dismiss those claims controlled by such precedent in the Federal District Court of Dallas during 2024.

We recognize interest and penalties relating to liabilities for unrecognized tax benefits as an element of tax expense. We have not recorded any penalties related to our uncertain tax positions as we believe that it is more likely than not that there will not be any assessment of penalties.

We are subject to U.S. and Canadian federal income tax, Oklahoma, Oregon, Kansas, New Mexico, Iowa, Arizona, Utah, Colorado and Nebraska income tax and to income tax of multiple other state and local jurisdictions. We have substantially concluded all state and local income tax matters for tax years through 2019. Other than the federal claim noted above and to the extent of the federal net operating loss carried back to 2015 from 2020, we have materially concluded all U.S. federal income tax matters for tax years through December 31, 2019. We are currently under audit with the Canada Revenue Agency for the 2018, 2019, and 2020 tax years and the IRS for the 2020 and 2021 tax years.

NOTE 16: Stockholders' Equity

In September 2022, our Board of Directors approved a $1.0 billion share repurchase program (the“September 2022 Share Repurchase Program”), which replaced all existing share repurchase programs at that time, authorizing us to repurchase common stock in the open market or through privately negotiated transactions. Privately negotiated repurchases from REH Company were also authorized under the September 2022 Share Repurchase Program, subject to REH Company’s interest in selling its shares and other limitations. As of August 15, 2023, we had repurchased $995.0 million under the September 2022 Share Repurchase Program.

On August 15, 2023, our Board of Directors approved a new $1.0 billion share repurchase program (the“August 2023 Share Repurchase Program”), which replaced all existing share repurchase programs, including the $5.0 million remaining authorization under the September 2022 Share Repurchase Program. The August 2023 Share Repurchase Program authorizes us to repurchase common stock in the open market or through privately negotiated transactions. Privately negotiated repurchases from REH Company are also authorized under the August 2023 Share Repurchase Program, subject to REH Company’s interest in selling its shares and other limitations. The timing and amount of share repurchases, including those from REH Company, will depend on market conditions and corporate, tax, regulatory and other relevant considerations. In addition, we are authorized by our Board of Directors to repurchase shares in an amount sufficient to offset shares issued under our compensation programs. The August 2023 Share Repurchase Program may be discontinued at any time by our Board of Directors.

On January 3, 2024, we repurchased 454,380 shares of our outstanding common stock from REH Company in a privately negotiated transaction under the August 2023 Share Repurchase Program and pursuant to the Stock Purchase Agreement, dated January 3, 2024 (the "January Stock Purchase Agreement"), between us and REH Company. The price paid by us under the January Stock Purchase Agreement was $ 55.02 per share resulting in an aggregate purchase price of $25.0 million. The purchase price was funded with cash on hand.

On February 8, 2024, we repurchased 1,061,946 shares of our outstanding common stock from REH Company in a privately negotiated transaction under the August 2023 Share Repurchase Program and pursuant to the Stock Purchase Agreement, dated February 8, 2024 (the "February Stock Purchase Agreement"), between us and REH Company. The price paid by us under the February Stock Purchase Agreement was $ 56.50 per share resulting in an aggregate purchase price of $60.0 million. The purchase price was funded with cash on hand. As of February 15, 2024, we had remaining authorization to repurchase up to $591.4 million under the August 2023 Share Repurchase Program.

The following table presents total open market and privately negotiated purchases of shares under our share repurchase programs for the years ended December 31, 2023 and 2022.

	Years Ended December 31,	
	2023	2022
Number of shares repurchased [1]	18,779,880	25,716,042
Cash paid for shares repurchased (in thousands)	$ 974,474	$ 1,313,006

(1) During the years ended December 31, 2023 and 2022, 15,515,302 and 14,407,274 shares, respectively, were repurchased for $810.6 million and $750.0 million, respectively, pursuant to privately negotiated repurchases from REH Company.

On December 14, 2022, we agreed to repurchase an aggregate of 1,000,000 shares of our outstanding common stock from a registered broker for an aggregate purchase price of $48.6 million (the "December 2022 Repurchase"). The purchase price was funded with cash on hand. The shares repurchased are held as treasury stock.

The December 2022 Repurchase was made in connection with the sale by REH Company of approximately 5,000,000 shares of common stock, inclusive of the 1,000,000 shares we repurchased, in an unregistered block trade permitted under applicable securities laws (such sale, the "Sale"). In connection with the Sale, REH Company agreed to customary "lock-up" restrictions that expired 60 days following the date of the Sale, subject to waiver by the broker and certain exceptions, including, but not limited to, privately negotiated sales or transfers of common stock to us from REH Company.

The December 2022 Repurchase was made pursuant to separate authorization from our Board of Directors and not as part of the September 2022 Share Repurchase Program, and accordingly, did not reduce the remaining authorization thereunder.

During the years ended December 31, 2023, 2022 and 2021, we withheld 332,741, 278,025, and 217,151 shares, respectively, of our common stock from certain employees in the amounts of $18.1 million, $16.5 million and $7.1 million, respectively. These withholdings were made under the terms of restricted stock unit and performance share unit agreements upon vesting, at which time, we concurrently made cash payments to fund payroll and income taxes on behalf of officers and employees who elected to have shares withheld from vested amounts to pay such taxes.

NOTE 17: Other Comprehensive Income (Loss)

The components and allocated tax effects of other comprehensive income (loss) are as follows:

	Before-Tax	Tax Expense (Benefit)	After-Tax
		(In thousands)	
Year Ended December 31, 2023			
Net change in foreign currency translation adjustment	$ 13,161	$ 2,760	$ 10,401
Net change in pension and other post-retirement benefit obligations	101	273	(172)
Other comprehensive income attributable to HF Sinclair stockholders	$ 13,262	$ 3,033	$ 10,229
Year Ended December 31, 2022			
Net change in foreign currency translation adjustment	$ (32,383)	$ (6,817)	$ (25,566)
Net unrealized gain on hedging instruments	326	67	259
Net change in pension and other post-retirement benefit obligations	789	166	623
Other comprehensive loss attributable to HF Sinclair stockholders	$ (31,268)	$ (6,584)	$ (24,684)
Year Ended December 31, 2021			
Net change in foreign currency translation adjustment	$ (13,336)	$ (2,793)	$ (10,543)
Net unrealized gain on hedging instruments	31	8	23
Net change in pension and other post-retirement benefit obligations	(457)	(186)	(271)
Other comprehensive loss attributable to HF Sinclair stockholders	$ (13,762)	$ (2,971)	$ (10,791)

The following table presents the statement of income line item effects for reclassifications out of accumulated other comprehensive income ("AOCI"):

AOCI Component	Gain (Loss) Reclassified From AOCI			Income Statement Line Item
	Years Ended December 31,			
	2023	2022	2021	
	(In thousands)			
Hedging instruments:				
Commodity price swaps	$ (3,236)	$ (5,288)	$ (19,239)	Sales and other revenues
	—	—	1,660	Operating expenses
	(3,236)	(5,288)	(17,579)	
	(784)	(1,282)	(4,430)	Income tax expense (benefit)
	(2,452)	(4,006)	(13,149)	Net of tax
Other post-retirement benefit obligations:				
Pension obligations	(1,378)	208	407	Other, net
	(334)	50	103	Income tax expense (benefit)
	(1,044)	158	304	Net of tax
Post-retirement healthcare obligations	3,859	3,440	3,328	Other, net
	935	834	839	Income tax expense (benefit)
	2,924	2,606	2,489	Net of tax
Retirement restoration plan	(11)	(39)	(39)	Other, net
	(3)	(9)	(10)	Income tax expense (benefit)
	(8)	(30)	(29)	Net of tax
Total reclassifications for the period	$ (580)	$ (1,272)	$ (10,385)	

Accumulated other comprehensive loss in the equity section of our consolidated balance sheets includes:

	Years Ended December 31,	
	2023	2022
	(In thousands)	
Foreign currency translation adjustment	$ (23,026)	$ (33,427)
Unrealized gain (loss) on pension obligations	619	(2,661)
Unrealized gain on post-retirement benefit obligations	10,623	14,075
Accumulated other comprehensive loss	$ (11,784)	$ (22,013)

NOTE 18: Pension and Post-retirement Plans

Certain Petro-Canada Lubricants Inc. ("PCLI") employees are participants in union and non-union pension plans, which are closed to new entrants. Effective June 30, 2022, we ceased to accrue additional benefits under these plans, at which time the plan was fully frozen. During 2023, we had partial settlements in the union and non-union pension plans. We expect the remaining benefits will be settled by the end of 2024. In addition, Sonneborn employees in the Netherlands have a defined benefit pension plan which was frozen and all plan participants became inactive in 2016. The plan assets are in the form of a third-party insurance contract that is valued based on the assets held by the insurer and insures a value which approximates the accrued benefits related to the plan's accumulated benefit obligation. At that time, a new plan was established to provide future indexation benefits to participants who had accrued benefits under the expiring arrangements.

The following table sets forth the changes in the benefit obligation and plan assets of our PCLI pension plans and Sonneborn Netherlands plans for the years ended December 31, 2023 and 2022.

	Years Ended December 31,	
	2023	2022
	(In thousands)	
Change in plans' benefit obligations		
Pension plans benefit obligation - beginning of period	$ 90,443	$ 120,414
Service cost	—	1,839
Interest cost	3,781	3,086
Actuarial gain	2,097	(25,605)
Benefits paid	(5,472)	(2,306)
Settlements	(24,090)	—
Transfer from other plans	3,849	164
Foreign currency exchange rate changes	2,032	(7,149)
Pension plans benefit obligation - end of year	$ 72,640	$ 90,443
Change in pension plans assets		
Fair value of plans assets - beginning of period	$ 87,466	$ 119,325
Return on plans assets	7,146	(26,218)
Employer contributions	1,324	3,486
Benefits paid	(4,697)	(2,306)
Transfer payments	3,849	164
Settlements	(24,090)	—
Foreign currency exchange rate changes	2,020	(6,985)
Fair value of plans assets - end of year	$ 73,018	$ 87,466
Funded status		
Over (Under)-funded balance	$ 378	$ (2,977)
Amounts recognized in consolidated balance sheets		
Intangibles and other	$ 1,149	$ —
Other long-term liabilities	(771)	(2,977)
	$ 378	$ (2,977)
Amounts recognized in accumulated other comprehensive loss		
Cumulative actuarial loss	$ (607)	$ (3,872)

The accumulated benefit obligation was $72.6 million and $90.4 million at December 31, 2023 and 2022, respectively, which are also the measurement dates used for our pension plans.

The following tables provide information regarding pension plans with a projected benefit obligation and accumulated benefit obligation in excess of the fair value of plan assets:

	December 31,	
	2023	2022
	(In thousands)	
Projected benefit obligation	$ 24,579	$ 90,443
Fair value of plan assets	$ 23,808	$ 87,466

	December 31,	
	2023	2022
	(In thousands)	
Accumulated benefit obligation	$ 24,579	$ 90,443
Fair value of plan assets	$ 23,808	$ 87,466

The weighted average assumption used to determine the end of period benefit obligation for the PCLI plans for the year ended December 31, 2023 were discount rates of 4.60% to 4.65%. The weighted average assumptions used to determine the end of period benefit obligation for the PCLI plans for the year ended December 31, 2022 were discount rates of 3.70% to 4.44%. For the years ended December 31, 2023 and 2022, the weighted average assumption used to determine end of period benefit obligations for Sonneborn were discount rates of 3.60% and 4.20%, respectively.

Net periodic pension expense consisted of the following components:

	Years Ended December 31,		
	2023	2022	2021
	(In thousands)		
Service cost - benefit earned during the period	$ —	$ 1,839	$ 4,455
Interest cost on projected benefit obligations	3,781	3,086	2,740
Expected return on plans assets	(3,140)	(3,223)	(3,031)
Amortization of gain	—	(208)	(407)
Settlements	1,378	—	—
Net periodic pension expense	$ 2,019	$ 1,494	$ 3,757

The components, other than service cost, of our net periodic pension expense are recorded in Other, net on our consolidated statements of income.

The following table presents the fair values of PCLI's pension plans' assets, by level within the fair value hierarchy, as of December 31, 2023 and 2022.

	December 31, 2023				December 31, 2022			
	Level 1	Level 2	Level 3	Total	Level 1	Level 2	Level 3	Total
	(In thousands)							
Fixed income	822	48,387	—	49,209	457	66,295	—	66,752
	$ 822	$ 48,387	$ —	$ 49,209	$ 457	$ 66,295	$ —	$ 66,752

See Note 6 for additional information on Level 1 and 2 inputs.

The expected long-term rate of return on plan assets is4.60% to 4.65% for the PCLI pension plans and is based on a target investment of100% in fixed income.

We expect to contribute $0.2 million to the Sonneborn pension plans in 2024. PCLI and Sonneborn pension plan benefit payments, which reflect expected future service, are expected to be paid as follows: $1.9 million in 2024, $0.8 million in 2025, $0.9 million in 2026, $0.9 million in 2027, $1.0 million in 2028 and $6.0 million in 2029 to 2033. Benefit payments expected to be paid in 2024 include the estimate of the net present value of all expected benefit payments to be paid out once the PCLI union and non-union pension plans windup has been finalized.

Post-retirement Healthcare Plans

We have post-retirement healthcare and other benefits plans that are available to certain of our employees who satisfy certain age and service requirements. These plans are unfunded and provide differing levels of healthcare benefits dependent upon hire date and work location. Not all of our employees are covered by these plans at December 31, 2023.

The following table sets forth the changes in the benefit obligation and plan assets of our post-retirement healthcare plans for the years ended December 31, 2023 and 2022:

	Years Ended December 31,	
	2023	2022
	(In thousands)	
Change in plans' benefit obligation		
Post-retirement plans' benefit obligation - beginning of year	$ 28,678	$ 34,816
Service cost	1,604	2,081
Interest cost	1,340	990
Benefits paid	(613)	(582)
Actuarial gain	(761)	(7,884)
Foreign currency exchange rate changes	249	(743)
Post-retirement plans' benefit obligation - end of year	$ 30,497	$ 28,678
Change in plan assets		
Fair value of plan assets - beginning of year	$ —	$ —
Employer contributions	603	572
Participant contributions	10	10
Benefits paid	(613)	(582)
Fair value of plan assets - end of year	$ —	$ —
Funded status		
Under-funded balance	$ (30,497)	$ (28,678)
Amounts recognized in consolidated balance sheets		
Accrued liabilities	$ (1,701)	$ (1,706)
Other long-term liabilities	(28,796)	(26,972)
	$ (30,497)	$ (28,678)
Amounts recognized in accumulated other comprehensive loss		
Cumulative actuarial gain	$ 8,020	$ 7,603
Prior service credit	8,069	11,550
Total	$ 16,089	$ 19,153

Benefit payments, which reflect expected future service, are expected to be paid as follows: $1.7 million in 2024; $2.5 million in 2025; $2.5 million in 2026; $2.5 million in 2027; $2.7 million in 2028; and $13.2 million in 2029 through 2033.

The weighted average assumptions used to determine end of period benefit obligations:

	December 31,	
	2023	2022
Discount rate	4.60% - 4.81%	4.95% - 5.10%
Current health care trend rate	6.00% - 6.75%	6.00% - 7.00%
Ultimate health care trend rate	4.00% - 4.00%	4.00% - 4.00%
Year rate reaches ultimate trend rate	2035 - 2041	2027 - 2041

Net periodic post-retirement credit consisted of the following components:

	Years Ended December 31,		
	2023	2022	2021
	(In thousands)		
Service cost – benefit earned during the year	$ 1,604	$ 2,081	$ 2,324
Interest cost on projected benefit obligations	1,340	990	782
Amortization of prior service credit	(3,481)	(3,472)	(3,481)
Amortization of (gain) loss	(378)	32	153
Net periodic post-retirement credit	$ (915)	$ (369)	$ (222)

The components, other than service cost, of our net periodic post-retirement credit are recorded in Other, net on our consolidated statements of income. Prior service credits are amortized over the average remaining effective period to obtain full benefit eligibility for participants.

Defined Contribution Plans
We have defined contribution plans that cover substantially all qualified employees in the U.S., Canada and the Netherlands. Our contributions are based on an employee's eligible compensation and years of service. We also partially match our employees' contributions. We expensed $ 80.8 million, $73.7 million and $45.0 million for the years ended December 31, 2023, 2022 and 2021, respectively, in connection with these plans.

NOTE 19: Contingencies and Contractual Commitments

We are a party to various litigation and legal proceedings which we believe, based on advice of counsel, will not either individually or in the aggregate have a materially adverse effect on our financial condition, results of operations or cash flows.

During the year ended December 31, 2023, we recognized a gain of $15.0 million, which is reflected in our Refining segment, from the settlement of a preservation of property claim related to winter storm Uri that occurred in the first quarter of 2021. Additionally, during the year ended December 31, 2022, we recognized a gain of $15.2 million, which is reflected in our Corporate and Other segment, from the settlement of our business interruption claim related to winter storm Uri that occurred in the first quarter of 2021.

Pursuant to the Business Combination Agreement, all pre-closing RINs obligations of REH Company's subsidiaries (which are now subsidiaries of HF Sinclair as a result of the HFC Transactions) remained with REH Company. REH Company was required to transfer to HF Sinclair the number of each applicable type of RIN required for REH Company to demonstrate compliance for any pre-closing obligations it retained by the deadlines set forth in the Business Combination Agreement. If REH Company did not deliver all the required RINs by the applicable deadline, then, within five days following the delivery of an invoice therefor, REH Company was required to pay to HF Sinclair the amount of all out-of-pocket costs and expenses incurred by HF Sinclair to comply with REH Company's pre-closing obligations prior to such deadline, including the price of any RINs purchased by HF Sinclair. In relation to this, 2,570,000 shares of HF Sinclair common stock, out of the purchase consideration paid to REH Company, were held in escrow to secure REH Company's RINs credit obligations under Section 6.22 of the Business Combination Agreement and were released in January 2024 upon their satisfaction of the RINs credit obligation relating thereto. The 5,290,000 HEP common units that were also held in escrow to secure REH Company's RINs credit obligations were released to REH Company in April 2023 upon their satisfaction of the RINs credit obligations relating thereto.

During 2017 and 2019, the EPA granted the Cheyenne Refinery and the refinery in Woods Cross, Utah (the "Woods Cross Refinery") each a one-year small refinery exemption from the Renewable Fuel Standard ("RFS") program requirements for the 2016 and 2018, respectively, compliance years. As a result, the Cheyenne Refinery's and Woods Cross Refinery's gasoline and diesel production are not subject to the Renewable Volume Obligation for the respective years. Upon each exemption granted, we increased our inventory of RINs and reduced our cost of products sold. On April 7, 2022, the EPA issued a decision reversing the grant of small refinery exemptions for our Woods Cross and Cheyenne refineries for the 2018 compliance year. On June 3, 2022, the EPA issued a decision reversing the grant of small refinery exemptions for our Woods Cross and Cheyenne refineries for the 2016 compliance year and denying small refinery exemption petitions for our Woods Cross and Cheyenne refineries for the 2019 and 2020 compliance years. Various subsidiaries of HollyFrontier are currently pursuing legal challenges to the EPA's decisions to reverse its grant of small refinery exemptions for the 2016 and 2018 compliance years. The first lawsuit, filed against the EPA on May 6, 2022 and currently pending before the U.S. Court of Appeals for the DC Circuit, seeks to have the EPA's reversal of our 2018 small refinery exemption petitions overturned. The second lawsuit, filed against the EPA on August 5, 2022 and currently pending before the U.S. Court of Appeals for the DC Circuit, seeks to have the EPA's reversal of our 2016 small refinery exemption petitions overturned and to have the EPA's denial of our 2019 and 2020 small refinery exemption petitions reversed. In addition, for both the 2016 and 2018 compliance years, pursuant to the June 2022 and April 2022 decisions, respectively, the EPA established an alternative compliance demonstration for small refineries pursuant to which the EPA is not imposing any obligations for the small refineries whose exemptions were reversed. On June 24, 2022, Growth Energy filed two lawsuits in the U.S. Court of Appeals for the DC Circuit against the EPA challenging the alternative compliance demonstration for the 2016 and 2018 compliance years. On July 25, 2022, various subsidiaries of HollyFrontier intervened on behalf of the EPA to aid the defense of the EPA's alternative compliance demonstration decision. It is too early to predict the outcome of these matters. We are unable to estimate the costs we may incur, if any, at this time.

We have been party to multiple proceedings before the Federal Energy Regulatory Commission ("FERC") challenging the rates charged by SFPP, L.P. ("SFPP") on its East Line pipeline facilities from El Paso, Texas to Phoenix, Arizona. In March 2018, FERC ruled that SFPP, as a master limited partnership, was prohibited from including an allowance for investor income taxes in the cost of service underlying its East Line rates. We reached a negotiated settlement with SFPP that provides for a payment to us of $51.5 million. FERC approved the settlement on December 31, 2020 subject to a rehearing period that resulted in a settlement effective date of February 2, 2021. Under the terms of the settlement agreement, SFPP made the $51.5 million payment to us on February 10, 2021 we recorded as "Gain on tariff settlement" on our consolidated statements of income for the year ended December 31, 2021.

Contractual Commitments

We have various long-term agreements (entered in the normal course of business) to purchase crude oil, natural gas, feedstocks and other resources to ensure we have adequate supplies to operate our refineries. The substantial majority of our purchase obligations are based on market prices or rates. These contracts expire in 2024 through 2028.

We also have long-term agreements with third parties for the transportation and storage of crude oil, natural gas and feedstocks to our refineries and for terminal and storage services that expire in 2024 through 2038. At December 31, 2023, the minimum future transportation and storage fees under transportation agreements having terms in excess of one year are as follows:

	(In thousands)
2024	$ 237,534
2025	241,403
2026	203,316
2027	203,484
2028	207,742
Thereafter	1,079,956
Total	$ 2,173,435

Transportation and storage costs incurred under these agreements totaled $200.5 million, $180.2 million and $160.5 million for the years ended December 31, 2023, 2022 and 2021, respectively.

NOTE 20: Segment Information

Our operations are organized into five reportable segments: Refining, Renewables, Marketing, Lubricants & Specialties and Midstream. Our operations that are not included in one of these five reportable segments are included in Corporate and Other. Intersegment transactions are eliminated on our consolidated financial statements and are included in Eliminations. Corporate and Other and Eliminations are aggregated and presented under the Corporate, Other and Eliminations column.

As a result of the Sinclair Transactions that closed on March 14, 2022, the operations of the Acquired Sinclair Businesses are reported in the Refining, Renewables, Marketing and Midstream segments.

The Refining segment represents the operations of our El Dorado, Tulsa, Navajo and Woods Cross refineries and HF Sinclair Asphalt Company LLC ("Asphalt"). Also, effective with our acquisition that closed on November 1, 2021, the Refining segment includes our Puget Sound Refinery, and effective with our acquisition that closed on March 14, 2022, includes our Parco and Casper refineries. Refining activities involve the purchase and refining of crude oil and wholesale marketing of refined products, such as gasoline, diesel fuel and jet fuel. These petroleum products are primarily marketed in the Mid-Continent, Southwest and Rocky Mountains extending into the Pacific Northwest geographic regions of the United States. Asphalt operates various asphalt terminals in Arizona, New Mexico and Oklahoma.

The Renewables segment represents the operations of our Cheyenne renewable diesel unit ("RDU"), which was mechanically complete in the fourth quarter of 2021 and operational in the first quarter of 2022, the pre-treatment unit at our Artesia, New Mexico facility, which was completed and operational in the first quarter of 2022 and the Artesia RDU, which was completed and operational in the second quarter of 2022. Also, effective with the Sinclair Transactions that closed on March 14, 2022, the Renewables segment includes the Sinclair RDU.

Effective with the Sinclair Transactions that closed on March 14, 2022, the Marketing segment represents branded fuel sales to Sinclair branded sites in the United States and licensing fees for the use of the Sinclair brand at additional locations throughout the country. The Marketing segment also includes branded fuel sales to non-Sinclair branded sites from legacy HollyFrontier agreements and revenues from other marketing activities. Our branded sites are located in several states across the United States with the highest concentration of the sites located in our West and Mid-Continent regions.

The Lubricants & Specialties segment represents PCLI production operations, located in Mississauga, Ontario, that includes lubricant products such as base oils, white oils, specialty products and finished lubricants, and the operations of our Petro-Canada Lubricants business that includes the marketing of products to both retail and wholesale outlets through a global sales network with locations in Canada, the United States and Europe. Additionally, the Lubricants & Specialties segment includes specialty lubricant products produced at our Tulsa refineries that are marketed throughout North America and are distributed in Central and South America and the operations of Red Giant Oil Company LLC, one of the largest suppliers of locomotive engine oil in North America. Also, the Lubricants & Specialties segment includes Sonneborn, a producer of specialty hydrocarbon chemicals such as white oils, petrolatums and waxes with manufacturing facilities in the United States and Europe.

The Midstream segment includes all of the operations of HEP, which owns and operates logistics and refinery assets consisting of petroleum product and crude oil pipelines, terminals, tankage, loading rack facilities and refinery processing units in the Mid-Continent, Southwest and Rocky Mountains geographic regions of the United States. The Midstream segment also includes 50% ownership interests in each of the Osage Pipeline, the Cheyenne Pipeline and Cushing Connect, and a25.12% ownership interest in the Saddle Butte Pipeline and a 49.995% ownership interest in the Pioneer Pipeline. Revenues from the Midstream segment are earned through transactions with unaffiliated parties for pipeline transportation, rental and terminalling operations as well as revenues relating to pipeline transportation services provided for our refining operations.

The accounting policies for our segments are the same as those described in the summary of significant accounting policies (see Note 1).

The following is a summary of the financial information of our reportable segments reconciled to the amounts reported in the consolidated financial statements.

	Refining	Renewables	Marketing	Lubricants & Specialties	Midstream	Corporate, Other and Eliminations	Consolidated Total
				(In thousands)			
Year Ended December 31, 2023							
Sales and other revenues:							
Revenues from external customers	$ 24,156,278	$ 781,309	$ 4,146,292	$ 2,762,767	$ 117,749	$ —	$ 31,964,395
Intersegment revenues	4,516,326	407,681	—	12,566	490,566	(5,427,139)	—
	$ 28,672,604	$ 1,188,990	$ 4,146,292	$ 2,775,333	$ 608,315	$ (5,427,139)	$ 31,964,395
Cost of products sold (exclusive of lower of cost or market inventory valuation adjustment)	$ 23,969,557	$ 1,080,919	$ 4,050,759	$ 2,005,853	$ —	$ (5,322,639)	$ 25,784,449
Lower of cost or market inventory valuation adjustment	$ 220,558	$ 49,861	$ —	$ —	$ —	$ —	$ 270,419
Operating expenses	$ 1,946,958	$ 109,056	$ —	$ 258,578	$ 222,631	$ (99,075)	$ 2,438,148
Selling, general and administrative expenses	$ 199,547	$ 5,117	$ 34,413	$ 164,311	$ 26,453	$ 68,399	$ 498,240
Depreciation and amortization	$ 468,001	$ 77,100	$ 24,599	$ 86,341	$ 101,028	$ 13,504	$ 770,573
Income (loss) from operations	$ 1,867,983	$ (133,063)	$ 36,521	$ 260,250	$ 258,203	$ (87,328)	$ 2,202,566
Earnings of equity method investments	$ —	$ —	$ —	$ —	$ 17,531	$ (162)	$ 17,369
Capital expenditures	$ 223,225	$ 18,154	$ 27,630	$ 37,431	$ 31,962	$ 47,011	$ 385,413

	Refining	Renewables	Marketing	Lubricants & Specialties	Midstream	Corporate, Other and Eliminations	Consolidated Total
				(In thousands)			
Year Ended December 31, 2022							
Sales and other revenues:							
Revenues from external customers	$ 30,379,696	$ 654,893	$ 3,911,922	$ 3,149,128	$ 109,200	$ —	$ 38,204,839
Intersegment revenues	4,033,213	360,606	—	9,472	438,280	(4,841,571)	—
	$ 34,412,909	$ 1,015,499	$ 3,911,922	$ 3,158,600	$ 547,480	$ (4,841,571)	$ 38,204,839
Cost of products sold (exclusive of lower of cost or market inventory valuation adjustment)	$ 28,270,195	$ 974,167	$ 3,845,625	$ 2,333,156	$ —	$ (4,743,130)	$ 30,680,013
Lower of cost or market inventory valuation adjustment	$ —	$ 52,412	$ —	$ —	$ —	$ —	$ 52,412
Operating expenses	$ 1,815,931	$ 111,974	$ —	$ 277,522	$ 210,623	$ (81,157)	$ 2,334,893
Selling, general and administrative expenses	$ 146,660	$ 3,769	$ 2,954	$ 168,207	$ 17,003	$ 87,892	$ 426,485
Depreciation and amortization	$ 405,065	$ 52,621	$ 17,819	$ 83,447	$ 96,683	$ 1,152	$ 656,787
Income (loss) from operations	$ 3,775,058	$ (179,444)	$ 45,524	$ 296,268	$ 223,171	$ (106,328)	$ 4,054,249
Loss of equity method investments	$ —	$ —	$ —	$ —	$ (260)	$ —	$ (260)
Capital expenditures	$ 162,280	$ 225,274	$ 9,275	$ 34,887	$ 38,964	$ 53,327	$ 524,007

	Refining	Renewables	Lubricants & Specialties	Midstream	Corporate, Other and Eliminations	Consolidated Total
			(In thousands)			
Year Ended December 31, 2021						
Sales and other revenues:						
Revenues from external customers	$ 15,734,870	$ —	$ 2,550,624	$ 103,646	$ 2	$ 18,389,142
Intersegment revenues	623,688	—	9,988	390,849	(1,024,525)	—
	$ 16,358,558	$ —	$ 2,560,612	$ 494,495	$ (1,024,523)	$ 18,389,142
Cost of products sold (exclusive of lower of cost or market inventory valuation adjustment)	$ 14,673,062	$ —	$ 1,815,802	$ —	$ (921,812)	$ 15,567,052
Lower of cost or market inventory valuation adjustment	$ (318,353)	$ 8,739	$ —	$ —	$ (509)	$ (310,123)
Operating expenses	$ 1,090,424	$ 55,353	$ 252,456	$ 170,524	$ (51,279)	$ 1,517,478
Selling, general and administrative expenses	$ 127,563	$ —	$ 170,155	$ 12,637	$ 51,655	$ 362,010
Depreciation and amortization	$ 334,365	$ 1,672	$ 79,767	$ 86,998	$ 737	$ 503,539
Income (loss) from operations	$ 451,497	$ (65,764)	$ 242,432	$ 224,336	$ (103,315)	$ 749,186
Earnings of equity method investments	$ —	$ —	$ —	$ 12,432	$ —	$ 12,432
Capital expenditures	$ 160,431	$ 510,836	$ 30,878	$ 88,336	$ 22,928	$ 813,409

Item 9. Changes in and Disagreements with Accountants on Accounting and Financial Disclosure

We have had no change in, or disagreement with, our independent registered public accountants on matters involving accounting and financial disclosure.

Item 9A. Controls and Procedures

Evaluation of disclosure controls and procedures. Our principal executive officer and principal financial officer have evaluated, as required by Rule 13a-15(b) under the Securities Exchange Act of 1934 (the "Exchange Act"), our disclosure controls and procedures (as defined in Rules 13a-15(e) and 15d-15(e)) under the Exchange Act as of the end of the period covered by this Annual Report on Form 10-K. Our disclosure controls and procedures are designed to provide reasonable assurance that the information we are required to disclose in the reports that we file or submit under the Exchange Act is accumulated and communicated to our management, including our principal executive officer and principal financial officer, or persons performing similar functions, as appropriate, to allow timely decisions regarding required disclosure and is recorded, processed, summarized and reported within the time periods specified in the Securities and Exchange Commission's rules and forms. Based upon the evaluation, our principal executive officer and principal financial officer have concluded that our disclosure controls and procedures were effective at the reasonable assurance level as of December 31, 2023.

Changes in internal control over financial reporting. There have been no changes in our internal control over financial reporting (as defined in Rule 13a-15(f) under the Exchange Act) that occurred during our last fiscal quarter that have materially affected or are reasonably likely to materially affect our internal control over financial reporting.

See Item 8 for "Management's Report on its Assessment of the Company's Internal Control Over Financial Reporting" and "Report of the Independent Registered Public Accounting Firm."

Item 9B. Other Information

There have been no events that occurred in the fourth quarter of 2023 that would need to be reported on Form 8-K that have not previously been reported.

Item 9C. Disclosure Regarding Foreign Jurisdictions that Prevent Inspections

Not applicable

PART III

Item 10. Directors, Executive Officers and Corporate Governance

The information required by Items 401, 405, 406 and 407(c)(3), (d)(4) and (d)(5) of Regulation S-K in response to this item will be set forth in our definitive proxy statement for the 2024 annual meeting of stockholders and is incorporated herein by reference.

Item 11. Executive Compensation

The information required by Items 402 and 407(e)(4) and (e)(5) of Regulation S-K in response to this item will be set forth in our definitive proxy statement for the 2024 annual meeting of stockholders and is incorporated herein by reference.

Item 12. Security Ownership of Certain Beneficial Owners and Management and Related Stockholder Matters

The equity compensation plan information required by Item 201(d) and the information required by Item 403 of Regulation S-K in response to this item will be set forth in our definitive proxy statement for the 2024 annual meeting of stockholders and is incorporated herein by reference.

Item 13. Certain Relationships and Related Transactions, and Director Independence

The information required by Items 404 and 407(a) of Regulation S-K in response to this item will be set forth in our definitive proxy statement for the 2024 annual meeting of stockholders and is incorporated herein by reference.

Item 14. Principal Accounting Fees and Services

The information required by Item 9(e) of Schedule 14A in response to this item will be set forth in our definitive proxy statement for the 2024 annual meeting of stockholders and is incorporated herein by reference.

PART IV

Item 15. Exhibit and Financial Statement Schedules

(a) Documents filed as part of this report

(1) Index to Consolidated Financial Statements

	Page in Form 10-K
Report of Independent Registered Public Accounting Firm	82
Consolidated Balance Sheets at December 31, 2023 and 2022	84
Consolidated Statements of Income for the years ended December 31, 2023, 2022 and 2021	85
Consolidated Statements of Comprehensive Income for the years ended December 31, 2023, 2022 and 2021	86
Consolidated Statements of Cash Flows for the years ended December 31, 2023, 2022 and 2021	87
Consolidated Statements of Equity for the years ended December 31, 2023, 2022 and 2021	88
Notes to Consolidated Financial Statements	89

(2) Index to Consolidated Financial Statements Schedules

All schedules are omitted since the required information is not present or is not present in amounts sufficient to require submission of the schedule, or because the information required is included in the consolidated financial statements or notes thereto.

(3) Exhibits filed or furnished, as applicable, as part of this Annual Report on Form 10-K are listed in the Index to Exhibits.

HF SINCLAIR CORPORATION
INDEX TO EXHIBITS

Exhibits are numbered to correspond to the exhibit table
in Item 601 of Regulation S-K

Exhibit Number	Description
2.1†	Sale and Purchase Agreement, dated as of May 4, 2021, by and between Equilon Enterprises LLC d/b/a Shell Oil Products US and HollyFrontier Puget Sound Refining LLC (incorporated by reference to Exhibit 2.1 of HollyFrontier Corporation's Current Report on Form 8-K filed May 4, 2021, File No. 1-03876).
2.2†	Waiver and Amendment to Sale and Purchase Agreement, dated as of October 31, 2021, by and among HollyFrontier Puget Sound Refining LLC and Equilon Enterprises LLC d/b/a Shell Oil Products US (incorporated by reference to Exhibit 2.2 of HollyFrontier Corporation's Quarterly Report on Form 10-Q for the quarterly period ended September 30, 2021, File No. 1-03876).
2.3†	Business Combination Agreement, dated as of August 2, 2021, by and among HollyFrontier Corporation, Hippo Parent Corporation, Hippo Merger Sub, Inc., The Sinclair Companies, and Hippo Holding LLC (incorporated by reference to Exhibit 2.1 of HollyFrontier Corporation's Current Report on Form 8-K filed August 3, 2021, File No. 1-03876).
2.4†	Amendment to Business Combination Agreement, dated as of March 14, 2022, by and among HollyFrontier Corporation, Hippo Parent Corporation, Hippo Merger Sub, Inc., The Sinclair Companies and Hippo Holding LLC (incorporated by reference to Exhibit 2.1 of Registrant's Quarterly Report on Form 10-Q for the quarterly period ended March 31, 2022, File No. 1-41325).
2.5	Contribution Agreement, dated as of August 2, 2021, by and among Holly Energy Partners, L.P., The Sinclair Companies, and Sinclair Transportation Company (incorporated by reference to Exhibit 2.1 of Holly Energy Partners, L.P.'s Current Report on Form 8-K dated August 3, 2021, File No. 1-32225).
2.6	Amendment to Contribution Agreement, dated as of March 14, 2022, by and among Holly Energy Partners, L.P., The Sinclair Companies, and Sinclair Transportation Company (incorporated by reference to Exhibit 2.1 of Holly Energy Partners, L.P.'s Quarterly Report on Form 10-Q for the quarterly period ended March 31, 2022, File No. 1-32225).
2.7†	Agreement and Plan of Merger, dated August 15, 2023, by and among HF Sinclair Corporation, Navajo Pipeline Co., L.P., Holly Apple Holdings LLC, Holly Energy Partners, L.P., HEP Logistics Holdings, L.P., and Holly Logistic Services, L.L.C. (incorporated by reference to Exhibit 2.1 of Registrant's Current Report on Form 8-K filed August 16, 2023, File No. 1-41325).
3.1	Amended and Restated Certificate of Incorporation of HF Sinclair Corporation (incorporated by reference to Exhibit 3.1 of Registrant's Current Report on Form 8-K12B filed March 14, 2022, File No. 1-41325).
3.2	Amended and Restated By-Laws of HF Sinclair Corporation (incorporated by reference to Exhibit 3.1 of Registrant's Current Report on Form 8-K filed February 6, 2024, File No. 1-41325).
4.1	Indenture, dated March 22, 2016, between HollyFrontier Corporation and Wells Fargo Bank, National Association (incorporated by reference to Exhibit 4.1 of HollyFrontier Corporation's Current Report on Form 8-K filed March 22, 2016, File No. 1-03876).
4.2	Supplemental Indenture, dated March 22, 2016, between HollyFrontier Corporation and Wells Fargo Bank, National Association (incorporated by reference to Exhibit 4.2 of HollyFrontier Corporation's Current Report on Form 8-K filed March 22, 2016, File No. 1-03876).
4.3	Second Supplemental Indenture, dated as of September 28, 2020, between HollyFrontier Corporation and Wells Fargo Bank, National Association (incorporated by reference to Exhibit 4.2 of HollyFrontier Corporation's Current Report on Form 8-K dated September 28, 2020, File No. 1-03876).
4.4	Third Supplemental Indenture, dated April 8, 2022, among HollyFrontier Corporation and Computershare Trust Company, N.A., as agent for Wells Fargo, Bank National Association (incorporated by reference to Exhibit 4.1 of Registrant's Current Report on Form 8-K filed April 8, 2022, File No. 1-41325).
4.5	Indenture, dated February 4, 2020, by and among Holly Energy Partners, L.P., Holly Energy Finance Corp., each of the Guarantors party thereto and U.S. Bank National Association (incorporated by reference to Exhibit 4.1 of Holly Energy Partners, L.P.'s Current Report on Form 8-K filed February 4, 2020, File No. 1-32225).

HF SINCLAIR CORPORATION
INDEX TO EXHIBITS

Exhibit Number	Description
4.6	Indenture, dated April 8, 2022, by and among Holly Energy Partners, L.P., Holly Energy Finance Corp., each of the Guarantors party thereto and U.S. Bank National Association (incorporated by reference to Exhibit 4.1 of Holly Energy Partners, L.P.'s Current Report on Form 8-K filed April 8, 2022, File No. 1-32225).
4.7	First Supplemental Indenture, dated as of May 22, 2022, among UNEV Pipeline, LLC, Holly Energy Partners, L.P. and Holly Energy Finance Corp. and the other Guarantors party thereto and U.S. Bank Trust Company, National Association, as trustee, related to Holly Energy Partners, L.P.'s 6.375% Senior Notes due 2027 (incorporated by reference to Exhibit 4.9 of Holly Energy Partner, L.P.'s Registration Statement on Form S-3 filed May 23, 2022, File No. 333-265154).
4.8	Second Supplemental Indenture, dated as of November 10, 2023, among Holly Energy Partners, L.P., Holly Energy Finance Corp., the other Guarantors party thereto and U.S. Bank Trust Company, National Association, as trustee, related to Holly Energy Partners, L.P.'s and Holly Energy Finance Corp.'s 6.375% Senior Notes due 2027 (incorporated by reference to Exhibit 4.1 of Registrant's Current Report on Form 8-K filed November 14, 2023, File No. 1-41325).
4.9	First Supplemental Indenture, dated as of March 14, 2022, among Sinclair Transportation Company LLC, Sinclair Logistics LLC, Sinclair Pipeline Company LLC, Holly Energy Partners, L.P. and Holly Energy Finance Corp. and the other Guarantors party thereto and U.S. Bank Trust Company, National Association, as trustee, related to Holly Energy Partners, L.P.'s 5.000% Senior Notes due 2028 (incorporated by reference to Exhibit 4.2 of Holly Energy Partners, L.P.'s Quarterly Report on Form 10-Q for the quarterly period ended March 31, 2022, File No. 1-32225).
4.10	Second Supplemental Indenture, dated as of May 22, 2022, among UNEV Pipeline, LLC, Holly Energy Partners, L.P. and Holly Energy Finance Corp. and the other Guarantors party thereto and U.S. Bank Trust Company, National Association, as trustee, related to Holly Energy Partners, L.P.'s 5.000% Senior Notes due 2028 (incorporated by reference to Exhibit 4.7 of Holly Energy Partner, L.P.'s Registration Statement on Form S-3 filed May 23, 2022, File No. 333-265154).
4.11	Third Supplemental Indenture, dated as of November 10, 2023, among Holly Energy Partners, L.P., Holly Energy Finance Corp., the other Guarantors party thereto and U.S. Bank Trust Company, National Association, as trustee, related to Holly Energy Partners, L.P.'s and Holly Energy Finance Corp.'s 5.000% Senior Notes due 2028 (incorporated by reference to Exhibit 4.2 of Registrant's Current Report on Form 8-K filed November 14, 2023, File No. 1-41325).
4.12	Indenture, dated as of April 27, 2022, among HF Sinclair Corporation and Computershare Trust Company, N.A., as trustee (incorporated by reference to Exhibit 4.1 of Registrant's Current Report on Form 8-K filed April 27, 2022, File No. 1-41325).
4.13	First Supplemental Indenture, dated as of April 27, 2022, among HF Sinclair Corporation and Computershare Trust Company, N.A., as trustee (incorporated by reference to Exhibit 4.2 of Registrant's Current Report on Form 8-K filed April 27, 2022, File No. 1-41325).
4.14	Second Supplemental Indenture, dated as of December 4, 2023, among HF Sinclair Corporation and Computershare Trust Company, N.A., as trustee (incorporated by reference to Exhibit 4.1 of Registrant's Current Report on Form 8-K filed December 4, 2023, File No. 1-41325).
4.15	Registration Rights Agreement, dated as of December 4, 2023, among HF Sinclair Corporation, BofA Securities, Inc., Wells Fargo Securities, LLC and the additional parties listed thereto (incorporated by reference to Exhibit 4.4 of Registrant's Current Report on Form 8-K filed December 4, 2023, File No. 1-41325).
4.16*	Description of Common Stock.
10.1†	Senior Unsecured Multi-Year Revolving Credit Agreement, dated as of April 27, 2022, among HF Sinclair Corporation, as borrower, MUFG Bank, Ltd., as administrative agent, and each of the financial institutions party thereto as lenders (incorporated by reference to Exhibit 10.1 of Registrant's Current Report on Form 8-K filed April 27, 2022, File No. 1-41325).
10.2	Third Amended and Restated Credit Agreement, dated July 27, 2017, among Holly Energy Partners, L.P. as borrower, Wells Fargo Bank, National Association, as administrative agent, an issuing bank and a lender, and certain other lenders party thereto (incorporated by reference to Exhibit 10.1 of Holly Energy Partners, L.P.'s Current Report on Form 8-K dated July 31, 2017, File No. 1-32225).
10.3¥	Amendment No. 1 to Third Amended and Restated Credit Agreement, dated April 30, 2021, among Holly Energy Partners, L.P., as borrower, certain of its affiliates, as guarantors, Wells Fargo Bank, National Association, as administrative agent, an issuing bank and a lender, and certain other lenders party thereto (incorporated by reference to Exhibit 10.1 of Holly Energy Partners, L.P.'s Current Report on Form 8-K dated May 3, 2021, File No. 1-32225).

HF SINCLAIR CORPORATION
INDEX TO EXHIBITS

Exhibit Number	Description
10.4¥	Amendment No. 2 to Third Amended and Restated Credit Agreement, dated August 15, 2022, among Holly Energy Partners, L.P., as borrower, certain of its affiliates, as guarantors, Wells Fargo Bank, National Association, as administrative agent, an issuing bank and a lender, and certain other lenders party thereto (incorporated by reference to Exhibit 10.1 of Holly Energy Partners, L.P.'s Current Report on Form 10-Q for the quarterly period ended September 30, 2022, File No. 1-32225).
10.5¥	Amendment No. 3 to Third Amended and Restated Credit Agreement, dated as of December 1, 2023, by and among Holly Energy Partners, LP., as borrower, HF Sinclair Corporation, as parent guarantor, Wells Fargo Bank, National Association, as administrative agent and an issuing bank, and the financial institutions party thereto as lenders (incorporated by reference to Exhibit 10.1 of Registrant's Current Report on Form 8-K filed December 1, 2023, File No. 1-41325).
10.6	Parent Guaranty Agreement, dated December 1, 2023, by and among HF Sinclair Corporation, as guarantor, in favor of Wells Fargo Bank, National Association, in its capacity as administrative agent (incorporated by reference to Exhibit 10.2 of Registrant's Current Report on Form 8-K filed December 1, 2023, File No. 1-41325).
10.7†	Stockholders Agreement, dated as of August 2, 2021, by and among Hippo Parent Corporation, The Sinclair Companies, and the stockholders set forth on Schedule I thereto, as may be amended from time to time (incorporated by reference to Exhibit 10.1 of HollyFrontier Corporation's Current Report on Form 8-K filed August 3, 2021, File No. 1-03876).
10.8	Unitholders Agreement, dated as of August 2, 2021, by and among Holly Energy Partners, L.P., Holly Logistic Services, L.L.C., Navajo Pipeline Co., L.P., The Sinclair Companies, and the unitholders set forth on Schedule I thereto, as may be amended from time to time (incorporated by reference to Exhibit 10.1 of Holly Energy Partners, L.P.'s Current Report on Form 8-K filed August 3, 2021, File No. 1-32225).
10.9	Stock Purchase Agreement, dated as of February 8, 2024, by and among HF Sinclair Corporation and REH Company (incorporated by reference to Exhibit 10.1 of Registrant's Current Report on Form 8-K filed February 9, 2024, File No. 1-41325).
10.10	Stock Purchase Agreement, dated as of January 3, 2024, by and among HF Sinclair Corporation and REH Company (incorporated by reference to Exhibit 10.1 of Registrant's Current Report on Form 8-K filed January 4, 2024, File No. 1-41325).
10.11	Stock Purchase Agreement, dated as of December 18, 2023, by and among HF Sinclair Corporation and REH Company (incorporated by reference to Exhibit 10.1 of Registrant's Current Report on Form 8-K filed December 19, 2023, File No. 1-41325).
10.12	Stock Purchase Agreement, dated as of October 17, 2023, by and among HF Sinclair Corporation and REH Company (incorporated by reference to Exhibit 10.1 of Registrant's Current Report on Form 8-K filed October 18, 2023, File No. 1-41325).
10.13	Stock Purchase Agreement, dated as of September 8, 2023, by and among HF Sinclair Corporation and REH Company (incorporated by reference to Exhibit 10.1 of Registrant's Current Report on Form 8-K filed September 11, 2023, File No. 1-41325).
10.14	Stock Purchase Agreement, dated as of August 22, 2023, by and among HF Sinclair Corporation and REH Company (incorporated by reference to Exhibit 10.1 of Registrant's Current Report on Form 8-K filed August 23, 2023, File No. 1-41325).
10.15	Stock Purchase Agreement, dated as of July 28, 2023, by and among HF Sinclair Corporation and REH Company (incorporated by reference to Exhibit 10.1 of Registrant's Current Report on Form 8-K filed July 31, 2023, File No. 1-41325).
10.16	Stock Purchase Agreement, dated as of March 2, 2023, by and among HF Sinclair Corporation and REH Company (incorporated by reference to Exhibit 10.1 of Registrant's Current Report on Form 8-K filed March 3, 2023, File No. 1-41325).
10.17†	Escrow Agreement, dated as of March 14, 2022, by and among The Sinclair Companies, HF Sinclair Corporation (f/k/a Hippo Parent Corporation), and American Stock Transfer & Trust Company, LLC (incorporated by reference to Exhibit 10.2 of Registrant's Schedule 13D filed March 24, 2022, File No. 5-93610).

HF SINCLAIR CORPORATION
INDEX TO EXHIBITS

Exhibit Number	Description
10.18	Transition Services Agreement, dated as of March 14, 2022, between HF Sinclair Corporation (f/k/a Hippo Parent Corporation), and The Sinclair Companies (incorporated by reference to Exhibit 10.1 of Registrant's Current Report on Form 8-K/A filed March 16, 2022, File No. 1-41325).
10.19	Assignment and Assumption Agreement entered into and effective as of March 14, 2022, by and between HollyFrontier Corporation and HF Sinclair Corporation (incorporated by reference to Exhibit 10.4 of Registrant's Current Report on Form 8-K12B filed March 14, 2022, File No. 1-41325).
10.20+	Letter Agreement between HF Sinclair Corporation and Michael C. Jennings dated February 15, 2023 (incorporated by reference to Exhibit 10.1 of Registrant's Current Report on Form 8-K filed February 15, 2023, File No. 1-41325).
10.21+	Successor Transition Agreement and Release of Claims between HF Sinclair Corporation, HollyFrontier Corporation and HF Sinclair Payroll Services, Inc. and Michael C. Jennings dated February 15, 2023 (incorporated by reference to Exhibit 10.2 of Registrant's Current Report on Form 8-K filed February 15, 2023, File No. 1-41325).
10.22+	HF Sinclair Corporation Amended and Restated 2020 Long Term Incentive Plan (formerly named the HollyFrontier Corporation 2020 Long Term Incentive Plan) (incorporated by reference to Exhibit 99.1 of Registrant's Registration Statement on Form S-8 filed March 21, 2022, File No. 333-263721).
10.23+	Amendment No. 1 to the HF Sinclair Corporation Amended and Restated 2020 Long Term Incentive Plan (incorporated by reference to Exhibit 4.4 of Registrant's Registration Statement on Form S-8 filed December 4, 2023, File No. 333-275877).
10.24+	HF Sinclair Corporation 2020 Long-Term Incentive Plan Sub-Plan for UK Employees (formerly named the HollyFrontier Corporation 2020 Long-Term Incentive Plan Sub-Plan for UK Employees) (incorporated by reference to Exhibit 10.12 of Registrant's Quarterly Report on Form 10-Q for the quarterly period ended March 31, 2022, File No. 1-41325).
10.25+	HF Sinclair Corporation Employee Form of Change in Control Agreement (incorporated by reference to Exhibit 10.2 of Registrant's Quarterly Report on Form 10-Q for the quarterly period ended March 31, 2022, File No. 1-41325).
10.26+	Form of Performance Share Unit Agreement (incorporated by reference to Exhibit 10.47 of HollyFrontier Corporation's Annual Report on Form 10-K for its fiscal year ended December 31, 2021, File No. 1-03876).
10.27+	Form of Performance Share Unit Agreement (incorporated by reference to Exhibit 10.13 of Registrant's Quarterly Report on Form 10-Q for the quarterly period ended March 31, 2022, File No. 1-41325).
10.28+	Form of Performance Share Unit Agreement. (incorporated by reference to Exhibit 10.59 of Registrant's Annual Report on Form 10-K for its fiscal year ended December 31, 2022, File No. 1-41325).
10.29+	Form of Restricted Stock Units (for employees) (incorporated by reference to Exhibit 10.16 of Registrant's Quarterly Report on Form 10-Q for the quarterly period ended March 31, 2022, File No. 1-41325).
10.30+	Form of Notice of Grant of Restricted Stock Units (for employees) (incorporated by reference to Exhibit 10.17 of Registrant's Quarterly Report on Form 10-Q for the quarterly period ended March 31, 2022, File No. 1-41325).
10.31+	Form of Restricted Stock Unit Agreement (for employees) (incorporated by reference to Exhibit 10.67 of Registrant's Annual Report on Form 10-K for its fiscal year ended December 31, 2022, File No. 1-41325).
10.32+	Form of Notice of Grant of Restricted Stock Units (for employees) (incorporated by reference to Exhibit 10.68 of Registrant's Annual Report on Form 10-K for its fiscal year ended December 31, 2022, File No. 1-41325).
10.33+	Form of Restricted Stock Unit Agreement (for employees) (incorporated by reference to Exhibit 10.55 of HollyFrontier Corporation's Annual Report on Form 10-K for its fiscal year ended December 31, 2021, File No. 1-03876).

HF SINCLAIR CORPORATION
INDEX TO EXHIBITS

Exhibit Number	Description
10.34+	Form of Notice of Grant of Restricted Stock Unit (for employees) (incorporated by reference to Exhibit 10.56 of HollyFrontier Corporation's Annual Report on Form 10-K for its fiscal year ended December 31, 2021, File No. 1-03876).
10.35+	HF Sinclair Corporation Executive Nonqualified Deferred Compensation Plan (formerly the HollyFrontier Corporation Executive Nonqualified Deferred Compensation Plan) (incorporated by reference to Exhibit 10.71 of Registrant's Annual Report on Form 10-K for its fiscal year ended December 31, 2022, File No. 1-41325).
10.36+	HF Sinclair Corporation Director's Stock Compensation Deferral Plan (formerly the HollyFrontier Corporation Director's Stock Compensation Deferral Plan) (incorporated by reference to Exhibit 10.72 of Registrant's Annual Report on Form 10-K for its fiscal year ended December 31, 2022, File No. 1-41325).
10.37+	HF Sinclair Corporation Form of Indemnification Agreement to be entered into with officers and directors of HF Sinclair Corporation and its subsidiaries (incorporated by reference to Exhibit 10.1 of Registrant's Quarterly Report on Form 10-Q for the quarterly period ended March 31, 2022, File No. 1-41325).
10.38	HF Sinclair Corporation Severance Pay Plan and Summary Plan Description and Form of Participation Agreement, dated as of November 8, 2022 (incorporated by reference to Exhibit 10.1 of Registrant's Current Report on Form 8-K filed November 9, 2022, File No. 1-41325).
10.39+*	Form of Restricted Stock Unit Agreement (for non-employee directors).
10.40+*	Form of Notice of Grant of Restricted Stock Units (for non-employee directors).
10.41+*	Form of Performance Share Unit Agreement.
10.42+*	Form of Restricted Stock Unit Agreement (for employees).
10.43+*	Form of Grant of Restricted Stock Units (for employees).
21.1*	Subsidiaries of Registrant.
23.1*	Consent of Independent Registered Public Accounting Firm.
31.1*	Certification of Chief Executive Officer under Section 302 of the Sarbanes-Oxley Act of 2002.
31.2*	Certification of Chief Financial Officer under Section 302 of the Sarbanes-Oxley Act of 2002.
32.1**	Certification of Chief Executive Officer under Section 906 of the Sarbanes-Oxley Act of 2002.
32.2**	Certification of Chief Financial Officer under Section 906 of the Sarbanes-Oxley Act of 2002.
97.1*	HF Sinclair Corporation Mandatory Clawback Policy, adopted as of December 1, 2023.
101++	The following financial information from Registrant's Annual Report on Form 10-K for its fiscal year ended December 31, 2023, formatted as inline XBRL (Inline Extensible Business Reporting Language): (i) Consolidated Balance Sheets, (ii) Consolidated Statements of Income, (iii) Consolidated Statements of Comprehensive Income, (iv) Consolidated Statements of Cash Flows, (v) Consolidated Statements of Equity, and (vi) Notes to the Consolidated Financial Statements. The instance document does not appear in the interactive data file because its XBRL tags are embedded within the inline XBRL document.
104++	Cover page Interactive Data File (formatted as inline XBRL and contained in exhibit 101).

* Filed herewith.
** Furnished herewith.
+ Constitutes management contracts or compensatory plans or arrangements.
++ Filed electronically herewith.
† Schedules and certain exhibits have been omitted pursuant to Item 601(b)(2) of Regulation S-K. The registrant agrees to furnish supplementally a copy of the omitted schedules and exhibits to the SEC upon request.
¥ Schedules and certain exhibits have been omitted pursuant to Item 601(a)(5) of Regulation S-K. The registrant agrees to furnish supplementally a copy of the omitted schedules and exhibits to the SEC upon request.

HF SINCLAIR CORPORATION
SIGNATURES

Pursuant to the requirements of Section 13 or 15(d) of the Securities Exchange Act of 1934, the registrant has duly caused this report to be signed on its behalf by the undersigned, thereunto duly authorized.

HF SINCLAIR CORPORATION
(Registrant)

Date: February 21, 2024

/s/ Timothy Go
Timothy Go
Chief Executive Officer and President

Pursuant to the requirements of the Securities Exchange Act of 1934, this report has been signed below by the following persons on behalf of the registrant and in the capacities and as of the date indicated.

Signature	Capacity	Date
/s/ Timothy Go Timothy Go	Chief Executive Officer, President and Director	February 21, 2024
/s/ Atanas H. Atanasov Atanas H. Atanasov	Executive Vice President and Chief Financial Officer (Principal Financial Officer)	February 21, 2024
/s/ Indira Agarwal Indira Agarwal	Vice President, Controller and Chief Accounting Officer (Principal Accounting Officer)	February 21, 2024
/s/ Franklin Myers Franklin Myers	Chairman of the Board	February 21, 2024
/s/ Anne-Marie N. Ainsworth Anne-Marie N. Ainsworth	Director	February 21, 2024
/s/ Anna C. Catalano Anna C. Catalano	Director	February 21, 2024
/s/ Leldon Echols Leldon Echols	Director	February 21, 2024
/s/ Manuel J. Fernandez Manuel J. Fernandez	Director	February 21, 2024
/s/ Rhoman J. Hardy Rhoman J. Hardy	Director	February 21, 2024
/s/ Jeanne M. Johns Jeanne M. Johns	Director	February 21, 2024
/s/ R. Craig Knocke R. Craig Knocke	Director	February 21, 2024
/s/ Robert J. Kostelnik Robert J. Kostelnik	Director	February 21, 2024
/s/ James H. Lee James H. Lee	Director	February 21, 2024
/s/ Ross B. Matthews Ross B. Matthews	Director	February 21, 2024
/s/ Norman J. Szydlowski Norman J. Szydlowski	Director	February 21, 2024